BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2026-2027

SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE

(BCADSN25301)

Submitted By:-

Name – Aman kr. pandey

Section – BCADS31

Semester – 5th

Program – BCA (DS & AI)

Course Name – Predictive Analytics

Roll Number – 1240258063

Trainer:-

Mr. Karan Raj

(IBM MSME)

Practical - 1

Definition: - This practical explains how to use the IBM SPSS Modeler user interface to create, edit, modify, and save streams. It also demonstrates how to work with different nodes such as Excel, Sample, Derive, Histogram, Statistics Export, Select, and Table nodes

Outcomes/Learning: - Learners will become familiar with the IBM SPSS Modeler user interface and will understand how to create and modify streams, connect different nodes, generate outputs, add comments, and use table output to generate a Select node.

Required Tool: - The required tool will be "**IBM SPSS Modeler"**.

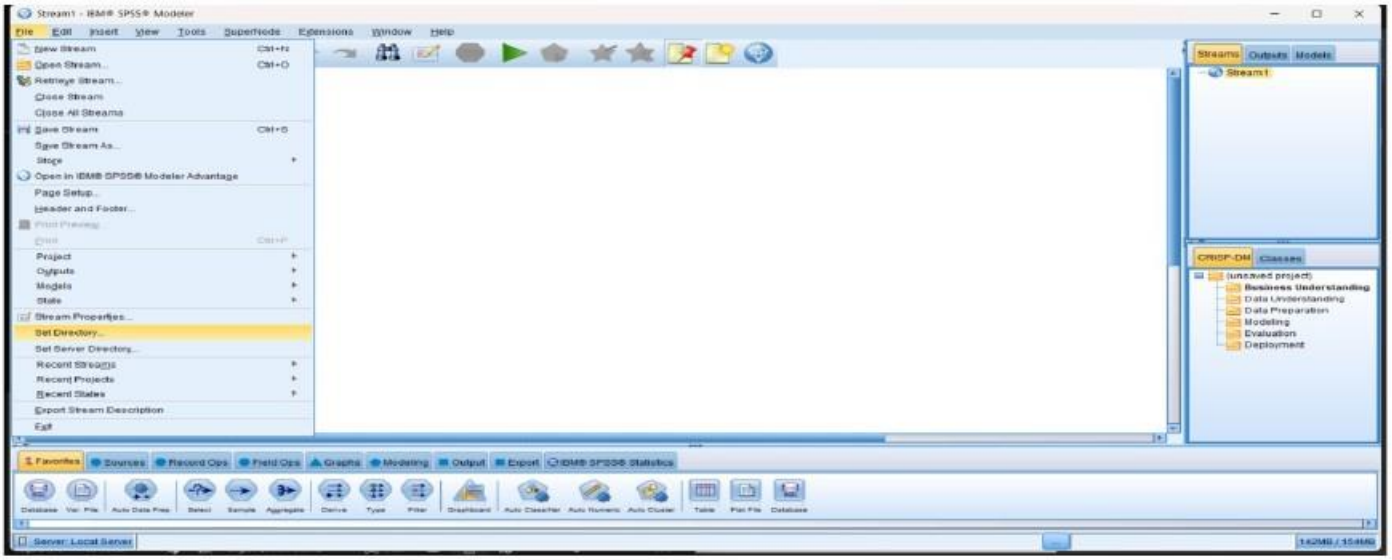
Purpose: - You want to become familiar with the IBM SPSS Modeler user interface that you will use to create and edit streams.

TASK – 1

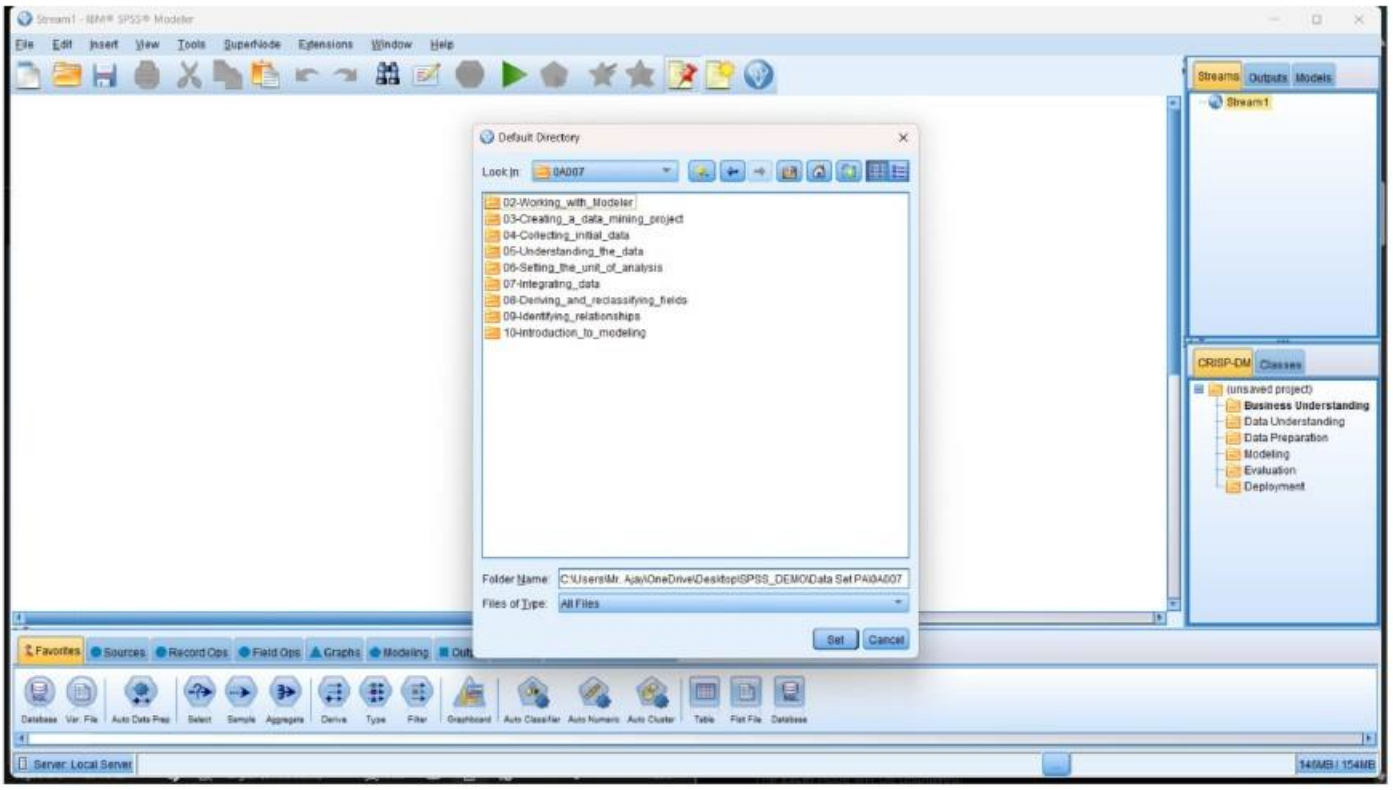
❖ Start IBM SPSS Modeler and set the working folder.

Definition: - This task explains how to open IBM SPSS Modeler and set the required working folder before creating a stream.

Step 1: - From the Start menu, click **All Programs**, click **IBM SPSS Modeler 18.0**, and then click **IBM SPSS Modeler 18.0**. The Welcome dialog box will appear. Click **Cancel** to close the Welcome dialog box.
From the File menu, click **Set Directory**.



Step 2: - Beside Look in, navigate to **C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.

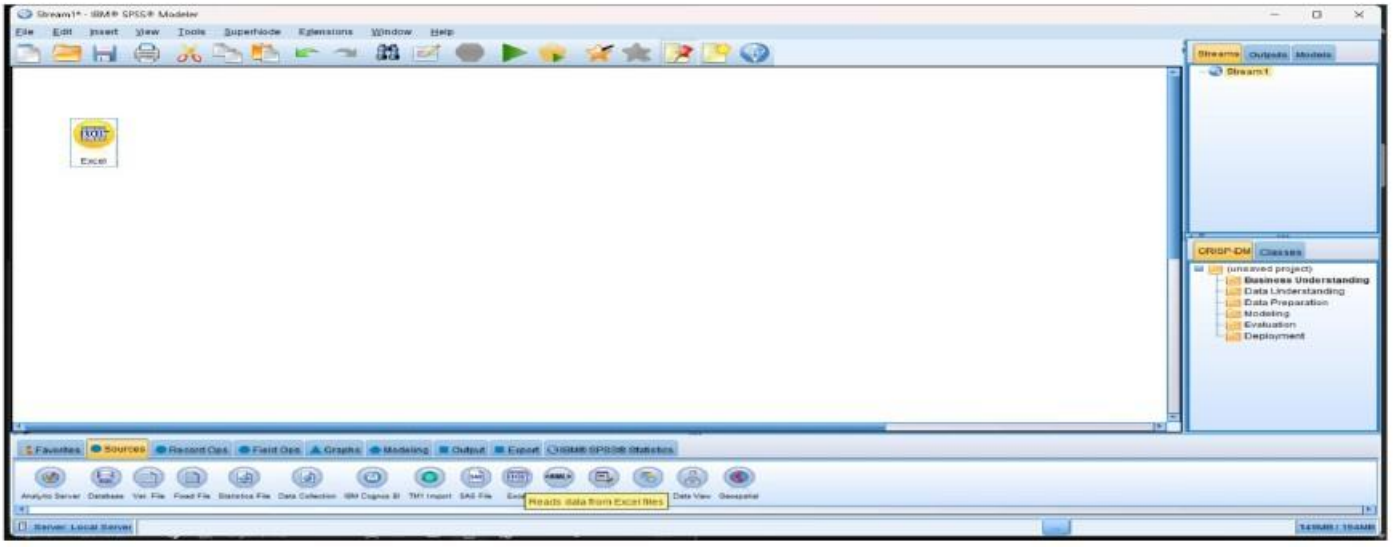


TASK – 2

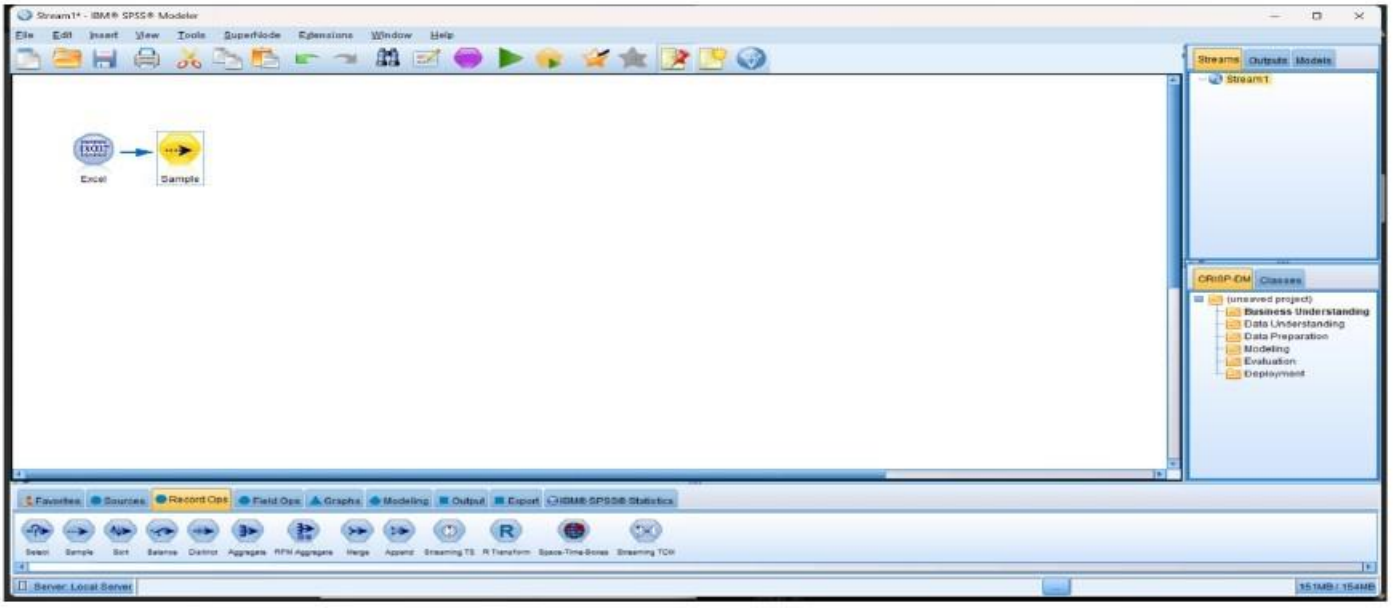
❖ Create a Stream

Definition: - This task demonstrates how to create a stream by adding different nodes to the stream canvas and connecting them. No actual data calculation is required because this task is performed as a dry run.

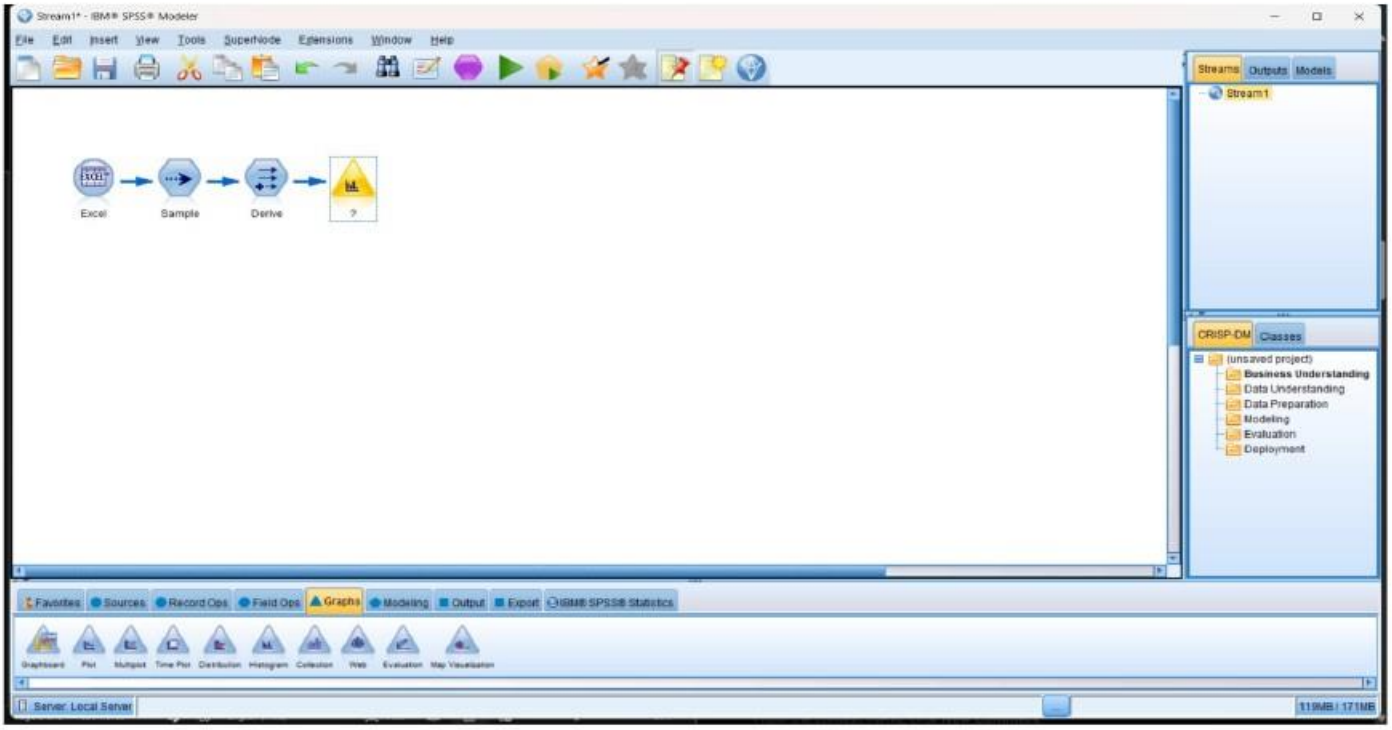
Step 1: Click the Sources palette; the "**Excel**" node will be displayed. Then double-click the Excel node. An Excel node will be placed on the stream canvas.



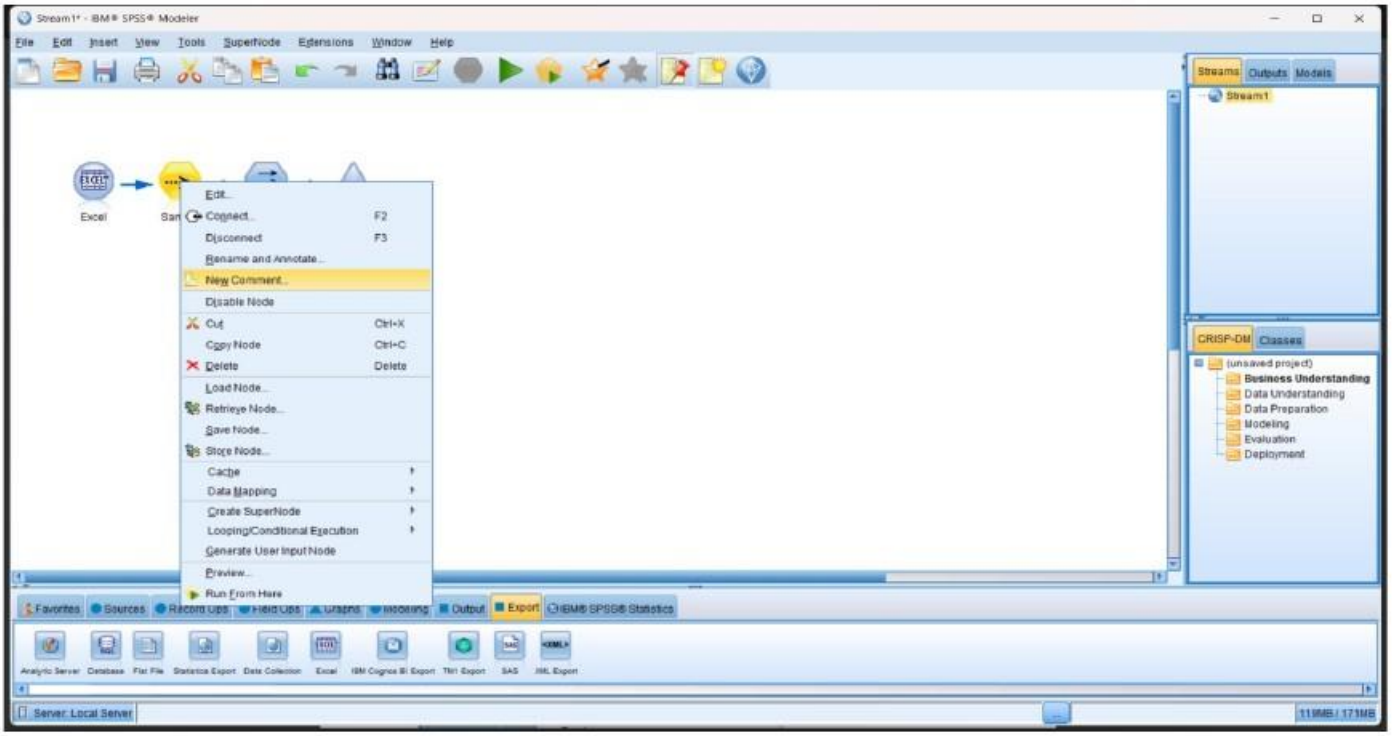
Step 2: Click the Record Ops palette and double-click the Sample node.
The Sample node will be automatically placed downstream from the Excel node. Right-click the connection between the Excel node and the Sample node and select Delete Connection. The connection between Excel and Sample will be restored.



Step 3: Click the **Field Ops** palette and double-click the **Derive** node. The Derive node will be placed downstream from the Sample node. Click the **Graphs** palette and double-click the **Histogram** node. The Histogram node will be placed downstream from the Derive node.



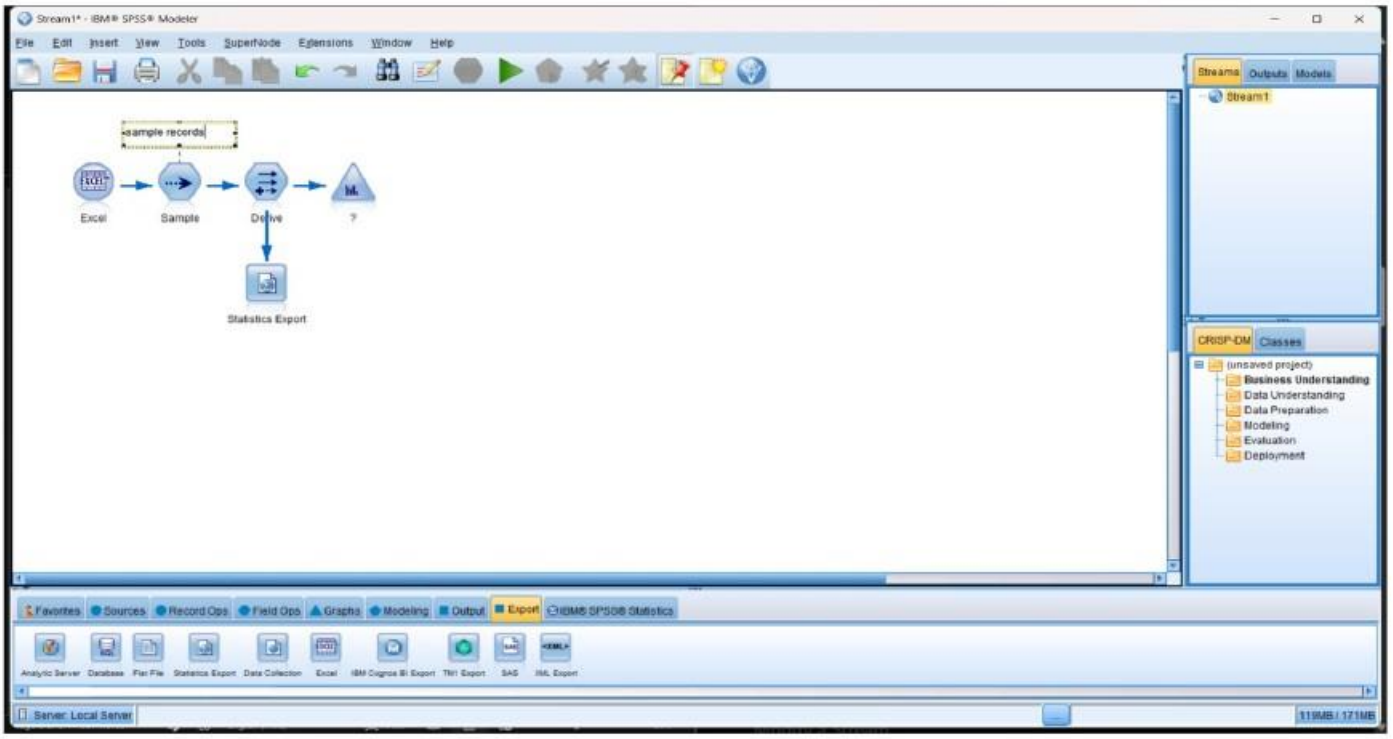
Step 4: Click the **Export** palette and double-click the **Statistics Export** node. The Statistics Export node will be placed downstream from the Derive node. On the stream canvas, right-click the **Sample** node.



Step 5: From the context menu, click **New Comment**.

Type: **sample records**

Click outside the comment box to remove focus. The stream is now ready for the next task.

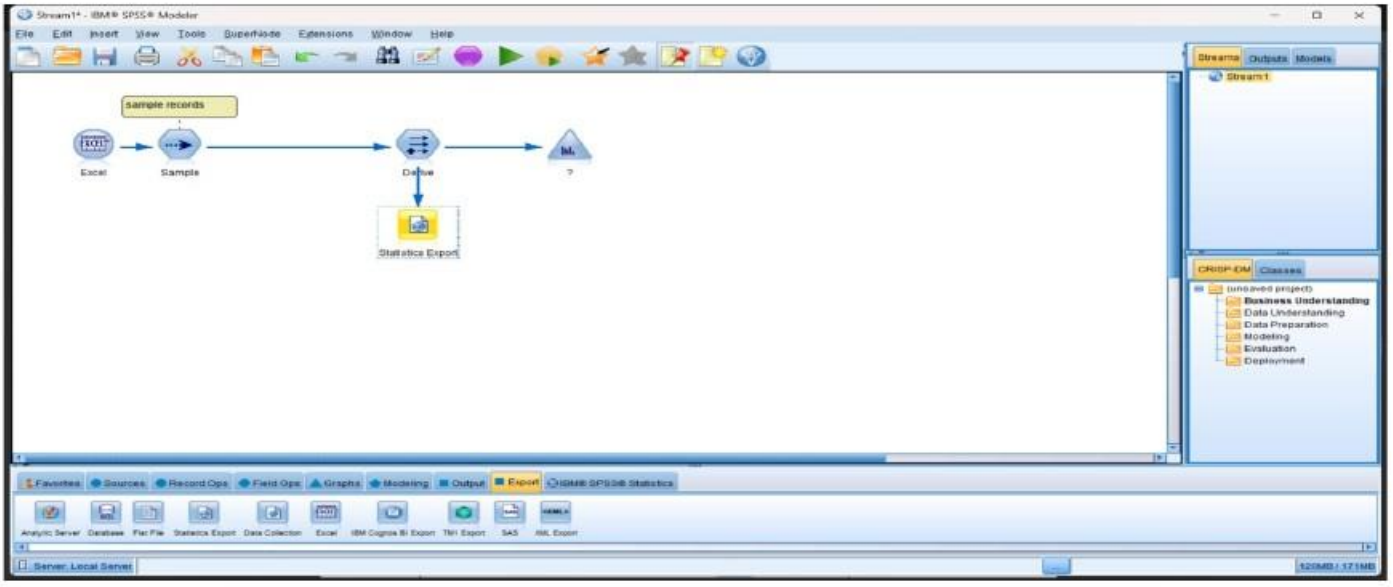


TASK – 3

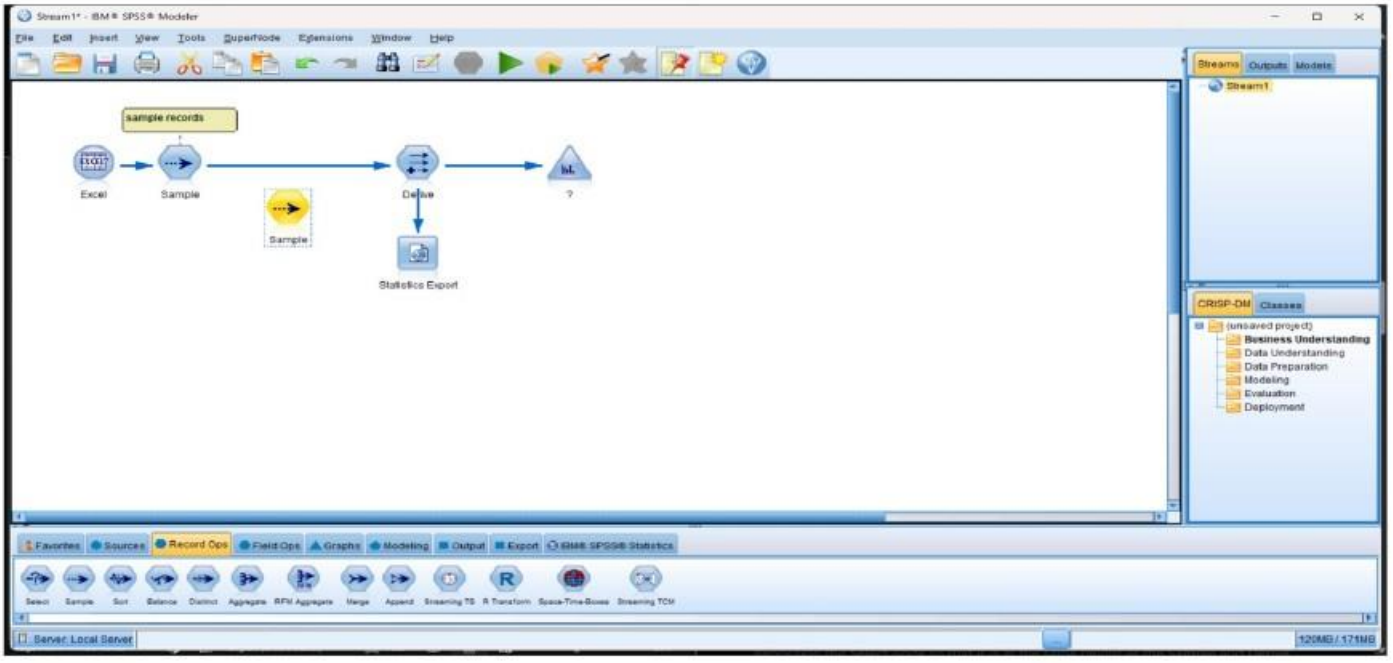
❖ Modify a Stream

Definition: - This task explains how to modify an existing stream by inserting a new **Select** node between the Sample and Derive nodes and disabling the Sample node when the stream is executed.

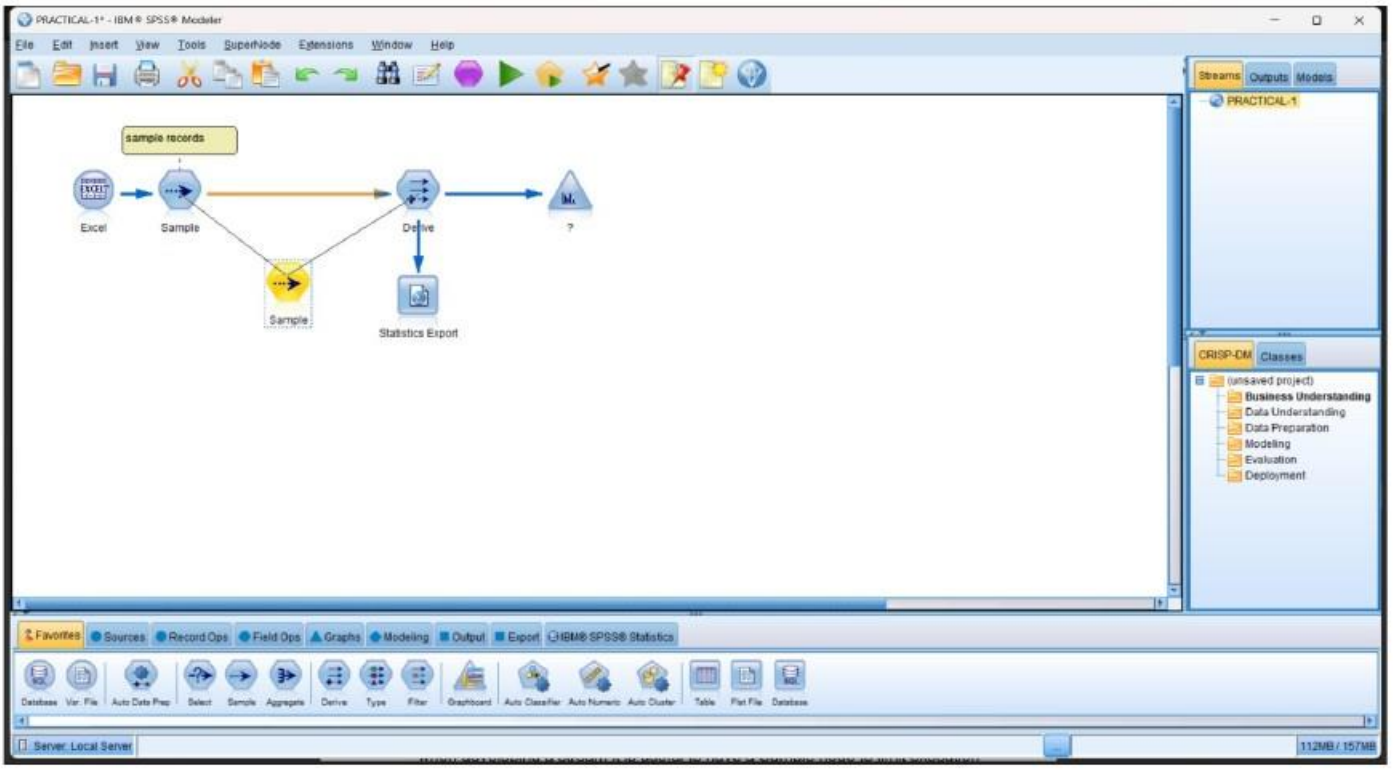
Step 1: Using the mouse, drag the **Select**, **Derive**, **Histogram**, and **Statistics Export** nodes so that the **Select** node can be inserted between the Sample and Derive nodes.



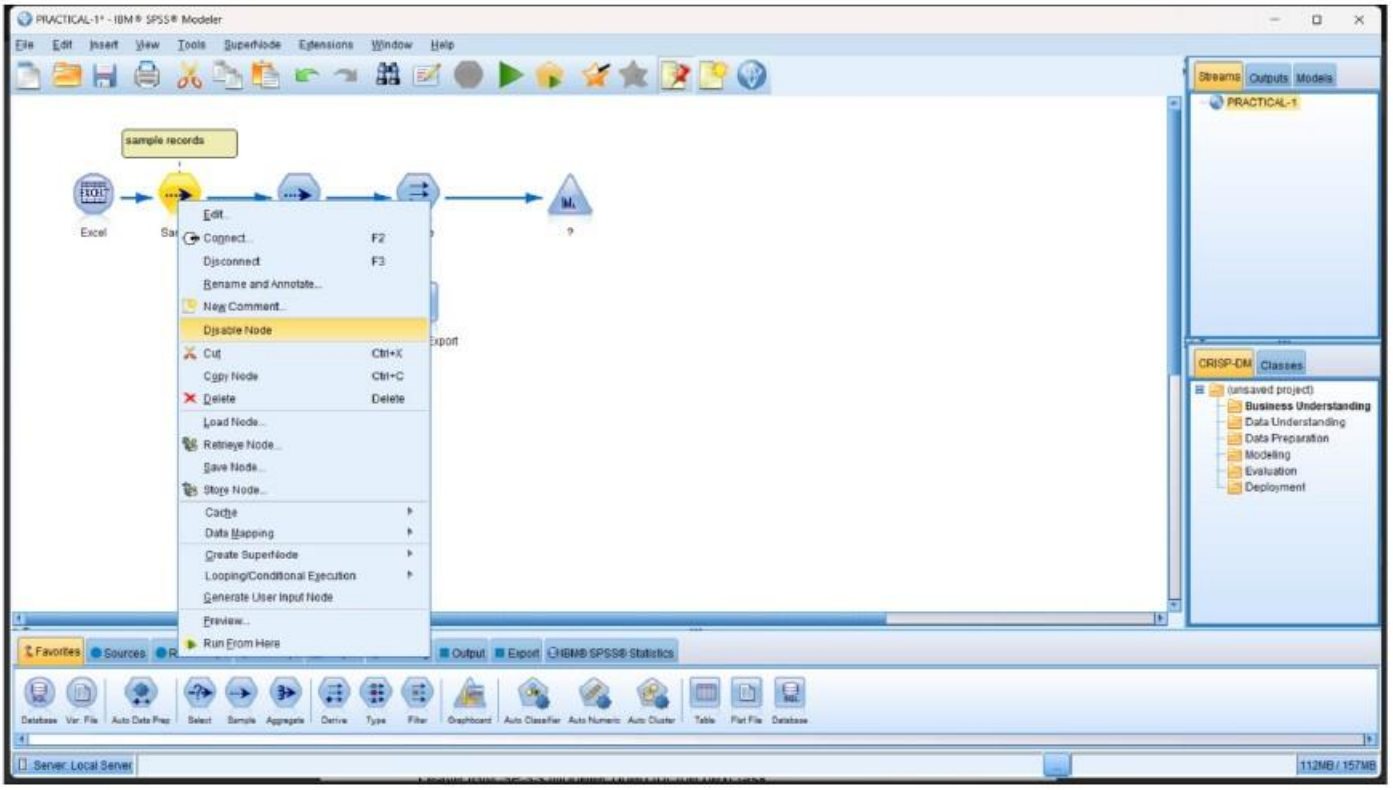
Step 2: From the **Record Ops** palette, click the **Select** node and drag it to the stream canvas below the connection from the Sample node to the Derive node.



Step 3: Click the connection from the Sample node to the Derive node, drag it over the Select node, as shown below. Reposition the **Select** node so that it is at the same height as the **Sample** and **Derive** nodes.



Step 4: Right-click the Sample node and click Disable Node from the context menu. The Sample node will become disabled.

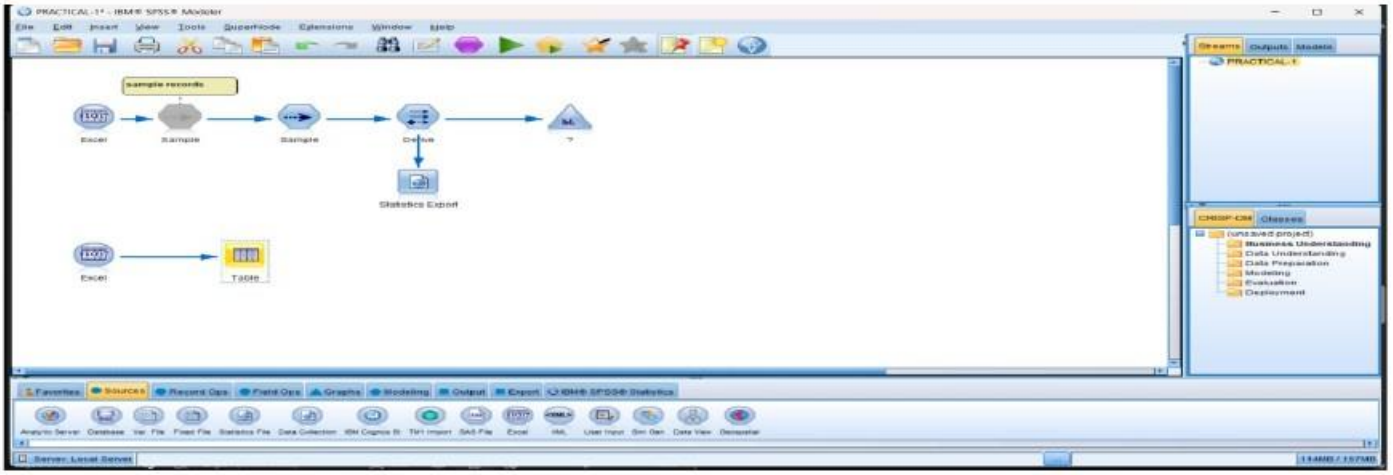


TASK – 4

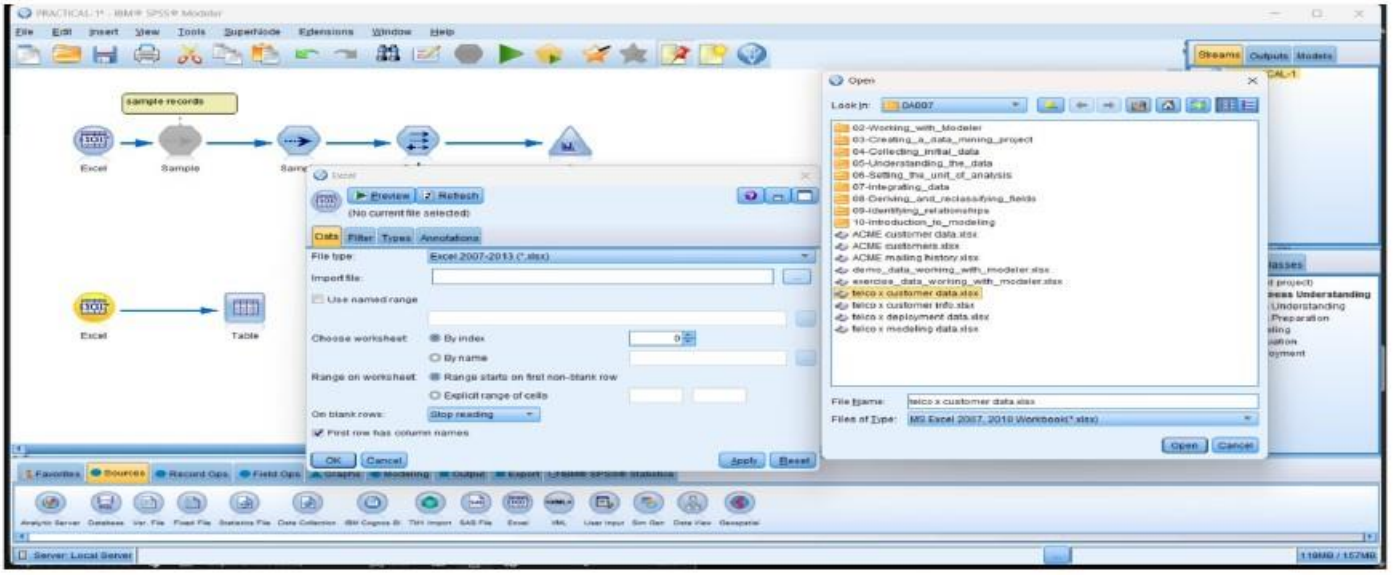
❖ Generate a Select Node from Table Output

Definition: - This task demonstrates how to generate a **Select node** from the output of a Table node. The generated Select node is used to select records having known data values.

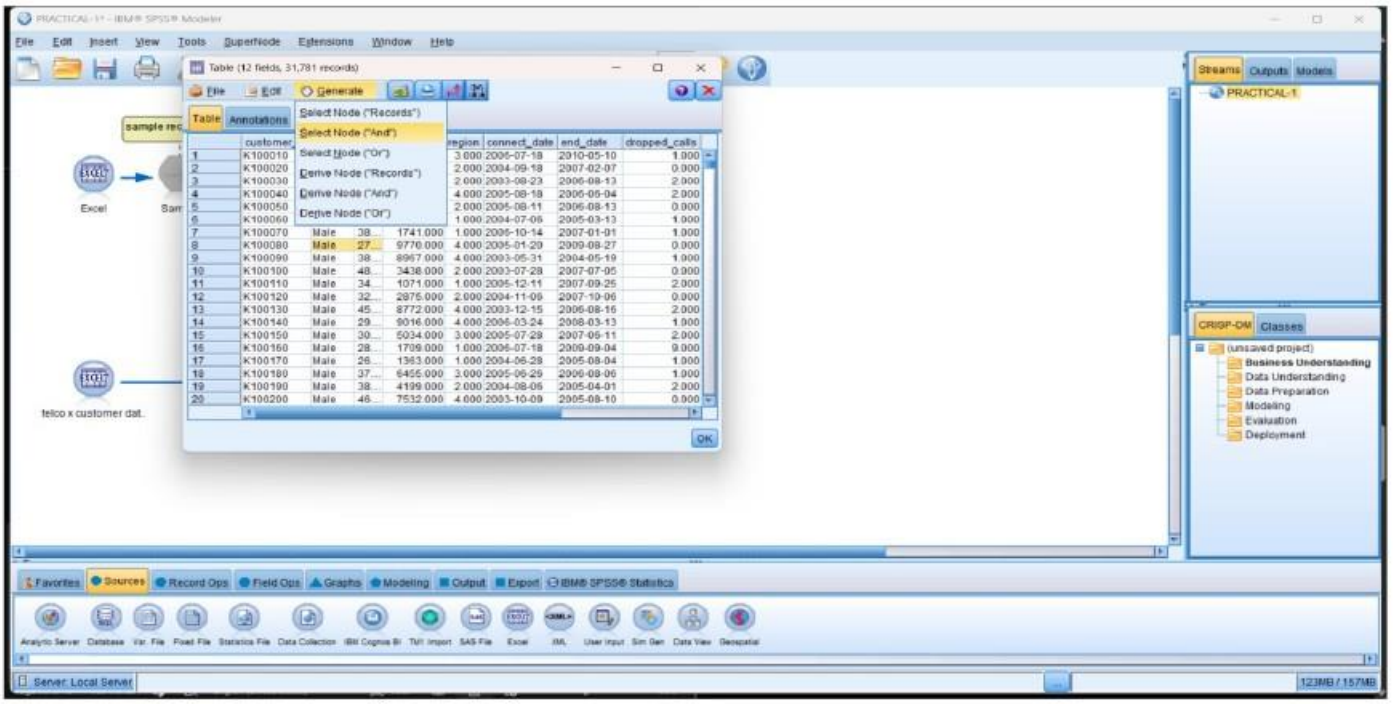
Step 1: Click on the **Sources palette** and double-click the **Excel node,** and again click the **Output palette** and then double-click the **Table node**. This places a **Table node** downstream(connect) from the **Excel node**.
The results appear as follows:



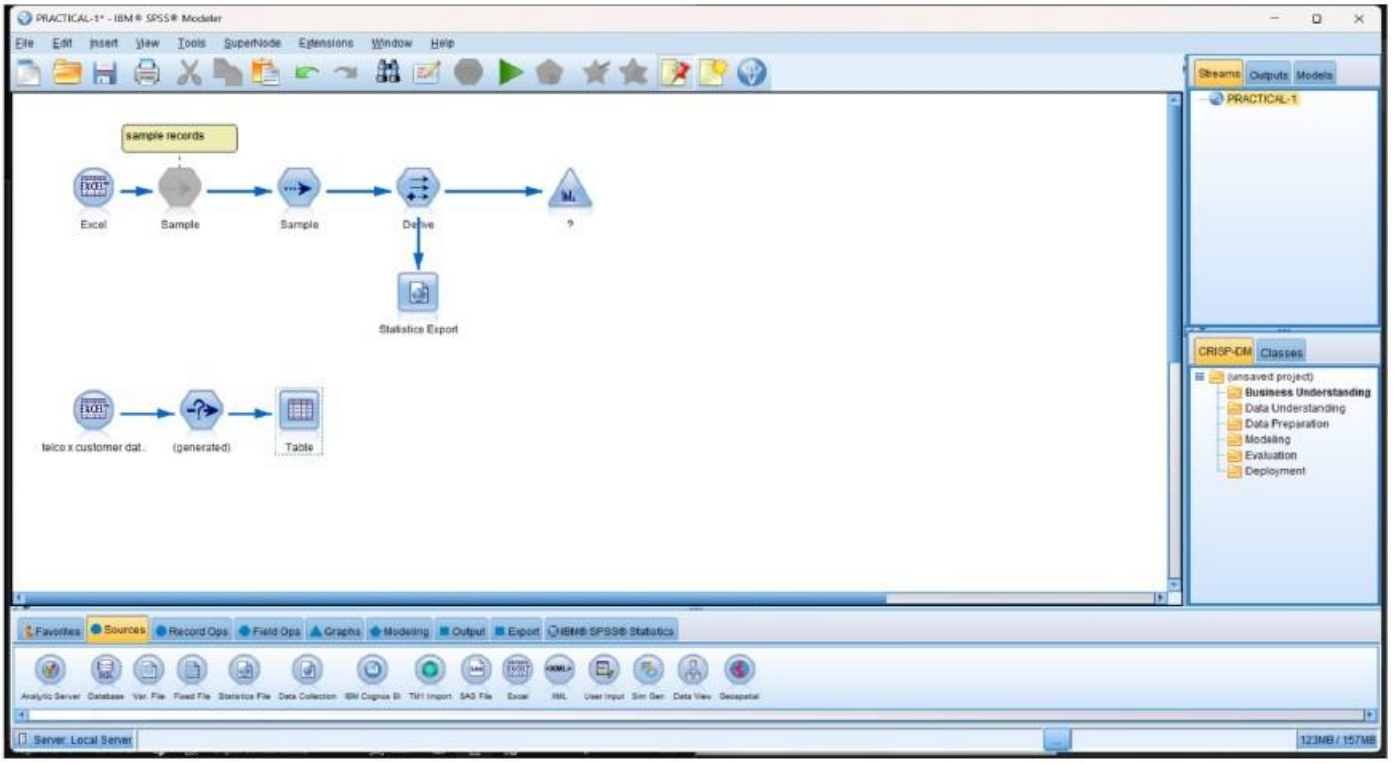
Step 2: Edit the Excel node. Set Import file to **telco x customer data.xlsx** from the dataset. (Like: - **C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007\telco x customer data.xlsx).** Click Open, then click OK.



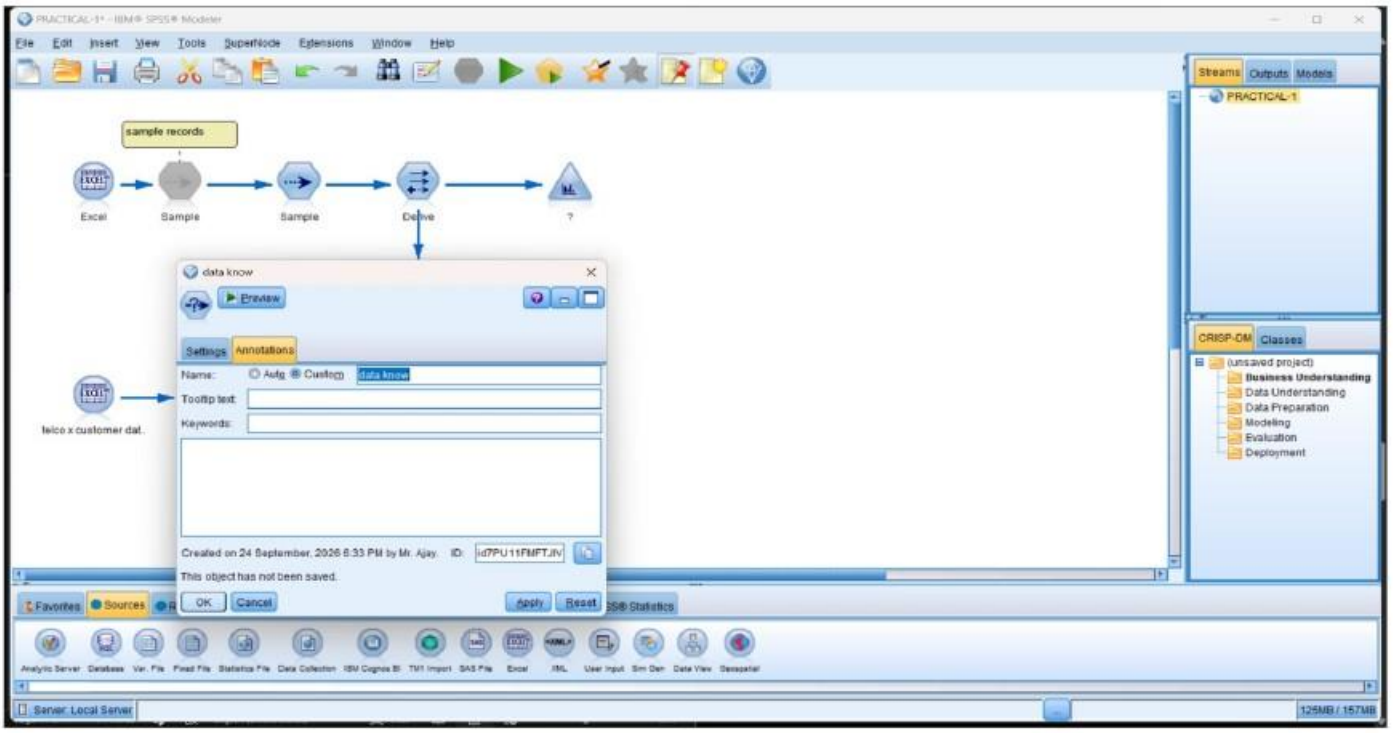
Step 3: Right-click the Table node, and then click Run. In the Eight row, click the Male value in the gender column to select it. From the menu, click **Generate > Select Node ("And")**.



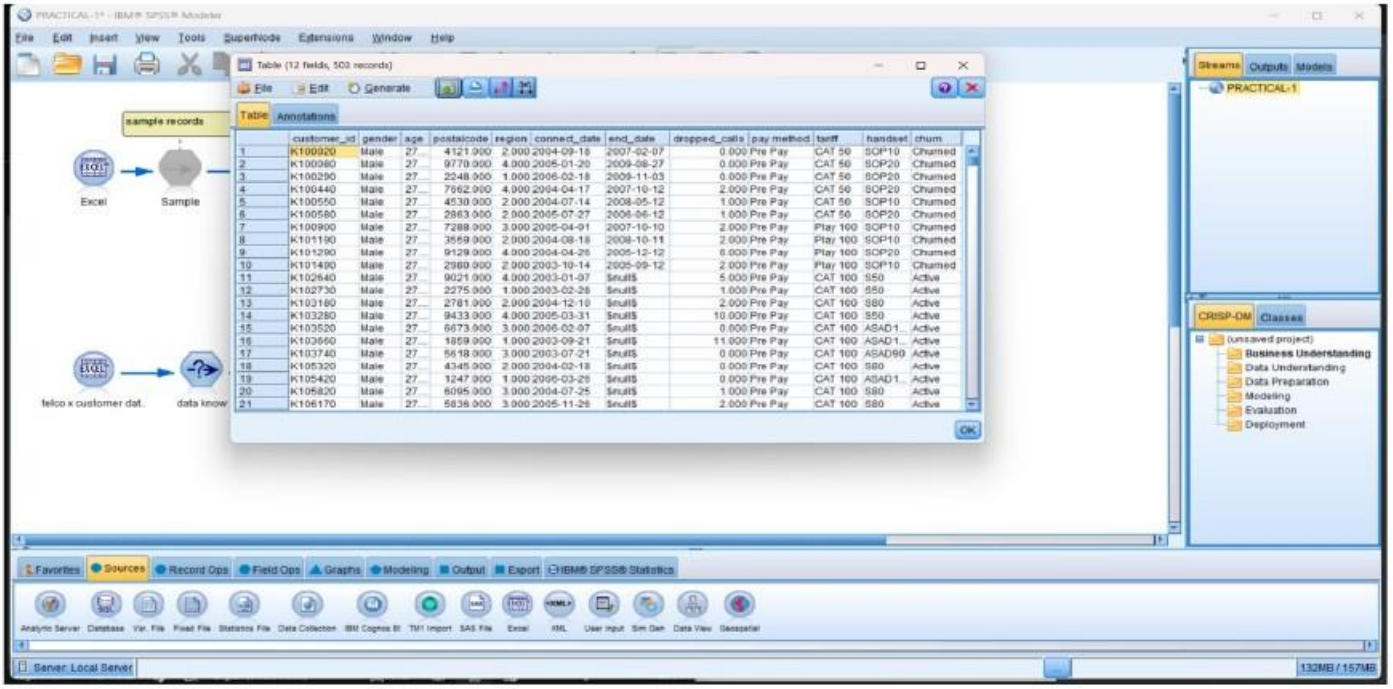
Step 4: Click OK, to close the Table output window. After that, drag the (generated) Select node below the Excel source node and the Table node, drag the existing connection over the Select node and then realign it with the other nodes. After that, the results appear as follows:



Step 5: Right-click the (generated) Select node. From the context menu, click Rename and Annotate. Beside Name, click Custom, if necessary. Replace the text (generated) with known data. Click OK, to close the Select dialogue box.

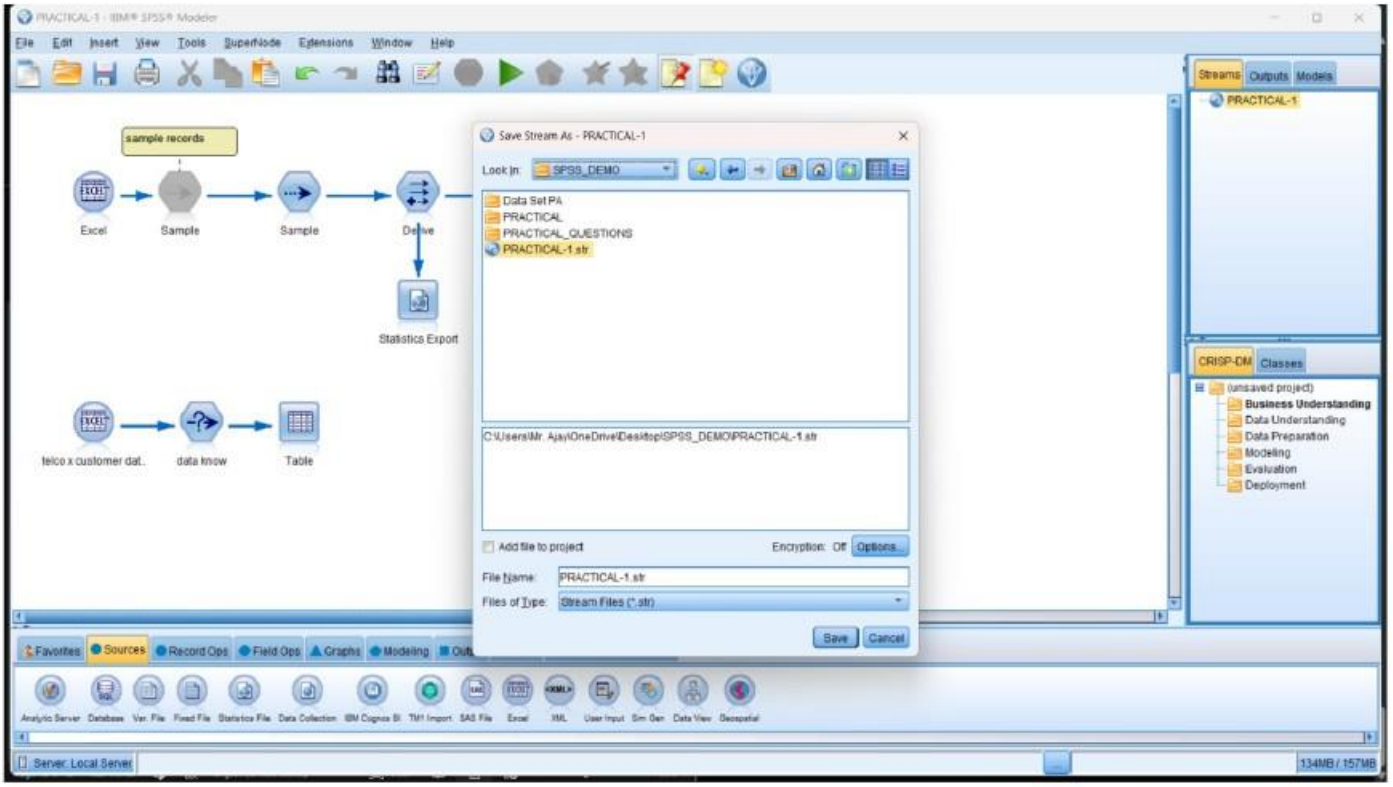


Step 6: For output of generated node. Right-click the Table node, and then click Run.



- The result appears as shown because a Select node was generated with the conditions **Gender = Male** and **Age = 27**. Therefore, only the records that satisfy both conditions are displayed in the output.

Step 7: To save the stream, go to the **File** menu and click **Save**, or press **Ctrl + S**. Choose the desired location/path.
Like: - (**C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, enter the file name you want, and then click **Save**.



Result: You have become familiar with IBM SPSS Modeler's user interface and created, edited, opened and saved streams.

Practical - 2

Definition: - This practical demonstrates how to import and examine data from Excel, Text, and IBM SPSS Statistics files using IBM SPSS Modeler.

Outcomes/Learning: - Learners will understand how to **import data, identify the unit of analysis, and set appropriate measurement levels** for different fields.

Purpose: - You work as a data miner for a telecommunications firm. You have to import data from various sources and examine the unit of analysis and the measurement levels of the fields.

TASK – 1

❖ Start IBM SPSS Modeler and set the working folder and import a Microsoft Excel File

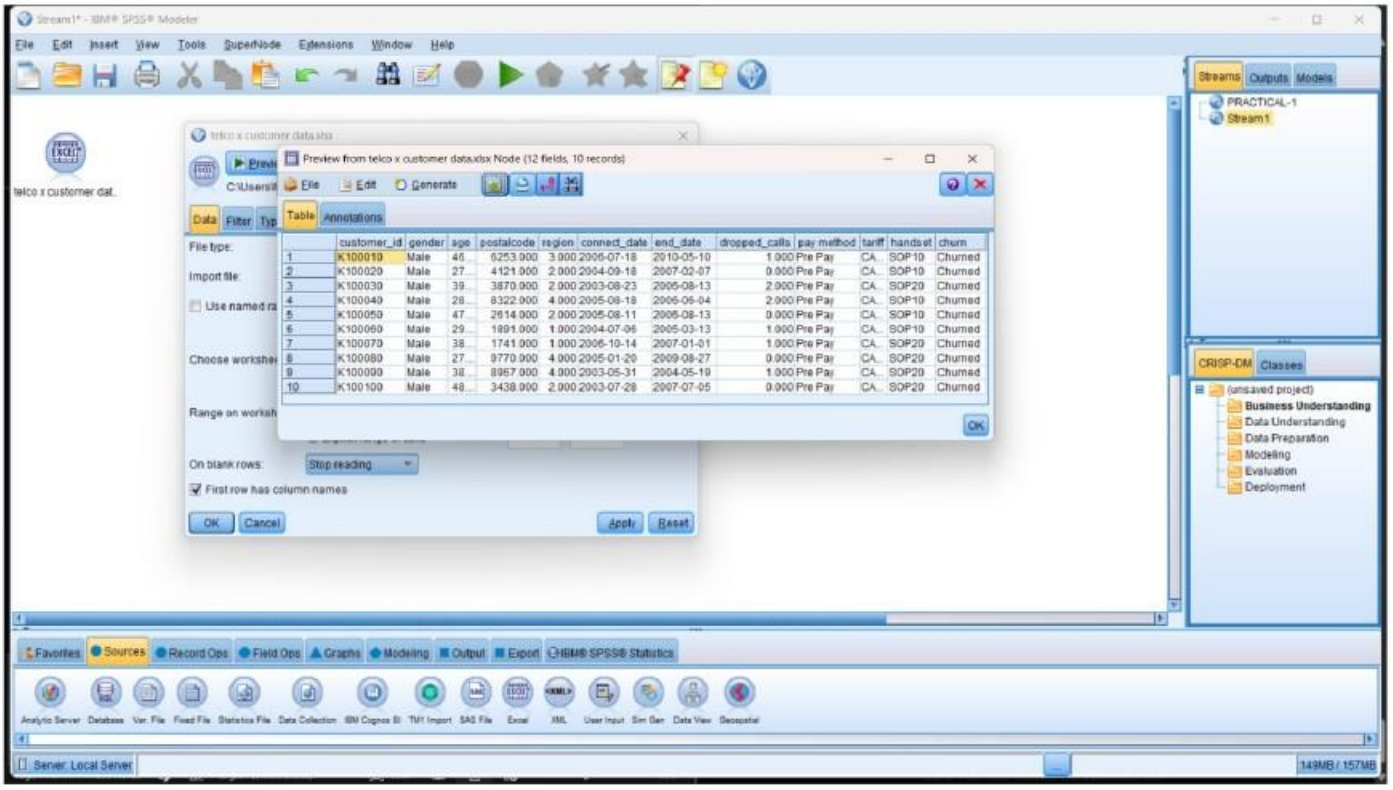
Definition: - This task explains how to **start IBM SPSS Modeler, set the working folder, and import an Excel file for data analysis**.

Step 1: First of all, from the File menu, click Set Directory.

- From the **Sources** palette, drag and drop an **Excel** node onto the stream canvas.
- Double-click the **Excel** node to open it.
- In the **Data** tab, make sure **Excel 2007 (*.xlsx)** is selected as the file type.
- Set the **Import file** to (telco x customer data.xlsx) from your file path.
- Like: - **C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007.**
- Then click OPEN.



Step 2: Check the **Preview** to confirm that the file has been imported correctly.



- After that, click **OK** to close the Preview window, and then click **OK** again to close the Excel dialogue box.

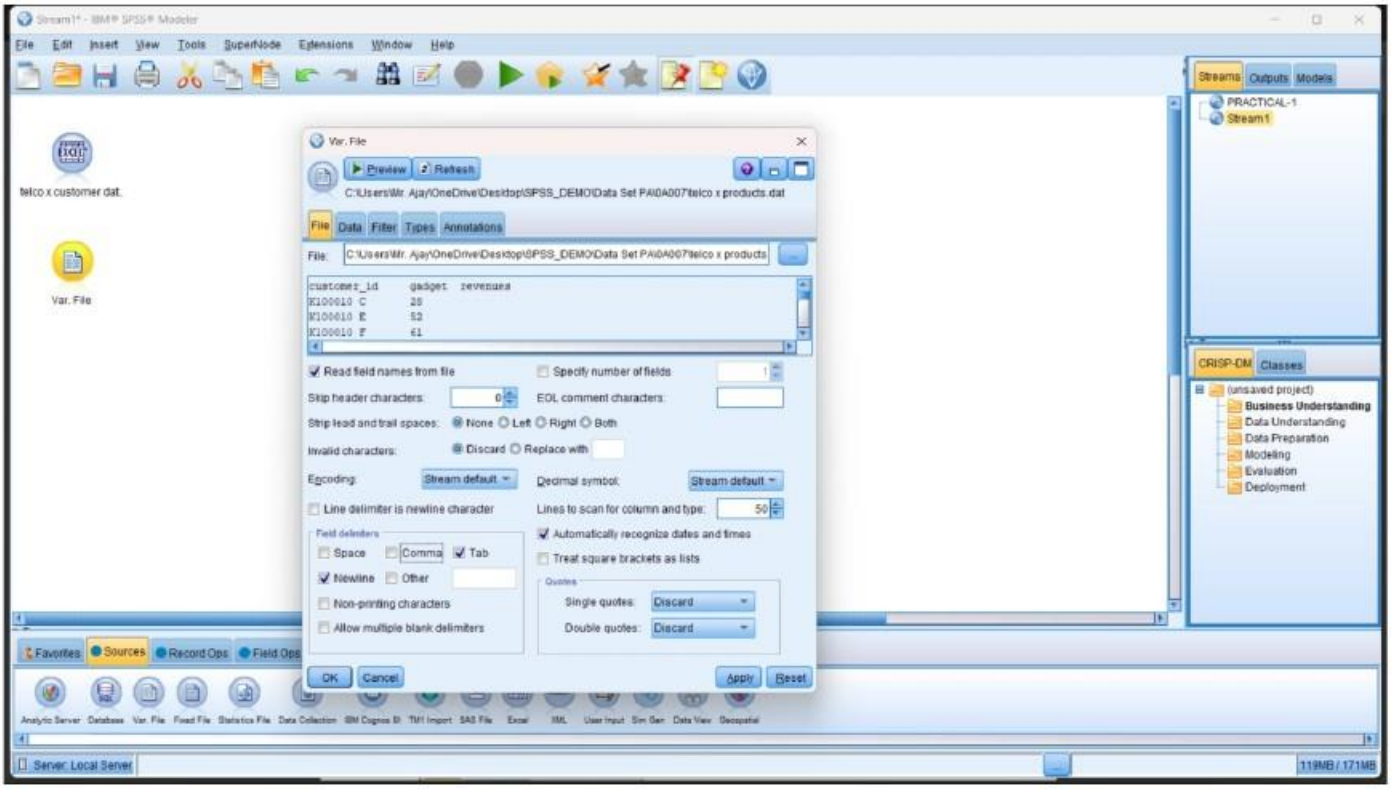
TASK – 2

❖ Import a Text File

Definition: - This task demonstrates how to import a text file into IBM SPSS Modeler and preview its data.

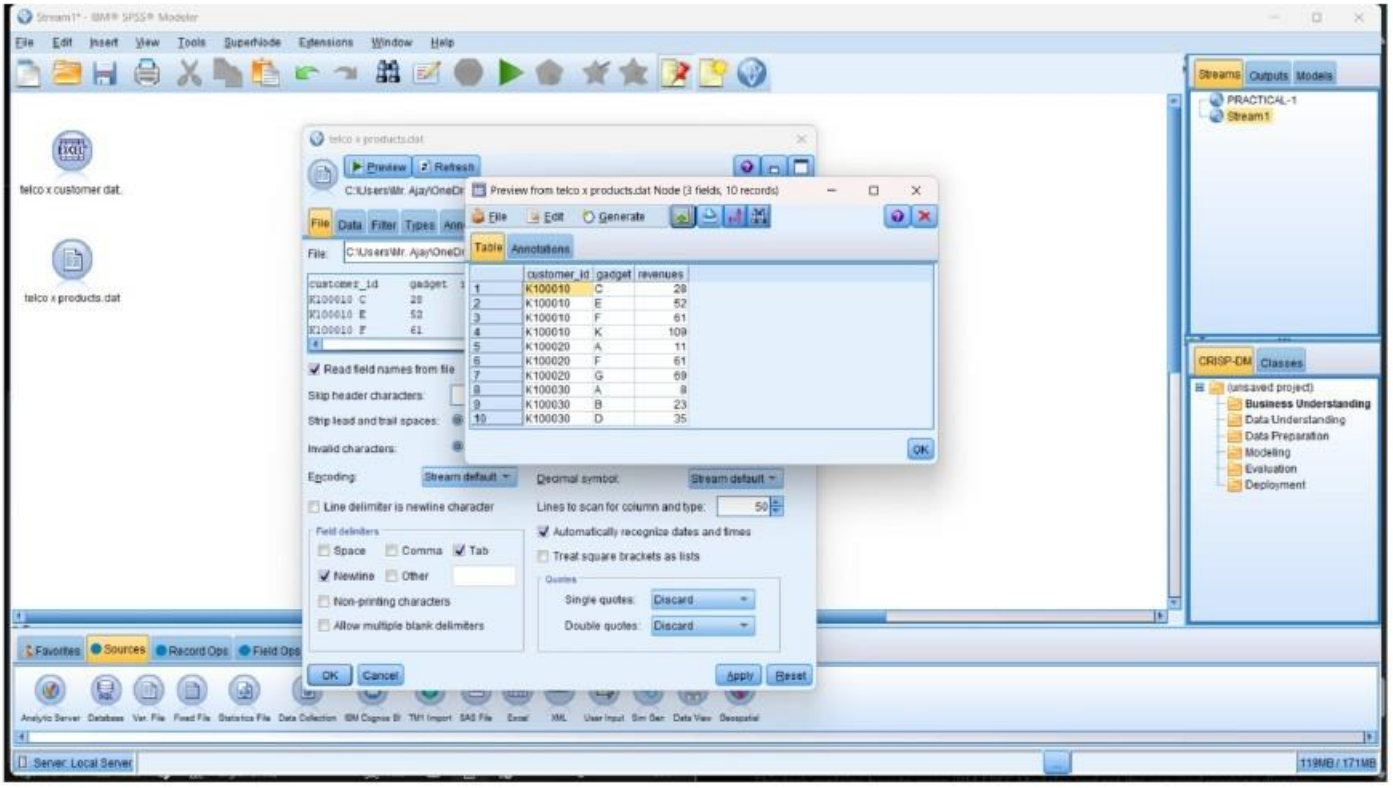
Step 1: To import a text file

- From the **Sources** palette, add a **Var. File** node to the stream canvas.
- Double-click the **Var. File** node to open it:
- Set the **Import file** to (telco x products.dat) from your file location/path.
- Like: - **C:\Users\Mr.**
- **Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**
- Make sure the option **Read field names from file** is enabled.
- Under **Field delimiters**, clear **Comma** and enable **Tab** delimiter.



Step 2: Click **Preview** to check the imported data.

Verify that the data is displayed correctly. The unit of analysis is now the combination of **customer_id and gadget**.



- Click **OK** to close the Preview window, and then click **OK** to close the Var. File dialog box.

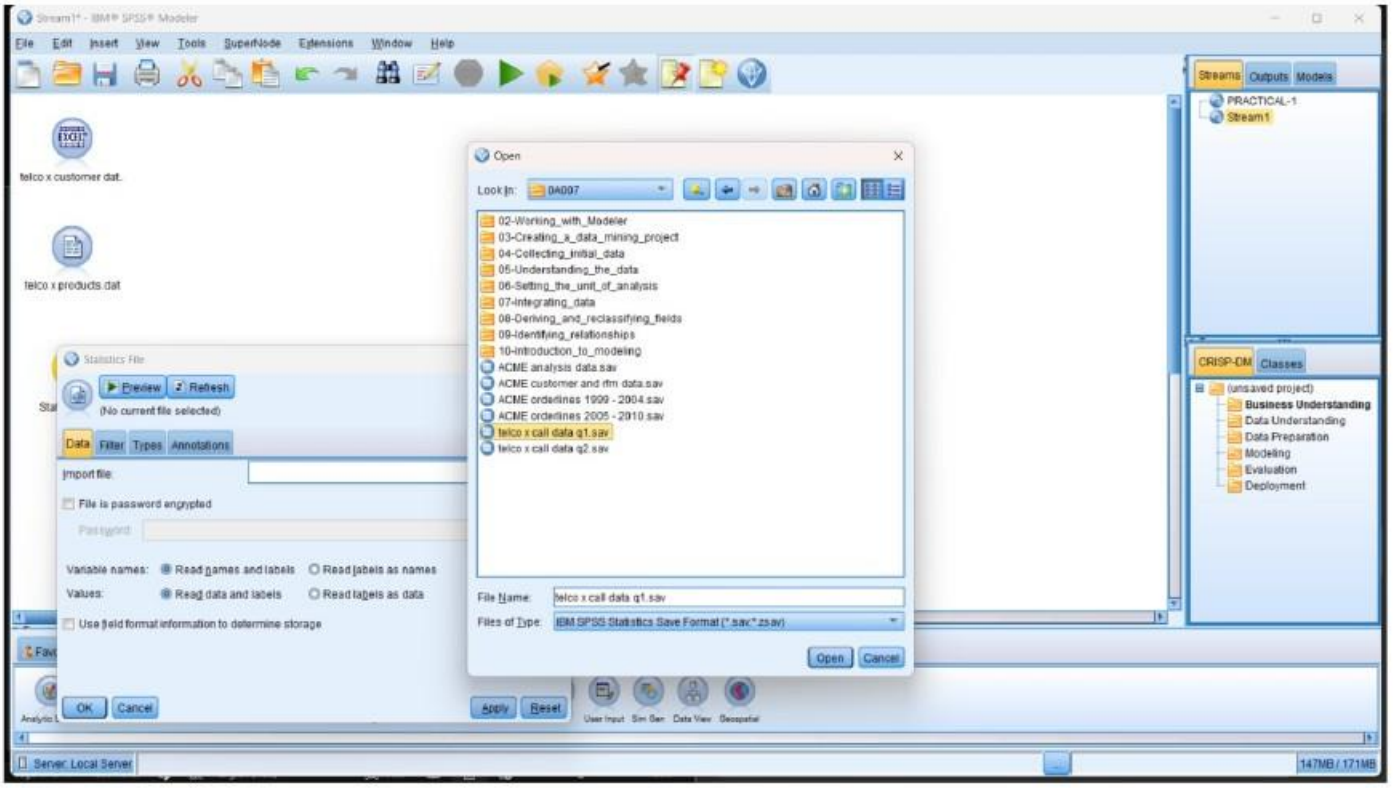
TASK – 3

❖ Import an IBM SPSS Statistics File

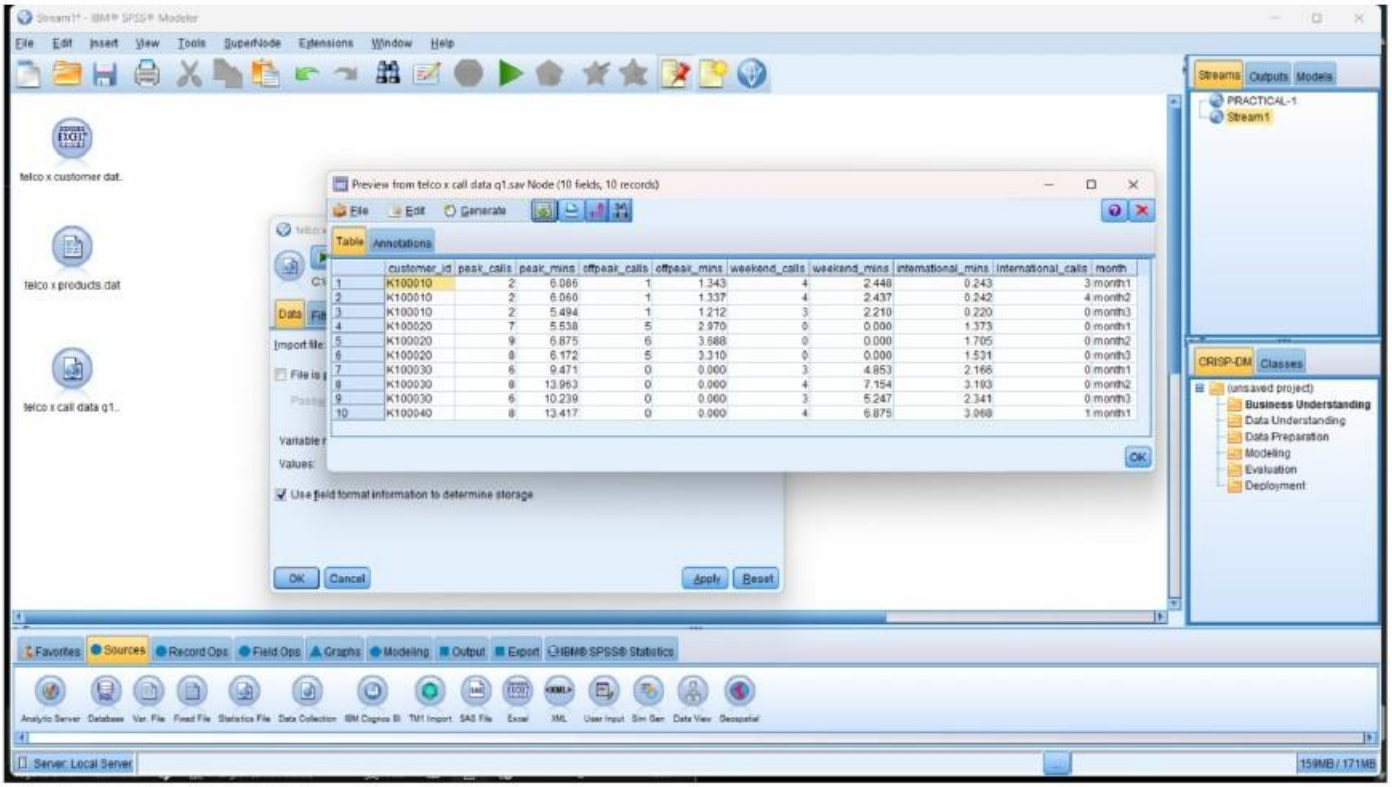
Definition: - This task demonstrates how to import an IBM SPSS Statistics (.sav) file into IBM SPSS Modeler and preview the data.

Step 1: To import a Statistics file

- From the **Sources** palette, add a **Statistics File** node to the stream canvas.
- Double-click the **Statistics File** node to open it:
- In the **Data** tab, set the **Import file** to (telco x call data ql .sav) from your file location/path.
- Like: - **C:\Users\Mr.**
- **Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**
- Enable the Use field format information to determine storage option.
- Click **Preview** to check the imported data.



Step 2: Enable the **"Use field format information to determine storage"** option and click **Preview** to check the imported data.



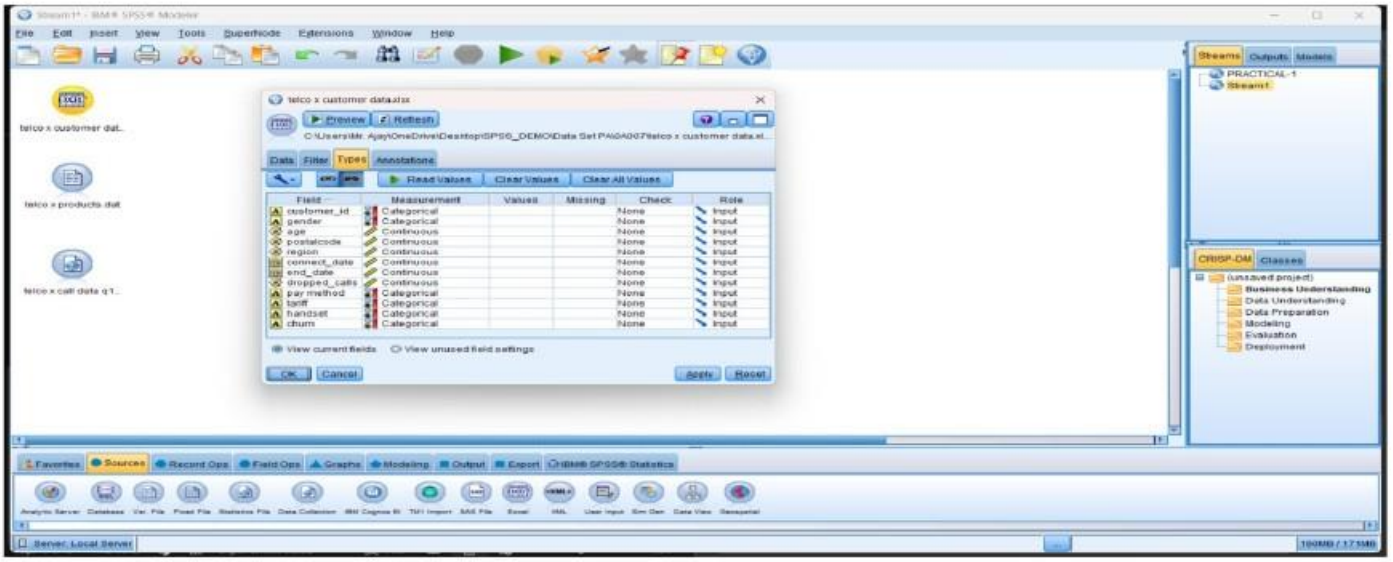
- Click **OK** to close the Preview window, and then click **OK** to close the Statistics File dialog box.

TASK – 4

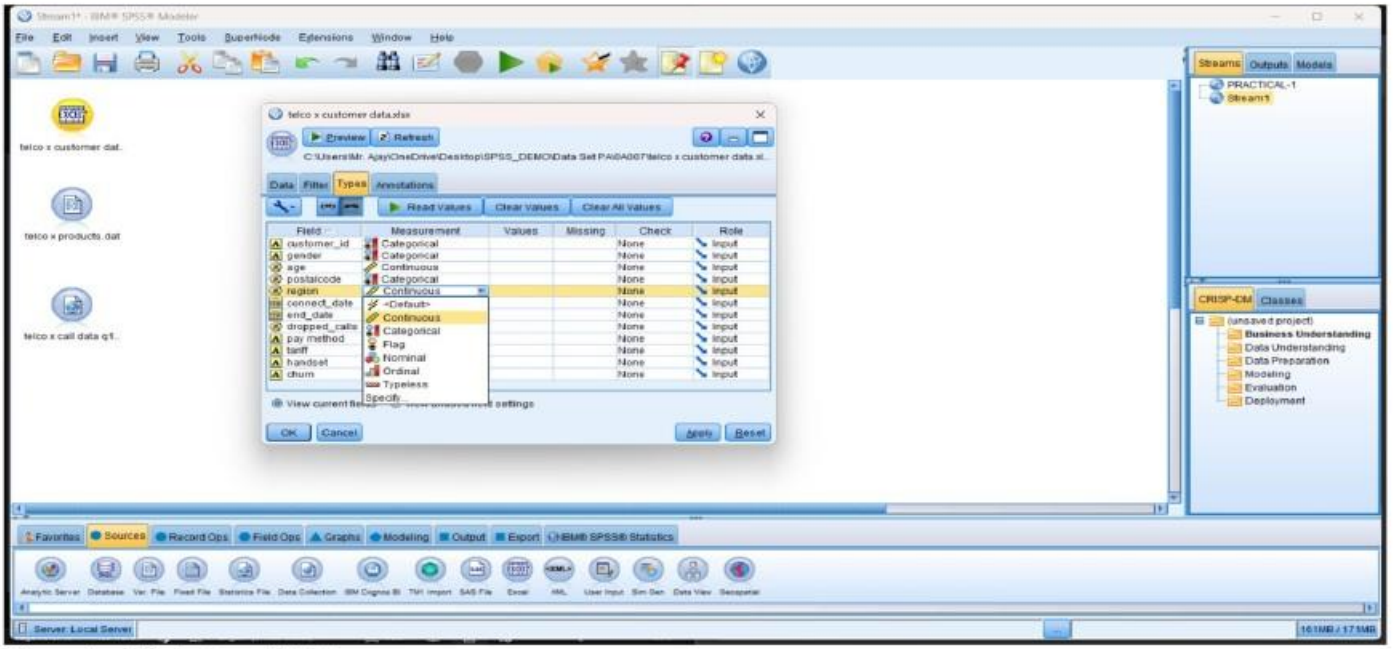
❖ Set the measurement levels of imported fields.

Definition: - You will check the measurement level for the fields imported from the Microsoft Excel file, telco x customer data.xlsx.

Step 1: Double-click on "**Excel node"** (telco x customer data.xlsx), and then click the **Types** tab.



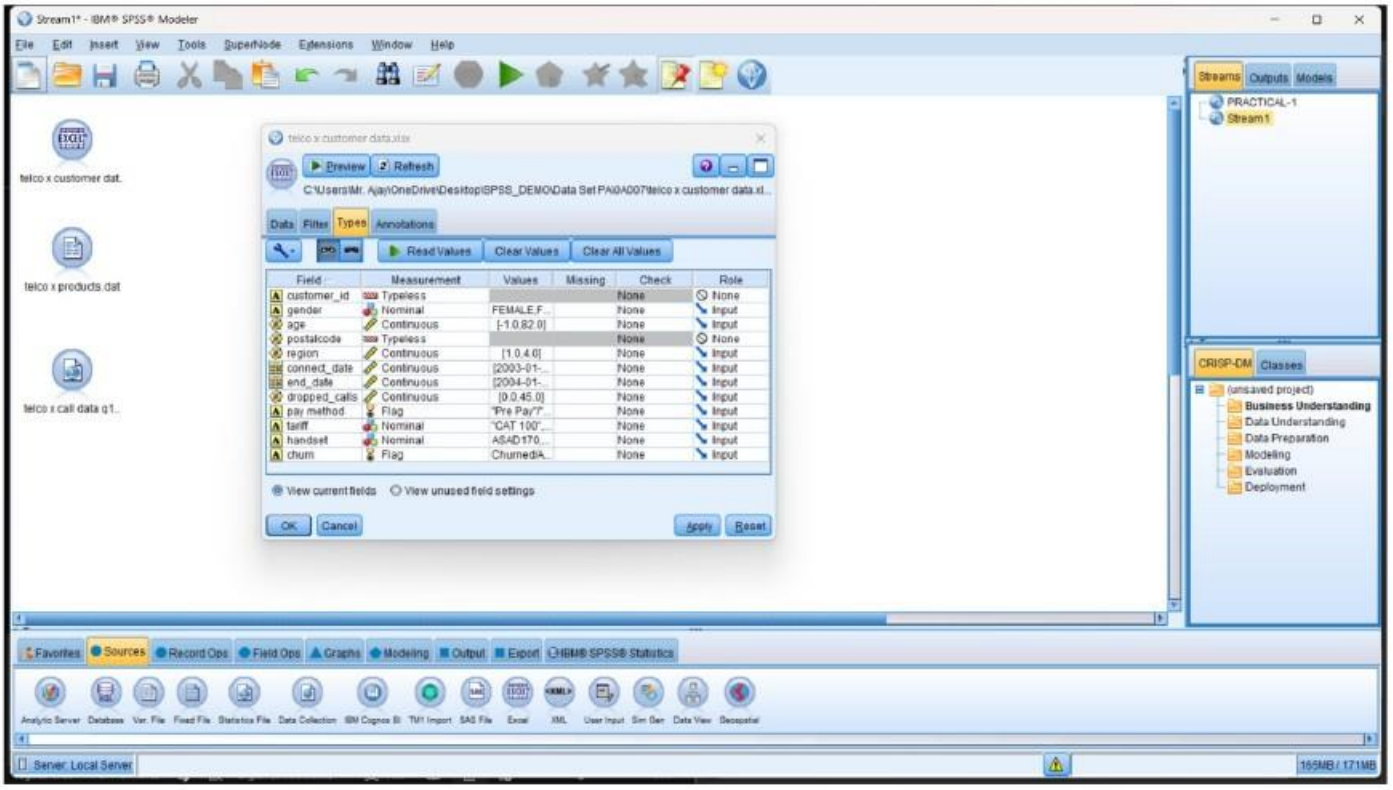
Step 2: In the Measurement column, click the cell for **postalcode**, and select **Categorical** from the dropdown; also click the cell for **region**, and select **Categorical** from the dropdown. The results appear as follows:



Step 3: Click **Read Values**.

A message box display, asking you if you want to read values for all fields.

Click **OK** to confirm. The results appear as follows



- Click **OK**, to close the Type dialog box. To save the stream, go to the **File** menu and click **Save**, or press **Ctrl + S**. Choose the desired location/path, enter the file name you want(like this), and then click **Save**.

Result: You have imported data from various data sources, and you examined the unit of analysis and the fields' measurement levels.

Practical - 3

Definition: - This practical demonstrates how to **audit data, identify data-quality issues, define valid values, and handle blank or undefined values** using IBM SPSS Modeler.

Outcomes/Learning: - Learners will understand how to **check data quality, identify invalid values, define valid ranges, and handle missing or blank values**.

Purpose: - You work as a data miner for a telecommunications firm. You have imported data, and you need to assess the quality of the data and define blanks where appropriate.

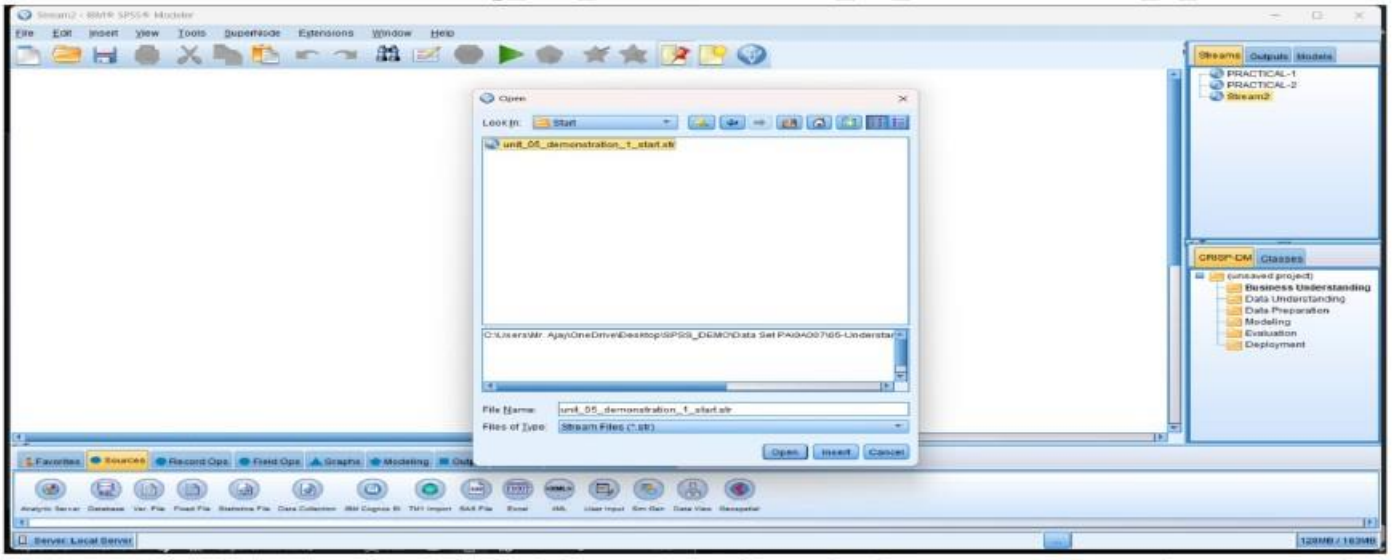
TASK – 1

❖ Start IBM SPSS Modeler and set the working folder, import a Microsoft Excel File and Open the Start Stream and audit the data.

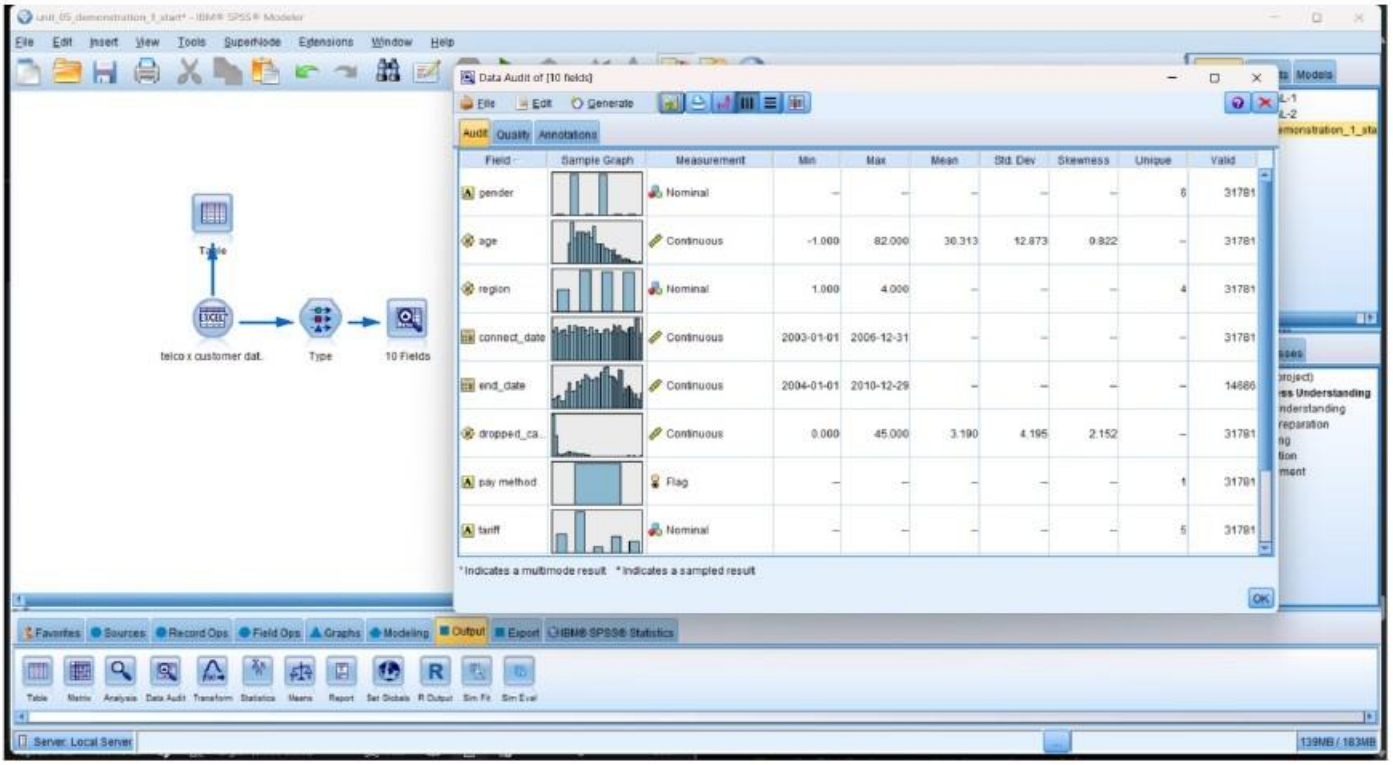
Definition: - This task explains how to **start IBM SPSS Modeler, set the working folder, and import an Excel file for data analysis**.

Step 1: First of all, from the File menu, click Set Directory.

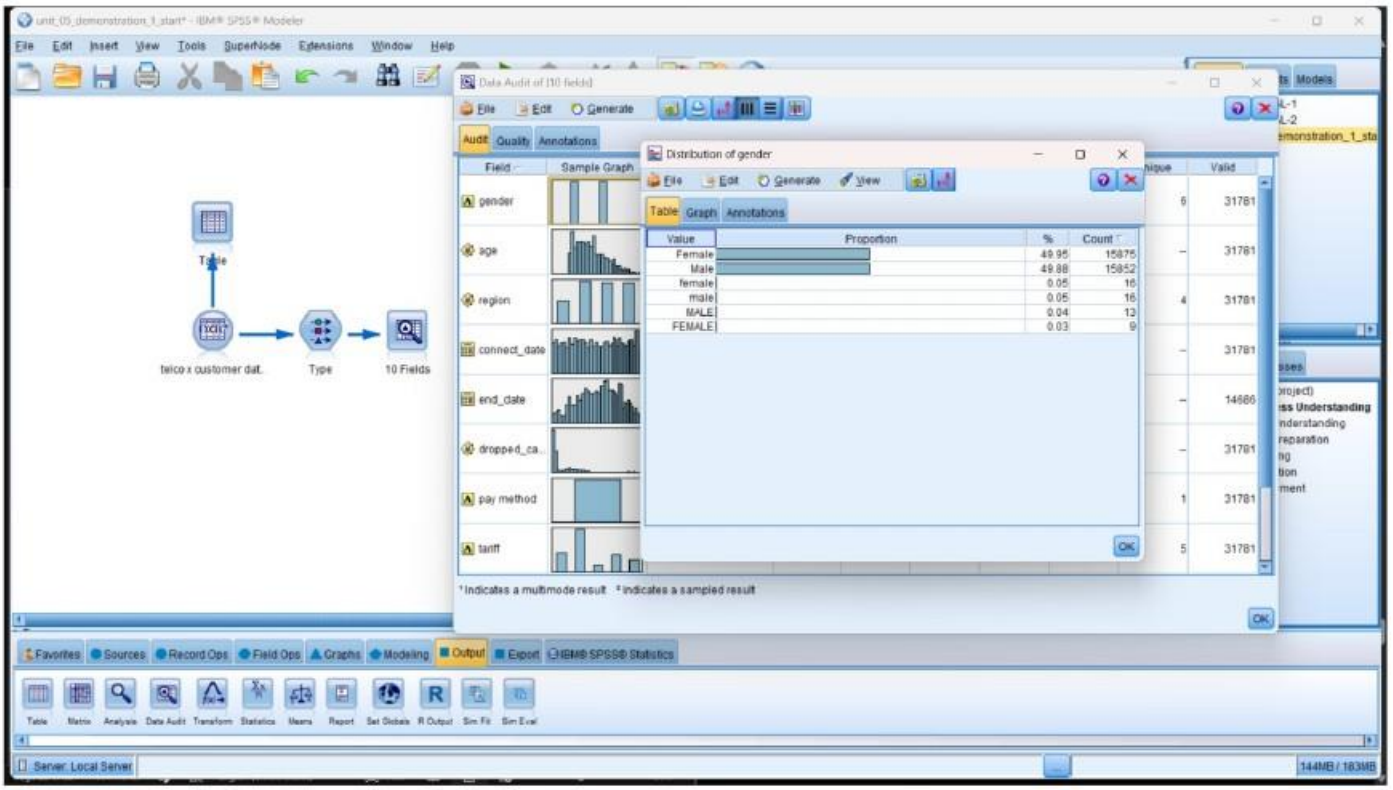
- From the **File** menu, click **Open Stream**
- Open **unit_05_demonstration_1_start.str**.
- For this, navigate to **OA007\05-Understanding_the_data\Start** and open the required dataset/stream file. Then click Open.
- Like this:- **C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007\05-Understanding_the_data\Start\unit_05_demonstration_1_start.str**



Step 2: From the **Output** palette, add a **Data Audit** node downstream from the **Type** node. Run the **Data Audit** node. They look like this.

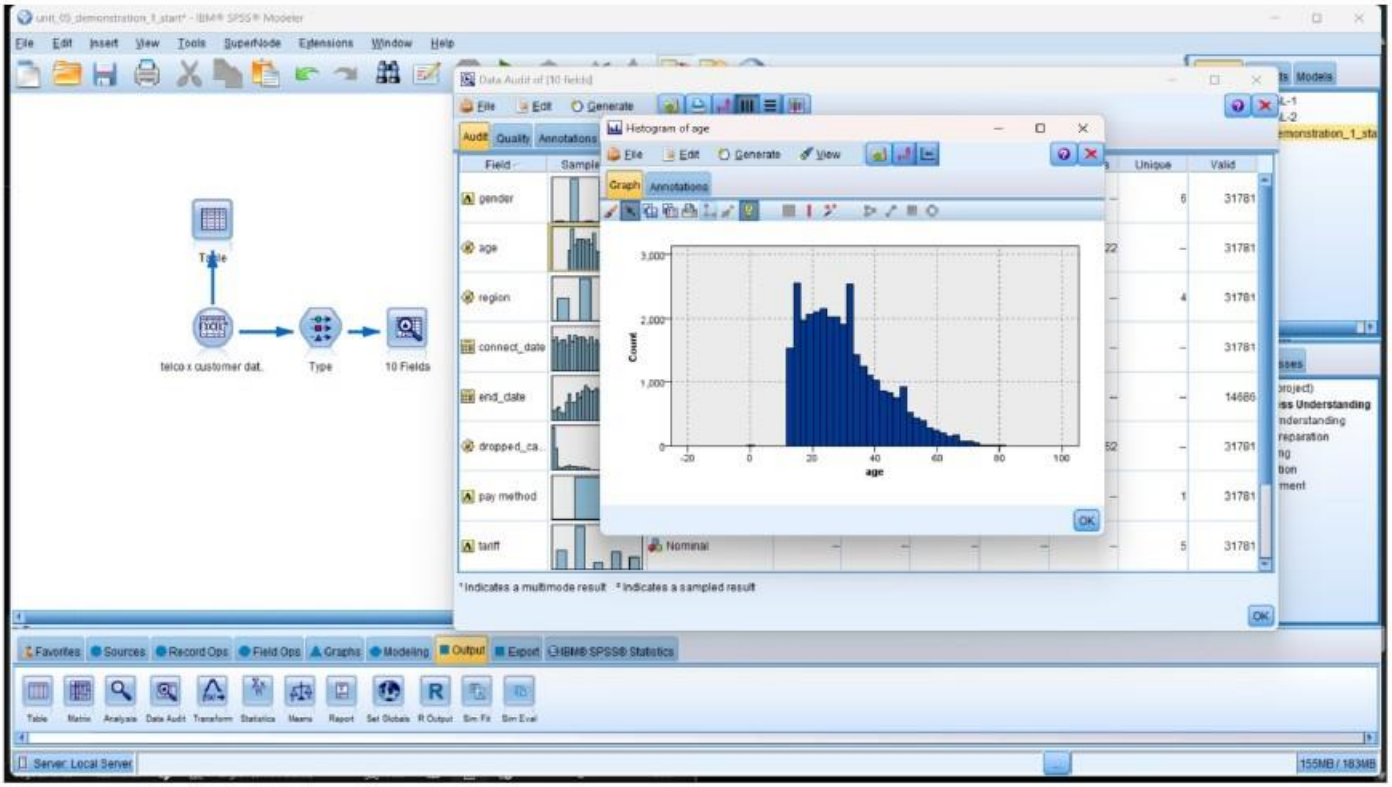


Step 3: Double-click the graph for gender. A Distribution output window opens. In the Distribution output window, click the Count column header twice to sort the values descending on frequency. The results appear as follows:



Step 4: Close the Distribution output and open the graph for age.

The little bar near O on the x-axis points to a few records with age -1.



- Click OK to close the Histogram output window, and then click OK to close the Data Audit output window.

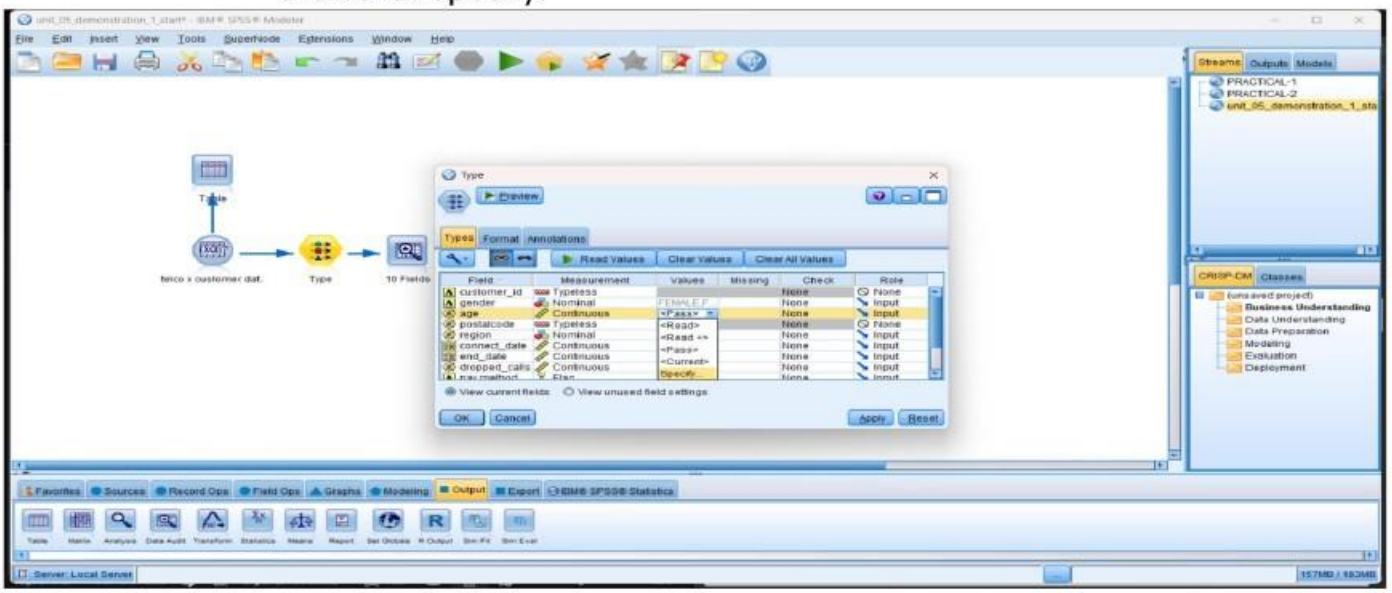
TASK – 2

❖ Define valid values and take action.

Definition: - This task demonstrates how to **define a valid range for age and specify an action for invalid values.**

Step 1: For this:

- Edit the **Type** node and click the **Types** tab.
- From the **File** menu, click **Open Stream**.
- In the Values column, click the age row (where it reads [-1.0, 82.0], and then click Specify.

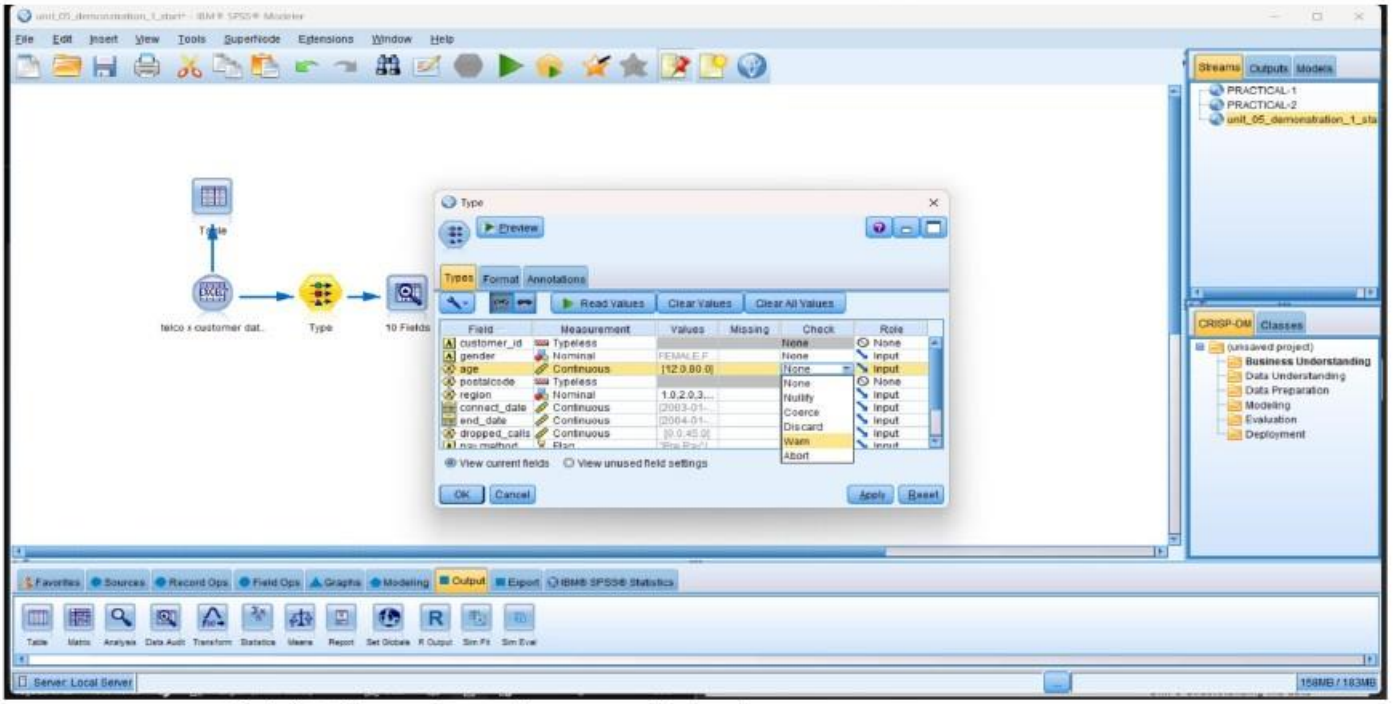


Step 2: After the Values dialog box opens. In the Values dialog box, click Specify values and labels, and then set Lower to 12 and Upper to 90. Like this.



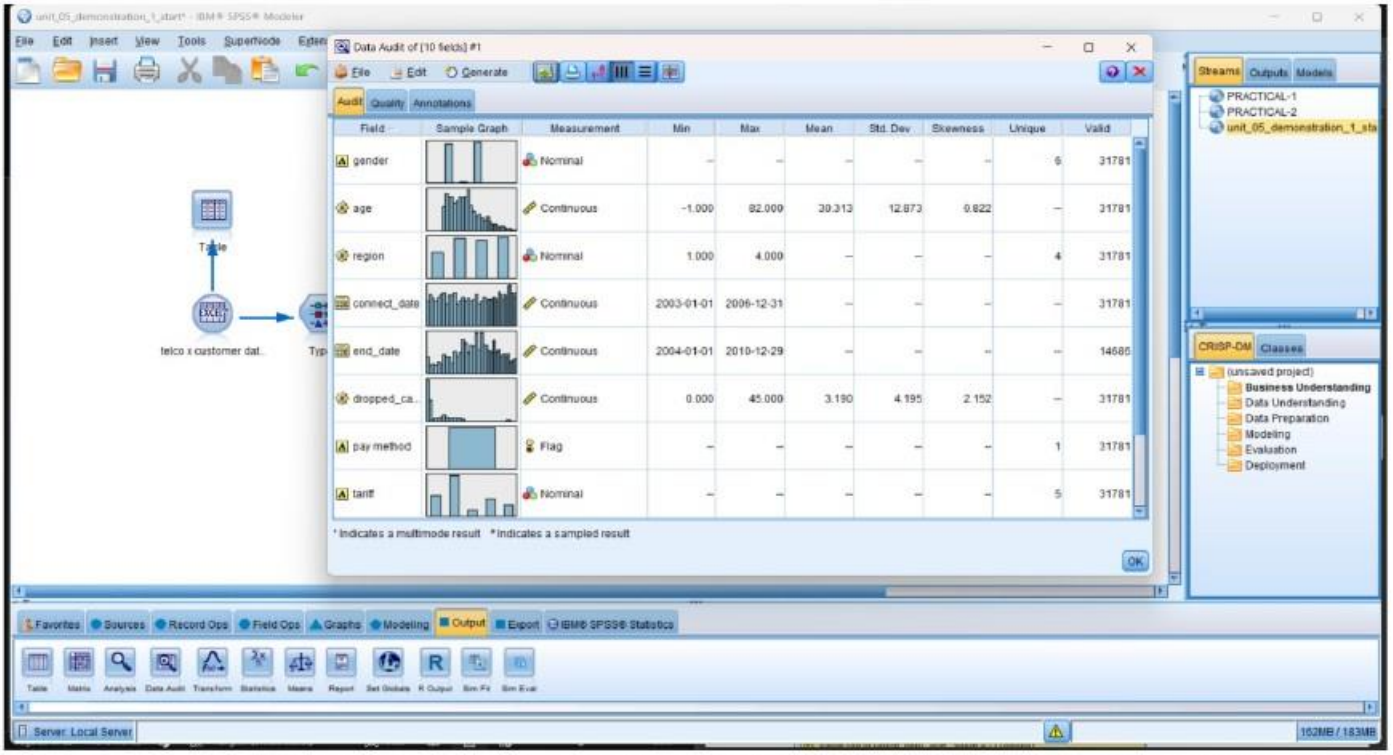
- Click OK, to close the Values dialog box.

Step 3: After that, in the Check column, click the age value, and select Warn. The results appear as follows:



- Click "**OK**", to close the Type dialog box.

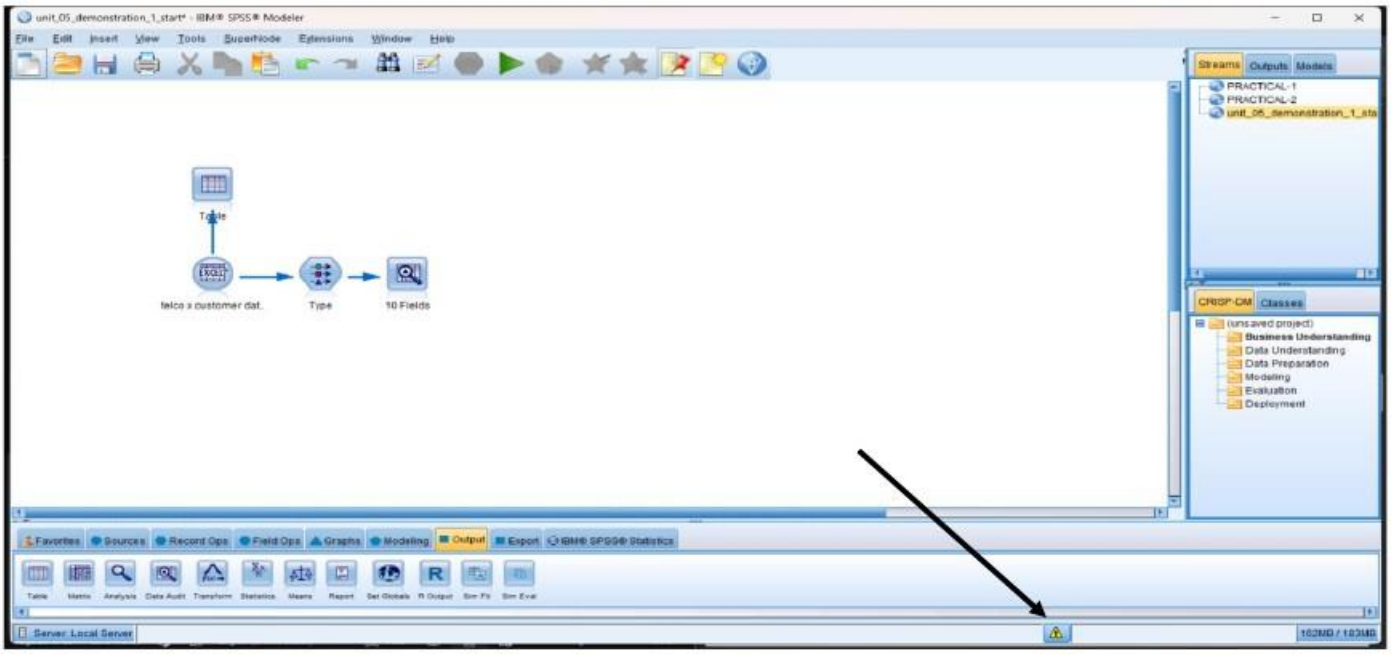
Step 4: Run the Data Audit node.



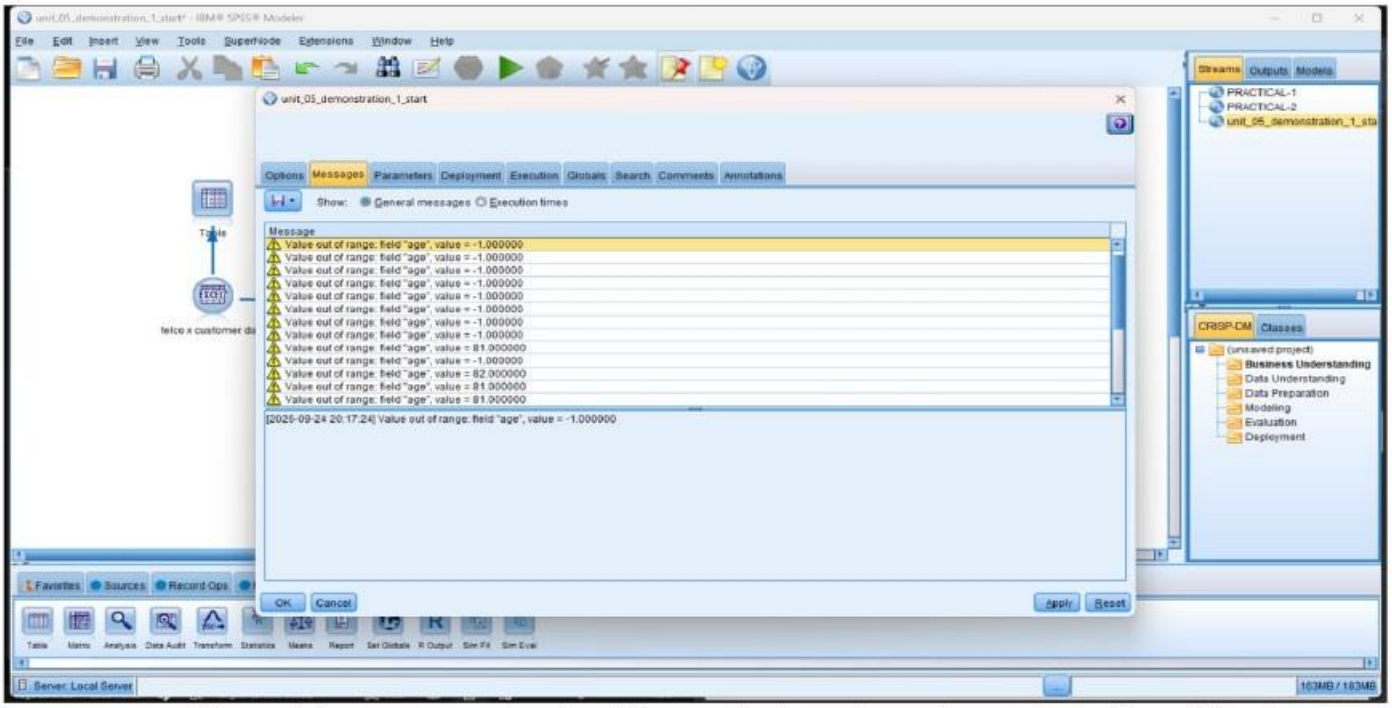
- The minimum age value is still reported as -1 , so it seems as if nothing has changed.
- Click "**OK**", to close the Data Audit output window.

Step 5: After that, at the bottom of the window, you will find a stream message like  .

The Show stream messages button( ) in the status bar shows an exclamation mark, indicating that you have new messages.



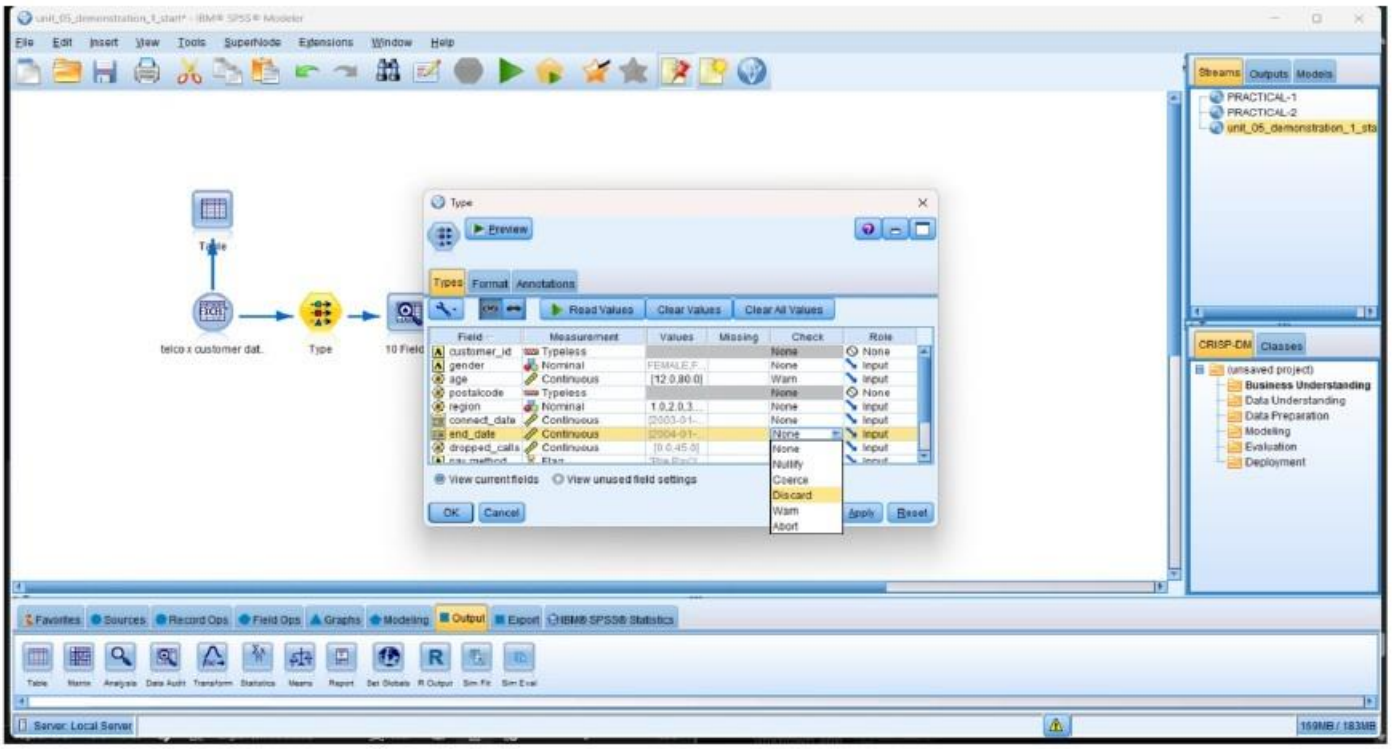
Step 6: Click the Show stream messages button. The results appear as follows:



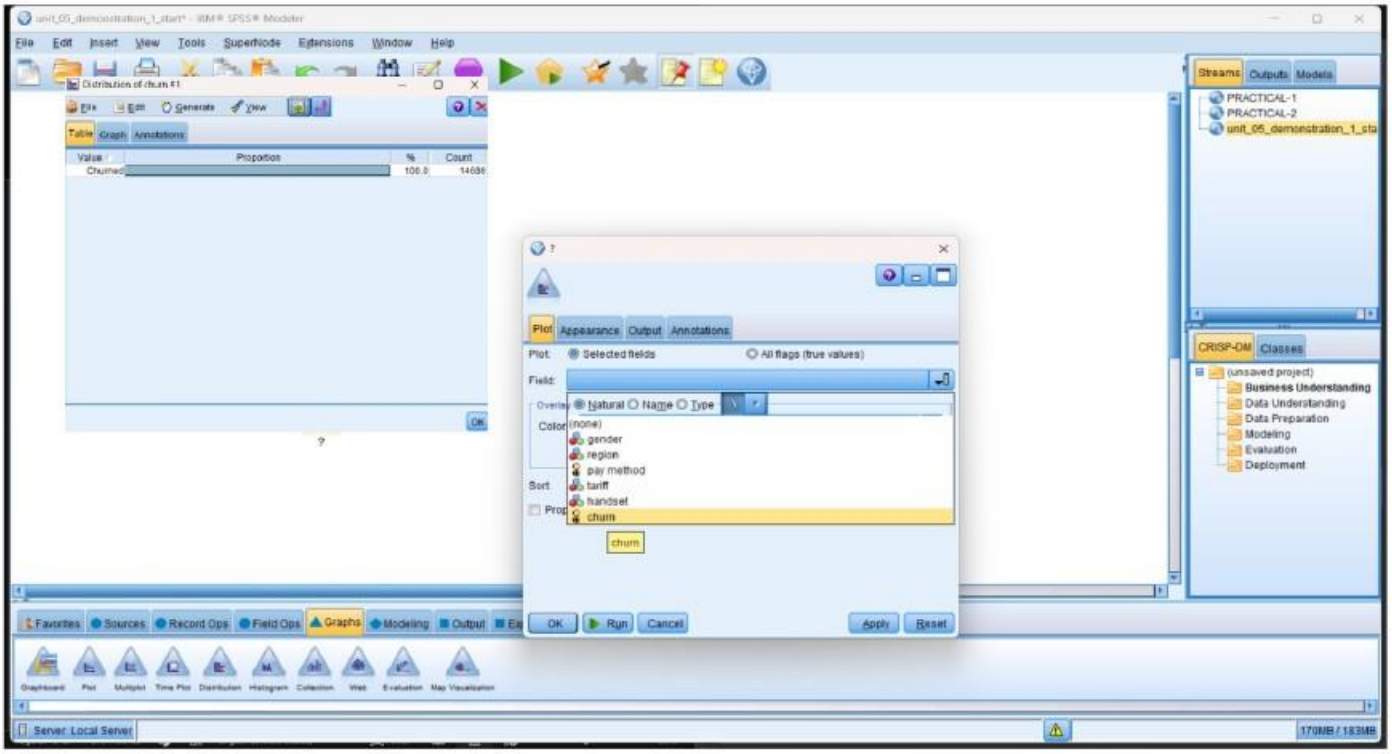
- The minimum age value is still reported as -1 , so it seems as if nothing has changed.
- As another example of dealing with missing values, you will discard records with an undefined ($null$) value for end_date.

Step 7: Edit the Type node, and in the Check column for end_date, set the action to

Discard.
You will examine the effect of removing these records.

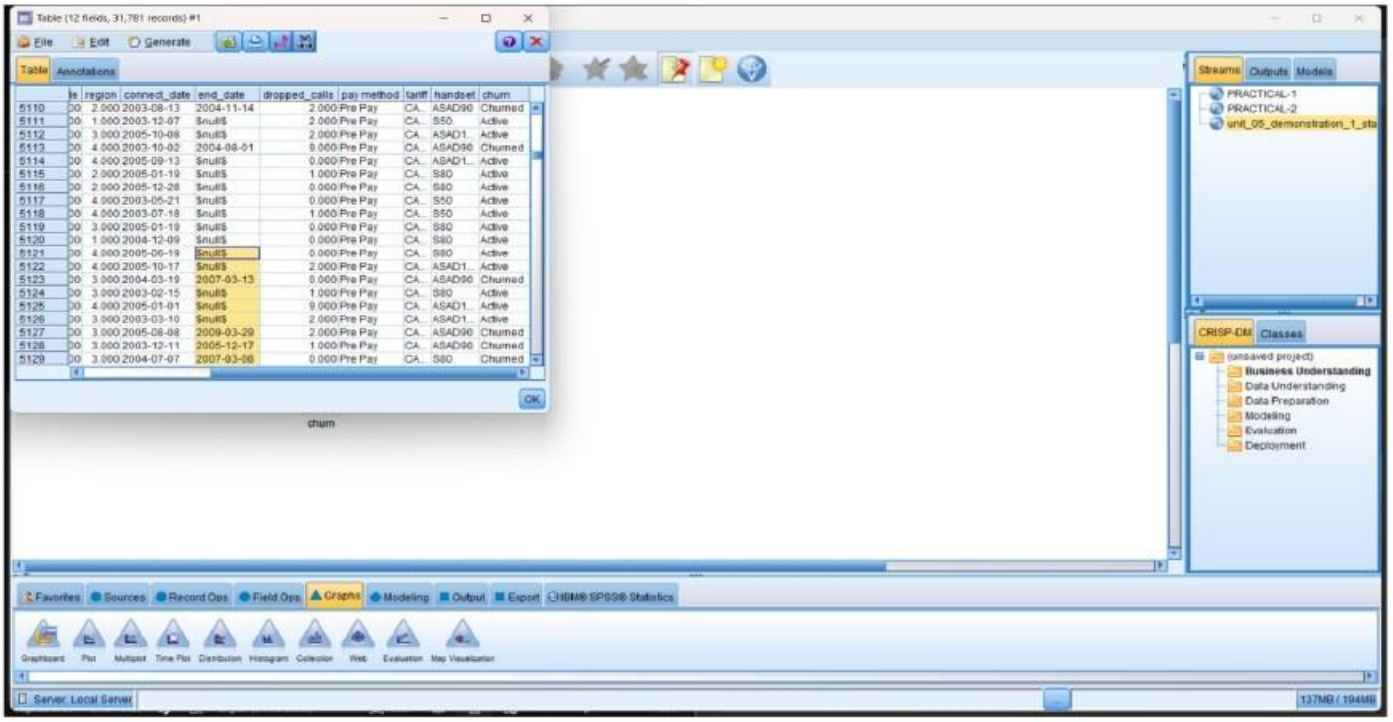


Step 8: From the Graphs palette, add a Distribution graph downstream from the Type node. Edit the Distribution node, click on Field option and click on chum and then click Run.



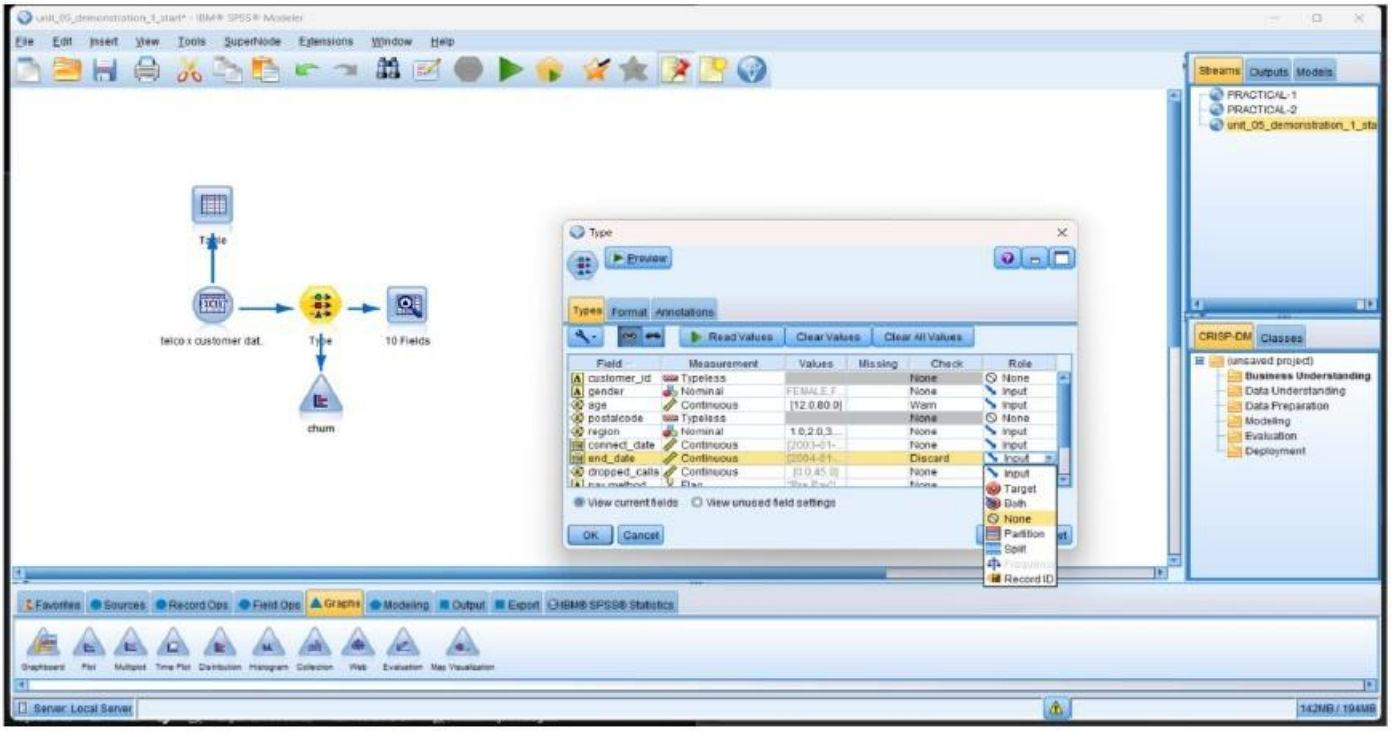
Step 9: After that run the Table node, and scroll to the right to display the end_date and

churn columns, and then scroll down to row 5121 The results appear as follows:



- Click "**OK**",to close the Table output.

Step 10: After that, double-click on the Type node, and in the Role column for end_date, set the action to **None**. Like this.



- Click "**OK**" to close the Type dialog box.

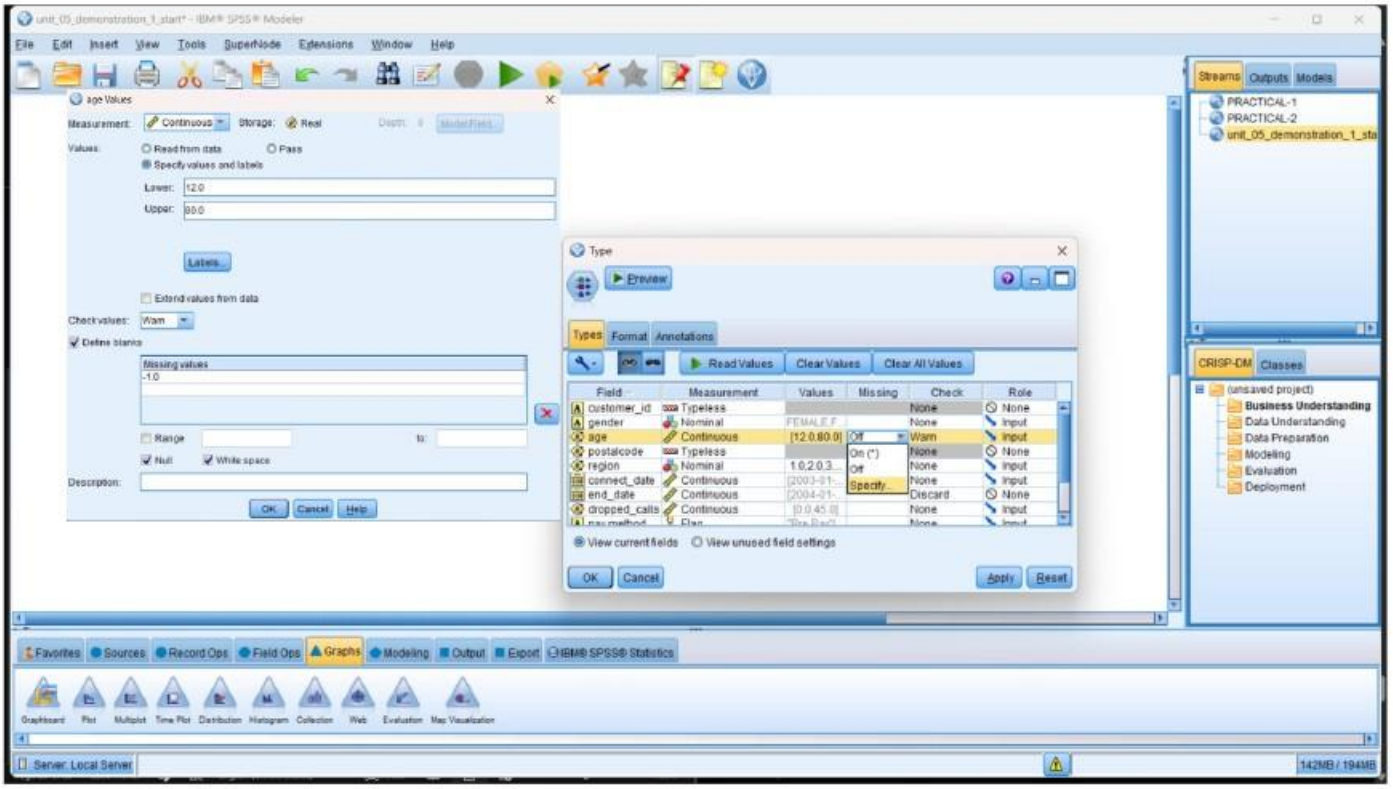
TASK – 3

❖ Declare Blank Values for Age.

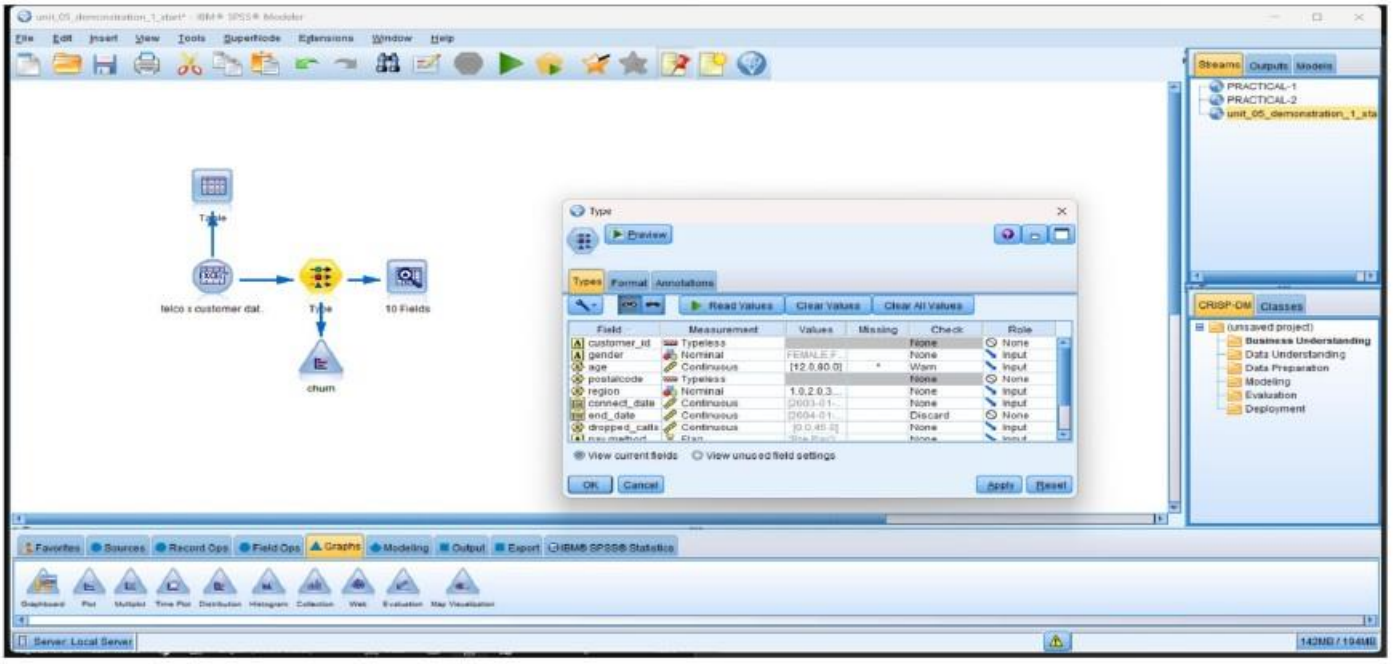
Definition: - This task demonstrates how to **declare a specific value as blank when it represents an unknown age**.

Step 1: For this:

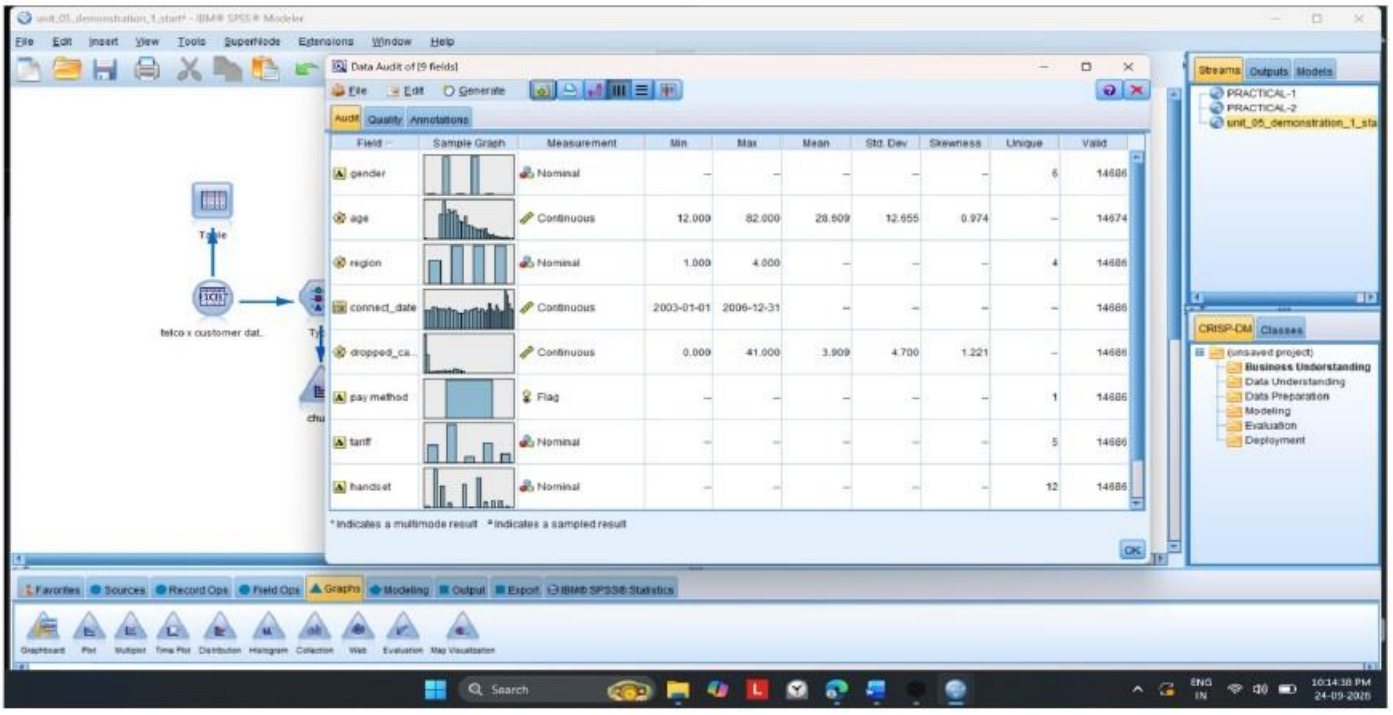
- Edit the **Type** node.
- In the **Missing** column, click the **age** cell, and then click **Specify**.
- After opening the box, enable the Define blanks check box. Under Missing values, type -1.
- Like this.



Step 2: After that, click **"OK"** to see it like this.



Step 3: After that, run the **Data Audit** node. The result appears as follows.



- The minimum value for age is 12, instead of -1 that you had in the previous task.
- Click OK, to close the Data Audit output window.
- In the bottom sidebar. The Show stream messages button does not show an exclamation mark  , which indicates that no out-of-range values are found.

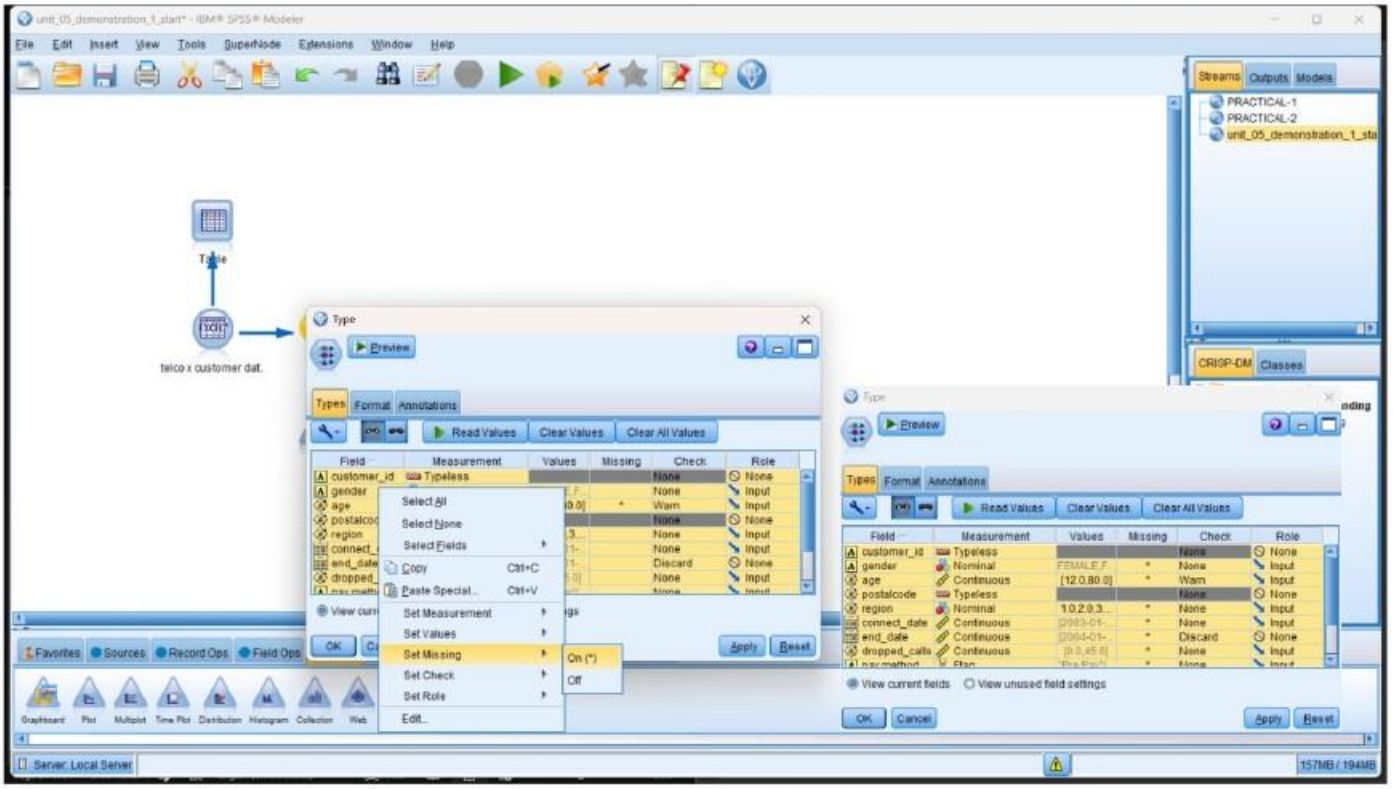
TASK – 4

❖ Declare Undefined Values as Blanks for All Fields.

Definition: - This task demonstrates how to **declare undefined or null values as blank values for all applicable fields**.

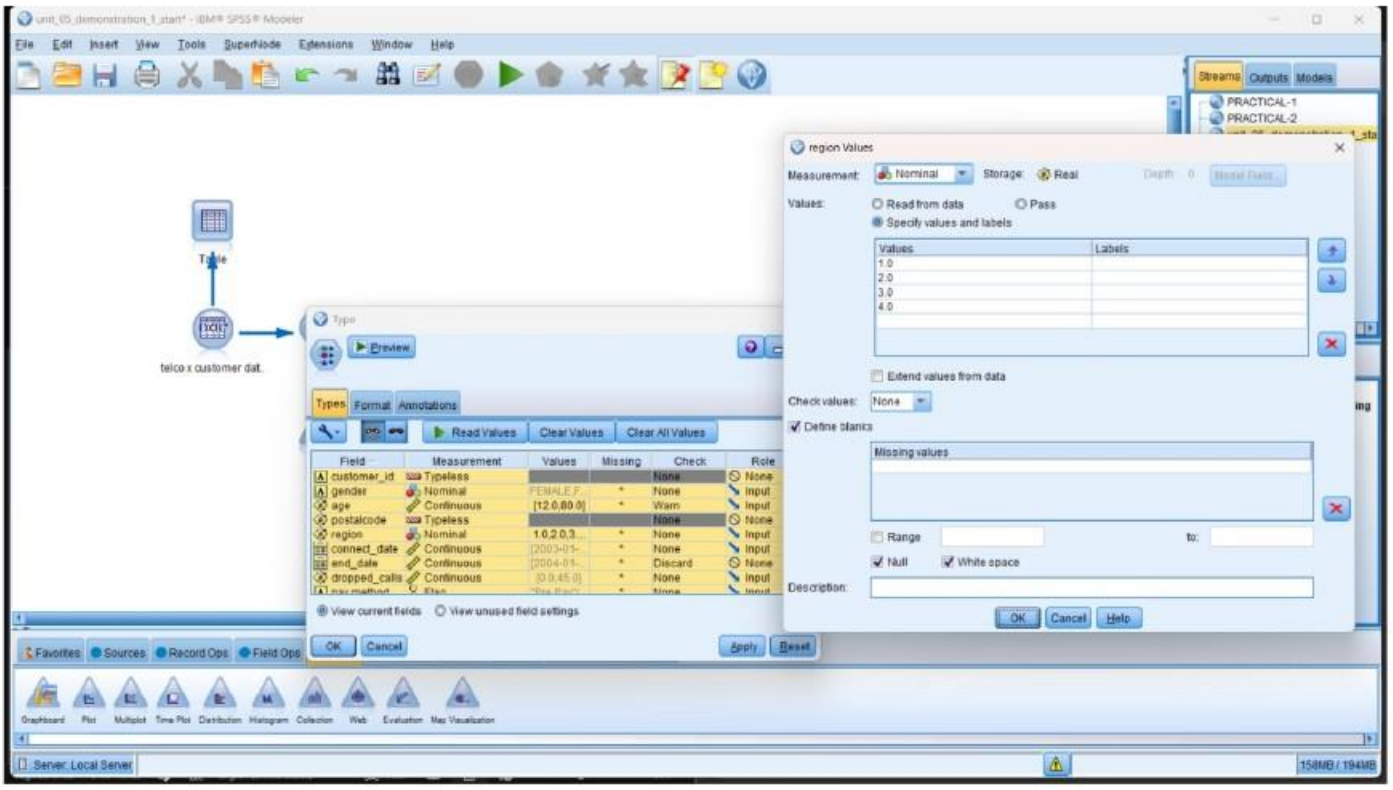
Step 1: For this:

- Edit the **Type** node.
- Right-click any field, and then click Select All from the context menu.
- Right-click any field, click Set Missing from the context menu, and then click on (*).
- The results appear as follows:



- All fields, except the typeless fields, have an asterisk in the Missing column, indicating that blanks are defined for each of them. To examine the difference from the situation before, you will look at the blank definitions for one of the fields.

Step 2: After that, in the Missing column, click the region cell, and then select Specify. The results appear as follows:



- **Define blanks** and **Null** options are enabled. This means **null, empty, or whitespace values are treated as blank values** and are not marked as invalid.
- Click OK to close the Values dialog box, and then click OK to close the Type node.
- This completes the demonstration for this unit. You will create a clean state for the exercise.

Step 3: Click **OK**, to close the Type dialog box. To save the stream, go to the **File** menu and click **Save As**. Choose the desired location/path, enter the file name, and then click **Save**.

Result: You have imported data, assessed its quality, and defined blanks as appropriate.

Practical – 4

Definition: - This practical demonstrates how to **identify duplicate records, remove them, and combine customer-level data using IBM SPSS Modeler**.

Outcomes/Learning: - Learners will understand how to **remove duplicates, aggregate data, and create one record for each customer**.

Purpose: - You work as a data miner for a telecommunications firm. Your job is to remove duplicate records in the customer dataset and transform a transactional dataset into one record per customer so you can merge the datasets later.

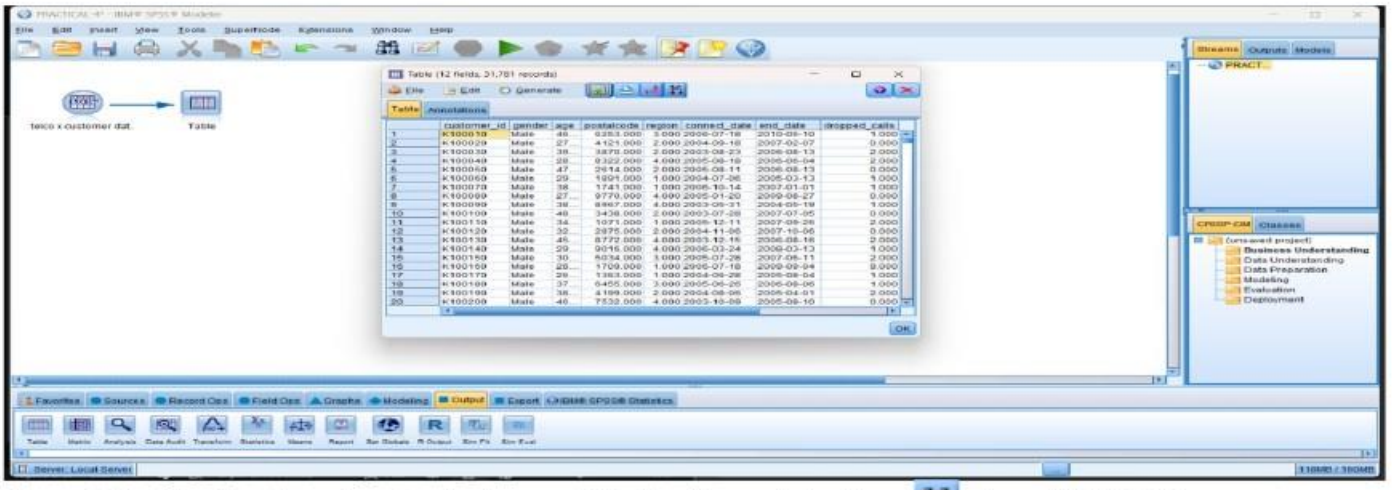
TASK – 1

❖ Start IBM SPSS Modeler and set the working folder, Import and Examine the Data

Definition: - This task explains how to **start IBM SPSS Modeler, set the working folder,** and **this demonstrates how to import customer data and check for duplicate records.**

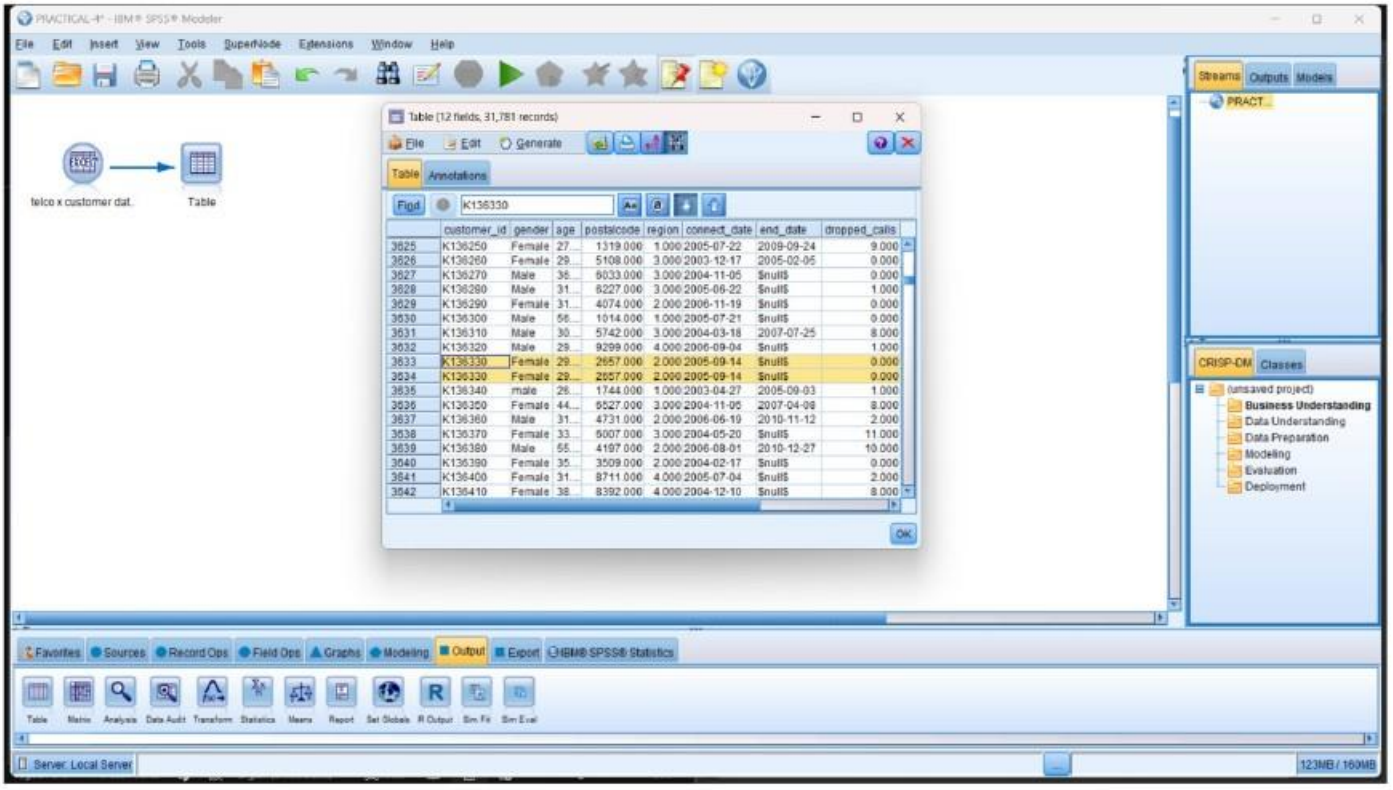
Step 1: First of all, from the File menu, click Set Directory.

- From the **Sources** palette, add an **Excel** node to the stream canvas.
- Double-click the **Excel** node to open it:
- Set the **Import file** to (telco x customer info.xlsx) from your file path.
- Like: - **C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007\telco x customer info.xlsx**
- After that, from the Output palette, add a Table node downstream from the Excel node, and then run the Table node.
- The Table output window opens like this.



Step 2: In the Table output window, click the **Search** button, type **K136330**, and then click Find button. Scroll down to a few records.

- The results appear as follows:
- Customer K136330 has two identical records.



- After that, click OK to close the Table output window.

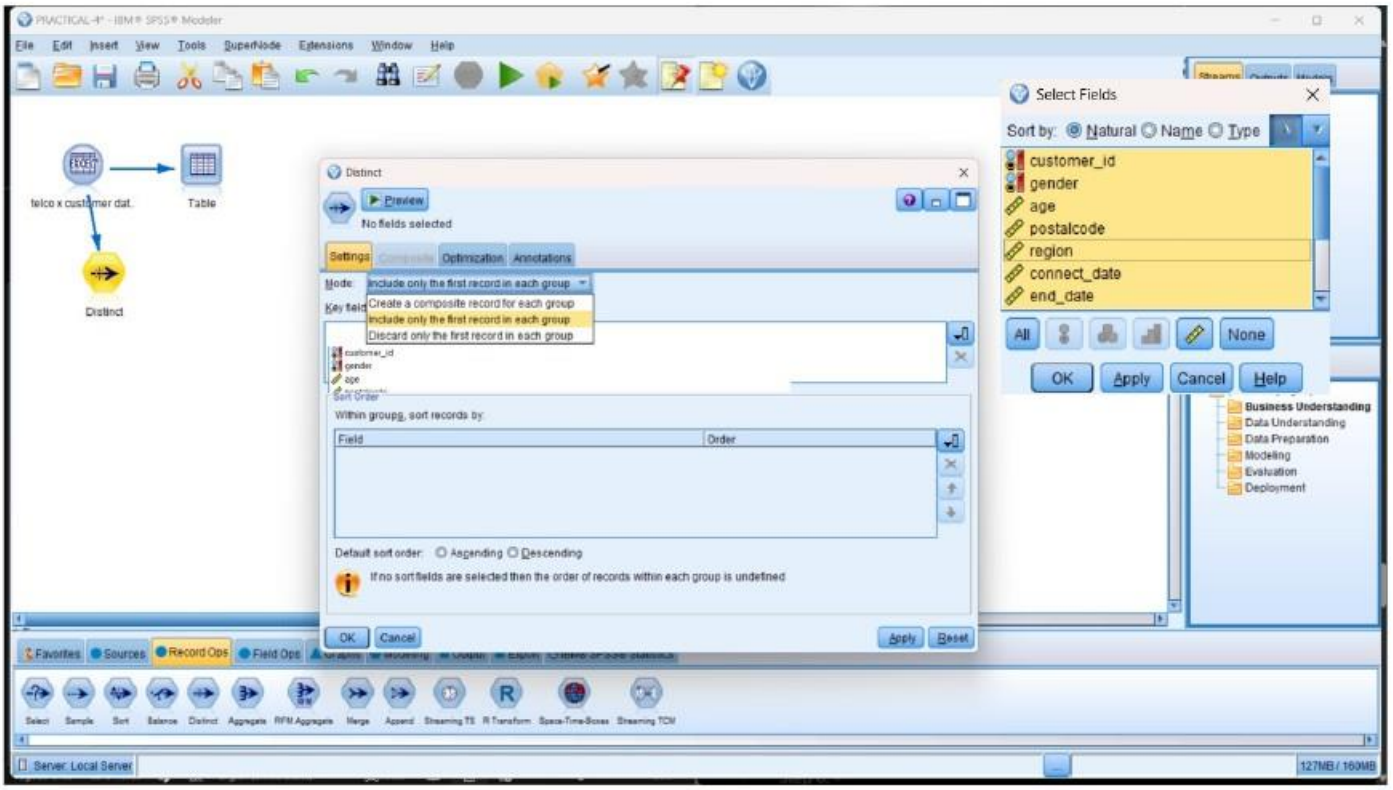
TASK – 2

❖ Use Distinct to remove duplicate records.

Definition: - This task demonstrates how to **remove duplicate records using Distinct node**.

Step 1: For this:

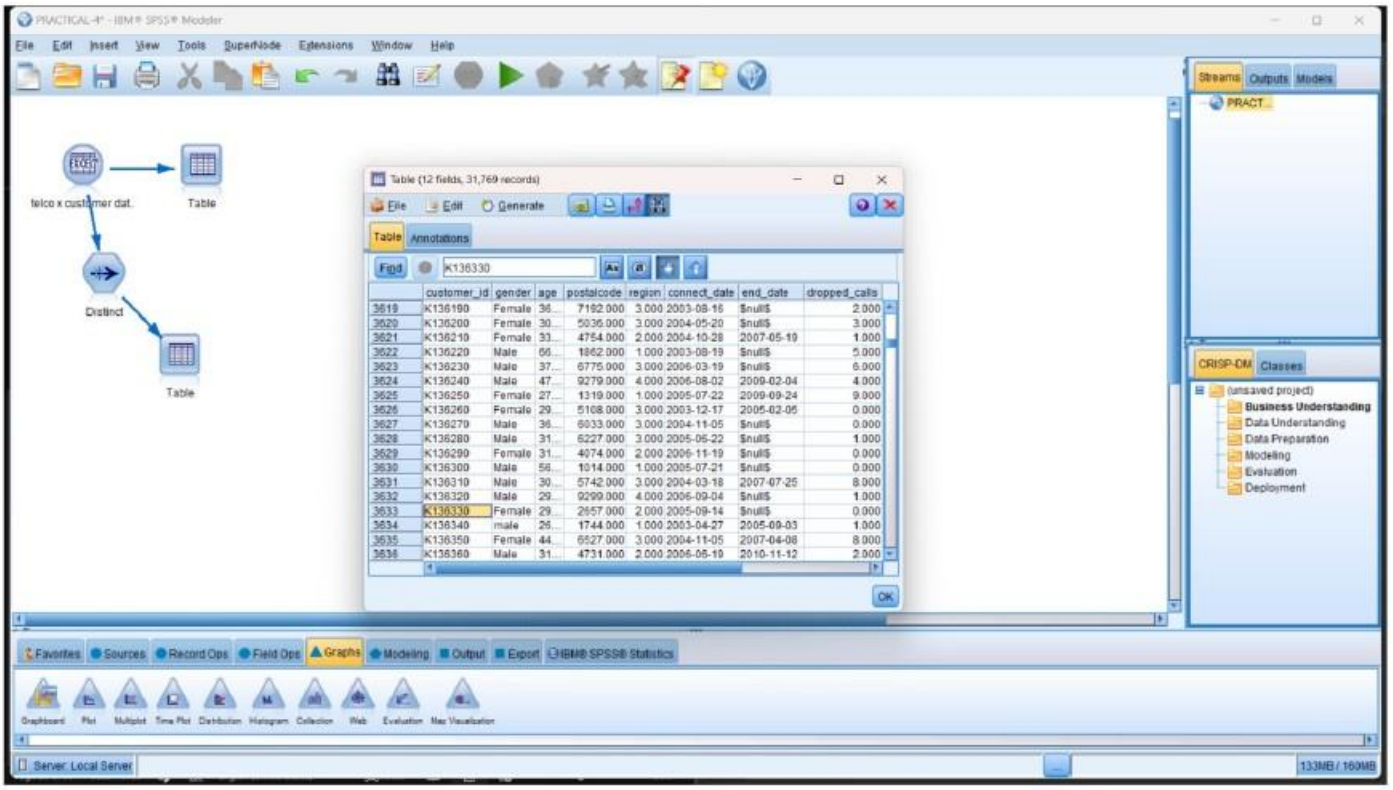
- From the **Record Ops** palette, add a **Distinct** node downstream from the Excel source node.
- Double-click to open the Distinct node.
- Besides Mode, select Include only the first record in each group from the list.
- Under Key fields for grouping, select all fields (records are duplicates if values for all fields are the same), and then click OK.
- The results appear as follows:



- Click **OK**, to close the Distinct dialog box.

Step 2: From the Output palette, add a Table node downstream from the Distinct node, and then run the Table node.

- After that, In the Table output window, click Search; type K136330 and then click Find.
- Scroll down a few records.
    - You will now find only one record for this customer.



- Click "**OK**", to close the Table output window.

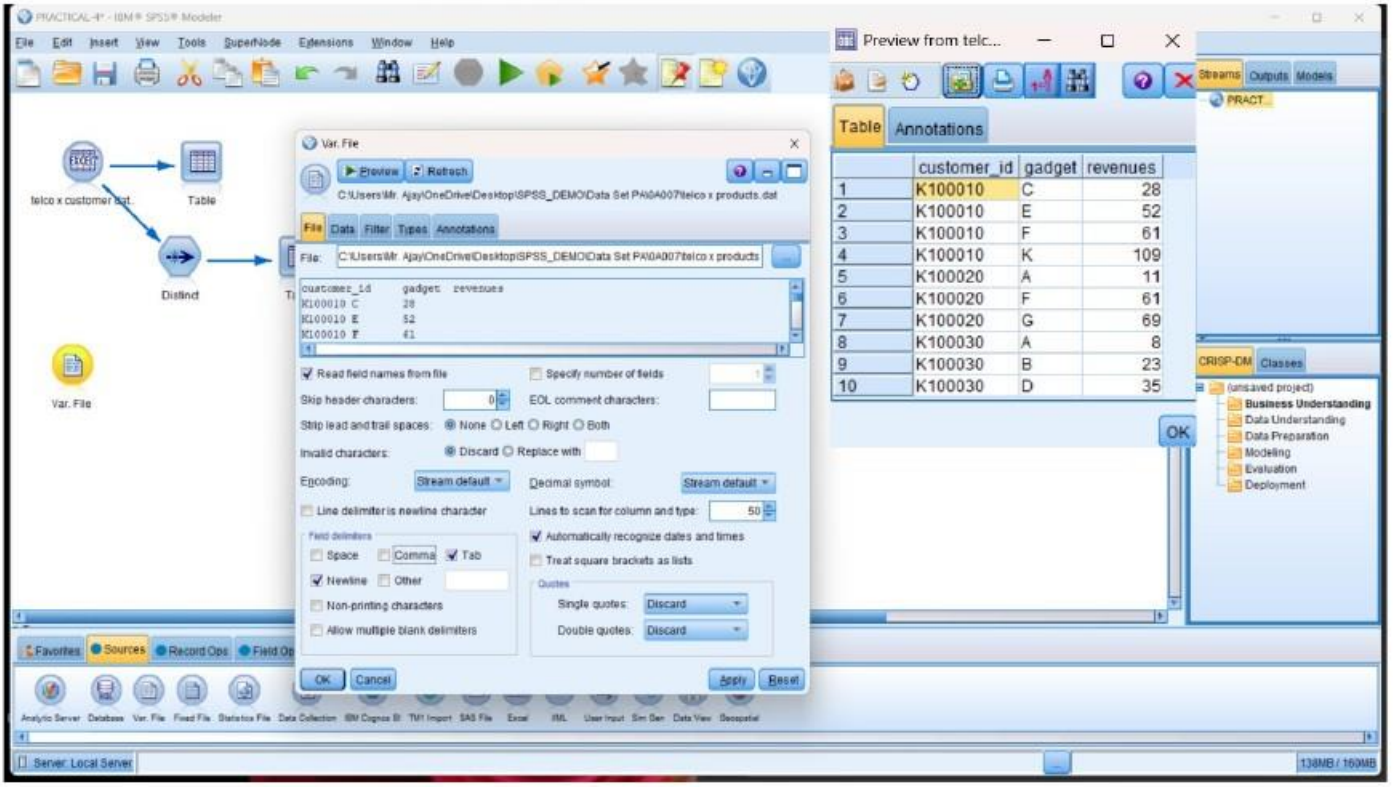
TASK – 3

❖ Obtain the Number of Gadgets Bought and Total Revenue per Customer

Definition: - This task demonstrates how to **calculate the number of gadgets and total revenue for each customer using the Aggregate node**.

Step 1: For this:

- From the **Sources** palette, drag and drop **Var. File** node onto the stream canvas.
- Double-click the **Var. File** node to open it.
- In the **File** options, click on the three dots on the right side.
- Set the **Import file** to (telco x product.dat) from your file path.
- Like: - **C:\Users\Mr. Aman\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007.**
- Click on the **Comma** box to deselect it as the delimiter, and select Tab as the delimiter.
- Click **Preview**.
- The results appear as follows:



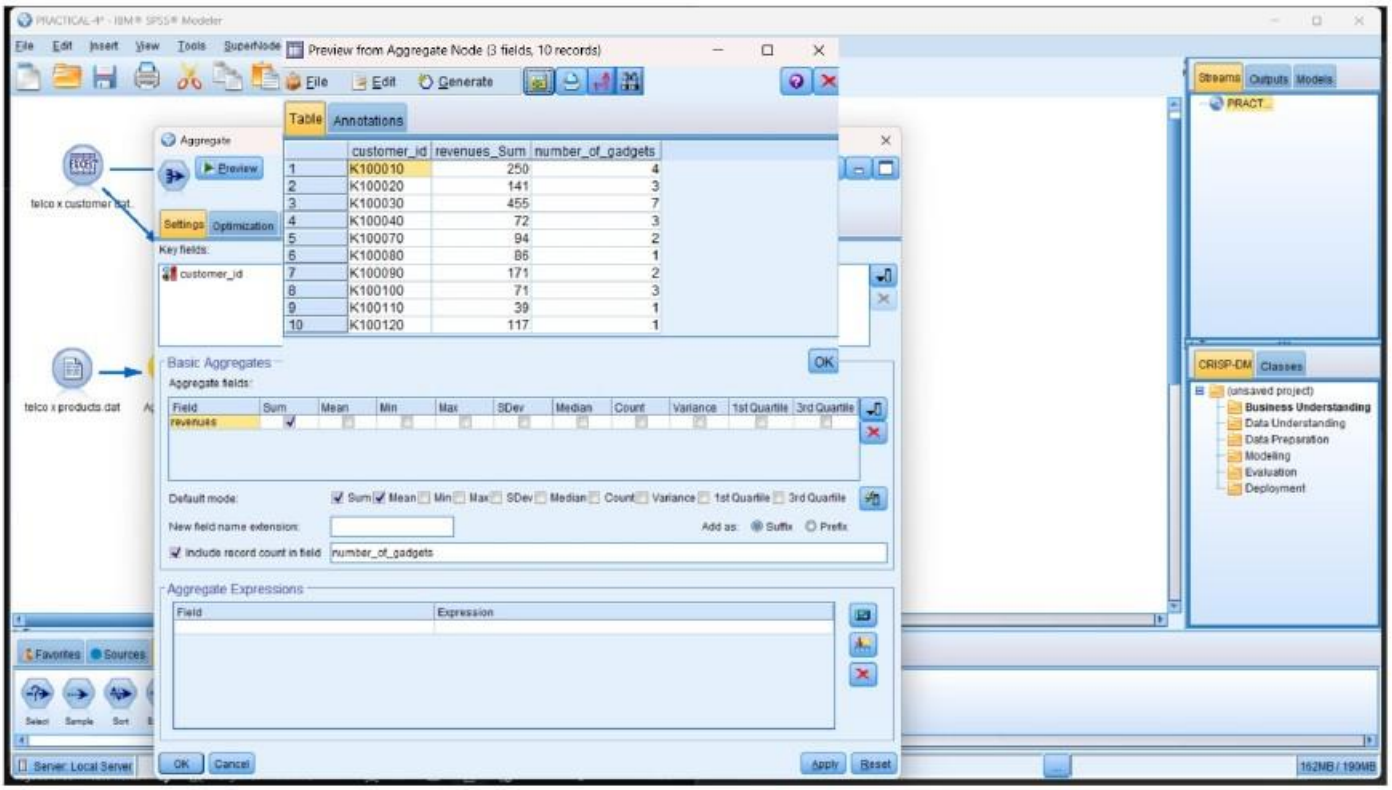
- A customer has as many records as he has gadgets.
- Click "**0K"**, to close the Preview output window.

Step 2: From the Record Ops palette, add an Aggregate node downstream from the Var. File node.

- After that, from the **Record Ops** palette, add an **Aggregate** node downstream from the **Var. File** node.
- Double-click to edit the **Aggregate** node.
- Under **Key fields**, click on , then select **customer_id**. Then click **"OK"**.
- In the **Basic Aggregates** section, click on , select **revenues**, click 0K,
- In **revenues**, select only the **Sum** as the aggregate statistic.
- After that, in **Include record count in field**, rename the field to

number_of_gadgets.

- After, Click **Preview**.
- After that, the results appear as follows:



- Click **OK**, to close the Preview output window.
- Click **OK**, to close the Aggregate dialog box.

TASK – 4

❖ Expand a Product Field into Flag Fields with One Record per Customer

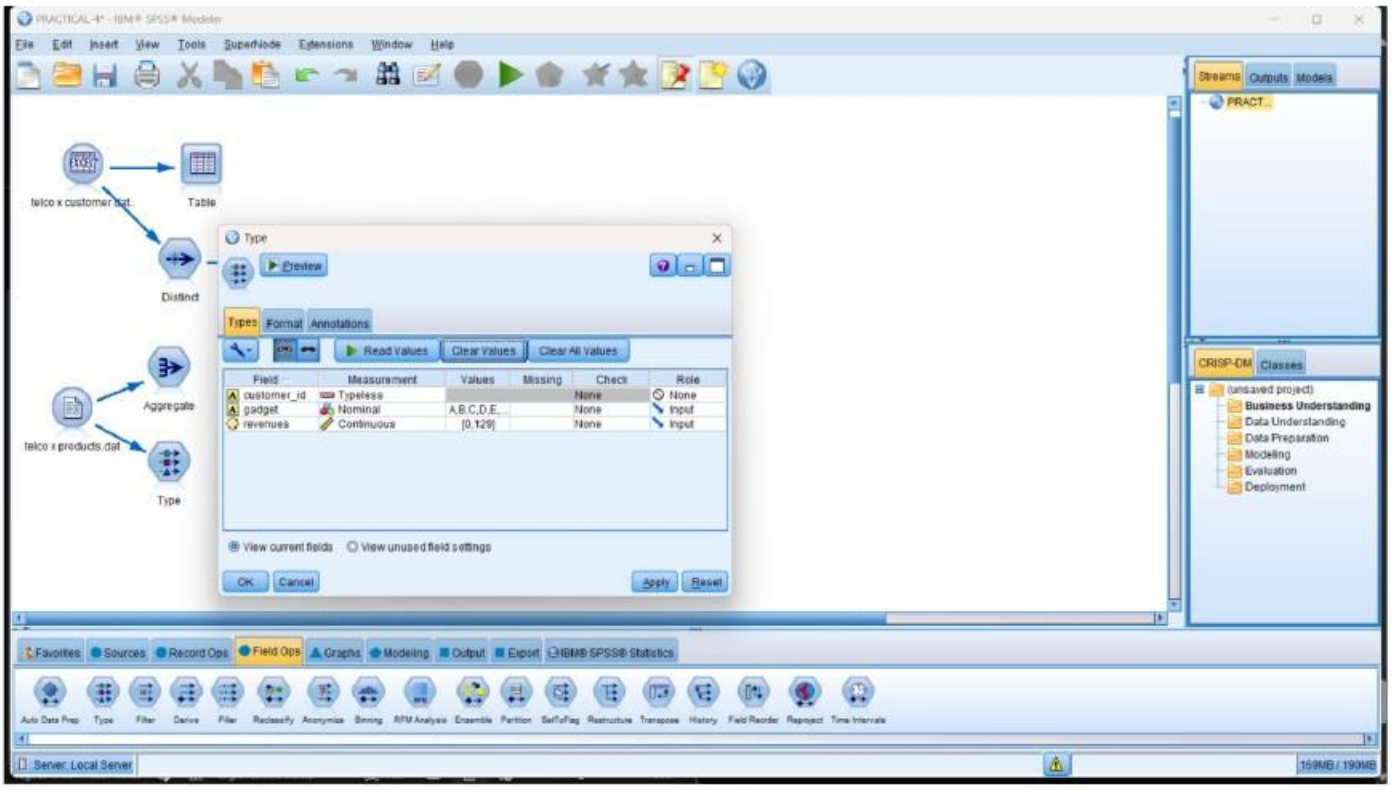
Definition: - This task demonstrates how to **convert gadget values into separate flag fields and create one record per customer using the SetToFlag node.**

Step 1: For this:

- From the **Field Ops** palette, drag and drop **Type** node downstream from the

Var. File node.

- Double-click to edit the **Type** node and click **Read Values**.



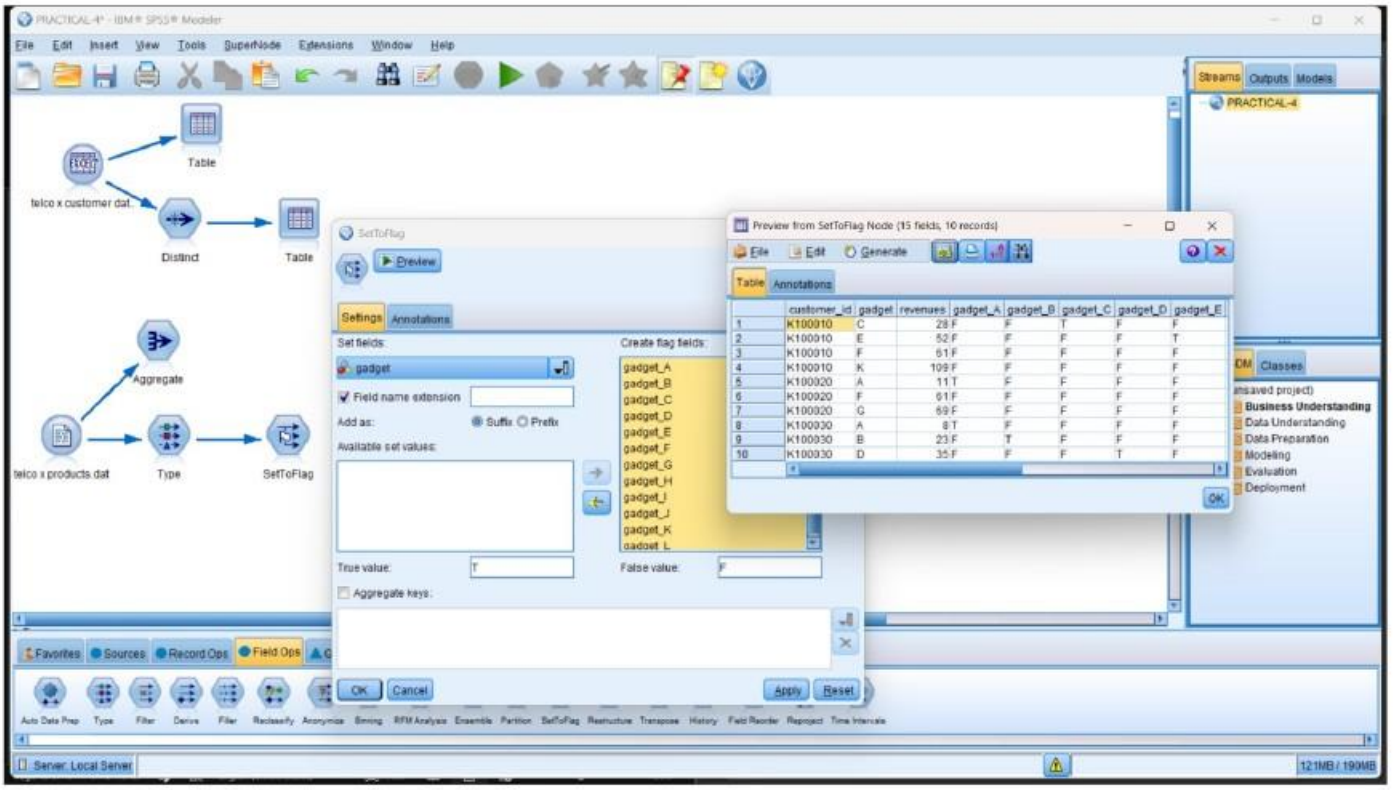
- Click **OK**, to close the Type dialog box.

Step 2: From the **Field Ops** palette, add a **SetToFlag** node downstream from the **Type node**.

- Double-click to edit the **Aggregate** node.
- Under **Set fields,** click and select **gadget**.
- Under **Available set values**, select all values and click  to mode them into the **Create flag fields** box.
    - At this point, the flag fields have been created, but there is not yet one record per customer. To verify that a customer still has

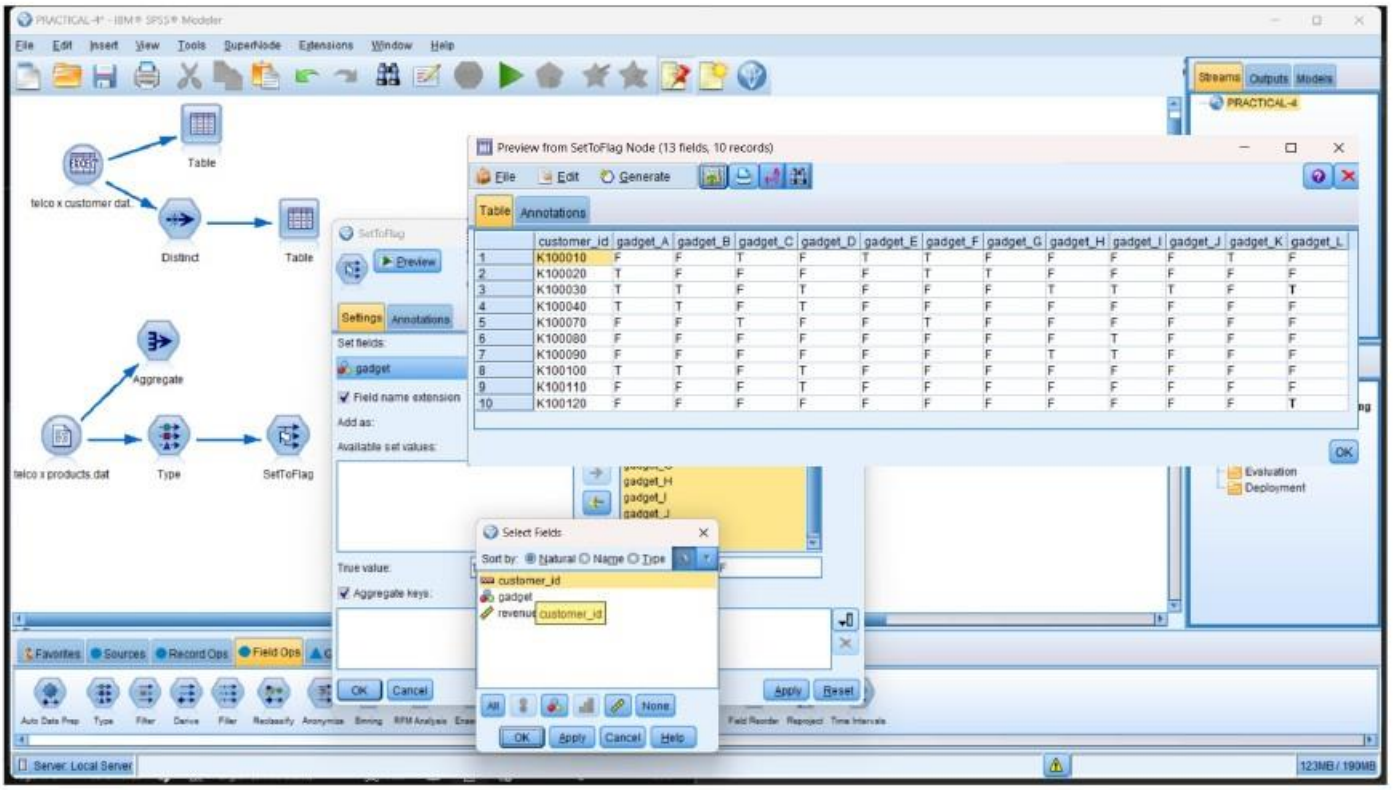
multiple records, you will preview the data.

- After that, click on **Preview**.
- The results appear as follows:
    - A customer still has as many records as he has gadgets one value is true for a record.



- Click **OK**, to close the Preview output window.

Step 3: In the bottom, click on the Check box of the **Aggregate keys** and then right side click on  and then double-click **customer_id** to select it as the field to aggregate on. And after that, Click Preview.



- Click **OK**, to close the Preview output window.
- Click **OK**, to close the SetToFlag dialog box.

Step 4: Click **OK**, to close the Type dialog box. To save the stream, go to the **File** menu and click **Save As**. Choose the desired location/path, enter the file name (like: PRACTICAL-4), and then click **Save**.

Result: You removed duplicate records in the customer dataset and transformed a transactional dataset into a dataset that has one record per customer. This enables you to merge these files in the next stage.

Practical - 5

Definition: - This practical explains how to use the IBM SPSS Modeler user interface to create, edit, modify, and save streams. It also demonstrates how to work with different nodes such as Excel, Sample, Derive, Histogram, Statistics Export, Select, and Table nodes

Outcomes/Learning: - Learners will become familiar with the IBM SPSS Modeler user interface and will understand how to create and modify streams, connect different nodes, generate outputs, add comments, and use table output to generate a Select node.

Required Tool: - The required tool will be "**IBM SPSS Modeler"**.

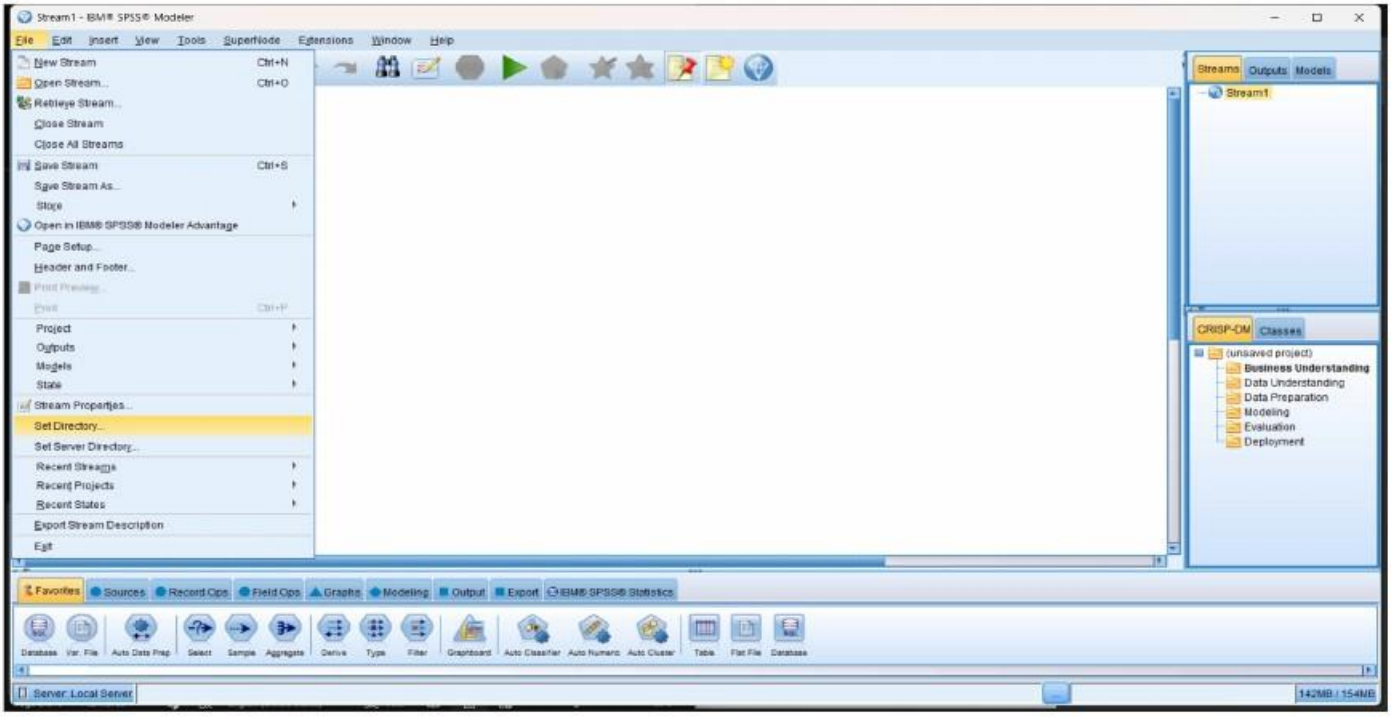
Purpose: - You work for a telecommunications firm and have to combine a number of datasets into a single dataset for analyses and modeling later.

TASK – 1

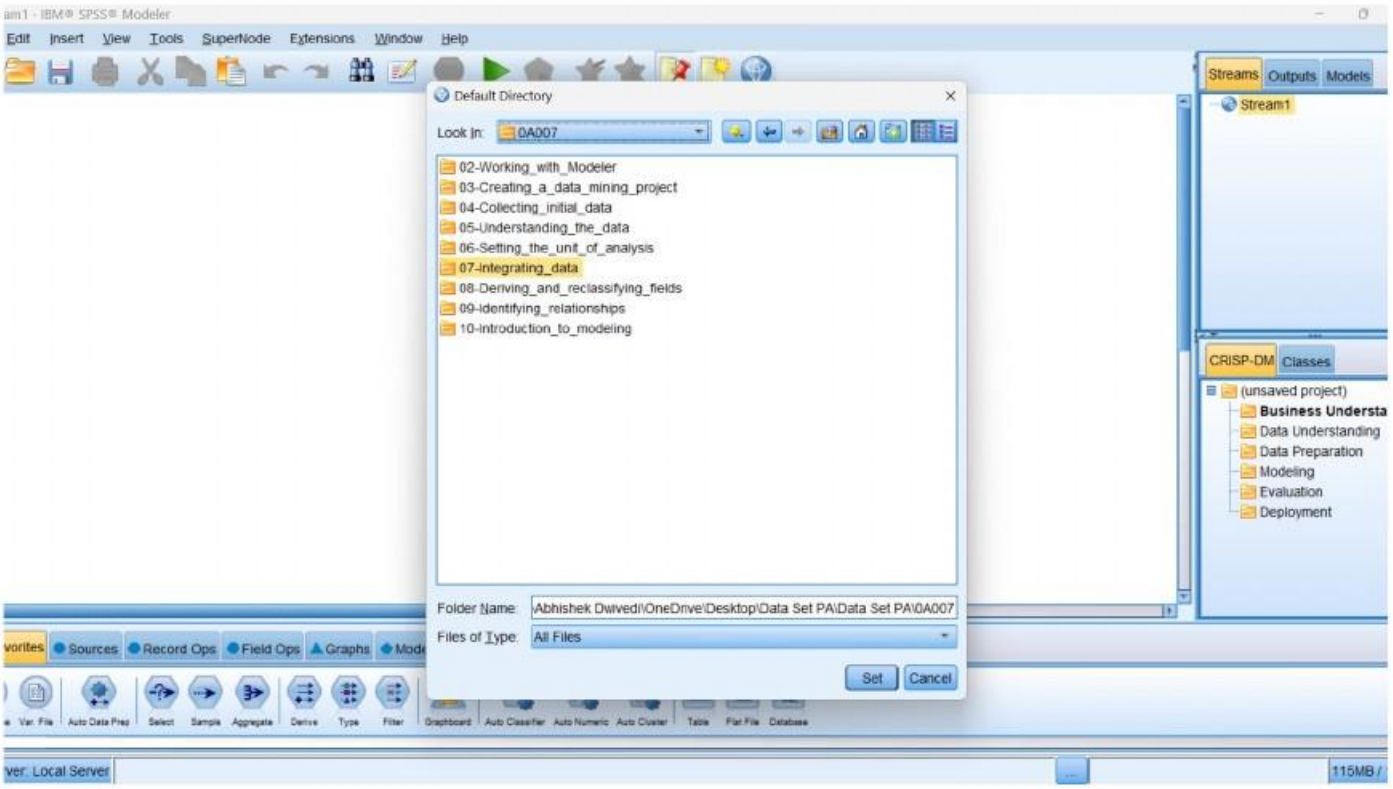
❖ Start IBM SPSS Modeler and set the working folder.

Definition: - This task explains how to open IBM SPSS Modeler and set the required working folder before creating a stream.

Step 1: - From the File menu, click **Set Directory**.



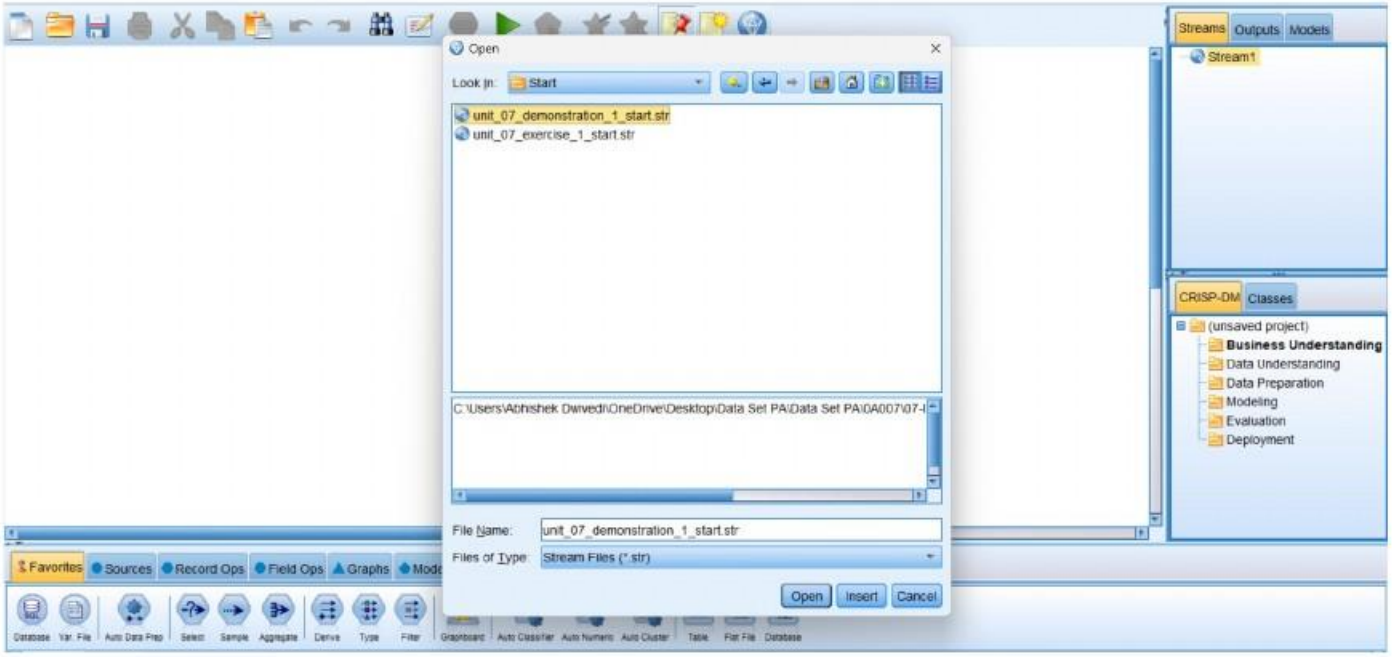
Step 2: - Beside Look in, navigate to **C:\Users\Mr. Abhishek\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.



TASK – 2

- **Open** the **stream**.

Definition: - From the **File** menu, click **Open Stream,** navigate to **C:\Training\0A007\07-Integrating_data\Start**, click **unit_07_demonstration_1_start.str** and then click **Open**.

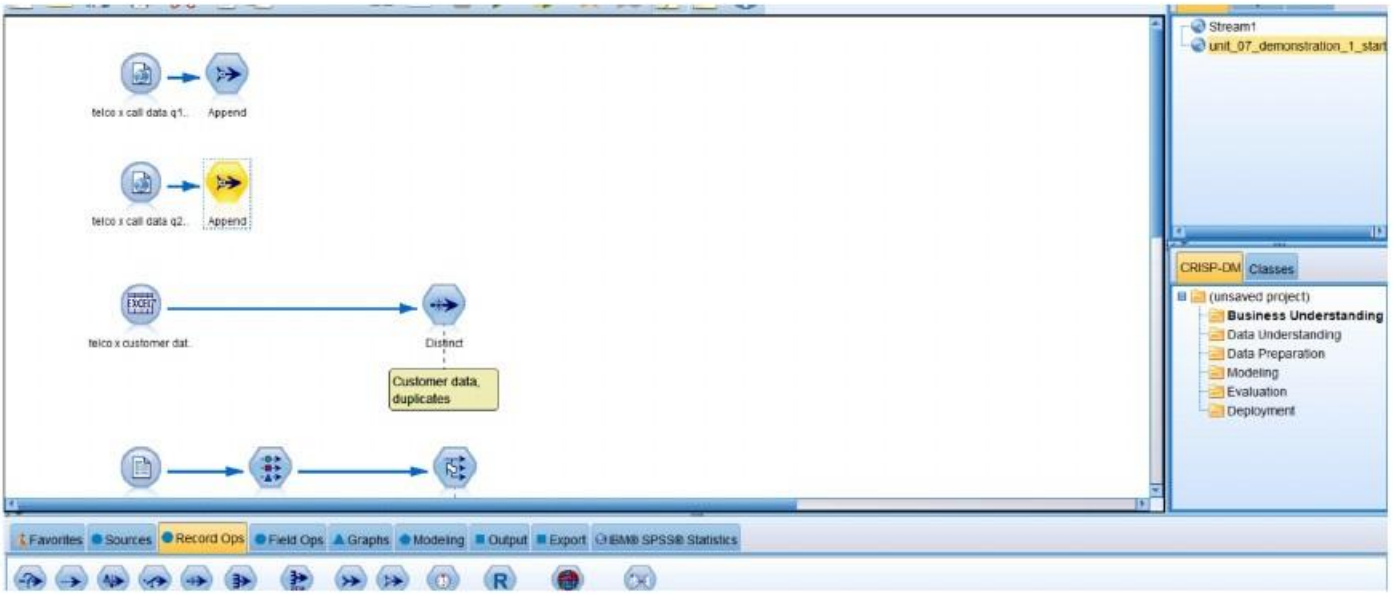


TASK – 3

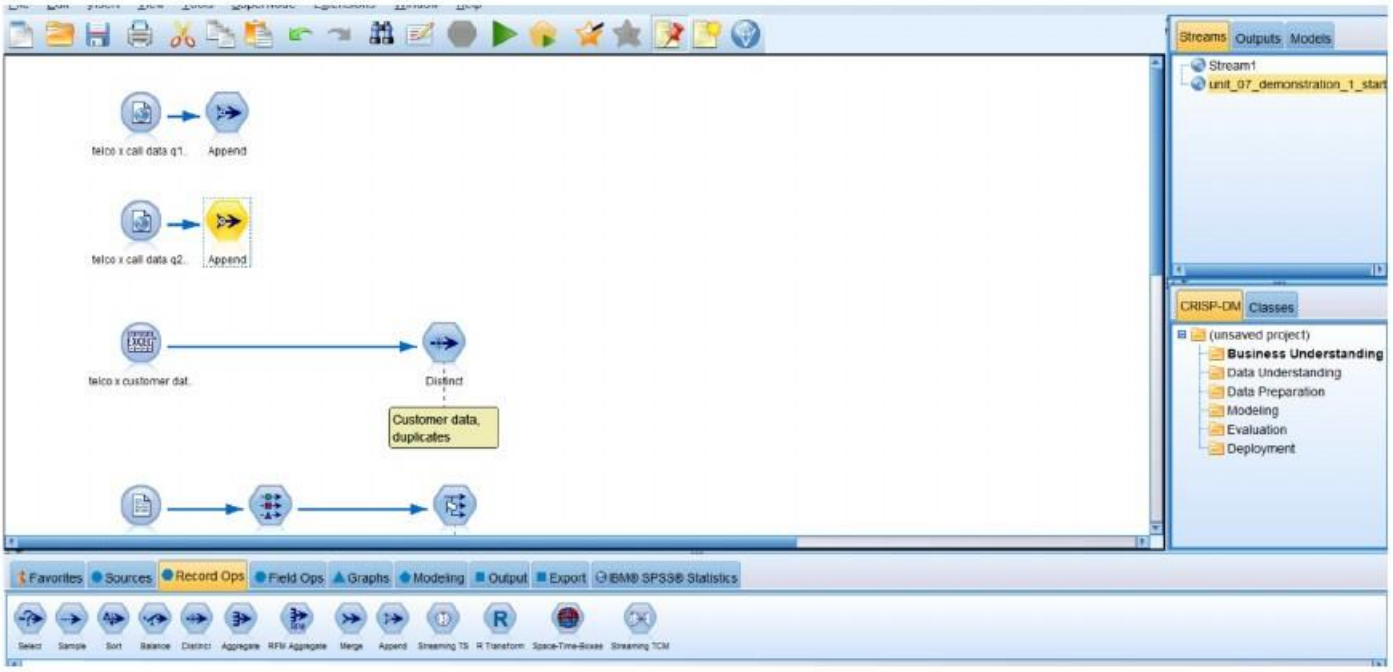
- Append records **from two datasets**.

Definition: - This task explains how to add an Append node from Records Ops.

Step 1: You will combine two IBM SPSS Statistics .sav files that store call detail records into a single dataset.
From the Record Ops palette, place an Append node on the stream canvas to the right of the two Statistics File sources nodes named telco x call data q1.sav and telco x call data q2.sav.

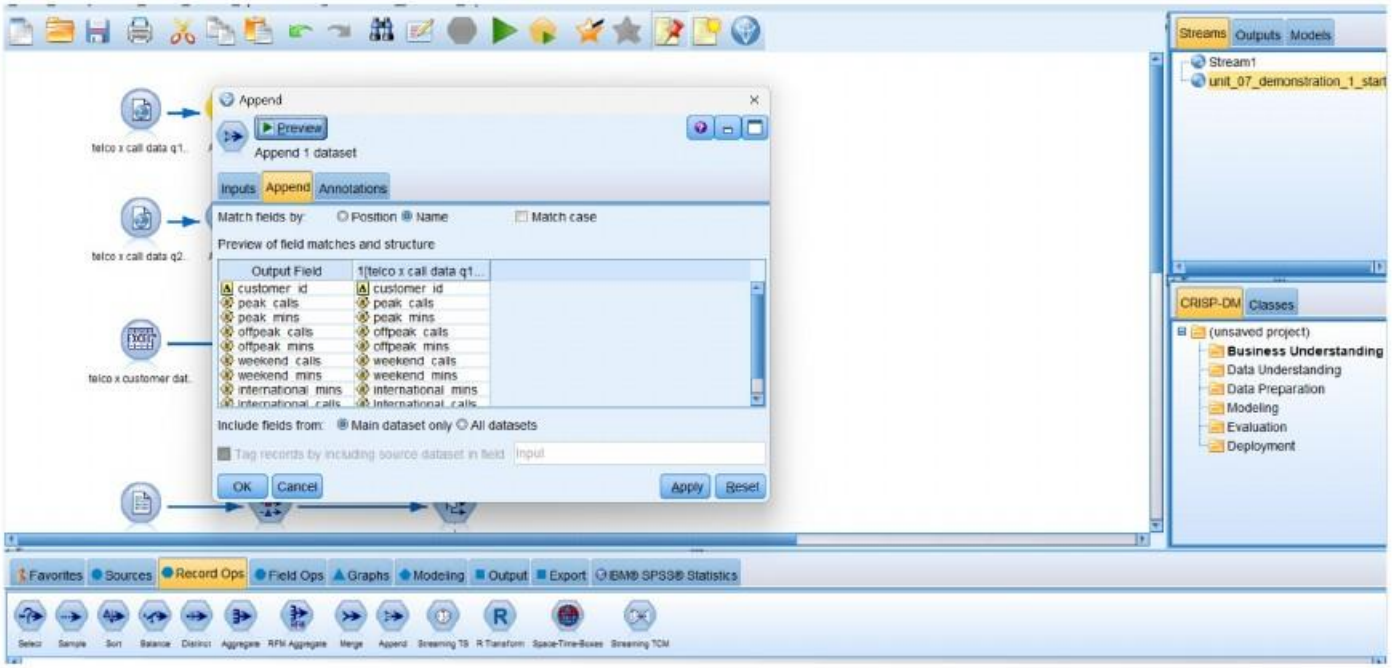


Step 2: Connect the **Statistics File** node named **telco** x **call data q1.sav** to the **Append Node**.

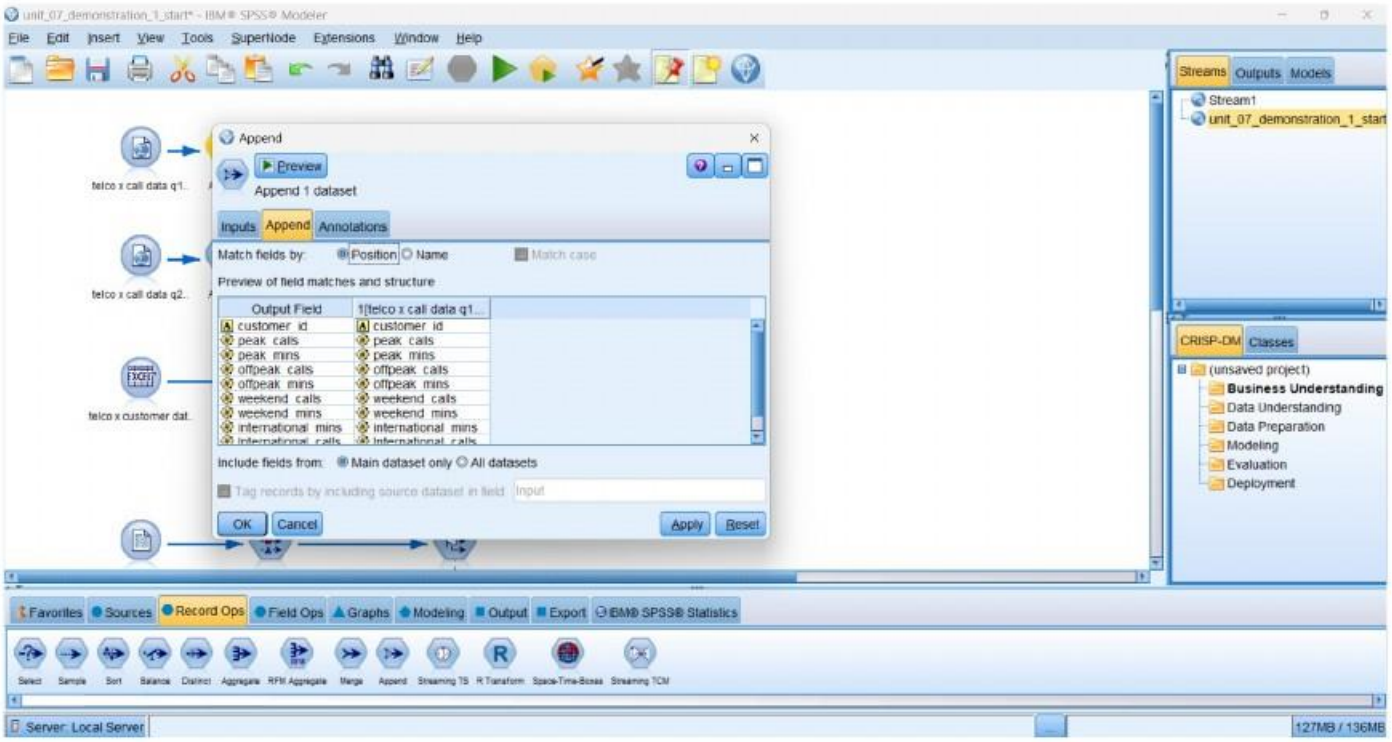


Step 3: Connect the **Statistics File** named **telco x call data q2.sav** to the **Append** node.

Step 4: Edit the **Append** node.



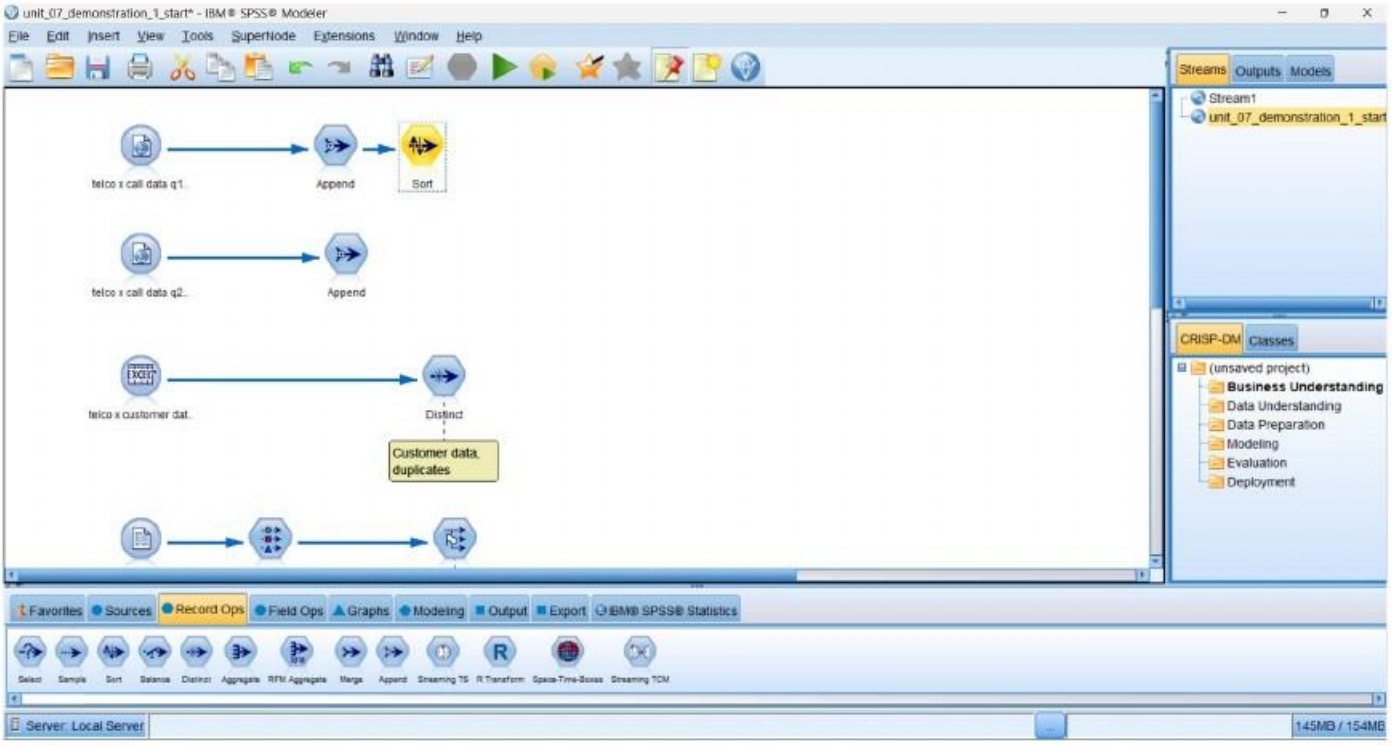
Step 5: Edit the **Append** node, and beside **Match fields by**, select **Position.**



Step 6: Enable the **Tag records by including source dataset in field** option.

Step 7: Rename the **Input** field to **quarter**.

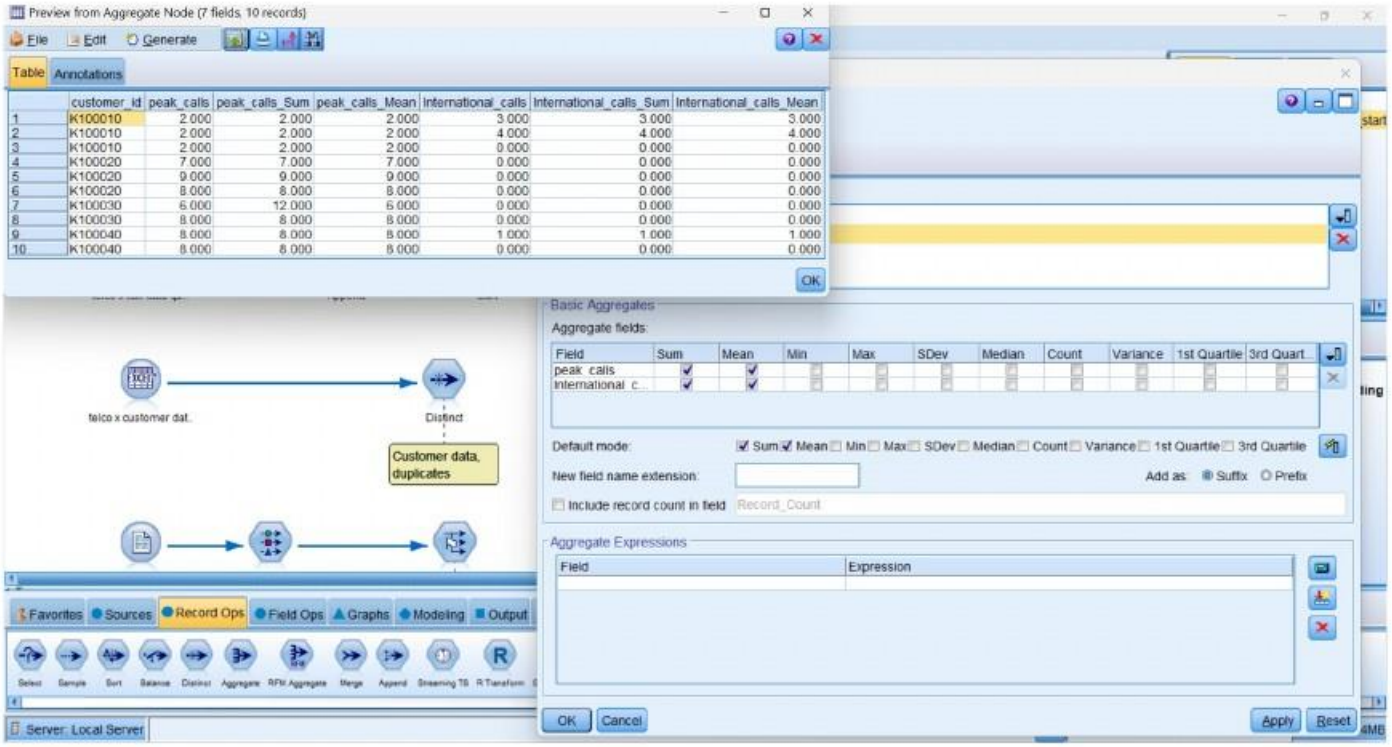
Step 8: When the Append node is processed, the first block of data that is output comes from the first dataset, followed by a block of data from the second dataset. To have a better view on the data, you will sort the data on customer_id and month. From the **Record Ops** palette, add a **Sort** node downstream from the **Append** node.



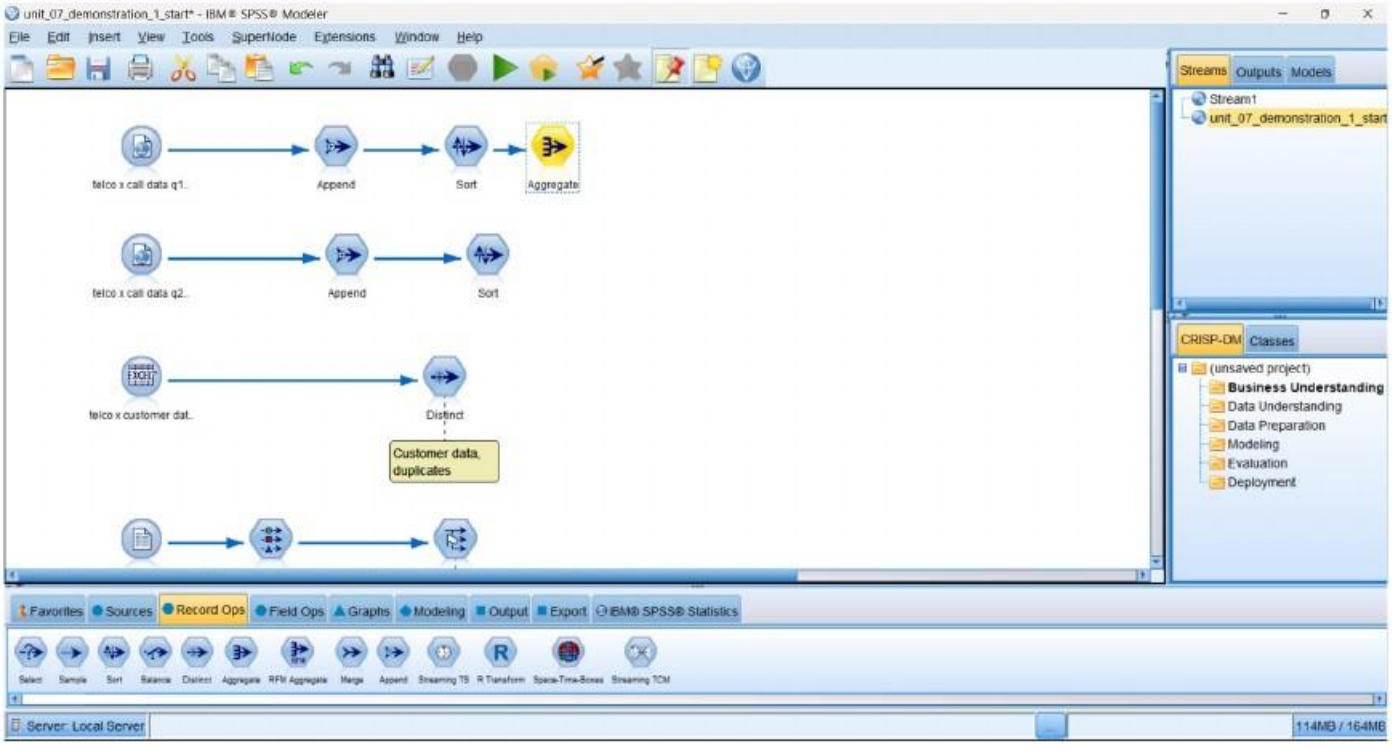
Step 9: Edit the **Sort** node, and under **Sort by**, Ctrl-click to select **customer_id** and **Month**.



Step 10: **Click Preview.**



Step 11: From the **Record Ops** palette, add an **Aggregate** node downstream from the **Sort** node.



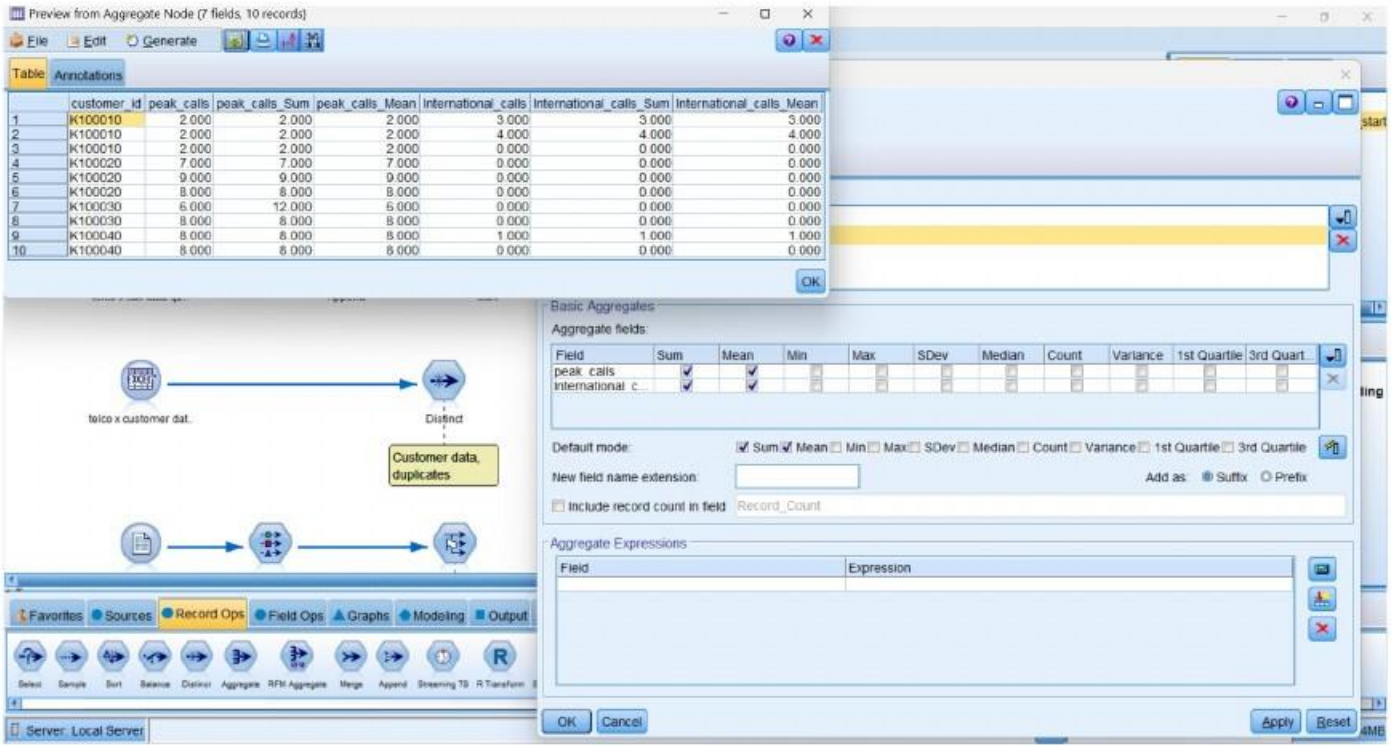
Step 11: Edit the **Aggregate** node, and under **Key fields**, select **customer_id**.



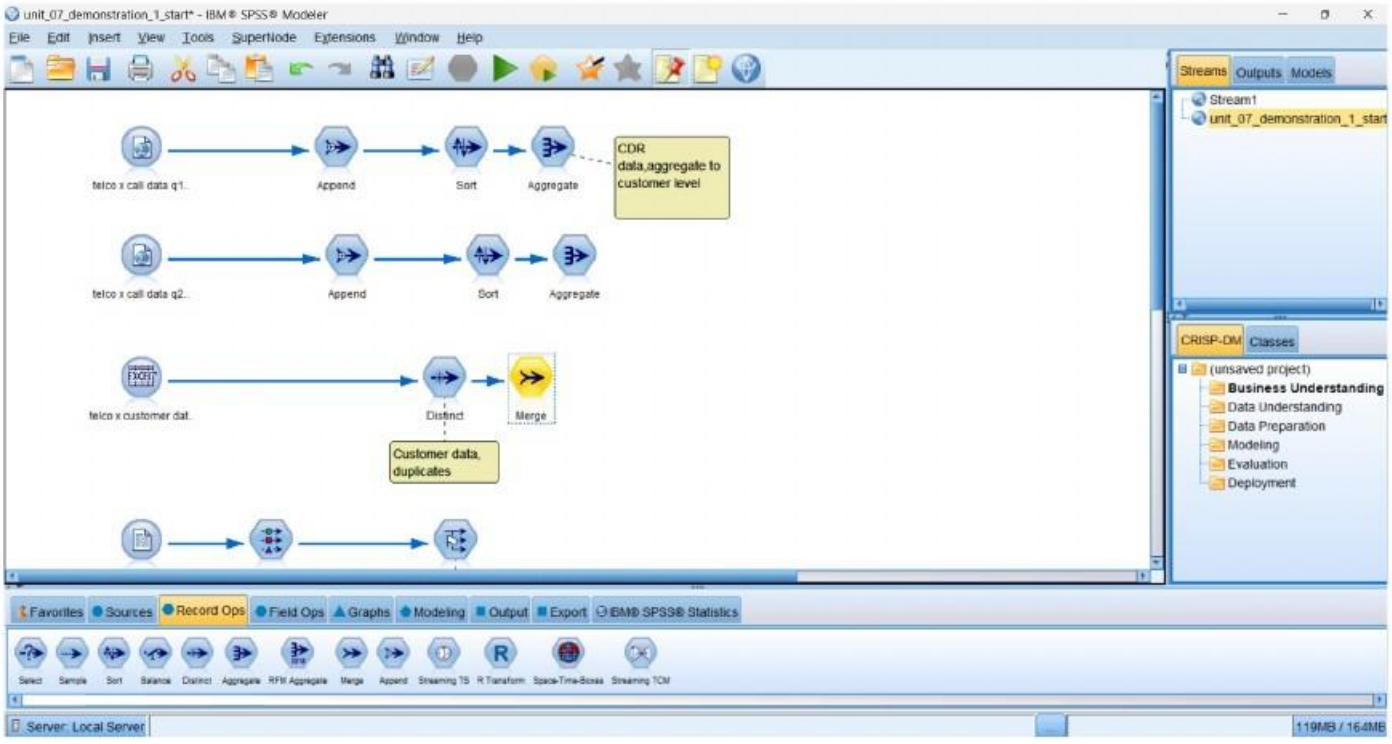
Step 12: Under **Aggregate fields**, click **peak_calls** and drag to **international_calls**, and then click **Mean** for any field, to deselect Mean for all selected fields.



Step 12: Click **Preview.**



Step 13: Add the following text as a comment to the **Aggregate** node: **CDR data, aggregated to customer level**.



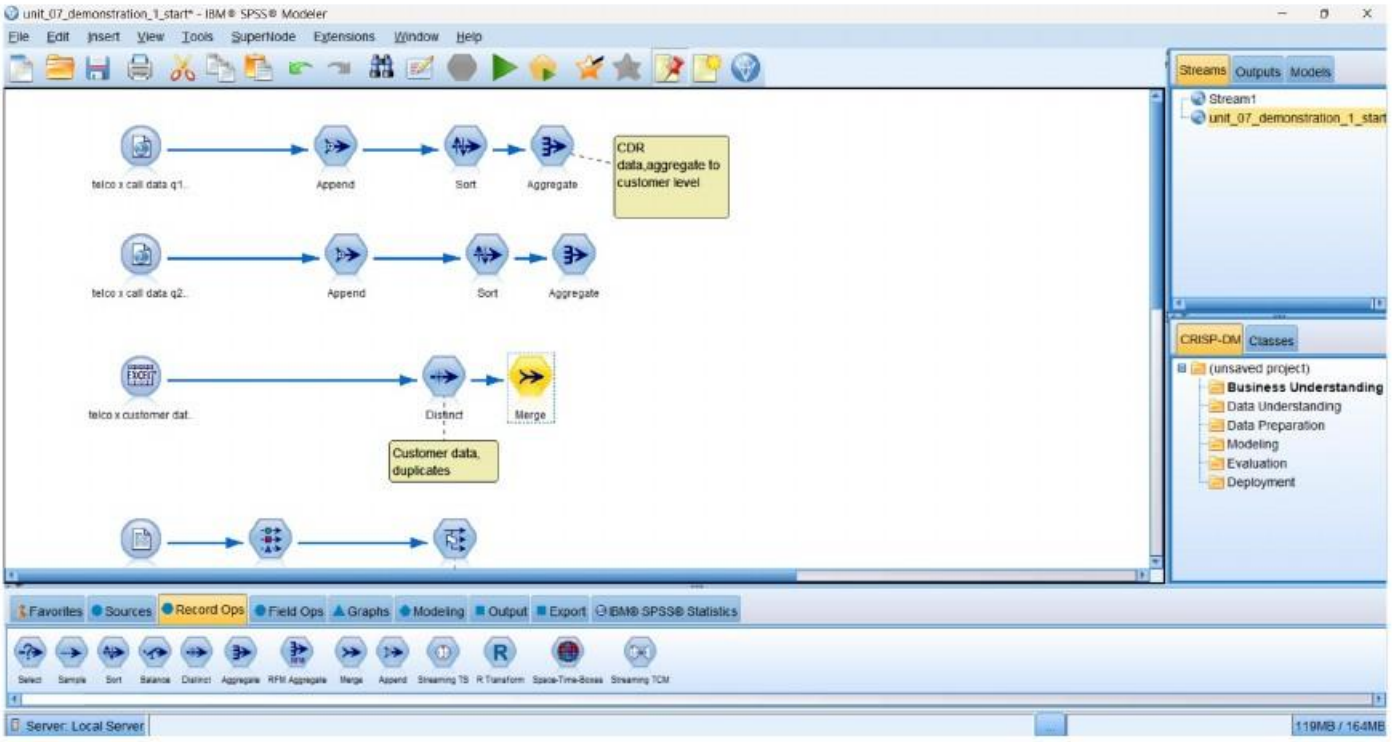
TASK – 4

❖ Merge fields from three datasets.

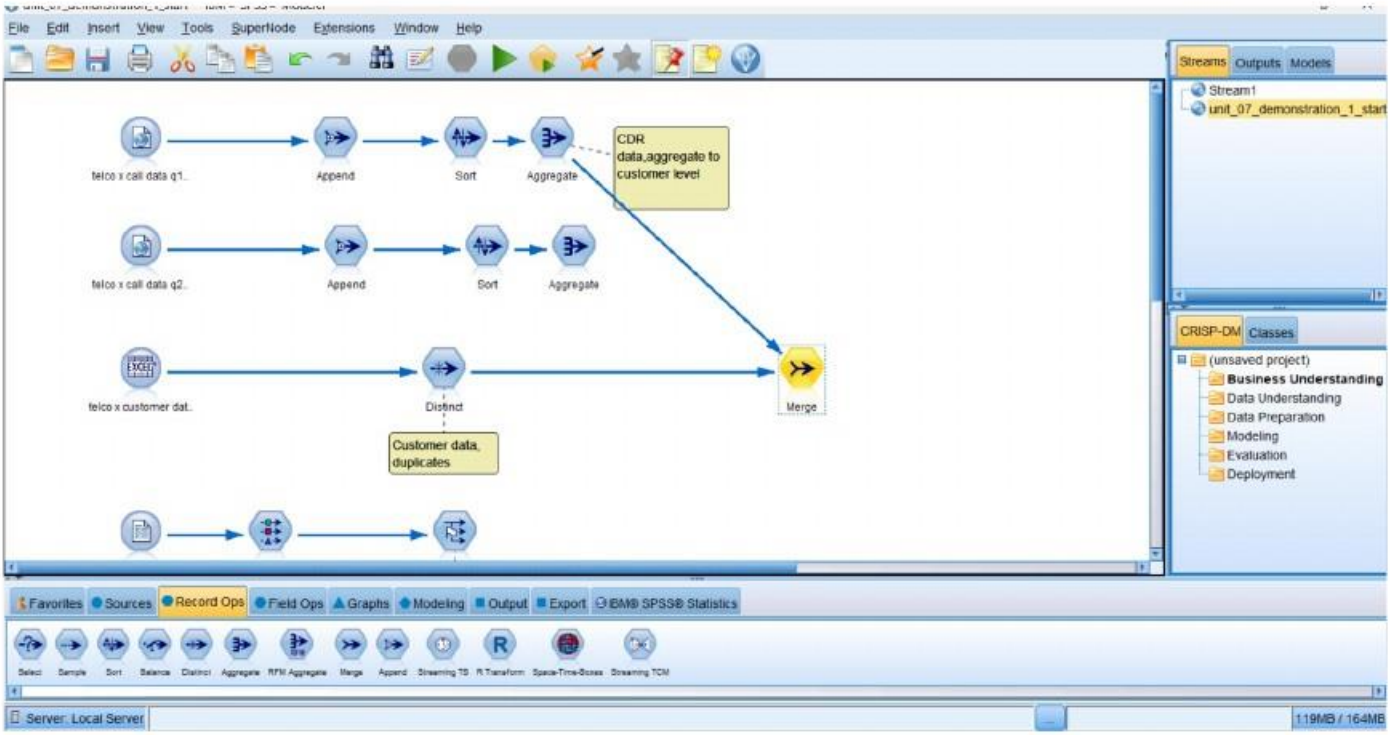
Definition: - Different datasets have to be merged:

- customer data (with duplicate records removed by using a Distinct node)
- call detail records (aggregated to a 6 month period in the previous task)
- product data (transformed into a dataset with one record per customer using a SetToFlag node).

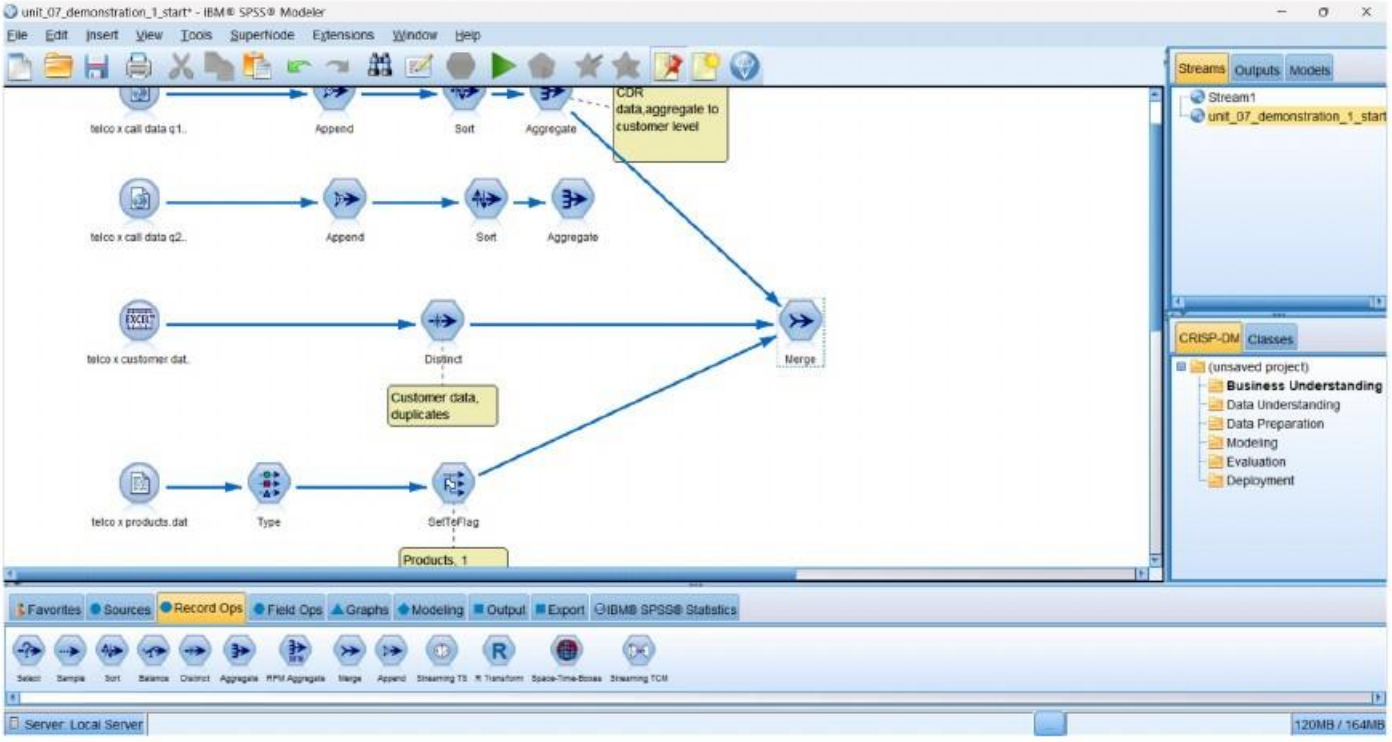
Step 1: From the **Record Ops** palette, drag a **Merge** node to the right of the **Distinct** node (do not connect the nodes - we want to explicitly connect the nodes to be in control of the order of connections).



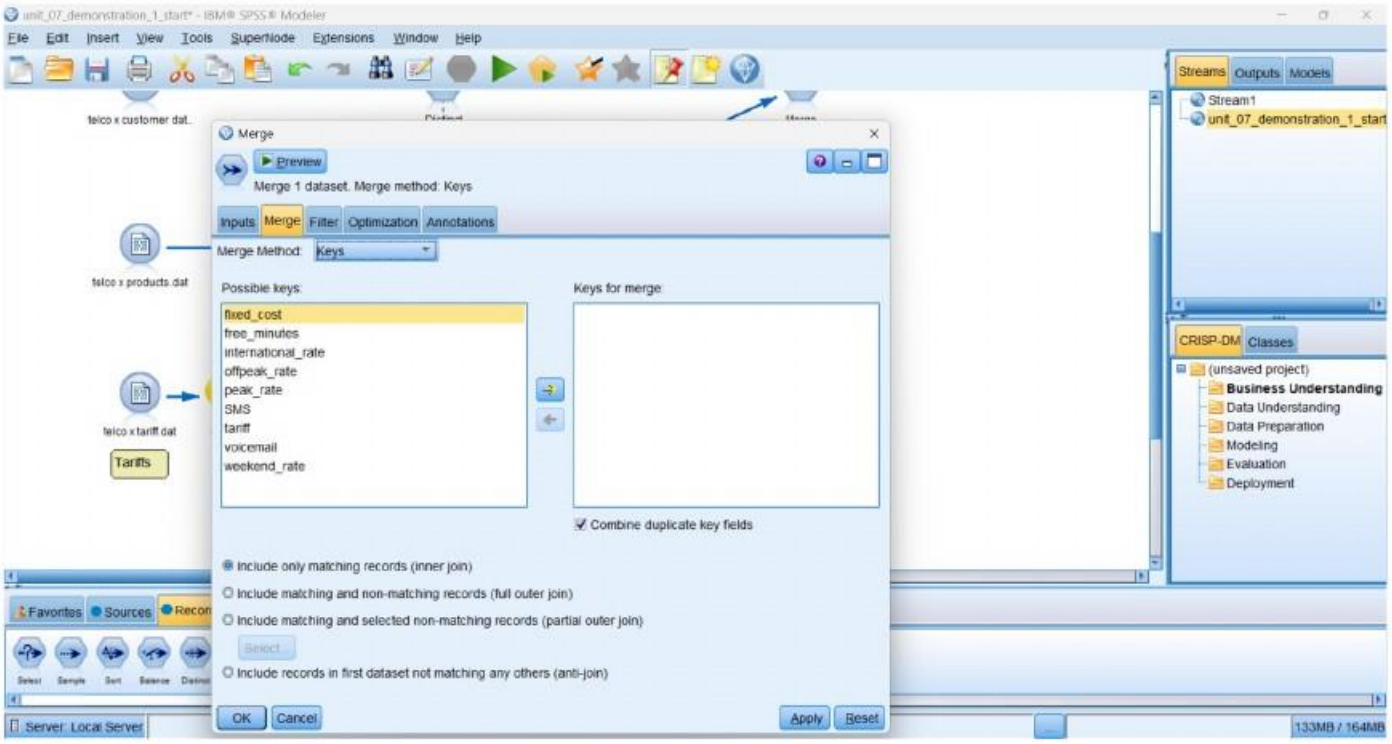
Step 2: Connect the **Aggregate** node to the **Merge** node.



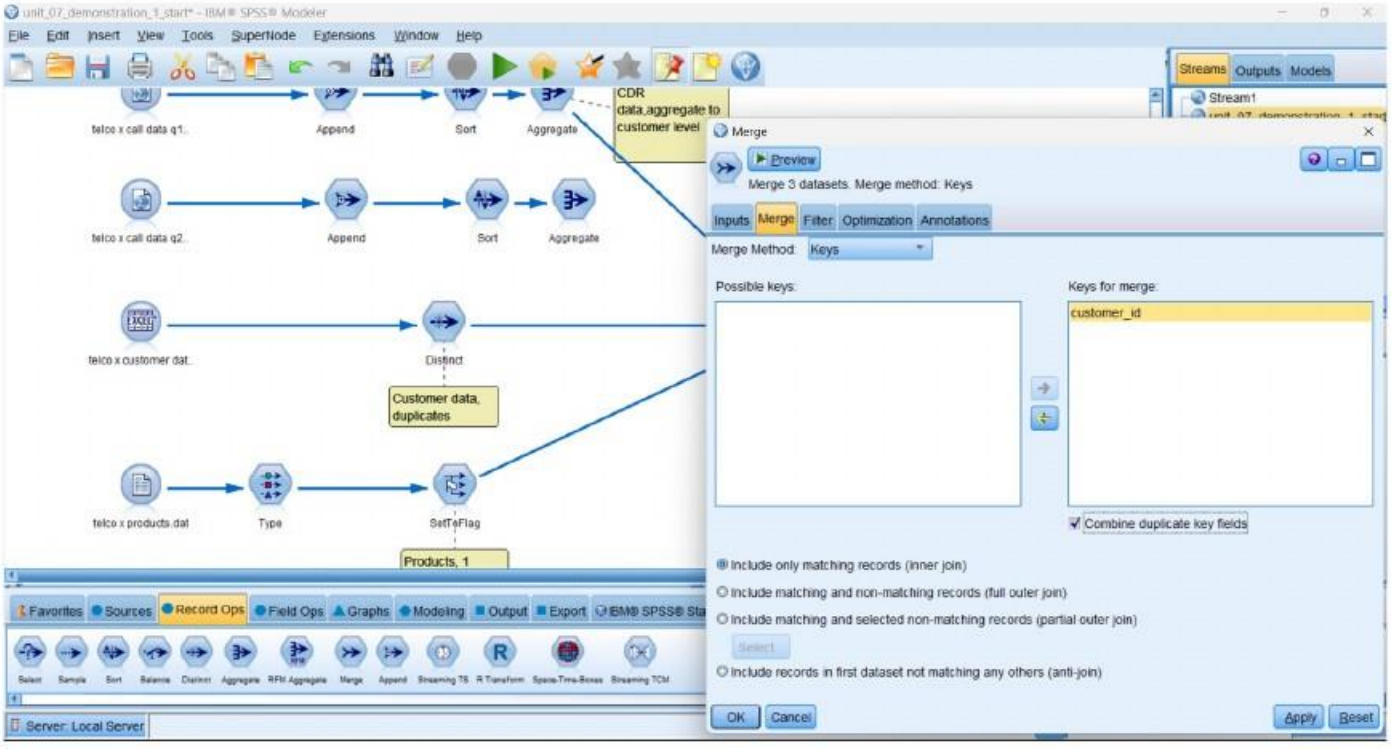
Step 3: Connect the **SetToFlag** node to the **Merge** node.



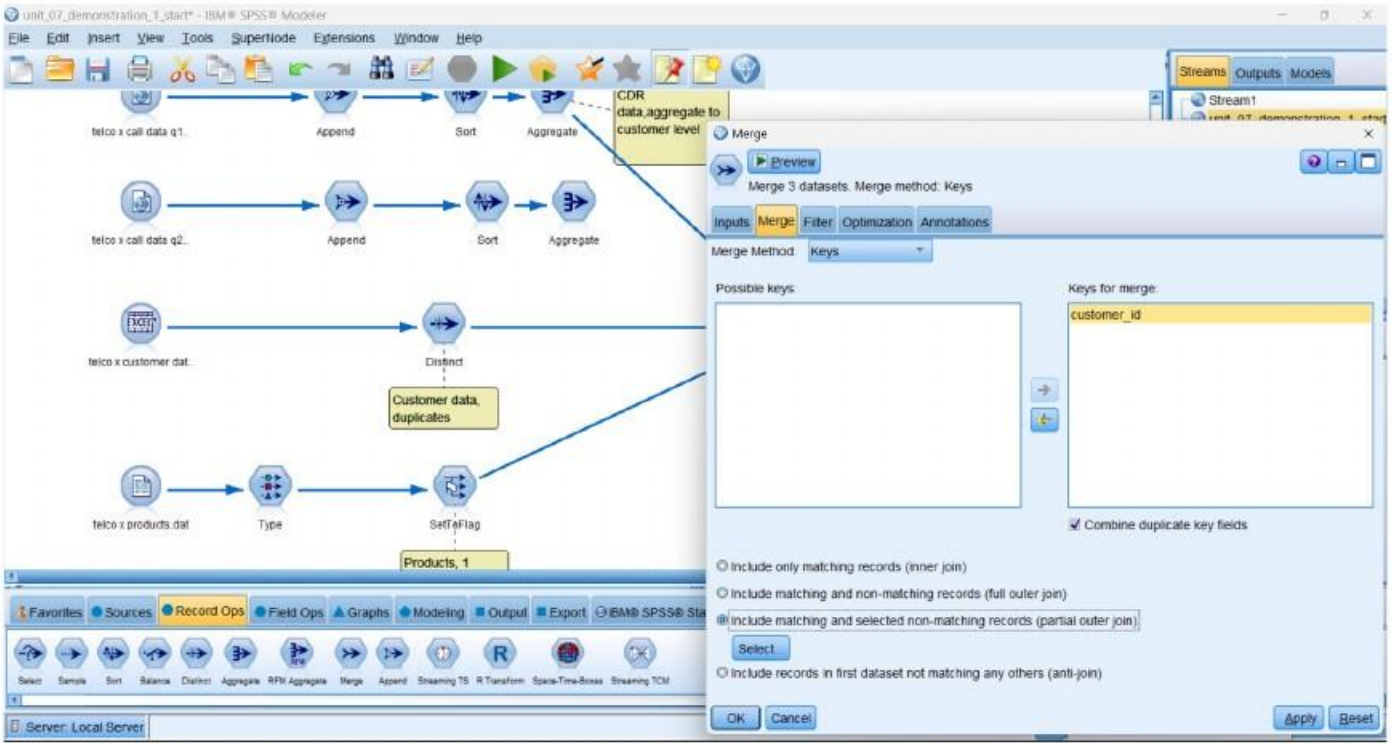
Step 4: For **Merge Method**, select **Keys**.



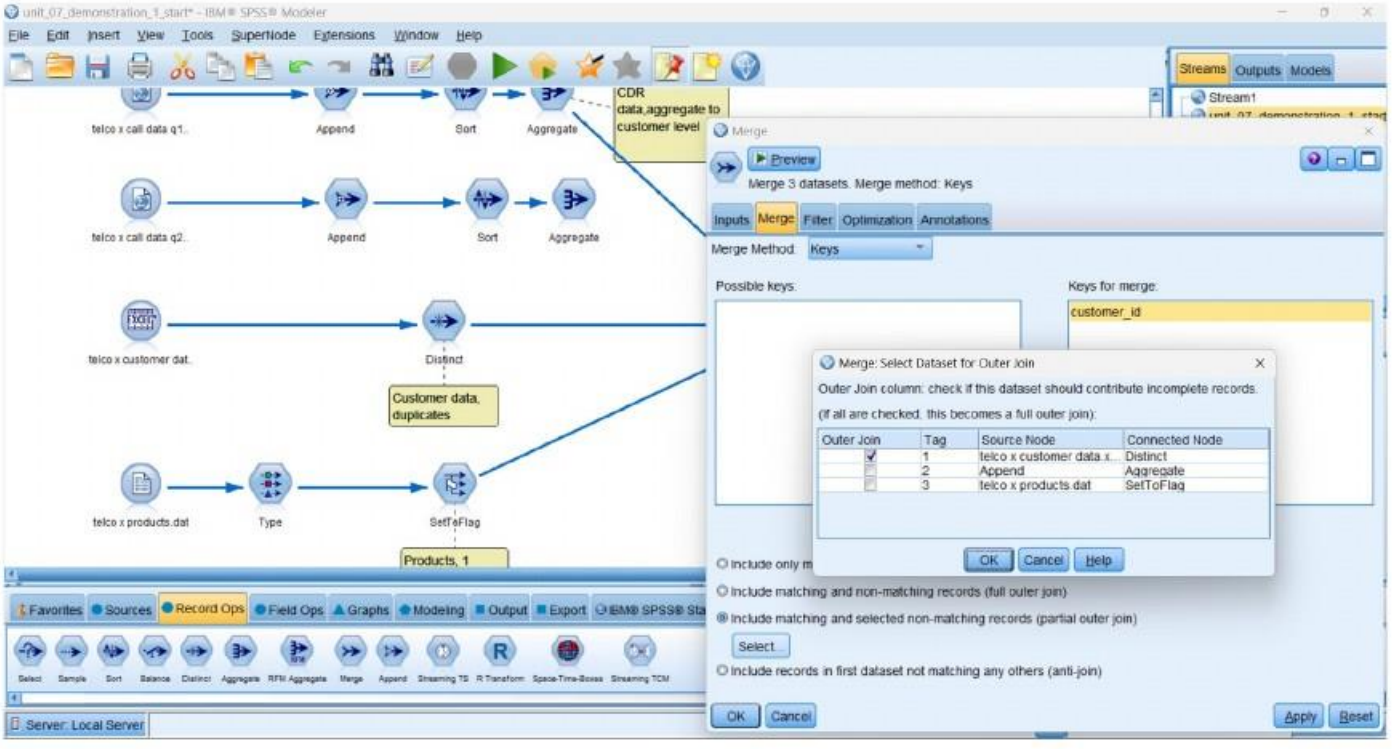
Step 5: Under **Possible keys,** click **customer_id**, and then click the **Add keys** button to move it into the **Keys for merge** box.



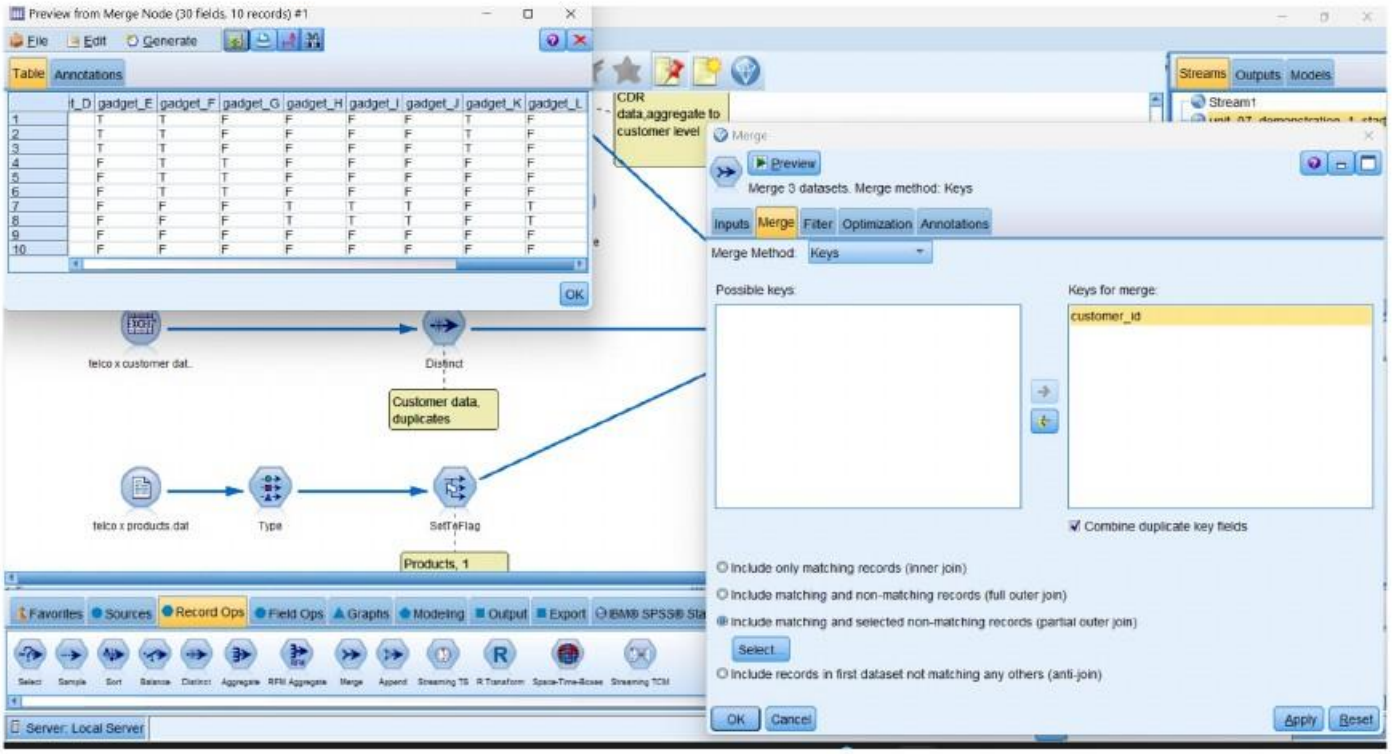
Step 6: Enable the **Include matching and selected non-matching records (partial outer join)** option.



Step 7: Click **Select**, ensure that **telco x customer data.xlsx** is the leading dataset.



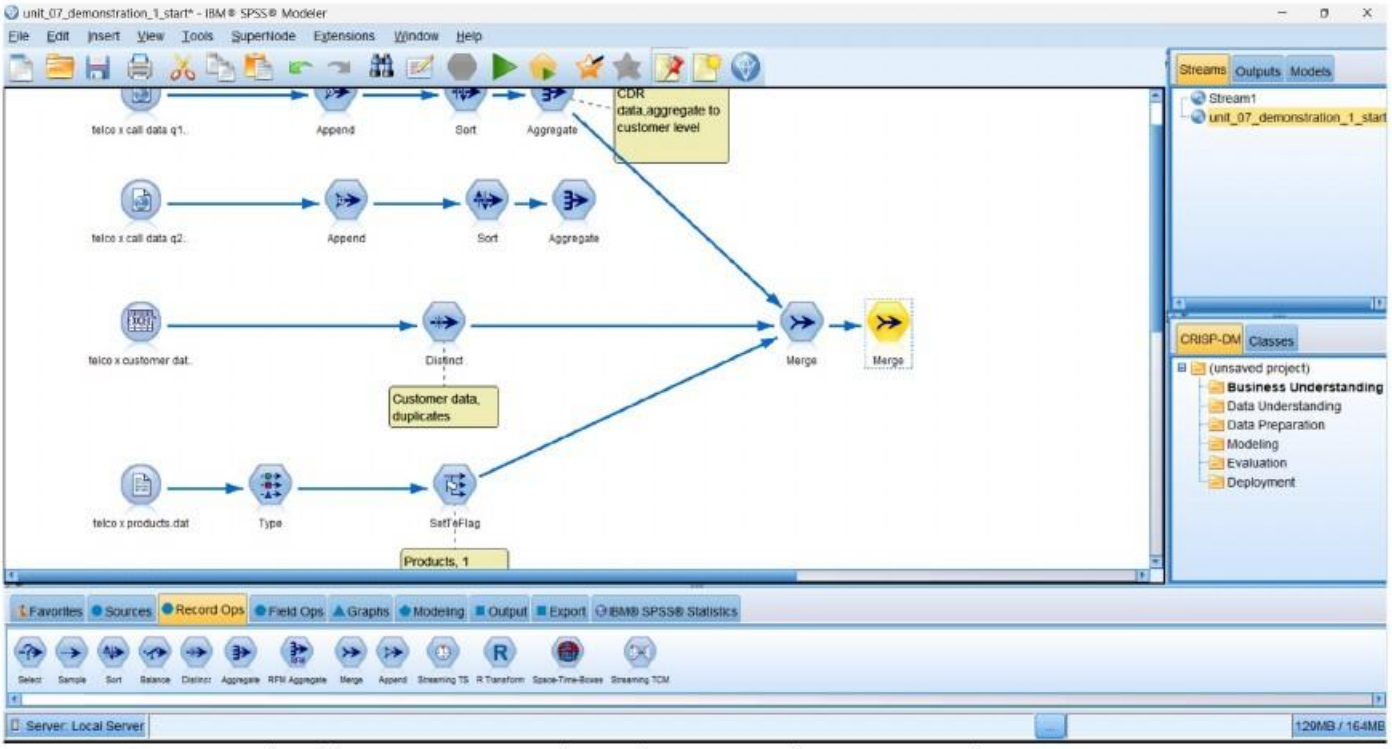
Step 8: Click **Preview**, and then scroll to the right, to the gadget fields.



TASK – 5

- **Enrich a dataset with** aggregated **data.**

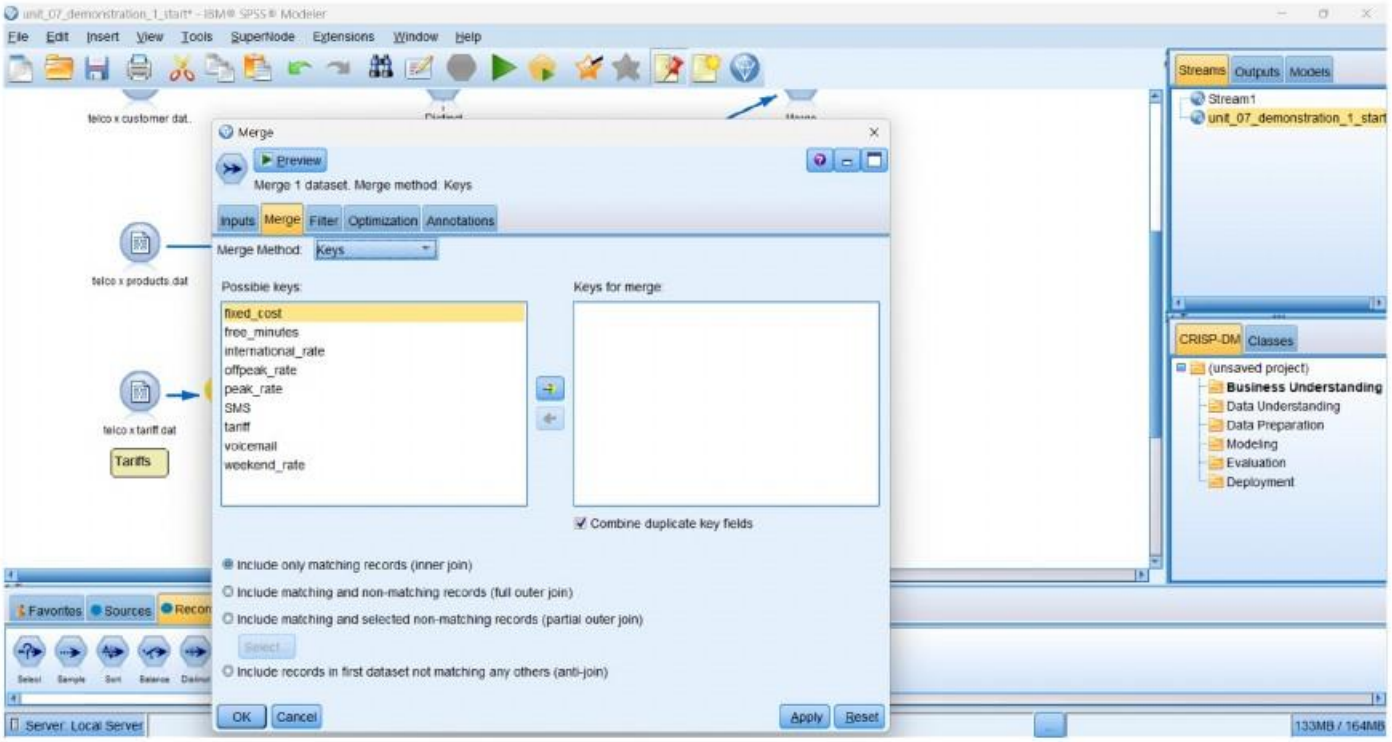
Step 1: From the **Record Ops** palette, place a second **Merge** node on the stream canvas, to the right of the data source **telco x tariff.dat** (do not connect it yet).



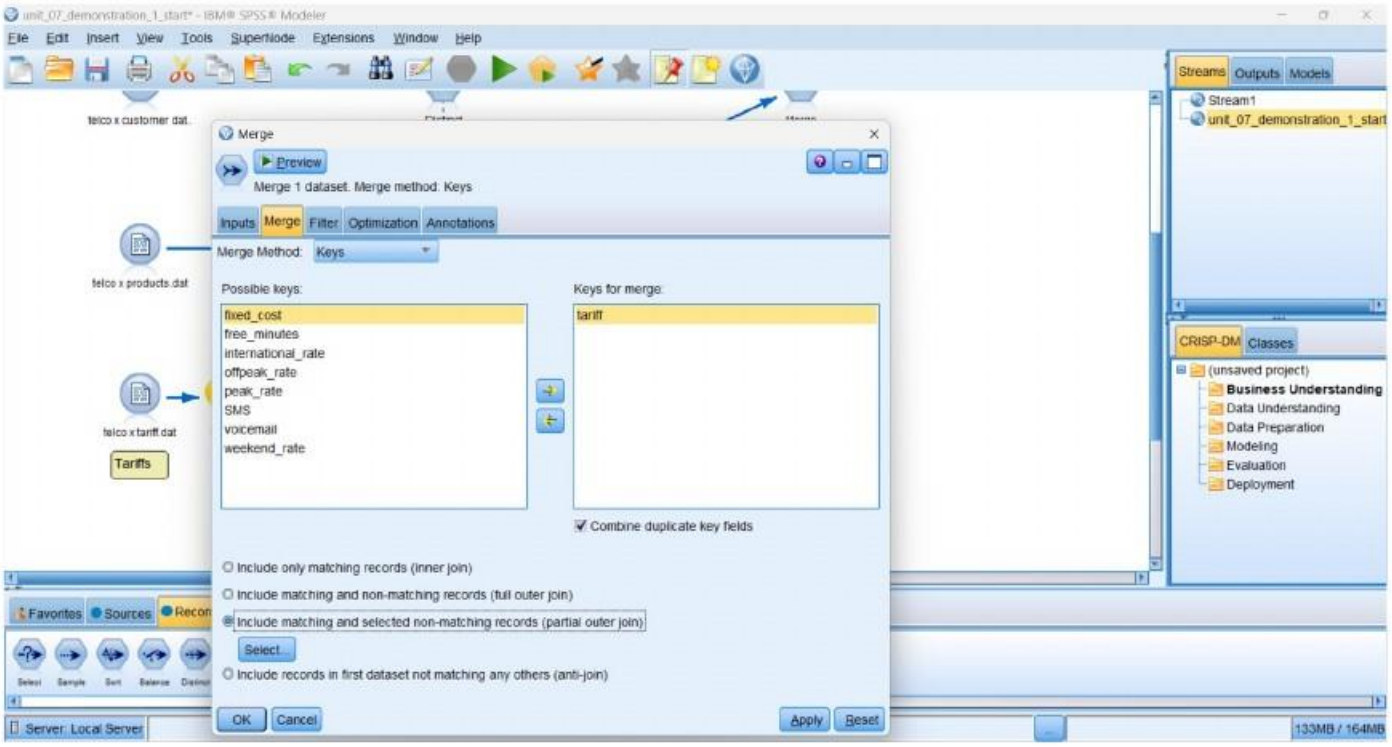
Step 2: Connect the **first Merge** node to the second **Merge** node.

Step 3: Edit the second **Merge** node.

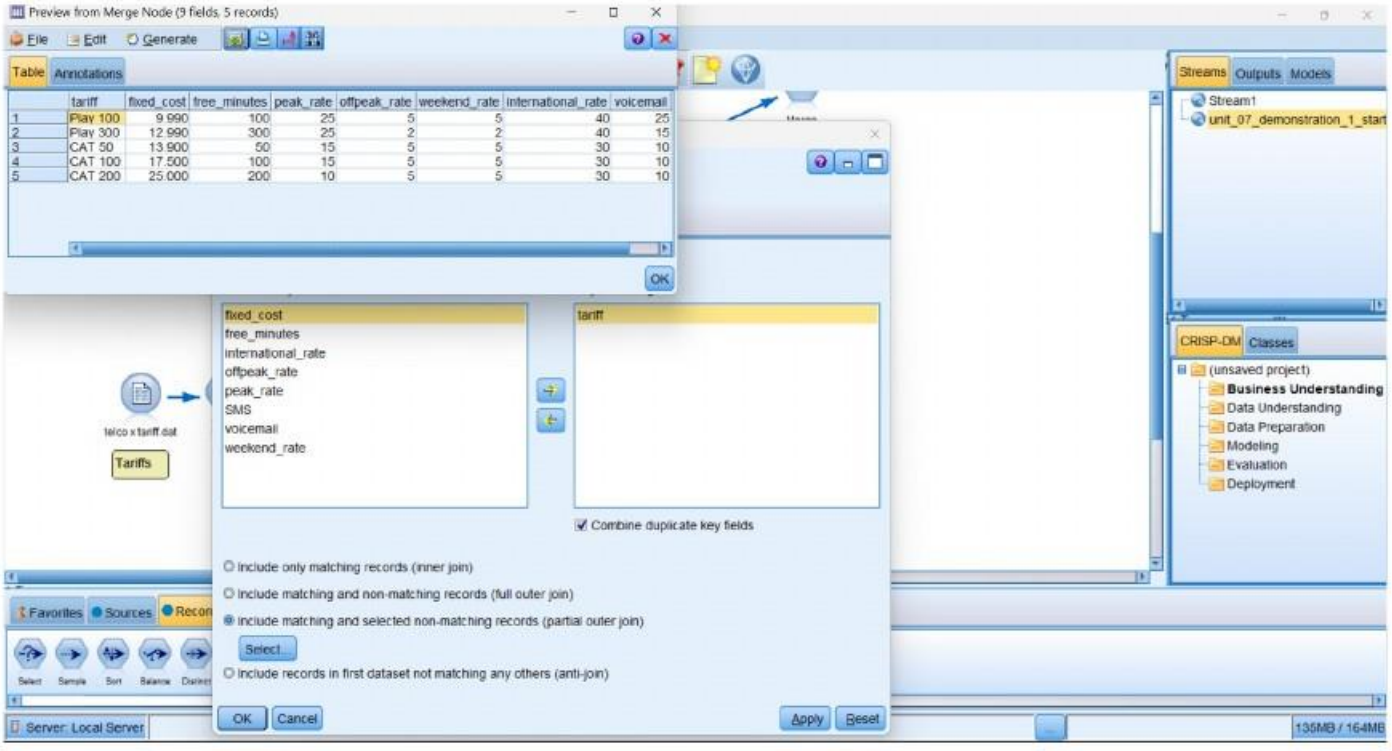
Step 4: For **Merge Method**, select **Keys**.



Step 4: Under **Possible keys,** click **tariff**, and then move it into the **Keys for merge** Box.



Step 5: Click **Preview**, and scroll to the right.



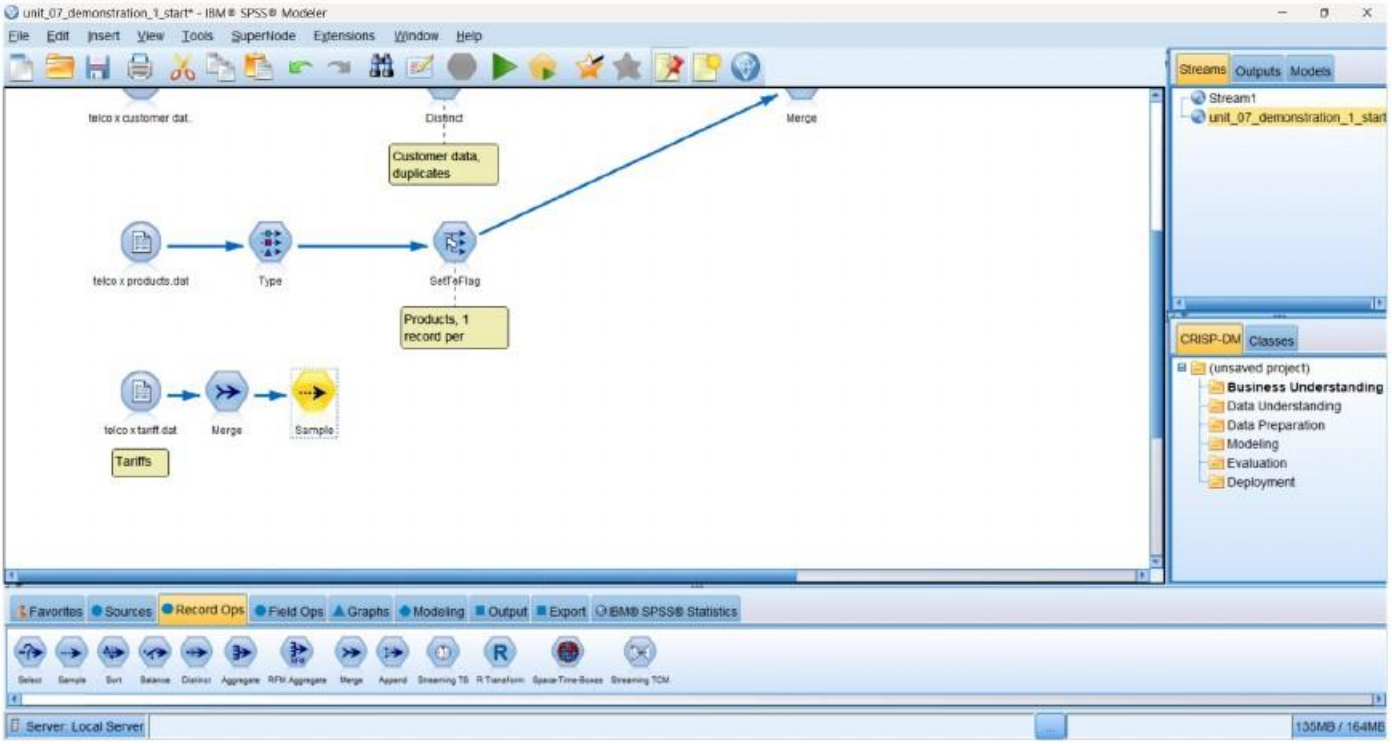
TASK – 6

- Sample **records.**

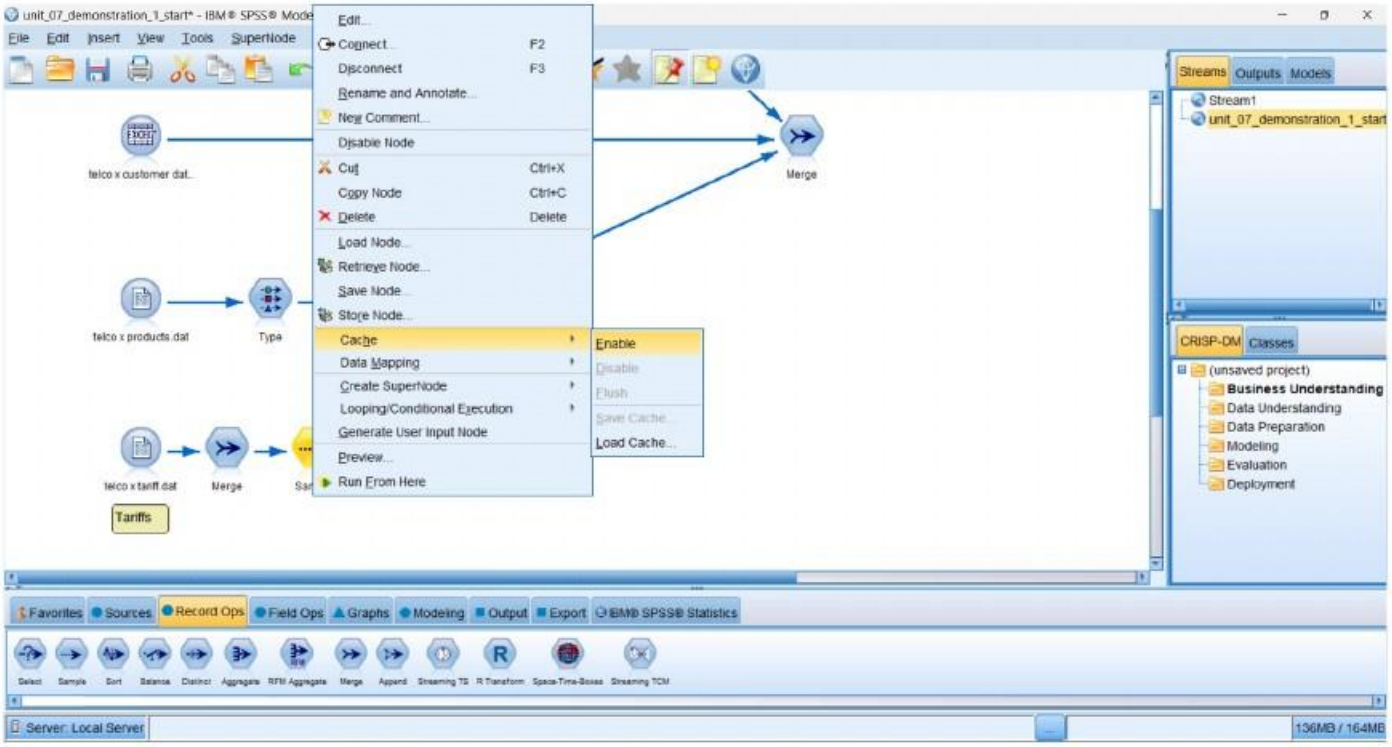
Definition: - From the combined dataset (storing data from four different sources), a random

sample is drawn of approximately 10%, and the data are cached. Furthermore, to ensure that the student samples the same records as are sampled in this demonstration, the value for the random seed will be fixed.In this task you will build from the previous stream.

Step 1: From the **Record Ops** palette, add a **Sample** node downstream from the second **Merge** node.

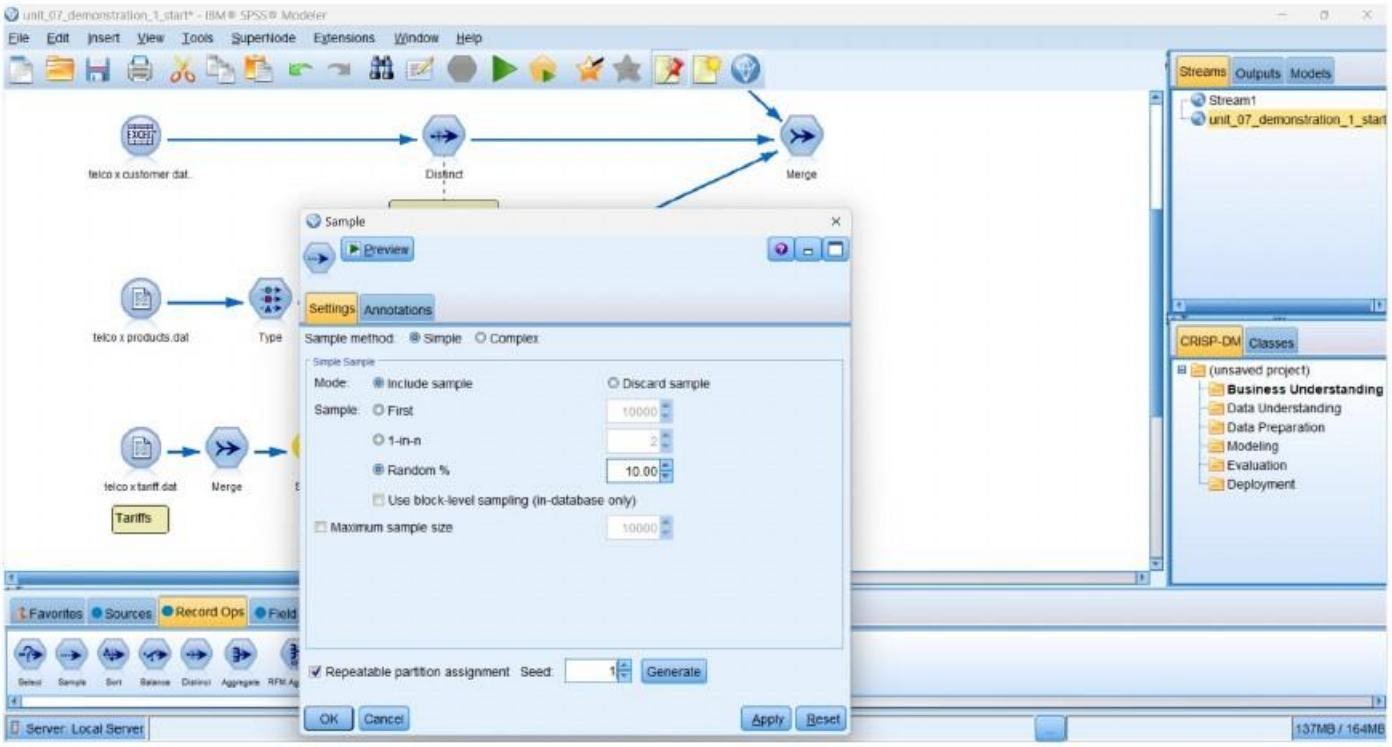


Step 2: Right-click the **Sample** node, click **Cache**, and then click **Enable** from the context menu.

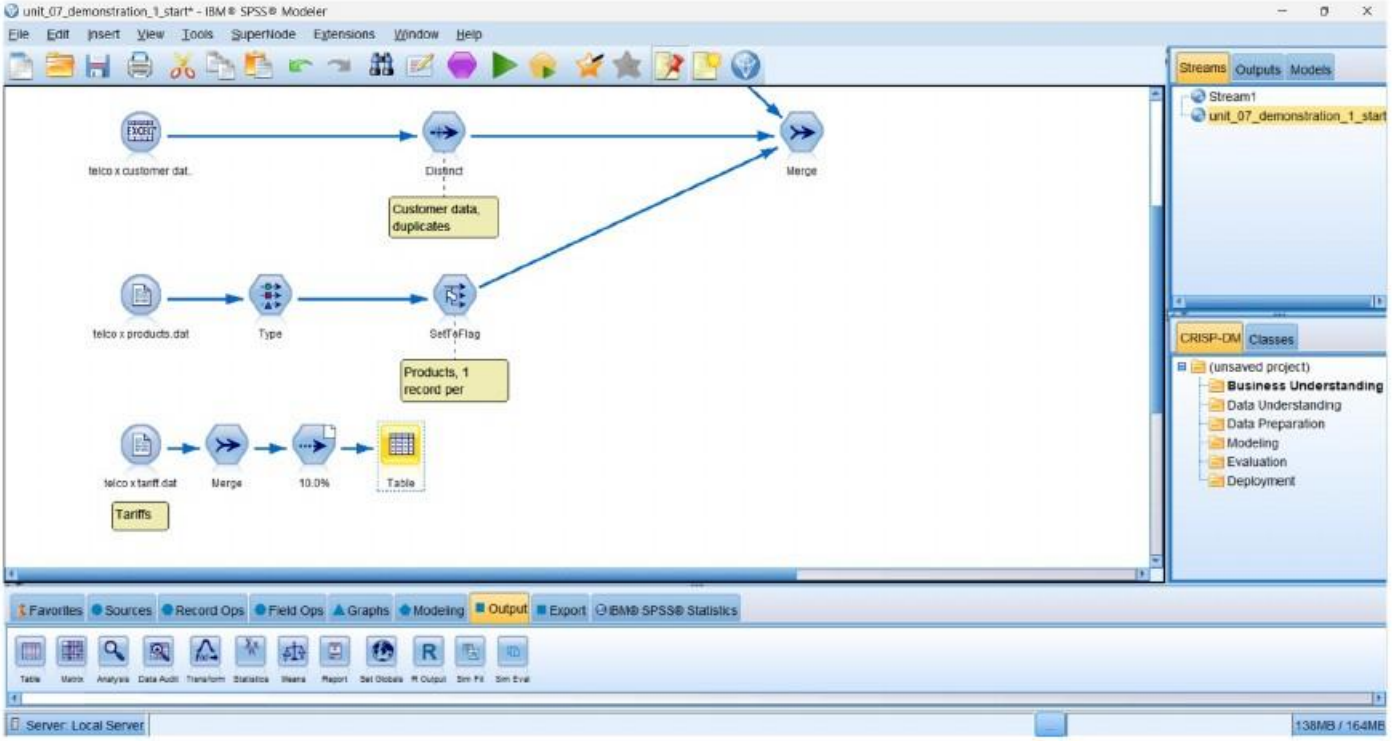


Step 3: Edit the **Sample** node.

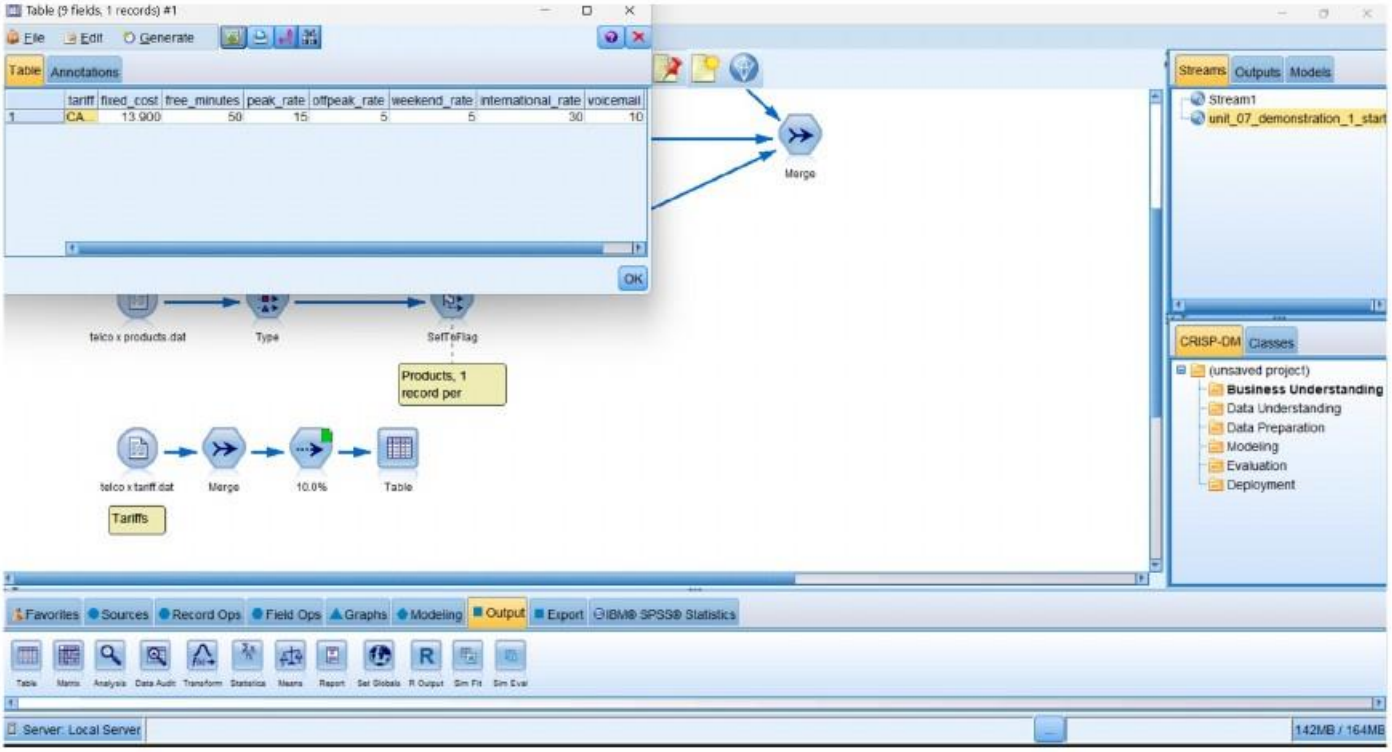
Step 4: Under **Sample**, select **Random %**, and change the associated value to **10.** Check the **Repeatable partition assignment** box.



Step 5: From the **Output** palette, add a **Table** node downstream from the **Sample** node.



Step 6: Run the **Table** node.



Practical - 6

Definition: - This practical explains how to use the IBM SPSS Modeler user interface to create, edit, modify, and save streams. It also demonstrates how to work with different nodes such as Excel, Sample, Derive, Histogram, Statistics Export, Select, and Table nodes

Outcomes/Learning: - Learners will become familiar with the IBM SPSS Modeler user interface and will understand how to create and modify streams, connect different nodes, generate outputs, add comments, and use table output to generate a Select node.

Required Tool: - The required tool will be "**IBM SPSS Modeler"**.

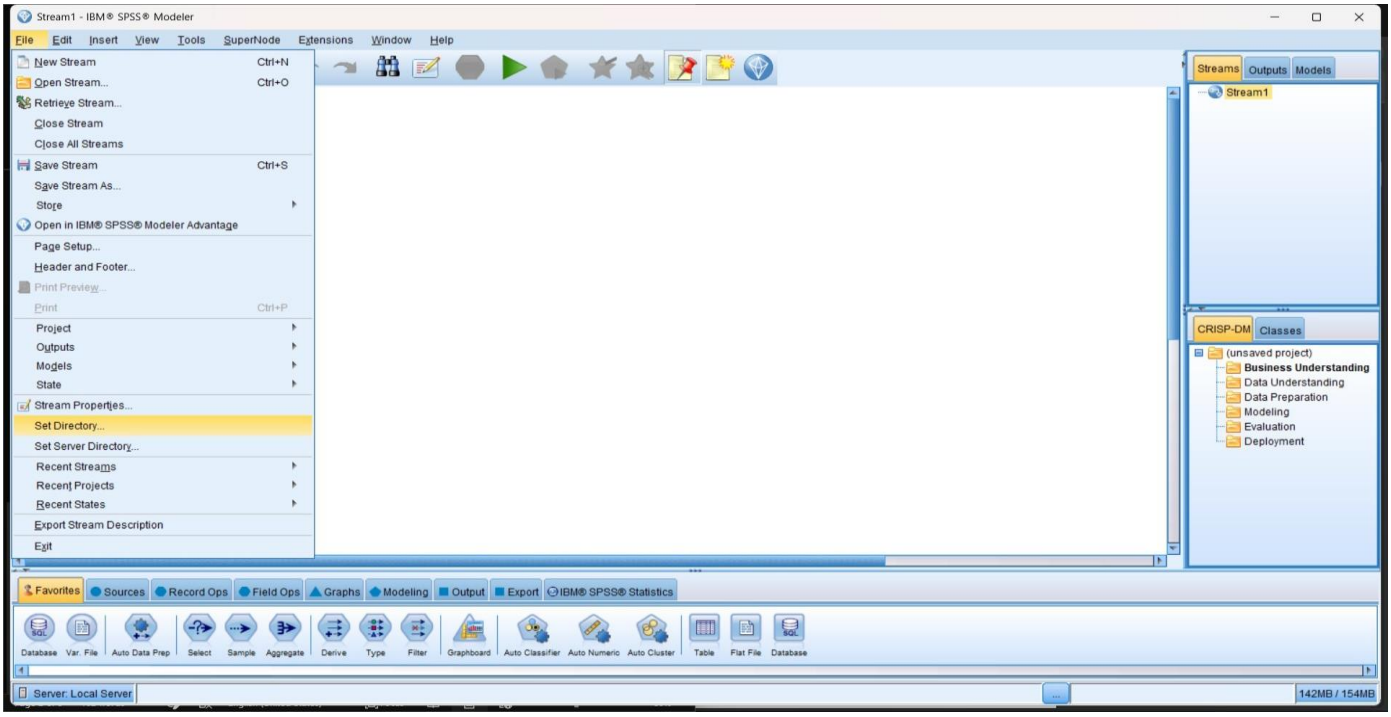
Purpose: - You work for a telecommunications firm and you need to cleanse and enrich a dataset in order to build models later.

TASK – 1

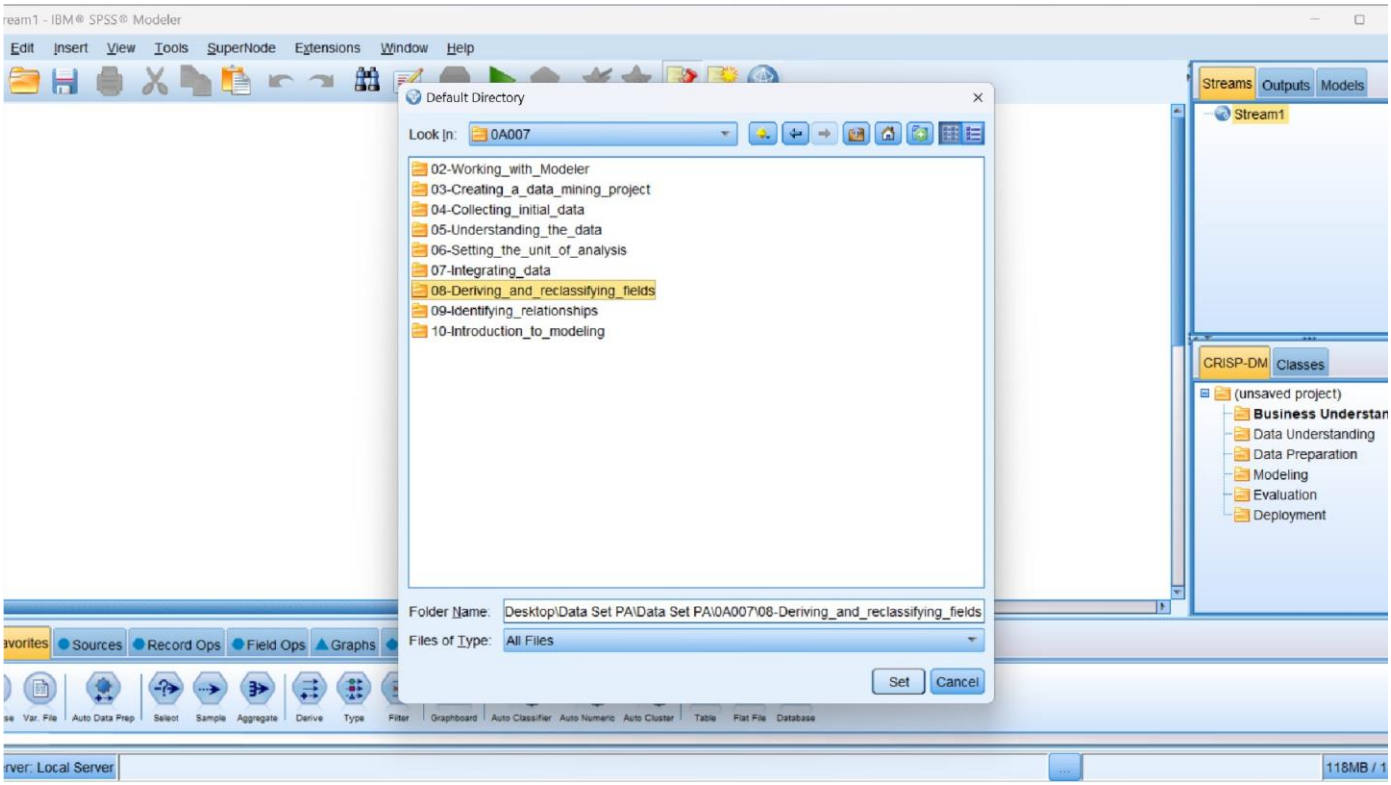
 Start IBM SPSS Modeler and set the working folder.

Definition: - This task explains how to open IBM SPSS Modeler and set the required working folder before creating a stream.

Step 1: - From the File menu, click **Set Directory**.



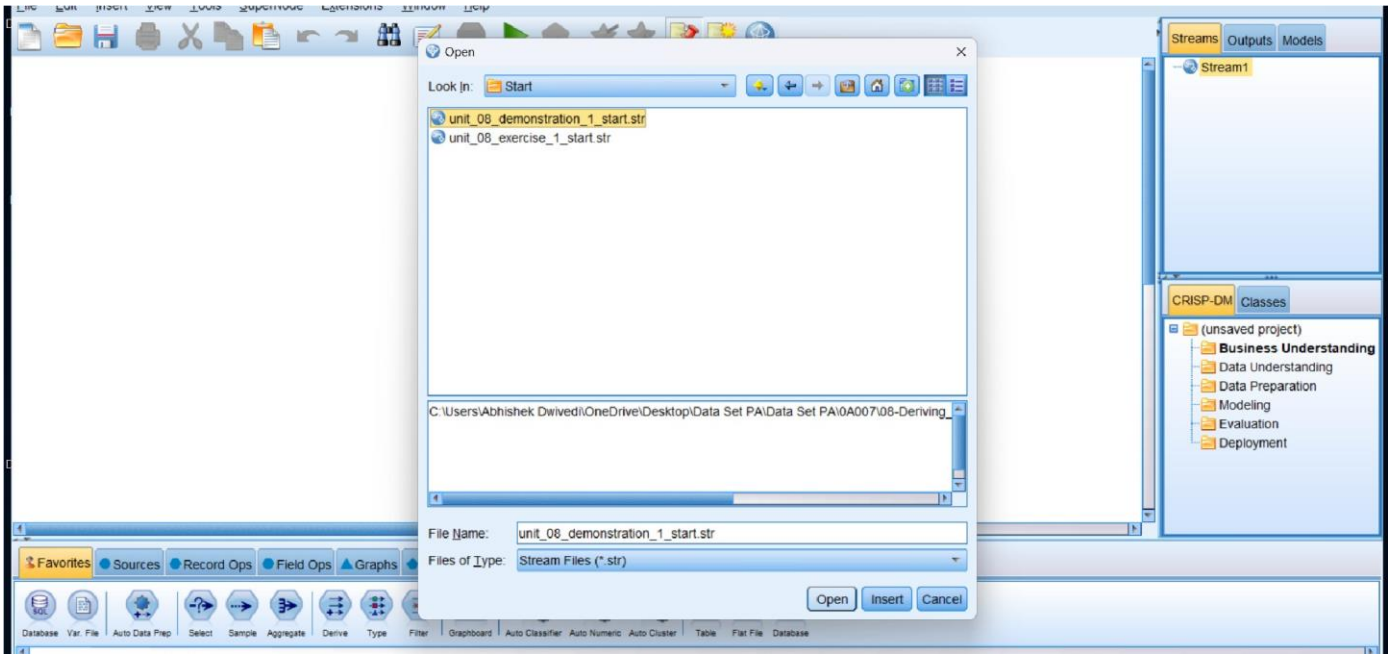
Step 2: - Beside Look in, navigate to **C:\Users\Mr. Abhishek\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.



TASK – 2

- **Open** the **stream**.

Definition: - From the **File** menu, click **Open Stream**, navigate to **C:\Training\0A007\08-Deriving_and_reclassifying_fields\Start**, click **unit_08_demonstration_1_start.str**, and then click **Open**.

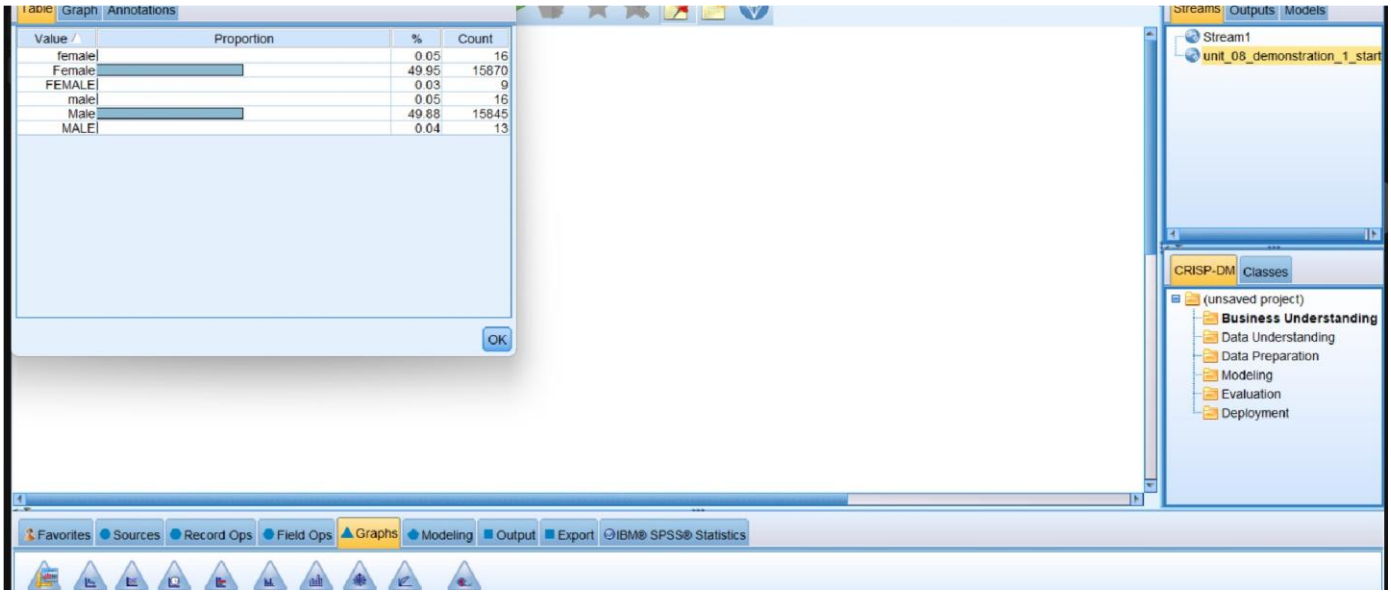


TASK – 3

- Cleanse data for gender.

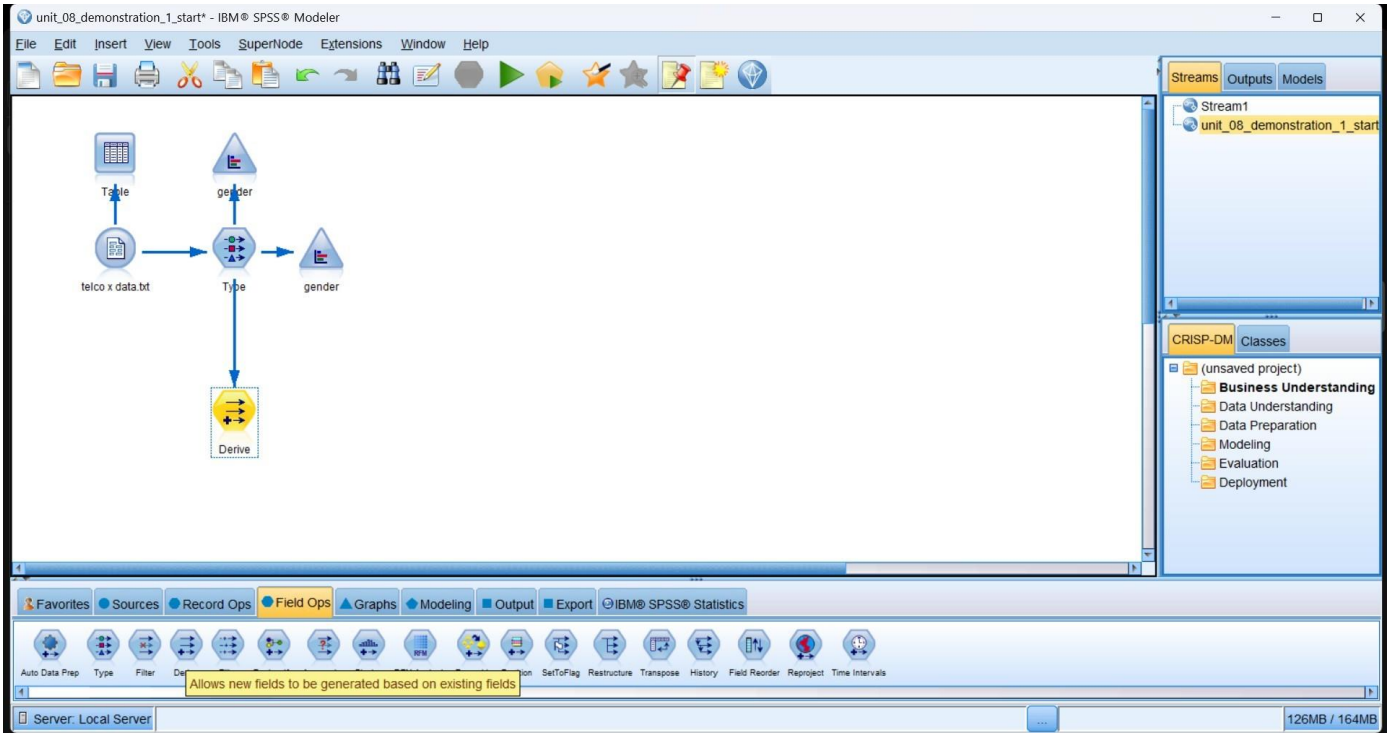
Definition: - This task explains how to Cleanse data for gender.

Step 1: Run the **Distribution** node for **gender**, downstream from the **Type** node.



Step 2: Close the **Distribution** output window.

Step 3: From the **Field Ops** palette, add a **Derive** node downstream from the **Type** node.



Step 4: Edit the **Derive** node.

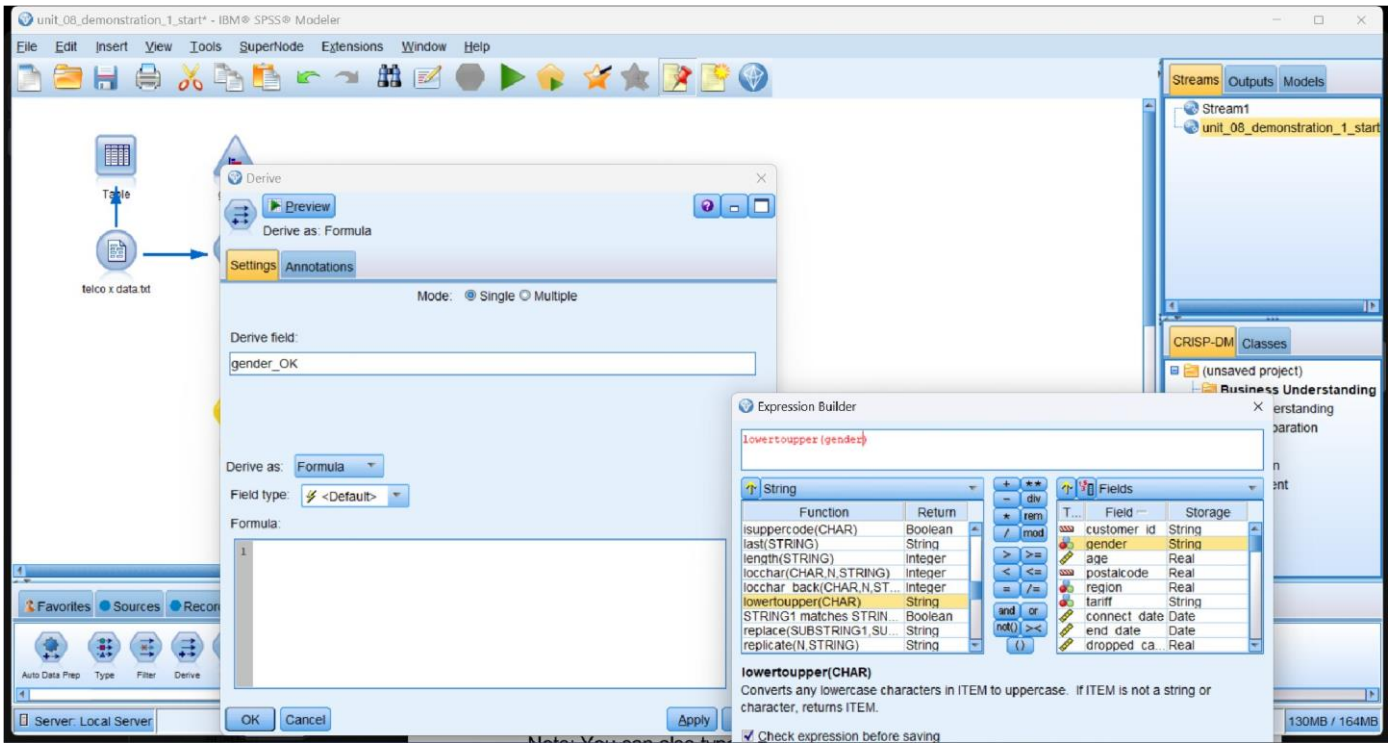
Step 5: In the **Derive field** text box, type **gender_OK**.

Step 6: Click the **Launch expression builder** button to launch the Expression Builder.

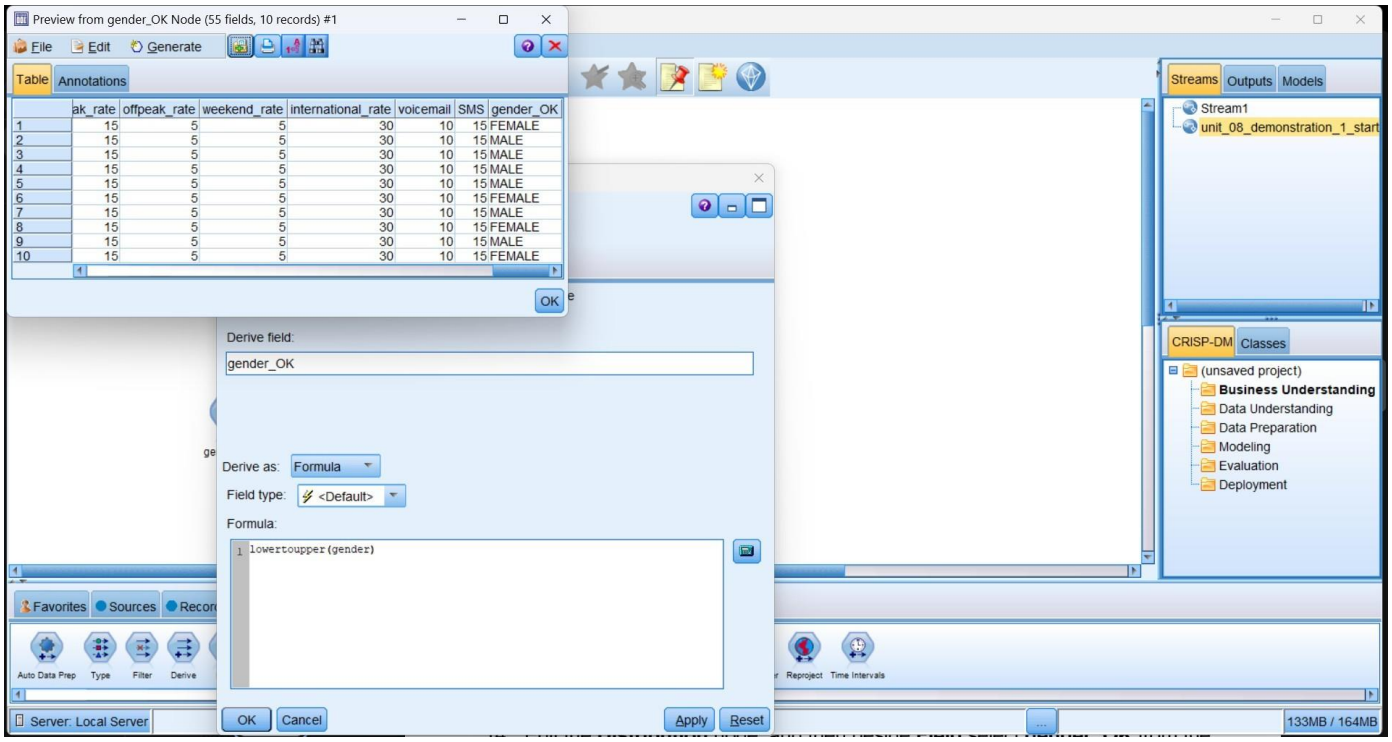
Step 7: On the left side, click **General Functions**, and select the **String** group of functions from the list.

Step 8: Under **Function**, select **lowertoupper**, and then click the **Insert** button to paste it into the expression area.

Step 9: In the right pane, under **Field**, double-click **gender**, to add it to the expression area.

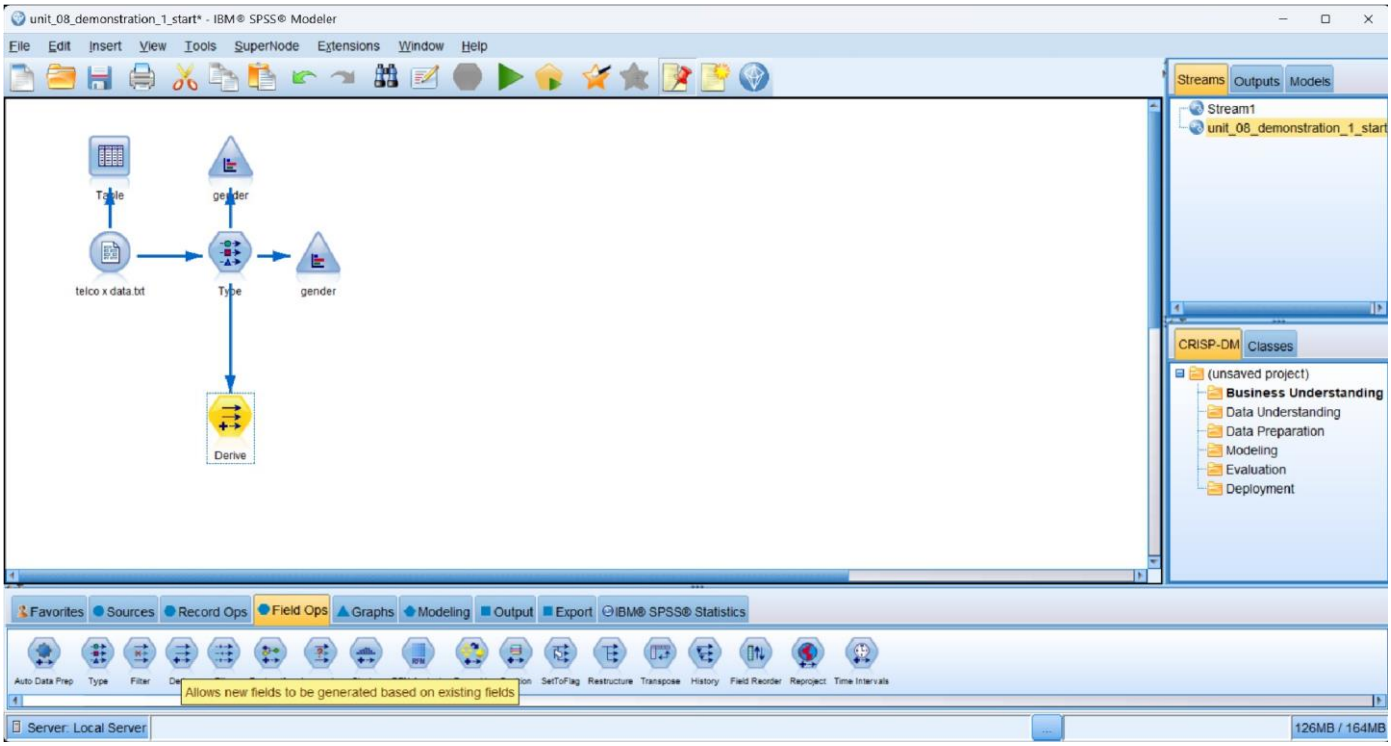


Step 10: Click **Preview**, and then scroll to the last field.

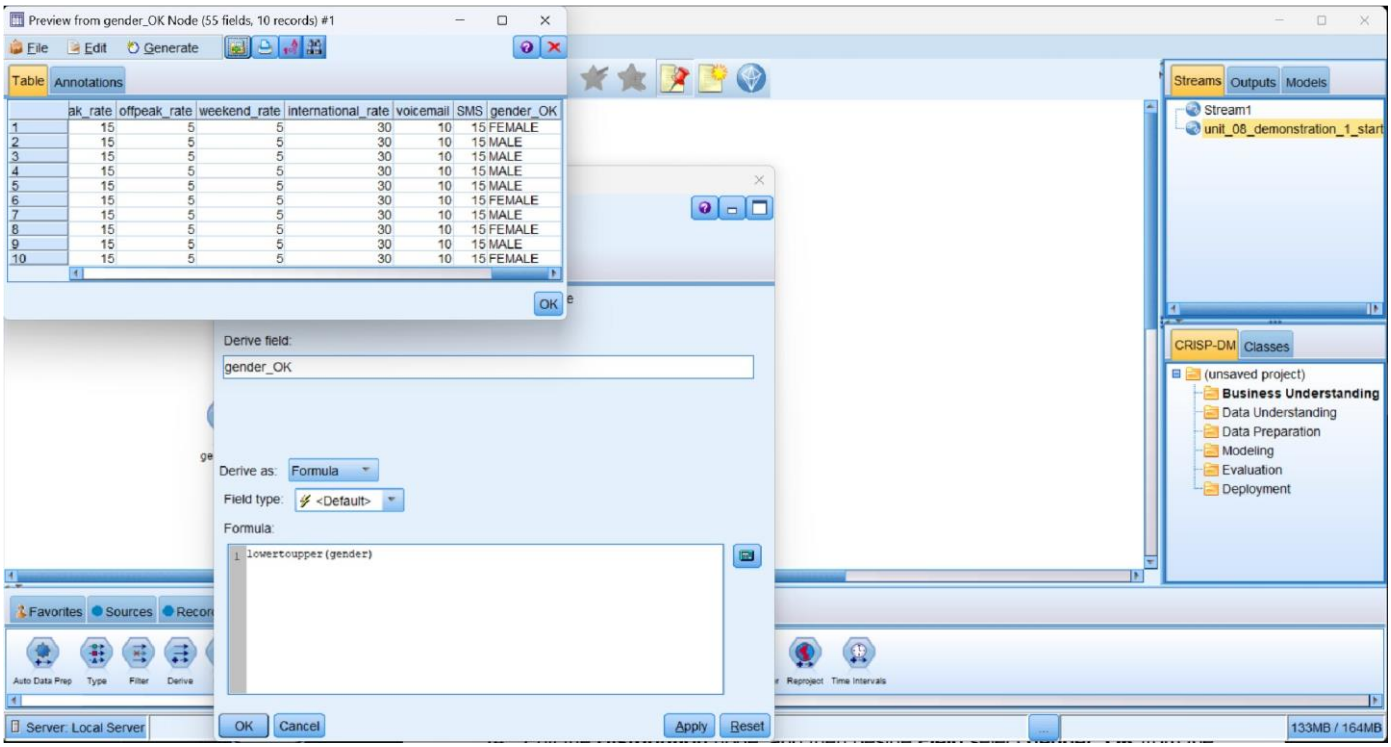


Step 11: From the **Graphs** palette, add a **Distribution** node, downstream from the **gender_OK** Derive node.

Step 12: Edit the **Distribution** node, and then beside **Field** select **gender_OK** from the list.



Step 13: Click **Run**.

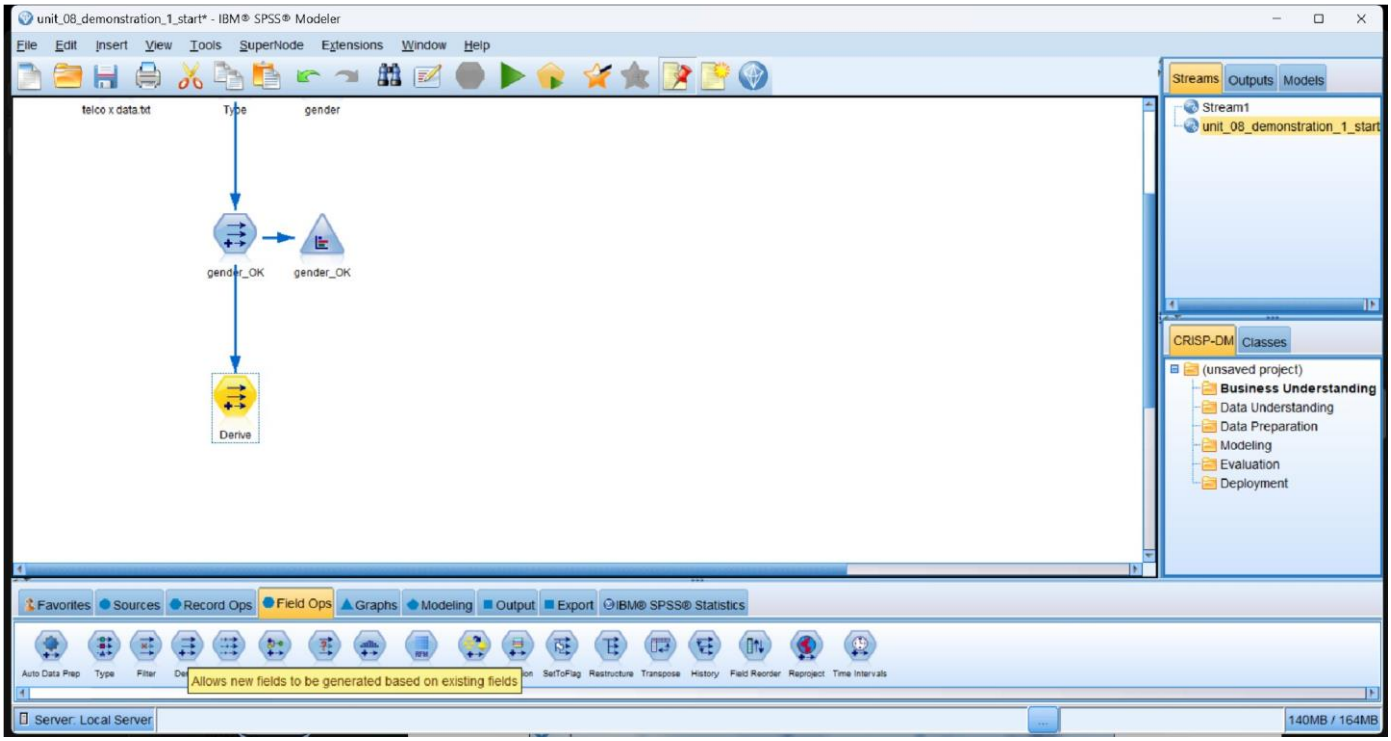


TASK – 4

- Derive bill for peak, offpeak and total bill.

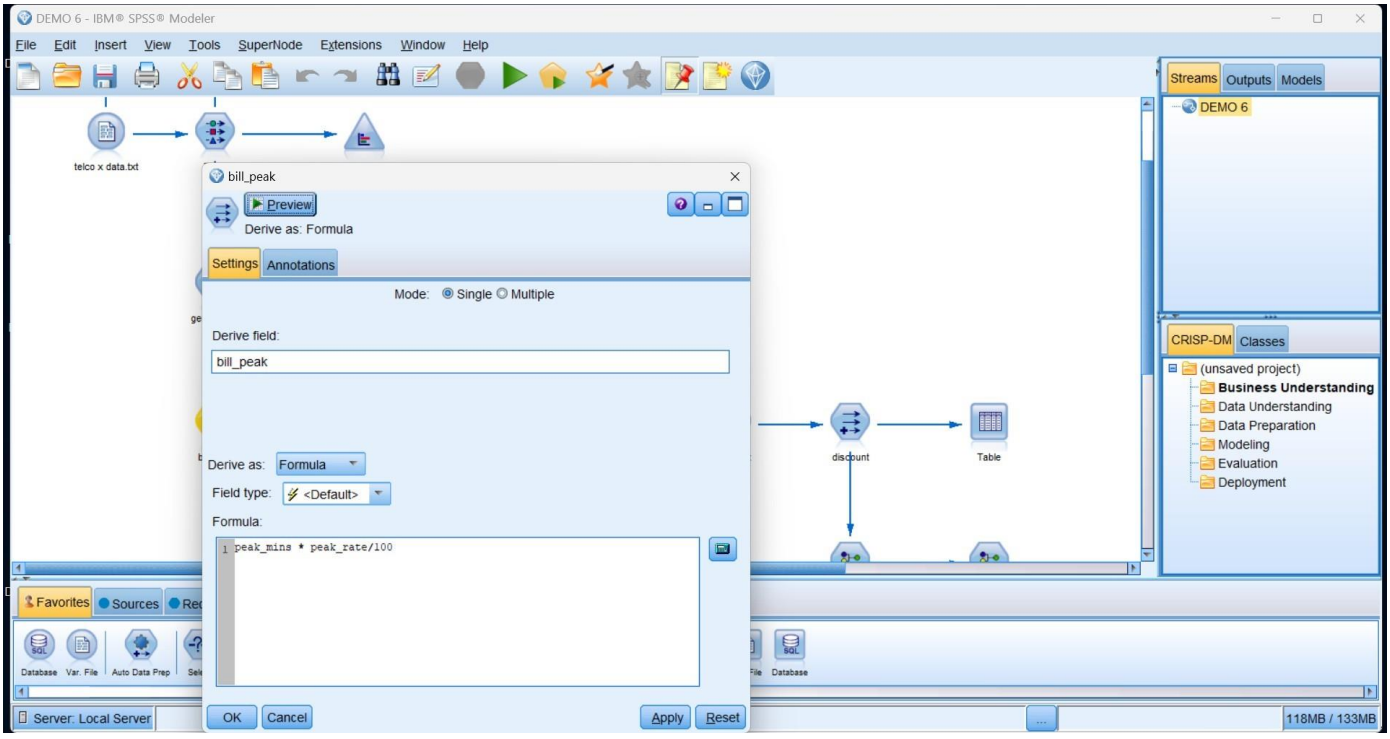
Definition: - You will first derive fields that give the bill for calling in the peak hours, and the bill for calling in the off-peak hours (these fields can be valuable predictors for churn). Each field is computed by multiplying the minutes by the corresponding rate. Rates are given in cents, so you will divide by 100.

Step 1: From the **Field Ops** palette, add a **Derive** node downstream from the **gender_OK** Derive node.

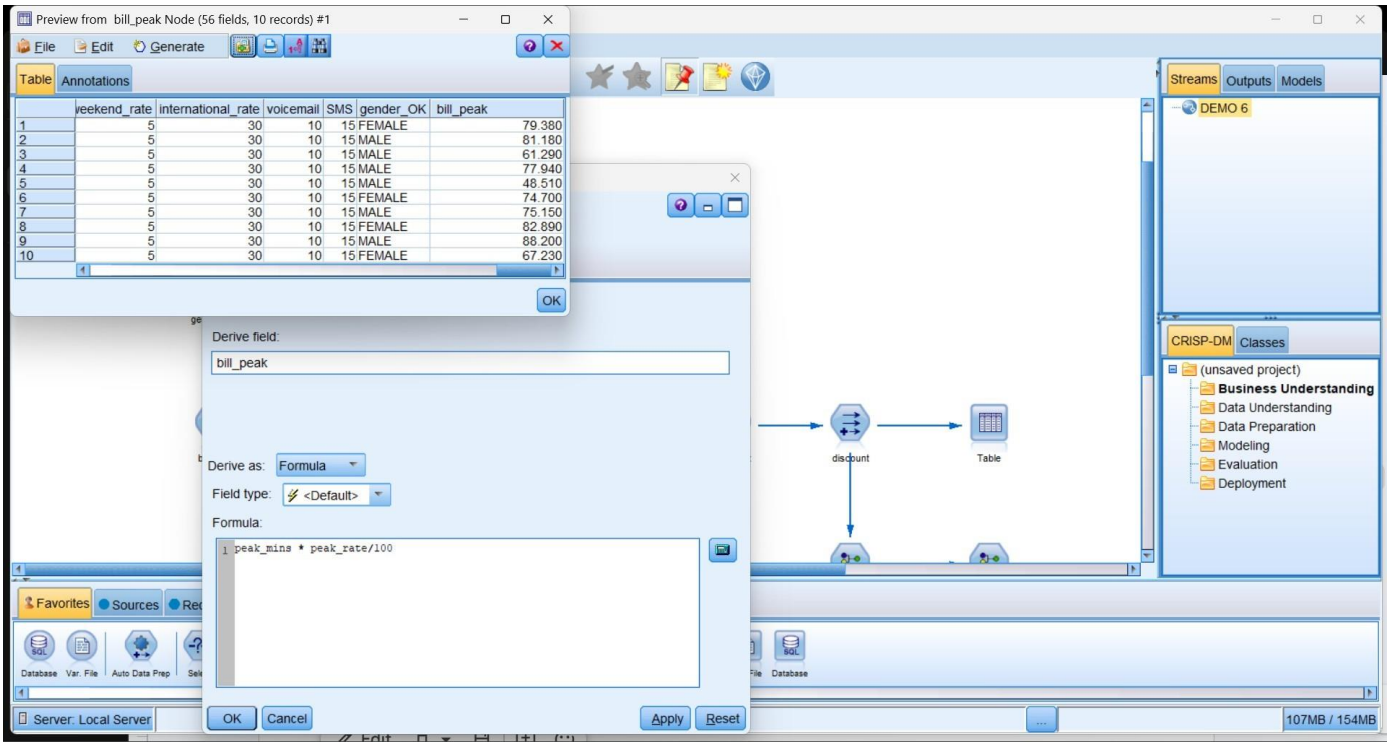


Step 2: Edit the **Derive** node as follows:

- Derive field: **bill_peak**.
- Derive as: **Formula**
- Expression Builder: **peak_mins * peak_rate/100**.

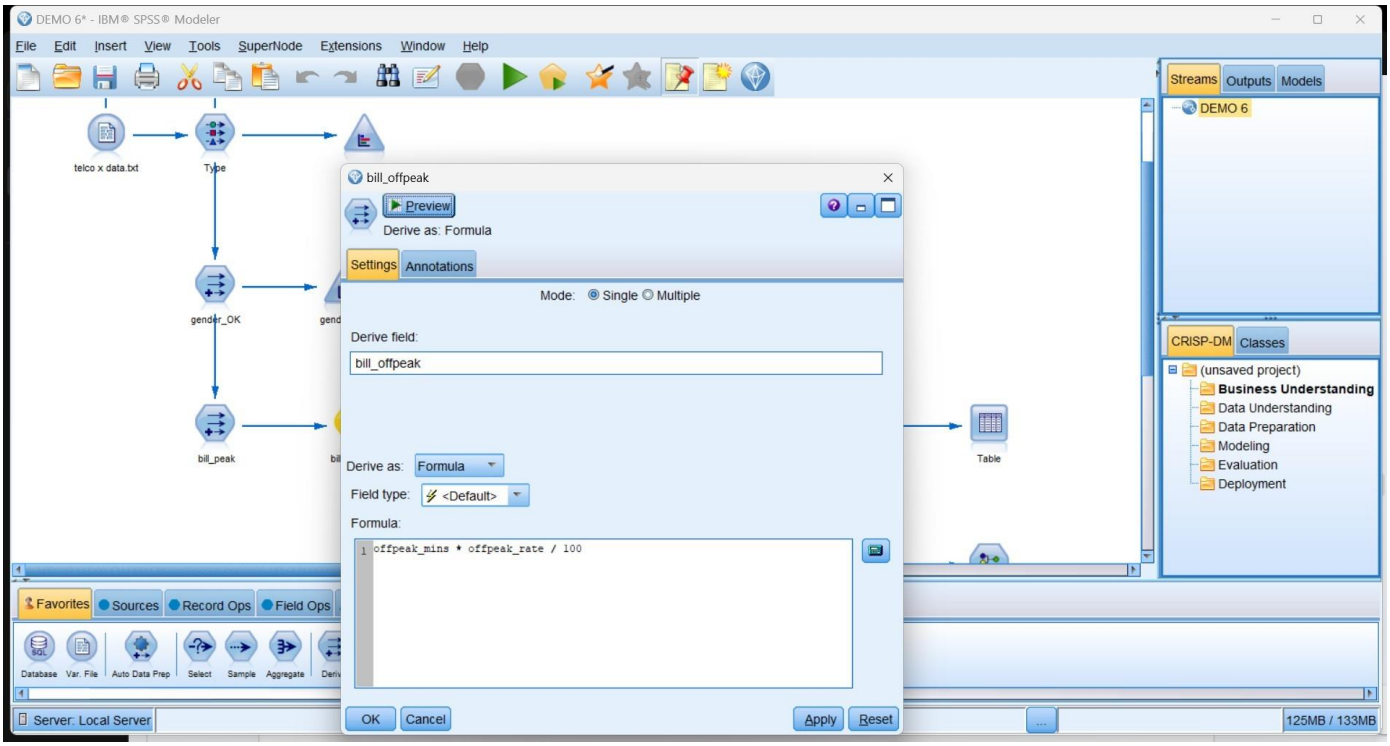


Step 3: Click **Preview**, and then scroll to the last field.



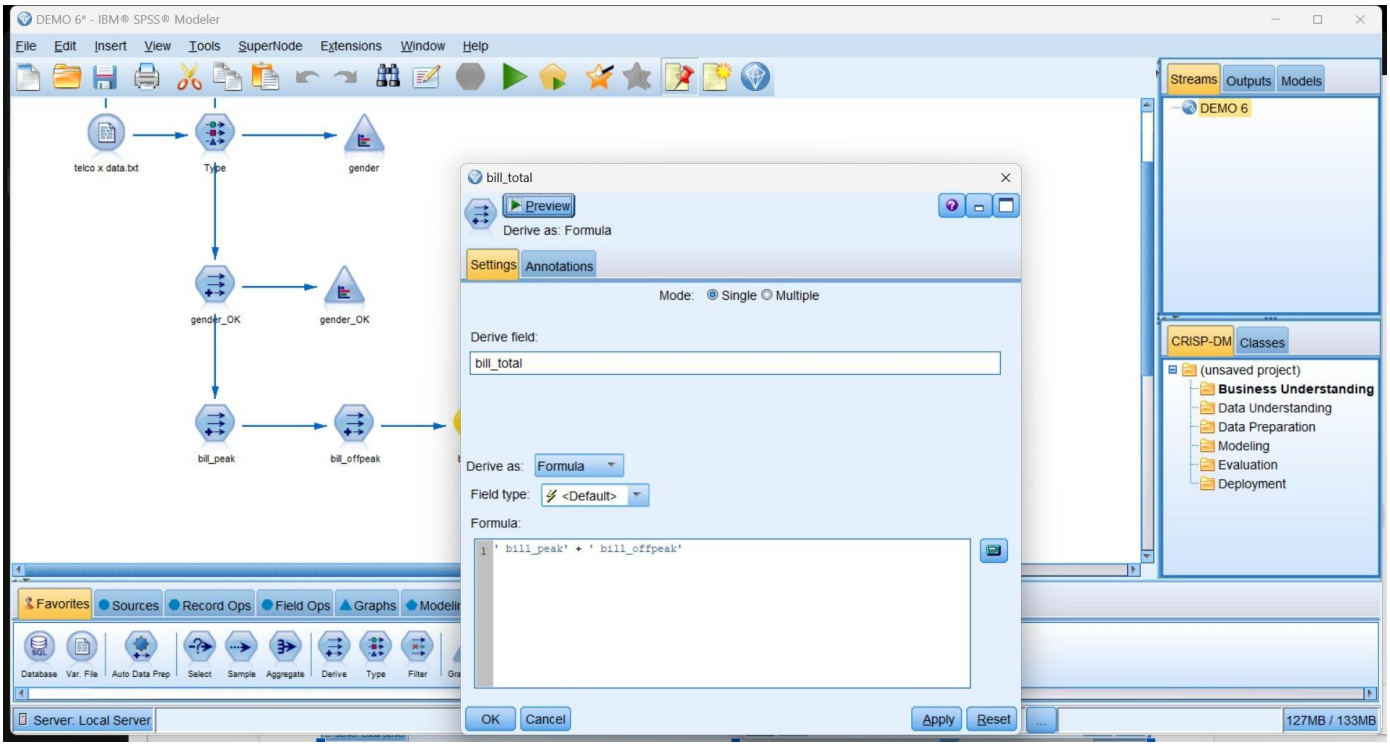
<u>Step 4:</u> Add a **Derive** node downstream from the **bill_peak** node, and edit it as follows:

- Derive field: **bill_offpeak**
- Derive as: **Formula** (the default)
- Expression: **offpeak_mins * offpeak_rate / 100**.

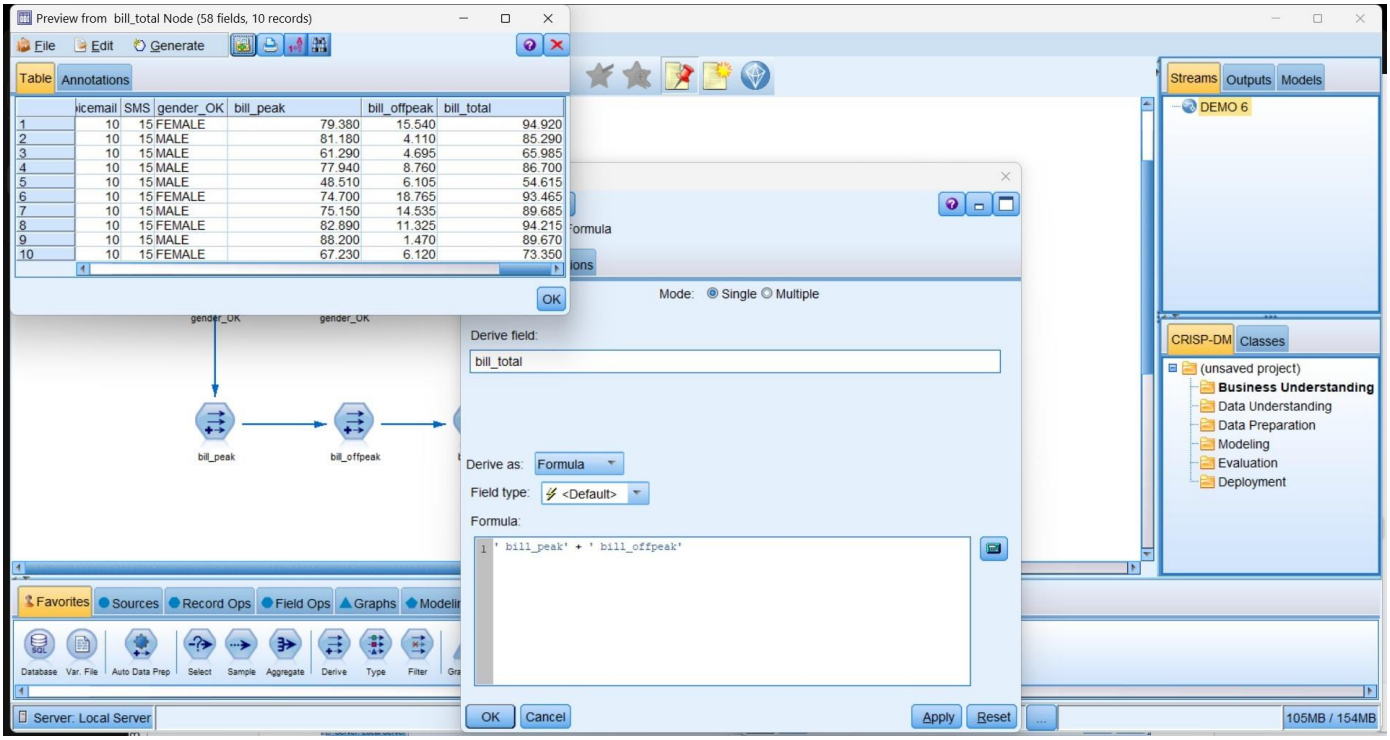


Step 5: Add a **Derive** node downstream from the **bill_offpeak** node, and edit it as follows:

- Derive field: **bill_total**.
- Derive as: **Formula** (the default)
- Expression **bill_peak + bill_offpeak**.



Step 6: Click **Preview**, and then scroll to the last field.



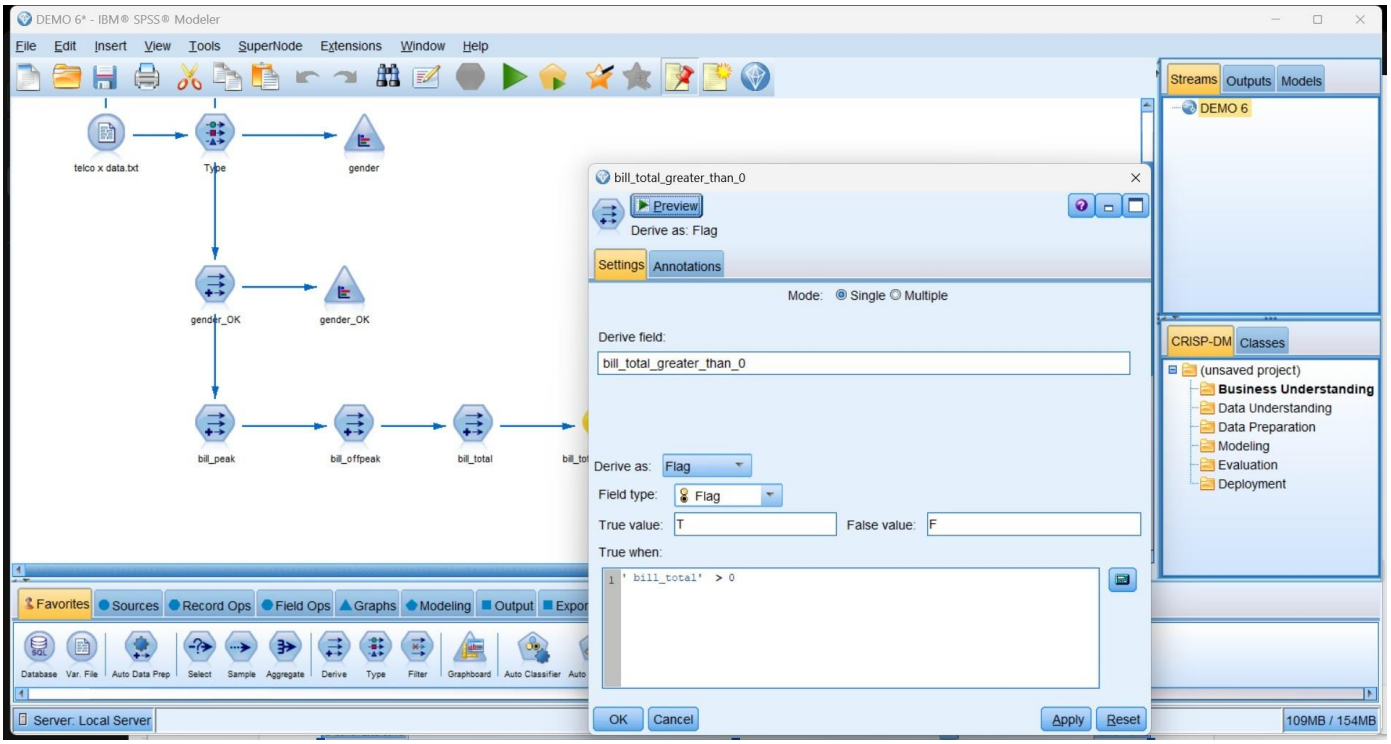
TASK – 5

- Derive more fields for modeling.

Step 1: Add a **Derive** node, downstream from the **bill_total** node, and edit it as follows: •

Derive field: **bill_total_greater_than_0**

- Derive as: **Flag**
- Field type: **Flag** (default value when the Derive type is Flag).
- True when: **bill_total > 0**



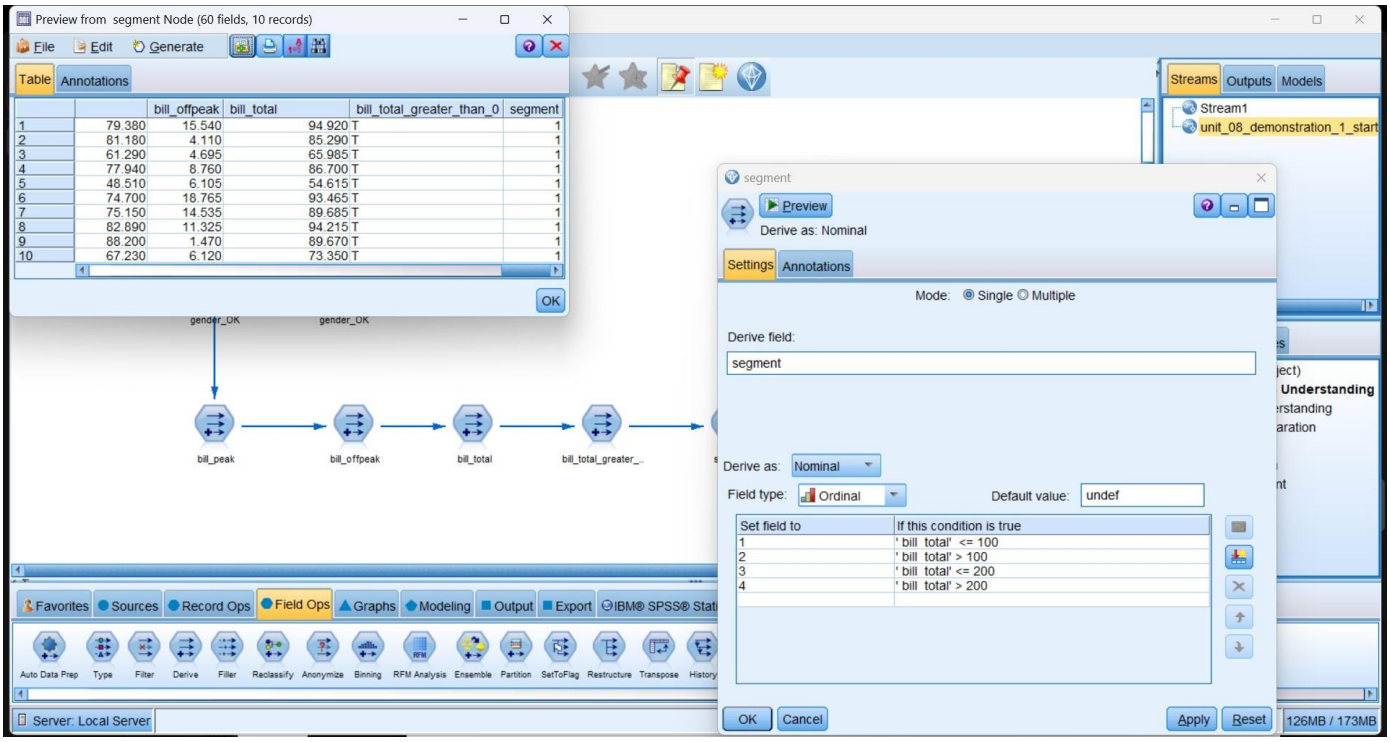
Step 2: Add a **Derive** node, downstream from the **bill_total_greater_than_0** node, and edit it as follows:

- Derive field: **segment**
- Derive as: **Nominal**
- Field type: **Ordinal**

Step 3: Under **Set field to**, enter: **1**, and under **If this condition is true**, enter: **bill_total <= 100.**

Step 4: Repeat the previous step for the following values and expressions: **2 bill_total > 100 and bill_total <= 200 3 bill_total > 200**.

Step 5: Beside **Default value**, type **undef**.

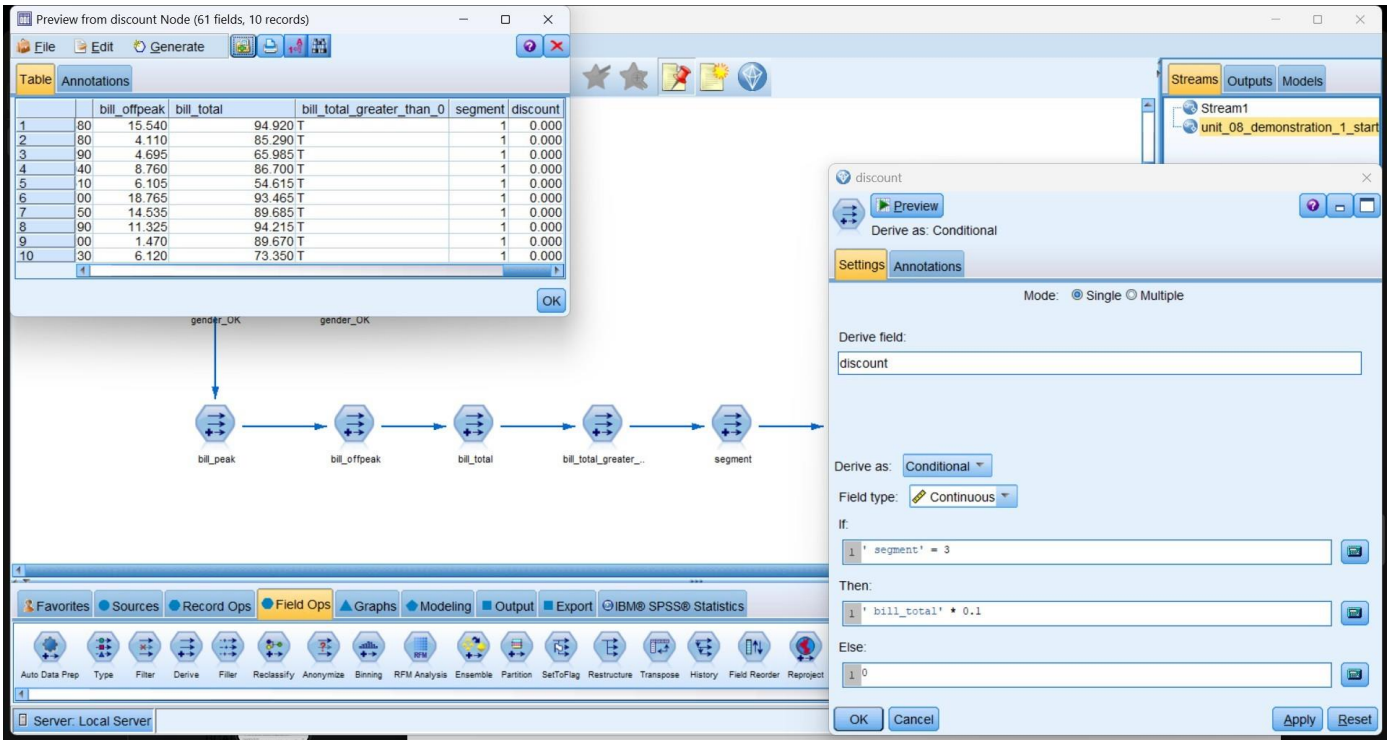


Step 6: Add a **Derive** node, downstream from **segment** node, and define it as follows: •

Derive field: **discount**.

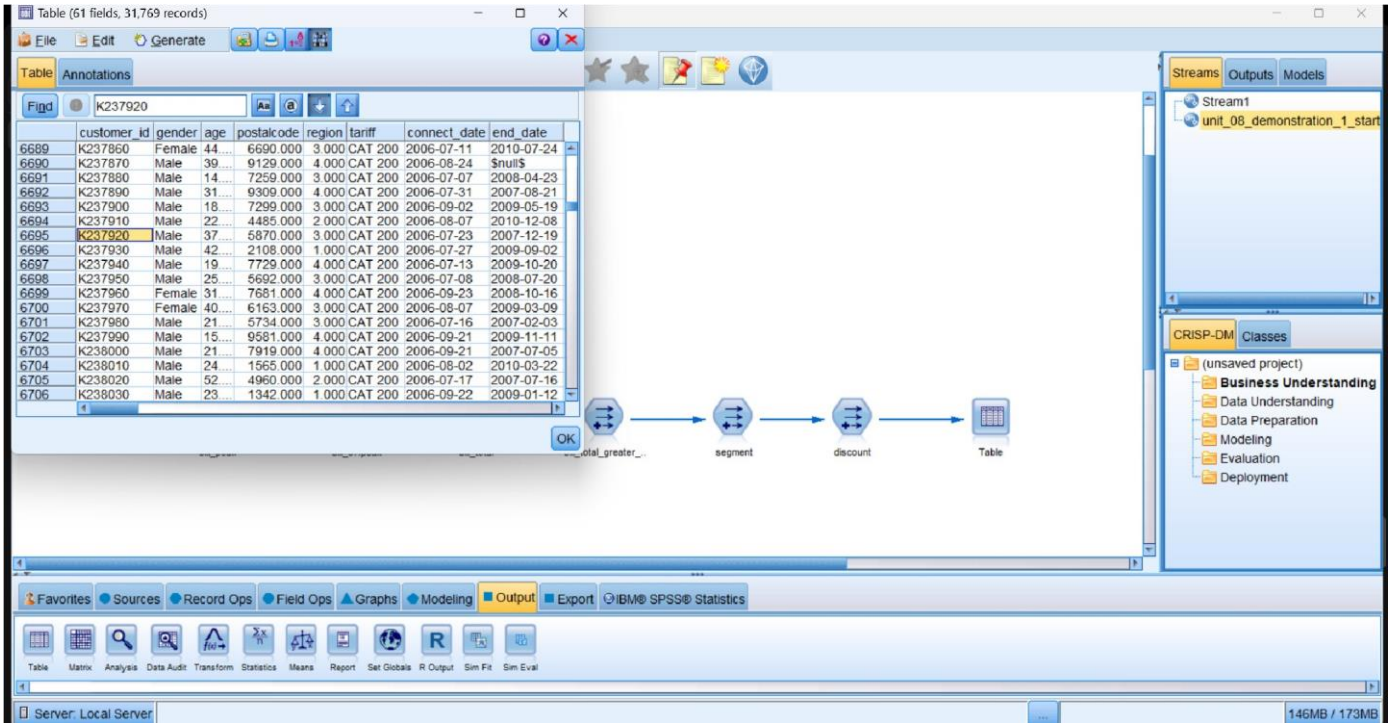
- Derive as: **Conditional**.
- Field type: **Continuous**
- If: **segment = 3**.
- Then: **bill_total * 0.1**

- Under **Else**, type **0**.



Step 7: From the **Output** palette, add a **Table** node, downstream from the **discount** node, and then run the **Table** node.

Step 8: In the **Table** output window, find **K237920**, and then scroll to the right.



TASK – 6

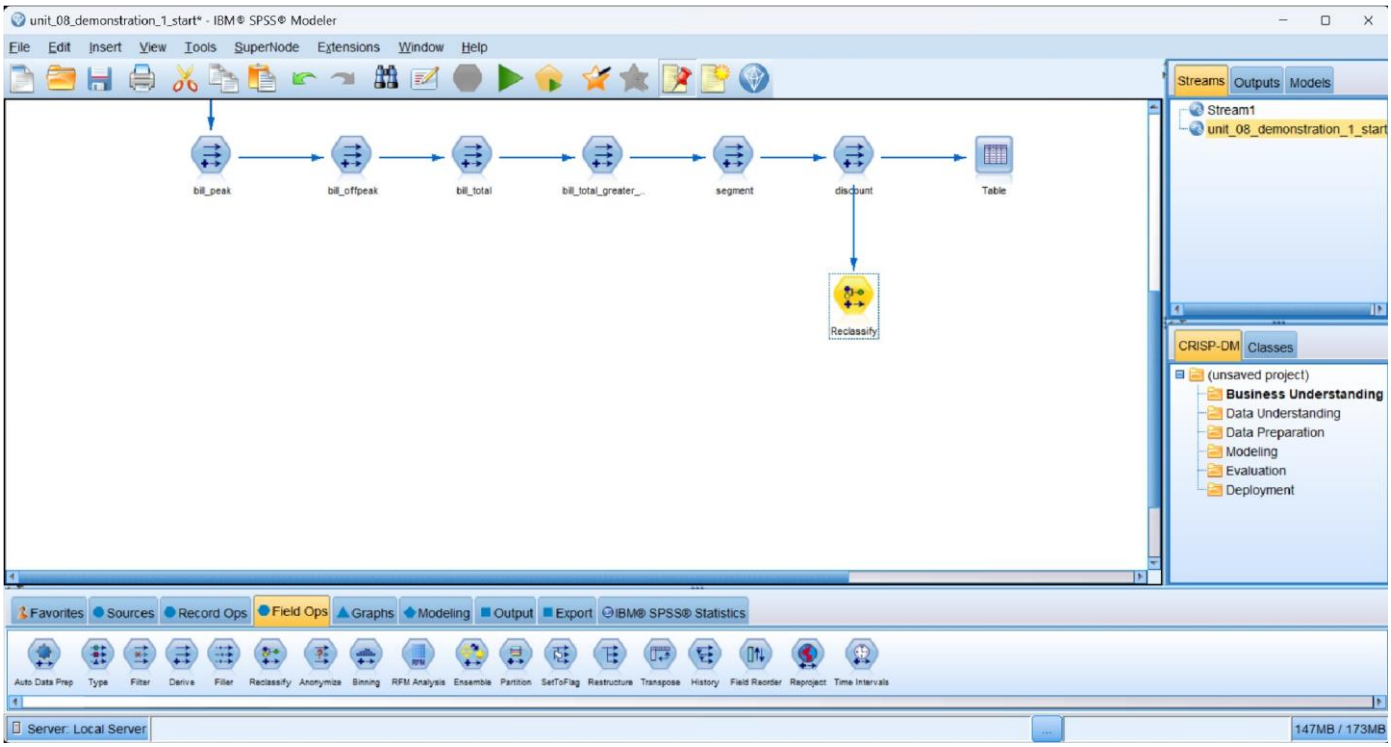
- Cleanse data and reclassify fields for modeling.

Definition: - In this task you will repair the different spellings used for gender. In the previous task, this was done by using the Derive node and adding a new field. In this task the Reclassify node will be used which has the advantage that the field's values can be overwritten.

Also, you will derive a new field recording the handsets into a (smaller) number of brands.

In this task you will build from the stream created in the previous task.

Step 1: From the **Field Ops** palette, add a **Reclassify** node, downstream from the **discount** node.



Step 2: Edit the **Reclassify** node.

Step 3: Beside **Reclassify into**, select **Existing field**.

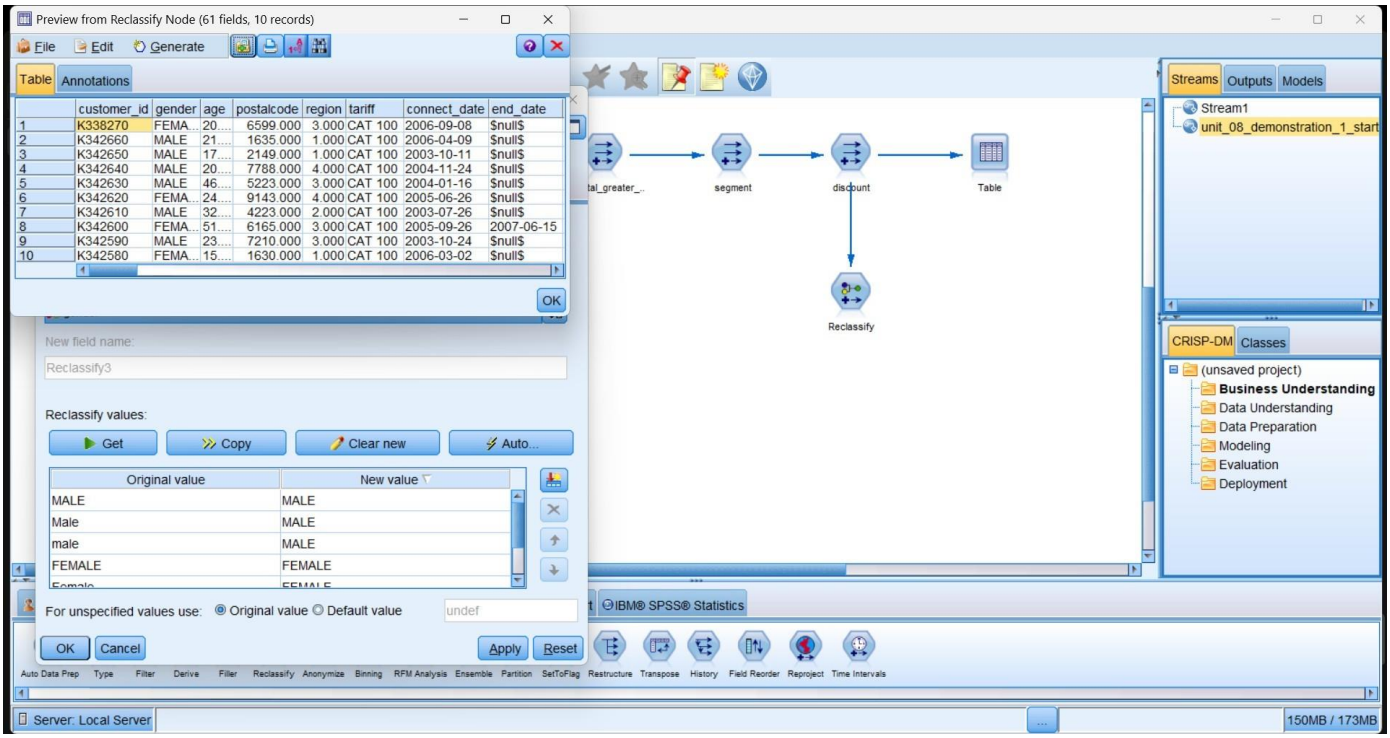
Step 4: Under **Reclassify field**, select **gender** from the list.

Step 5: Click the **Get** button, to populate the **Original value** column.

Step 6: Click the **Copy** button, to populate the **New value** column.

Step 7: Ensure that the **New value** column values are in upper case.

Step 8: Click the **Preview** button.



Step 9: From the **Field Ops** palette, add a **Reclassify** node, downstream from the first **Reclassify** node, and edit it as follows: • Reclassify field: **handset**

• New field name: **brand**.

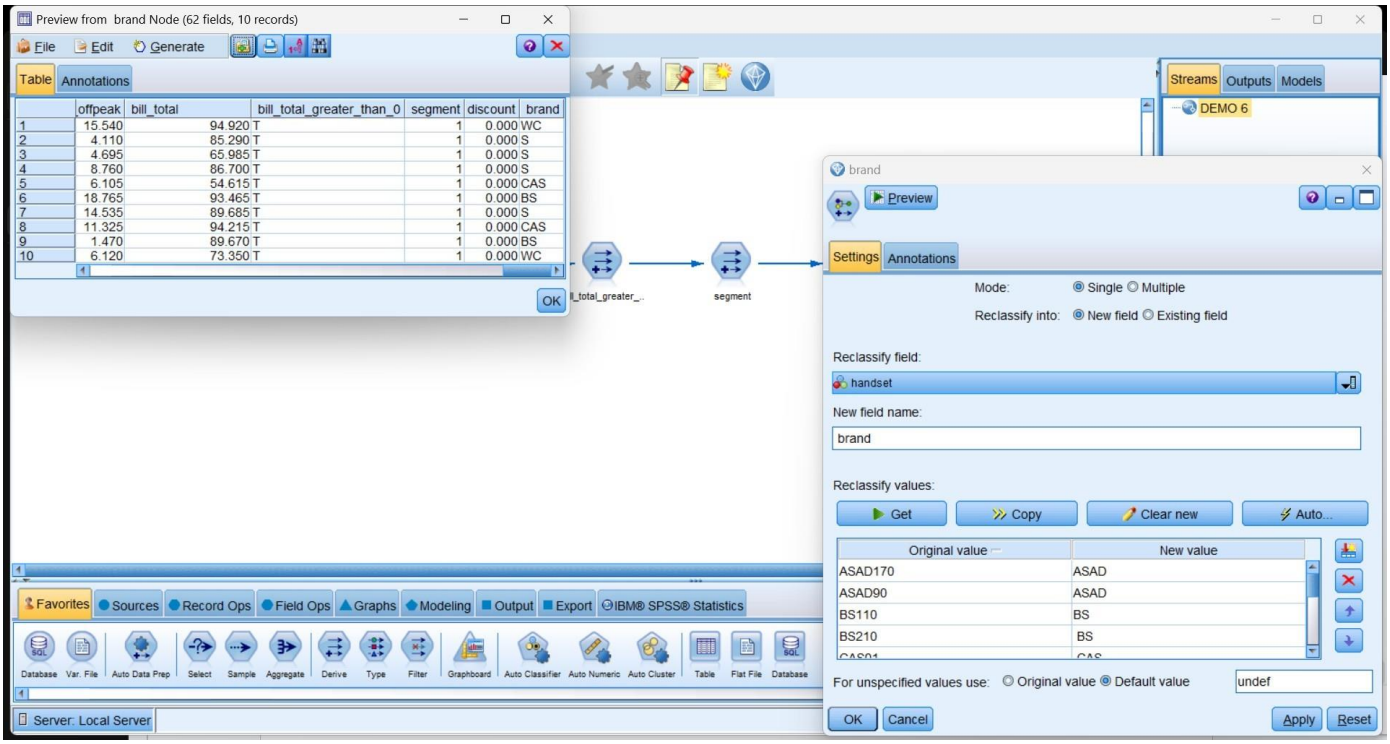
Step 10: Click the **Get** button to populate the **Original values** column.

Step 11: For the first new brand name, type **ASAD** in the **New value** column.

Step 12: For the second new brand name, type **ASAD**, or pick **ASAD** from the drop down list.

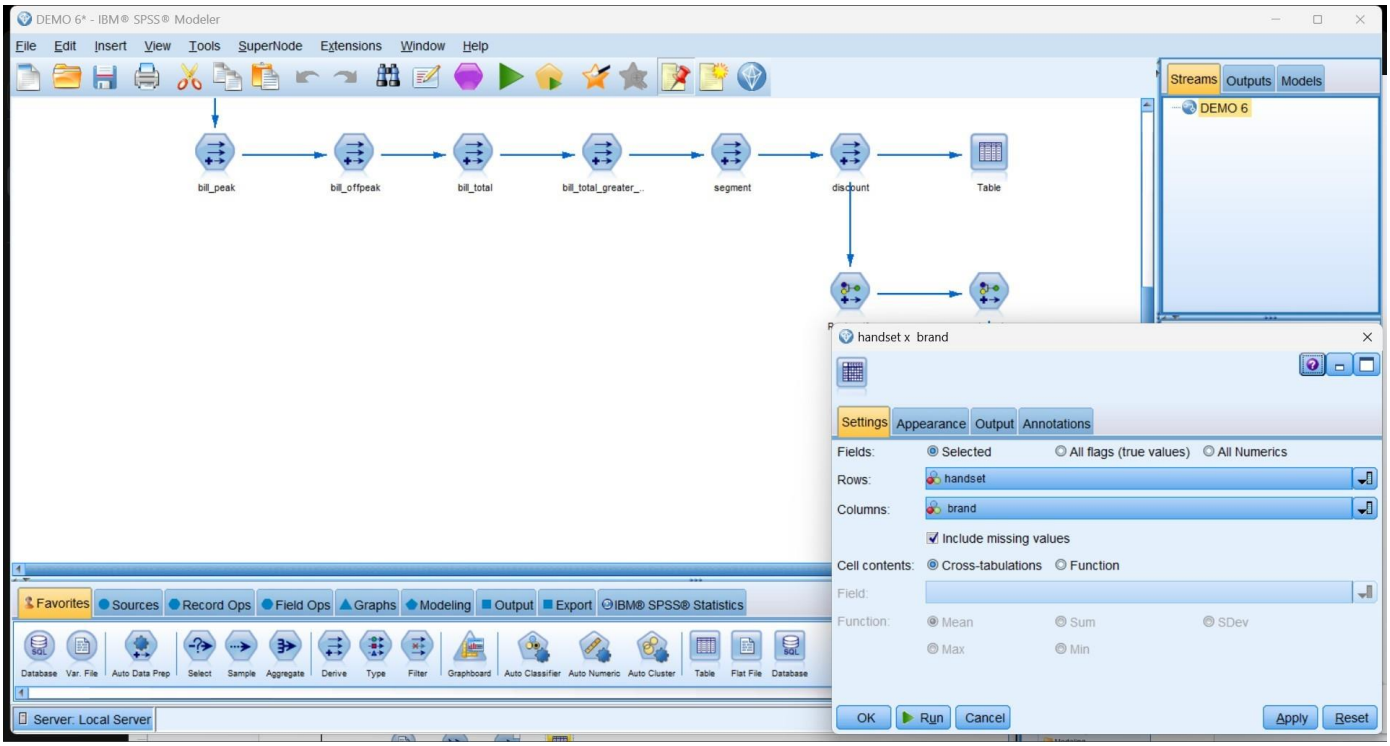
Step 13: Repeat for the other brands, using only the letters from the **Original value** as the **New value**.

Step 14: Beside **For unspecified values use**, click **Default value**.

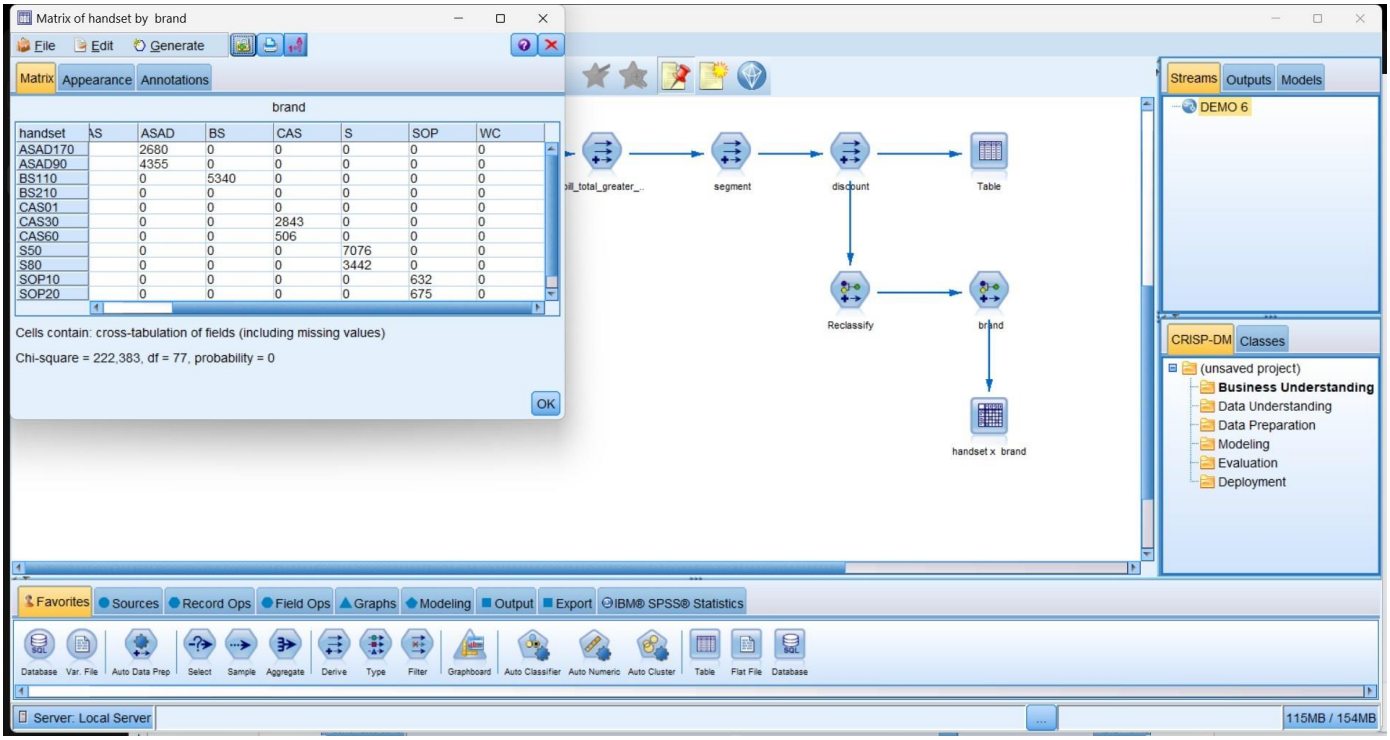


Step 15: From the **Output** palette, add a **Matrix** node, downstream from the **brand** Reclassify node, and define it as follows:

- Rows: **handset**
- Columns: **brand**



Step 16: Click **Run**.



Results:

You have cleansed and enriched the data, to build better models later.

Practical - 7

Definition: - This practical explains how to use the IBM SPSS Modeler user interface to create, edit, modify, and save streams. It also demonstrates how to work with different nodes such as Excel, Sample, Derive, Histogram, Statistics Export, Select, and Table nodes

Outcomes/Learning: - Learners will become familiar with the IBM SPSS Modeler user interface and will understand how to create and modify streams, connect different nodes, generate outputs, add comments, and use table output to generate a Select node.

Required Tool: - The required tool will be "**IBM SPSS Modeler"**.

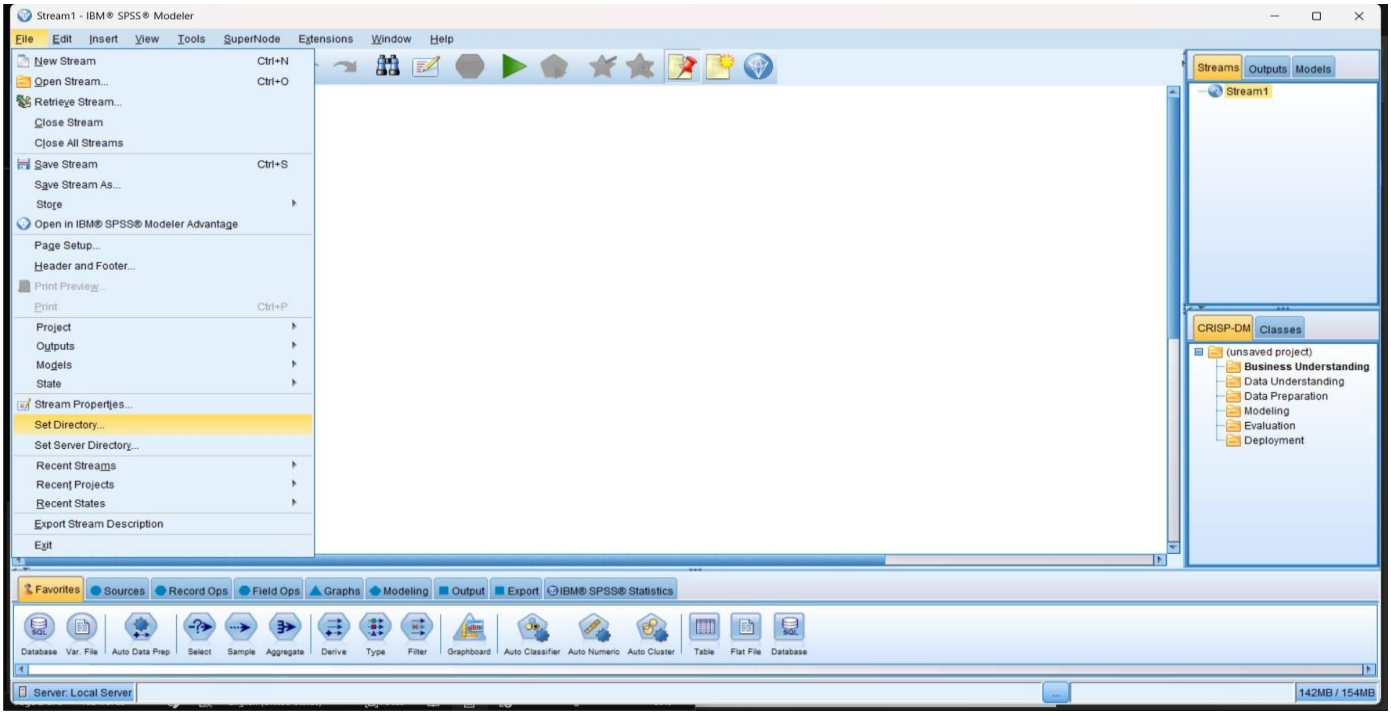
Purpose: - As a data miner in a telecommunications firm, you want to answer questions such as: Is churn related to handset? Is churn related to the number of dropped calls? And how strong is the relationship between number of gadgets purchased and revenues?

TASK – 1

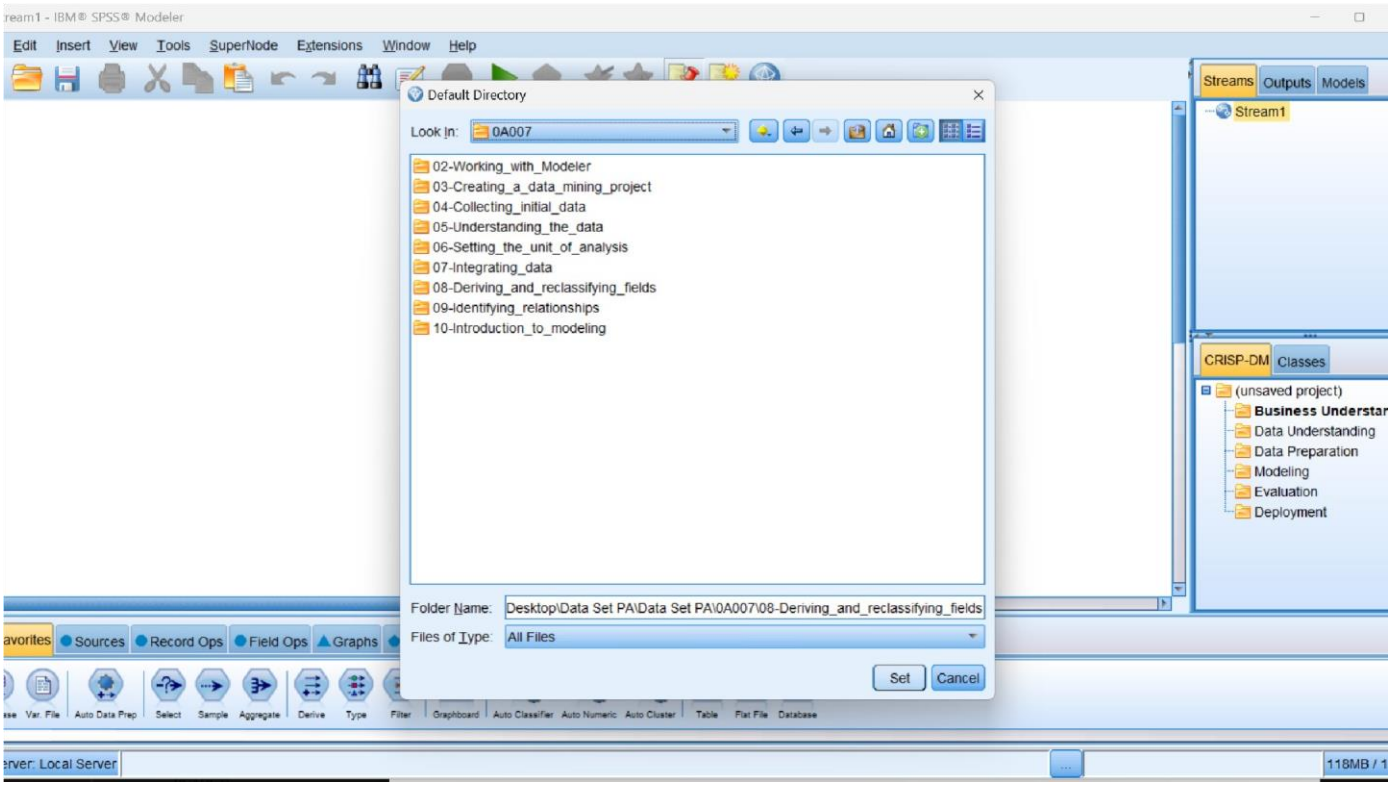
- Start IBM SPSS Modeler and set the working folder.

Definition: - Identify relationships in the telecommunications data

Step 1: - From the File menu, click **Set Directory**.



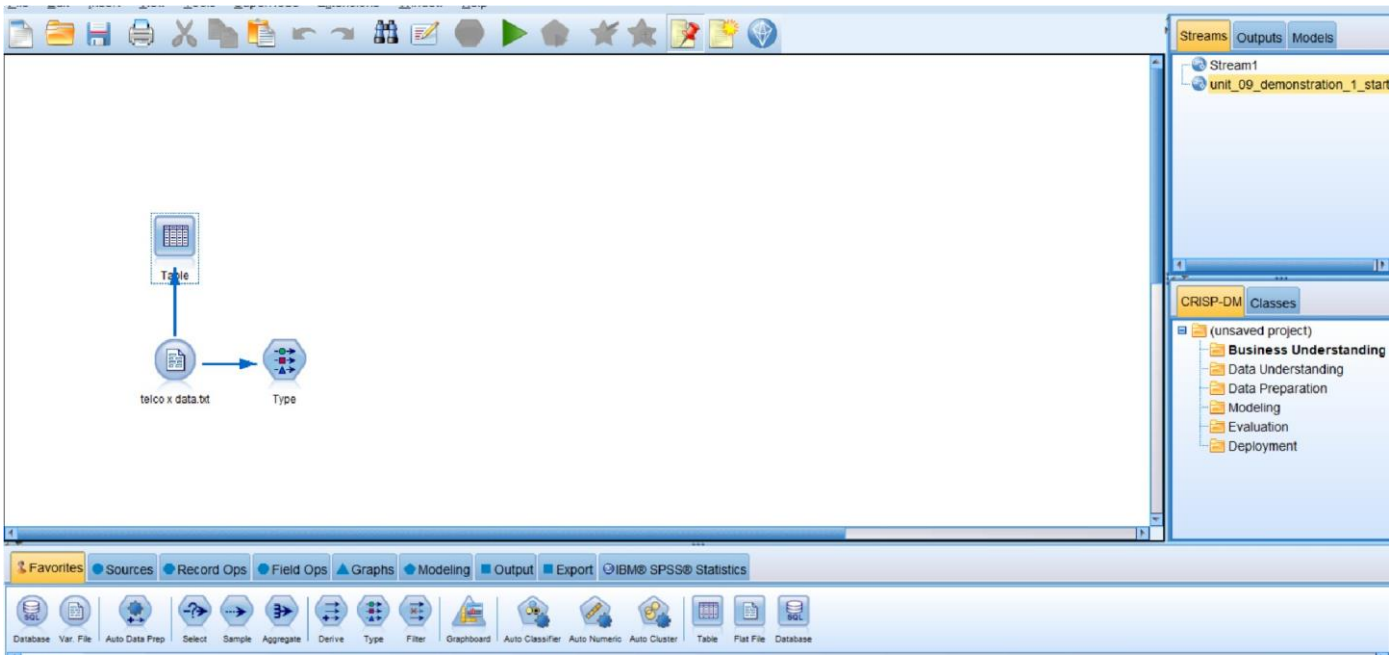
Step 2: - Beside Look in, navigate to **C:\Users\Mr. Abhishek\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.



TASK – 2

- **Open** the **stream**.

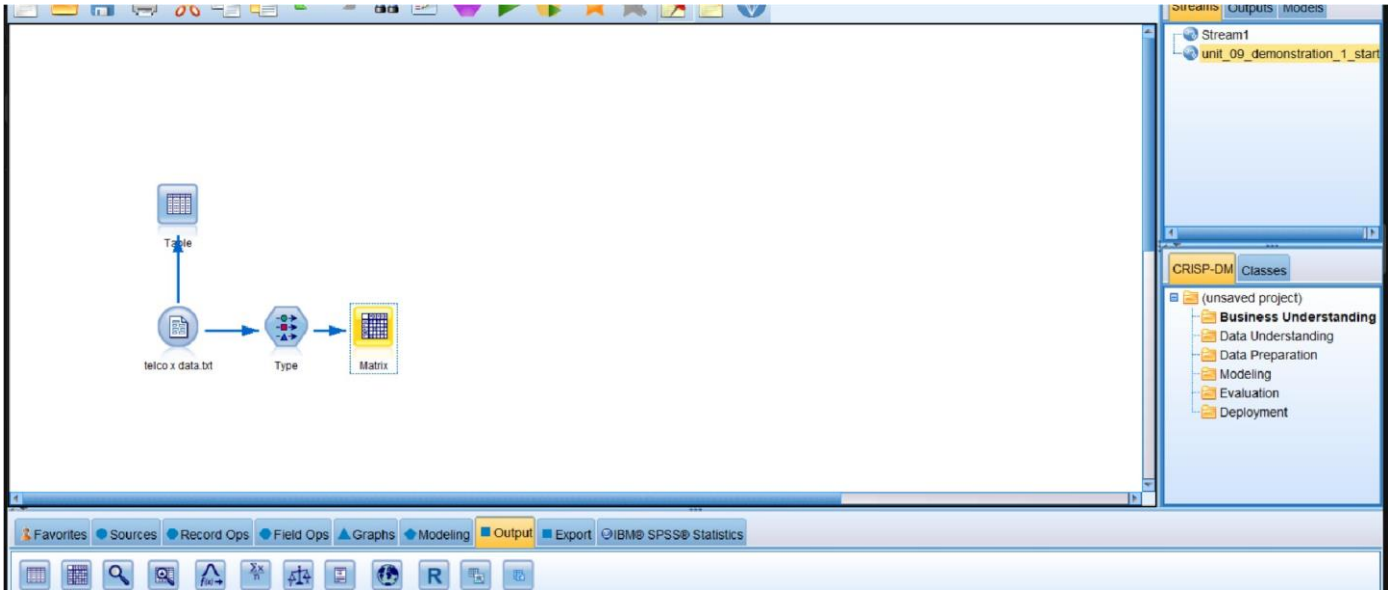
Definition: - From the **File** menu, click **Open Stream**, navigate to **C:\Training\0A007-Integrating_data\09-Identifying_relationships\Start**, click **unit_09_demonstration_1_start.str**, and then click **Open**.



TASK – 3

- **Examine the relationship** between **churn and handset**.

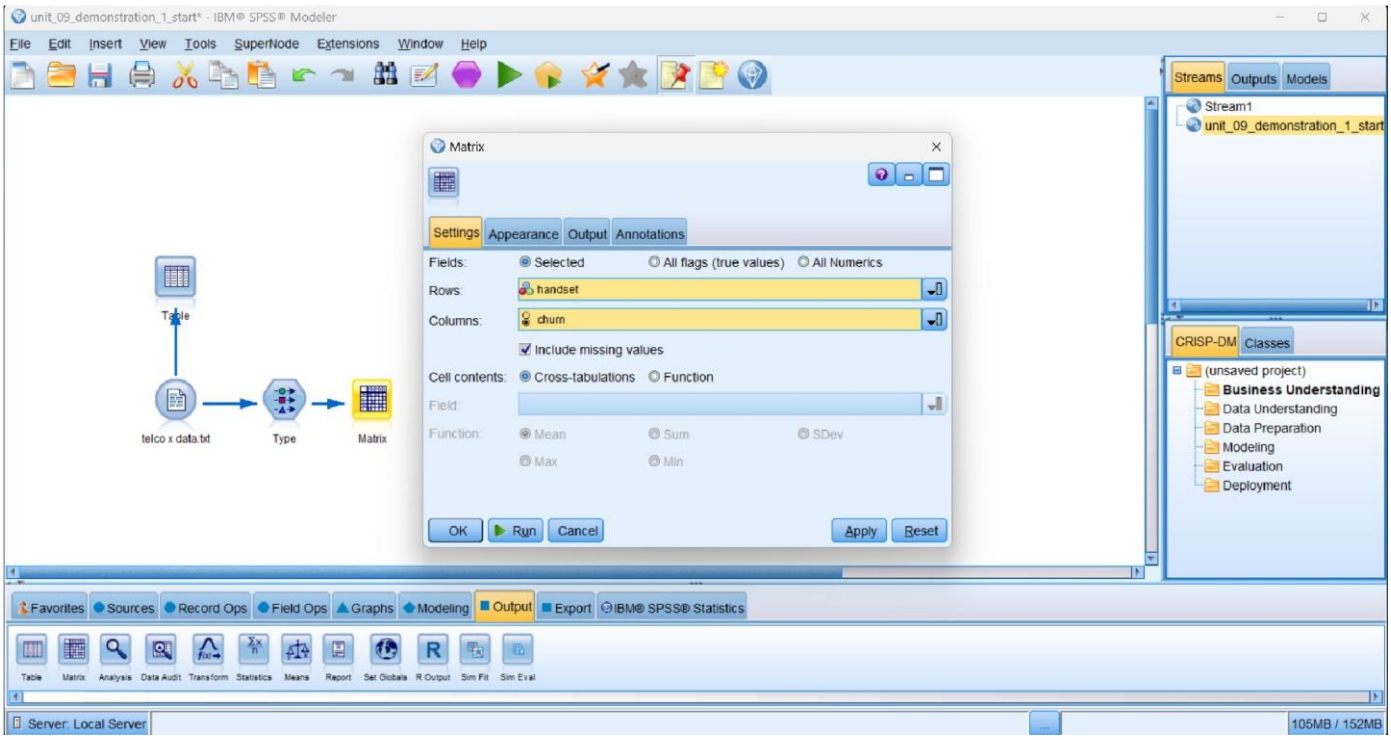
Step 1: From the **Output** palette, add a **Matrix** node downstream from the **Type** node.



Step 2: Edit the **Matrix** node.

Step 3: Beside **Rows**, select **handset** from the list.

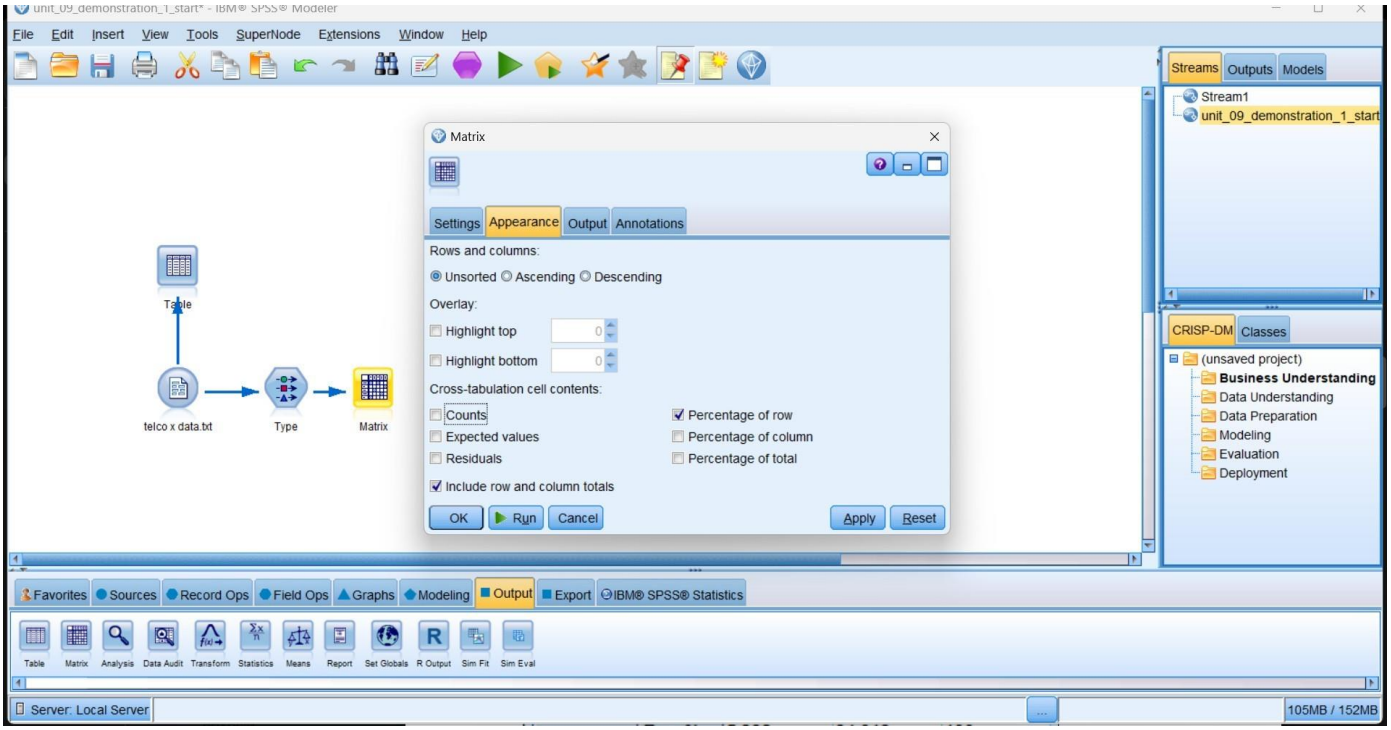
Step 4: Beside **Columns**, select **churn** from the list.



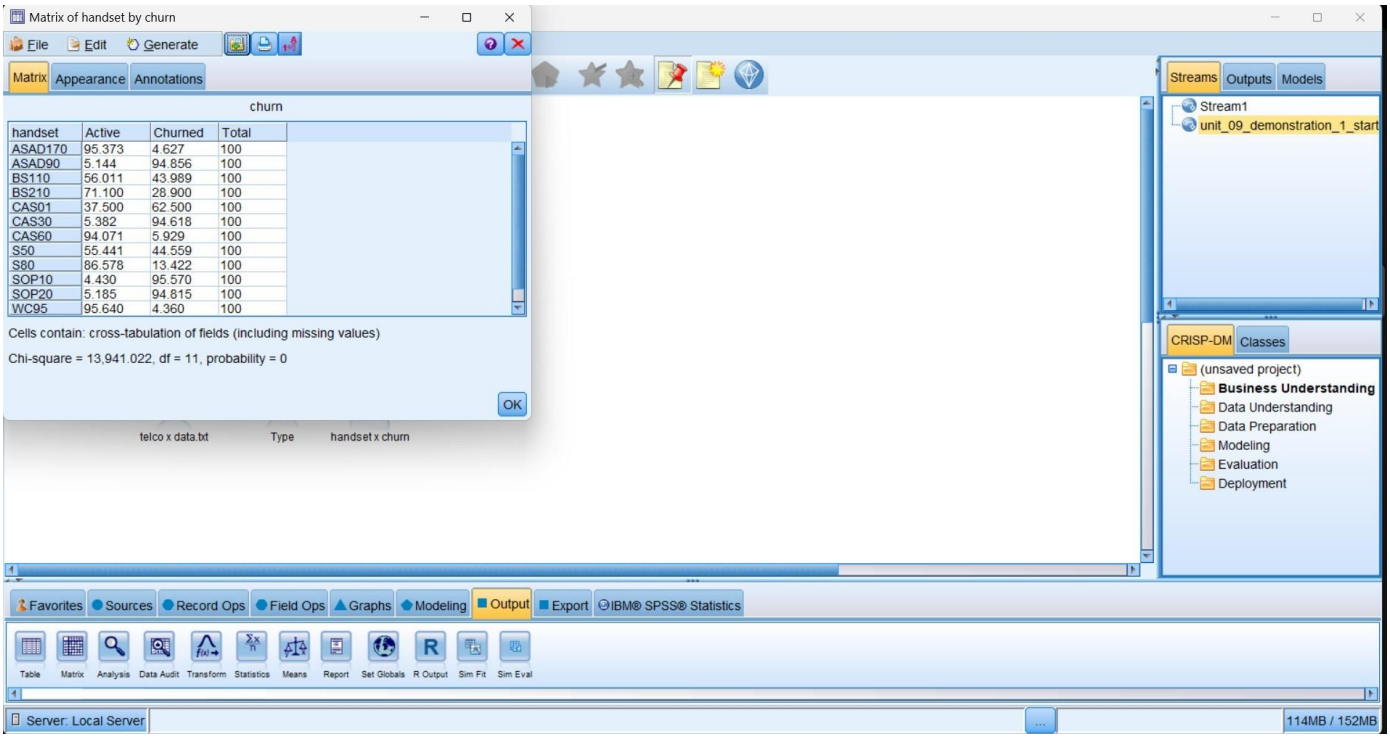
Step 5: Click the **Appearance** tab.

Step 6: Select the **Percentage of row** option.

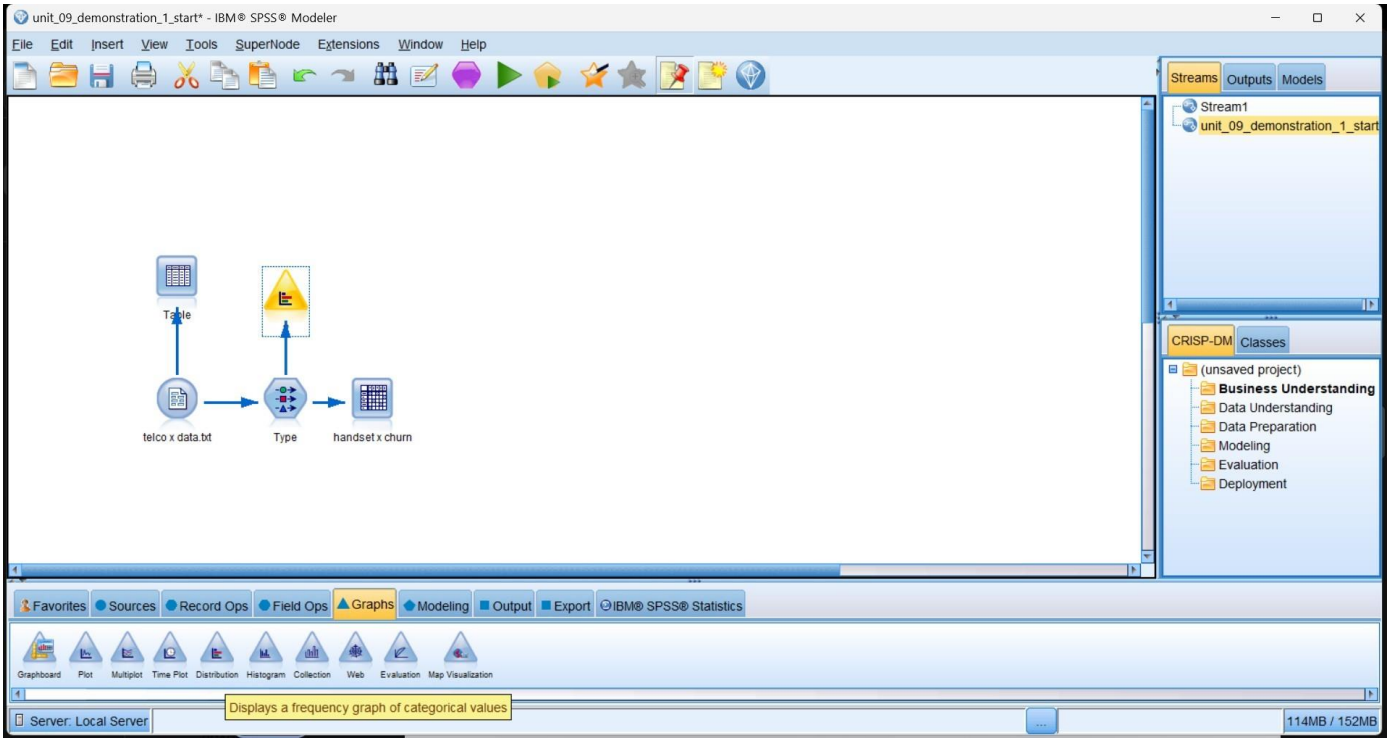
Step 7: Select the **Include row and column totals** option.



Step 8: Click **Run**.



Step 9: From the **Graphs** palette, add a **Distribution** node downstream from the **Type** node.

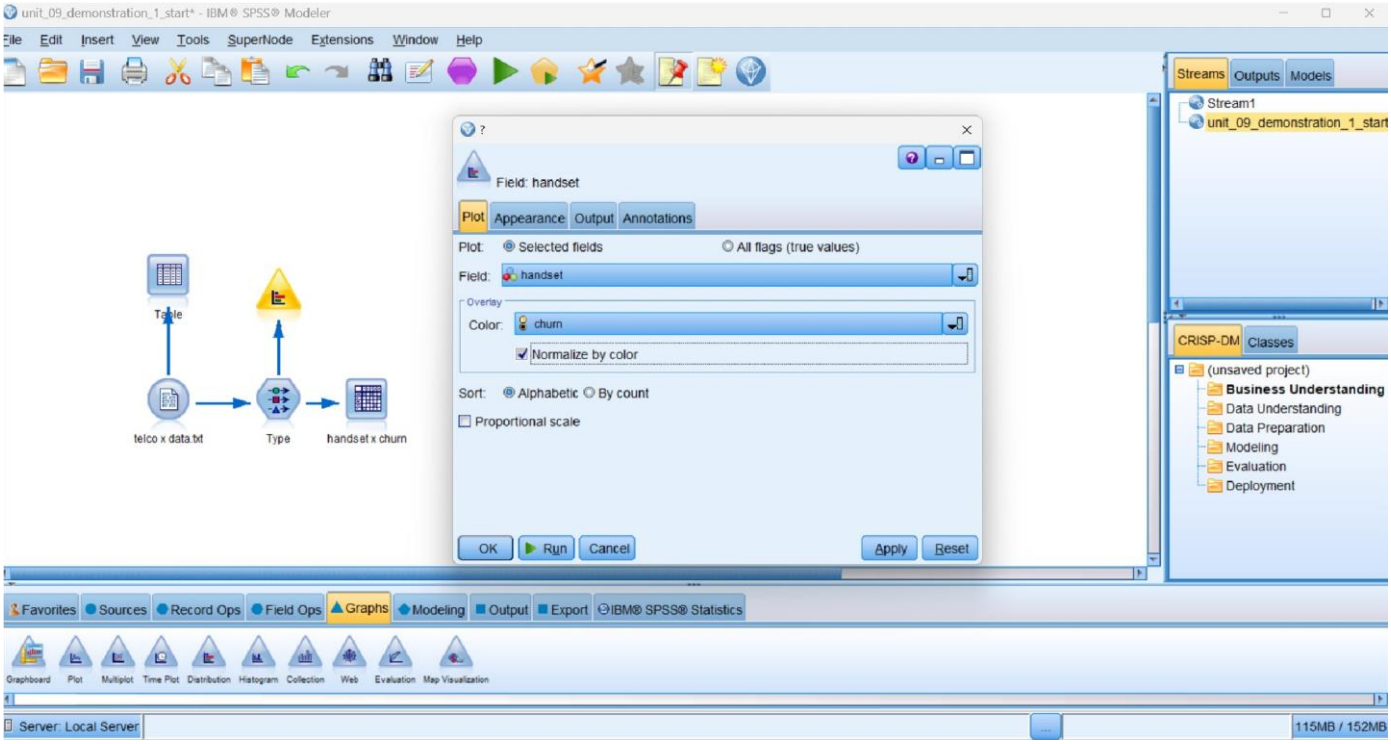


Step 10: Edit the **Distribution** node.

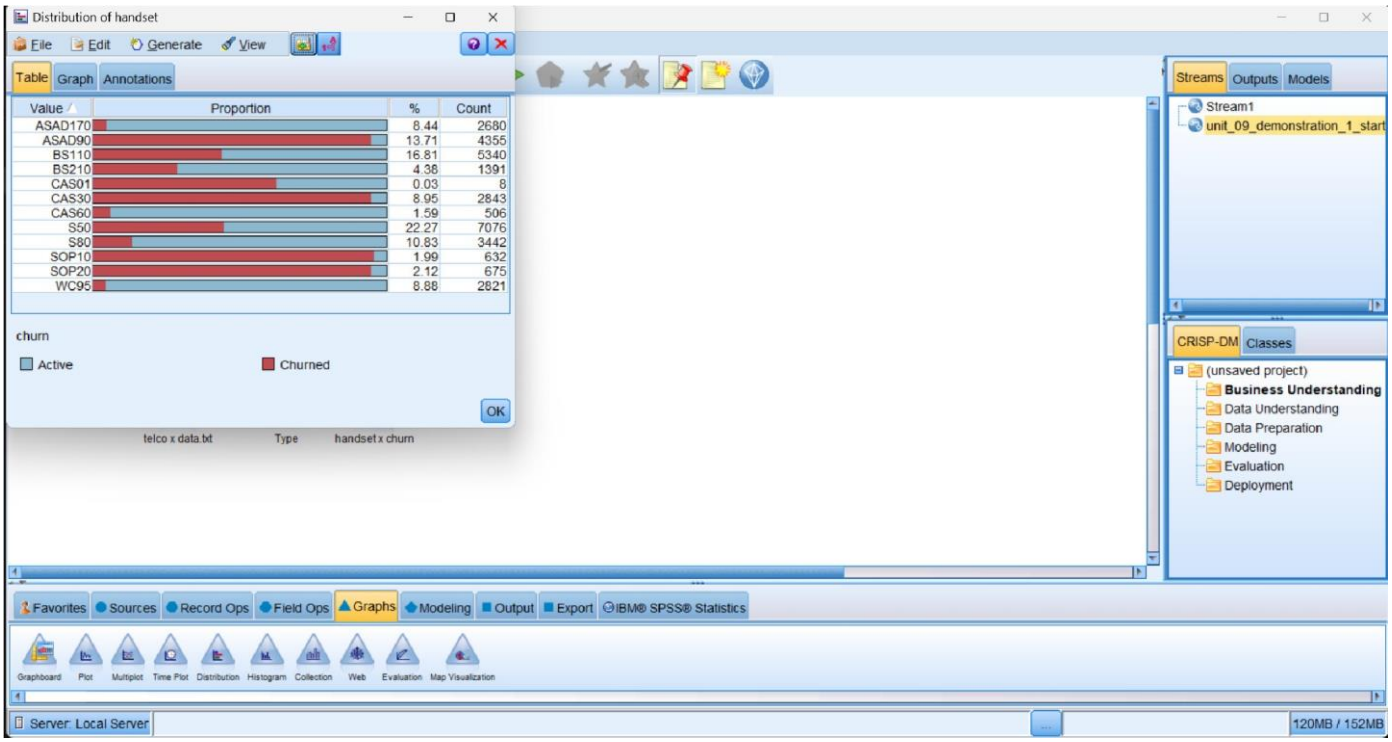
Step 11: Beside **Field**, select **handset**.

Step 12: Beside **Color**, select **churn**.

Step 13: Enable the **Normalize by color** check box.



Step 14: Click **Run.**

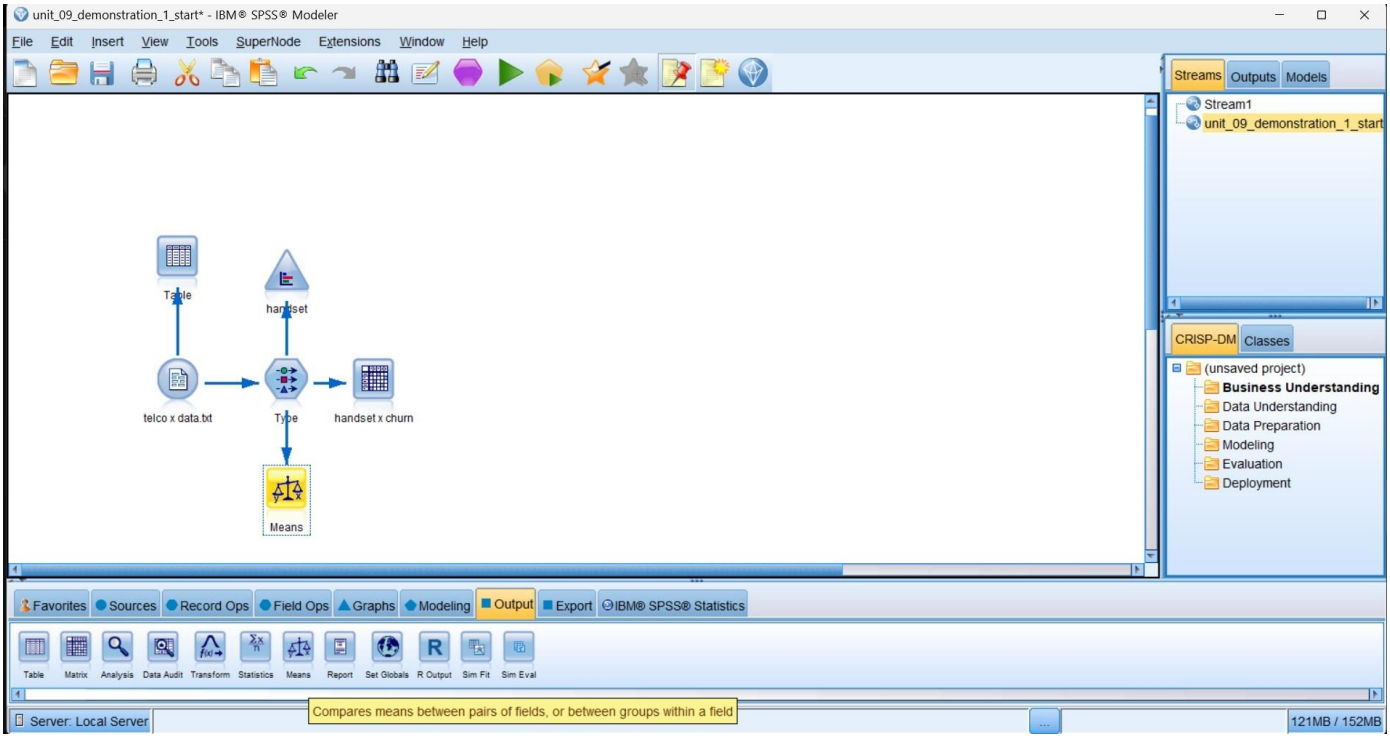


TASK – 4

- **Examine the relationship** between **churn and number** of **dropped calls**.

Definition: - In this task you will see if the number of dropped calls is related to churn. The first field is continuous, the second field is categorical. Therefore you will use a Means node and a Histogram node.

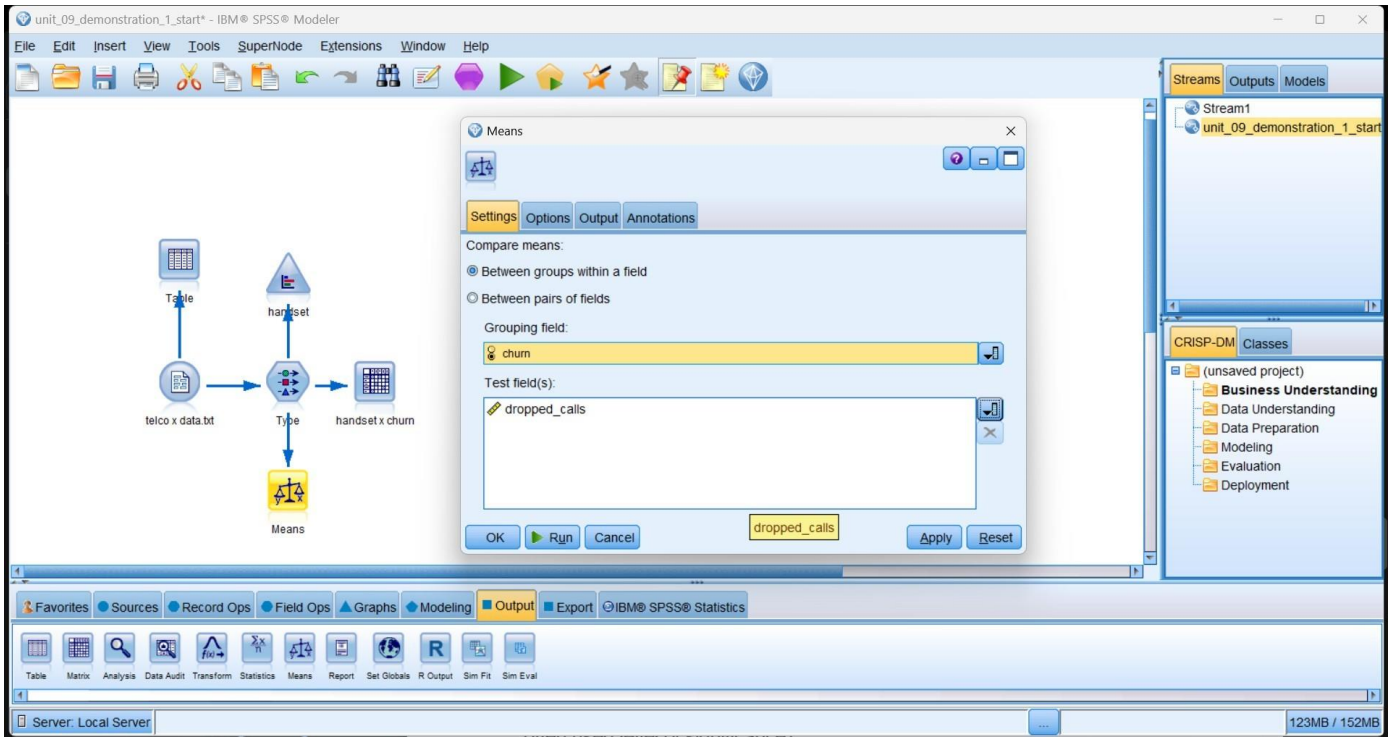
Step 1: From the **Output** palette, add a **Means** node downstream from the **Type** node.



Step 2: Edit the **Means** node.

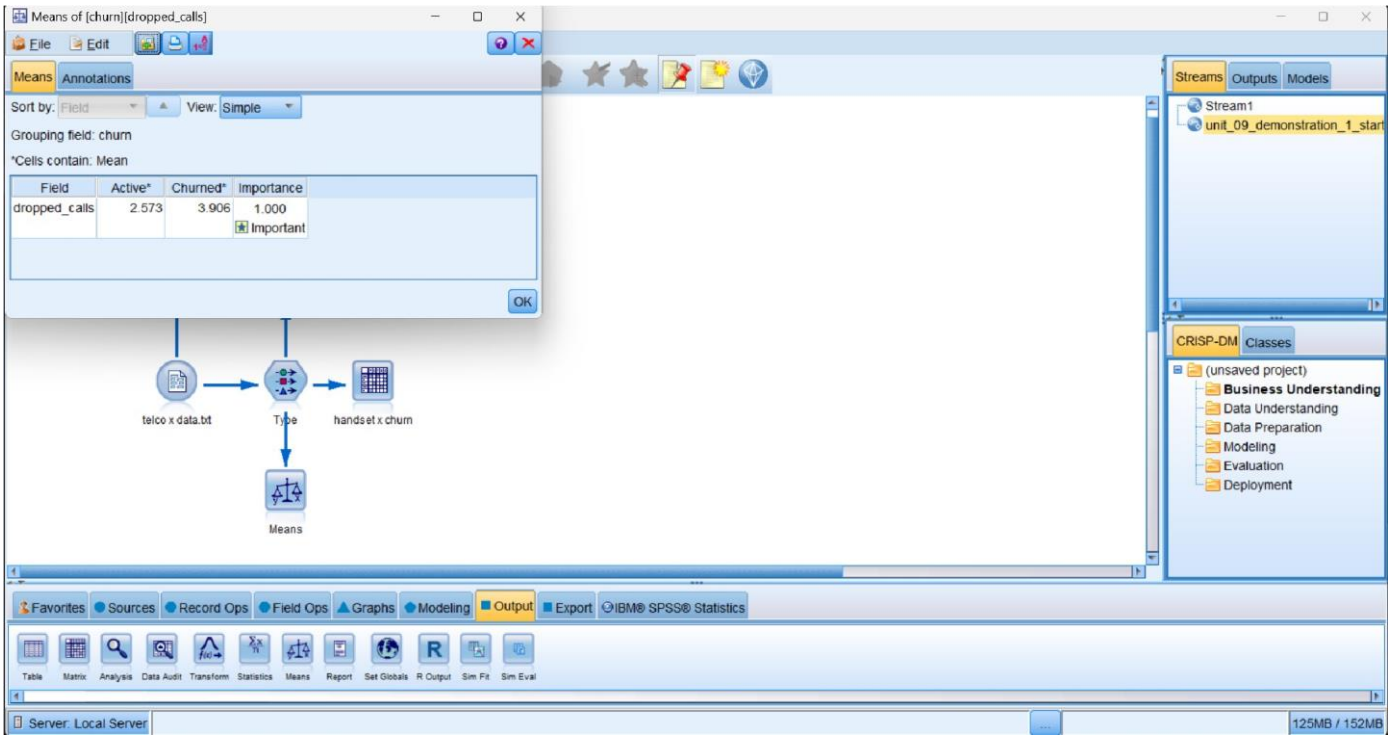
Step 3: For **Grouping** field, select **churn**.

Step 4: For **Test field(s)**, select **dropped_calls**.

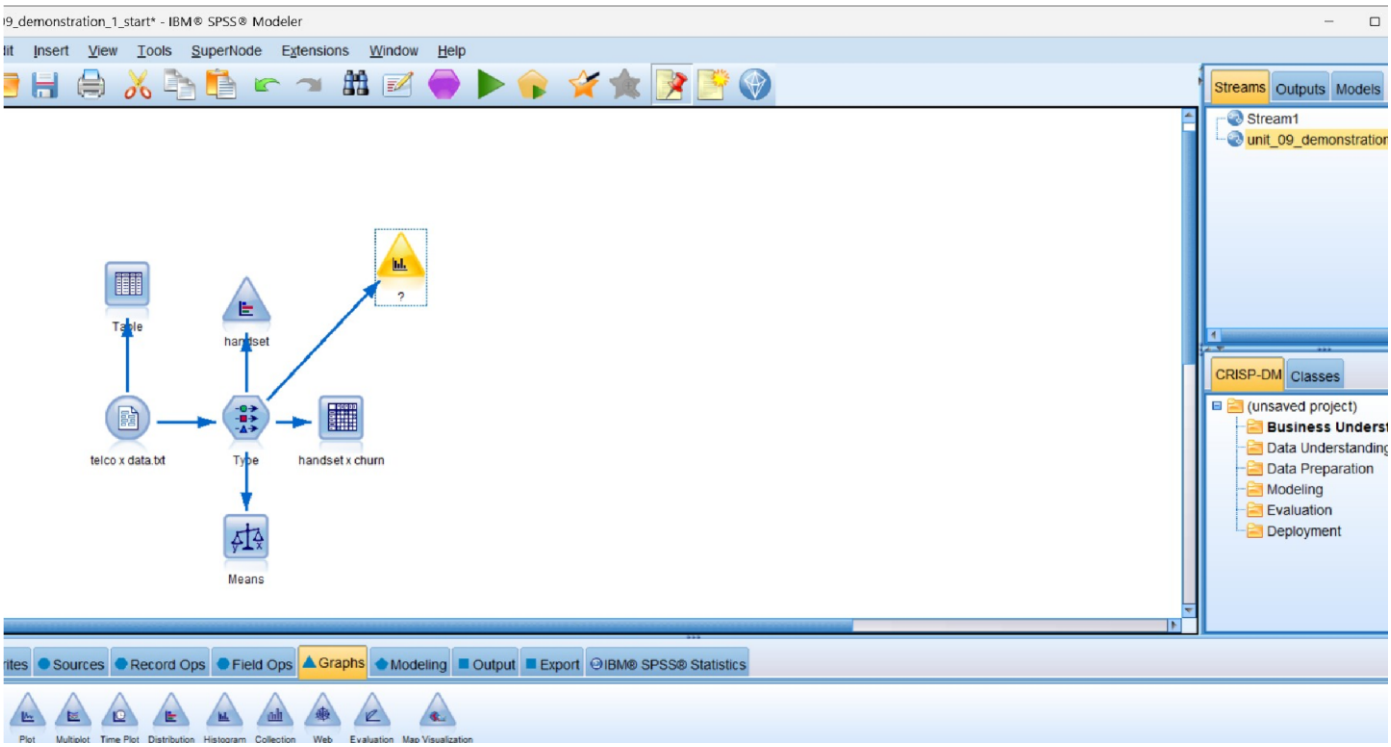


Step 5: Click the **Options** tab.

Step 6: Click **Run.**



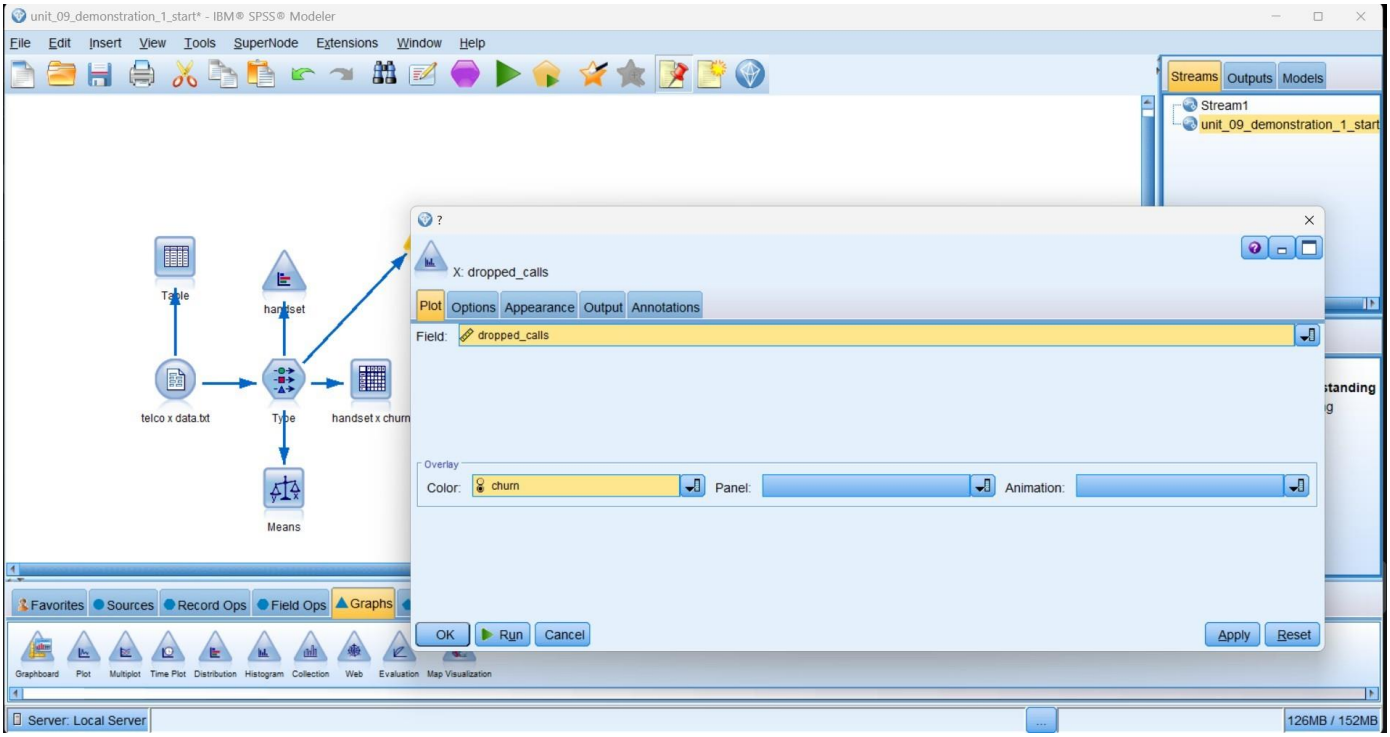
Step 7: From the **Graphs** palette, add a **Histogram** node downstream from the **Type** node.



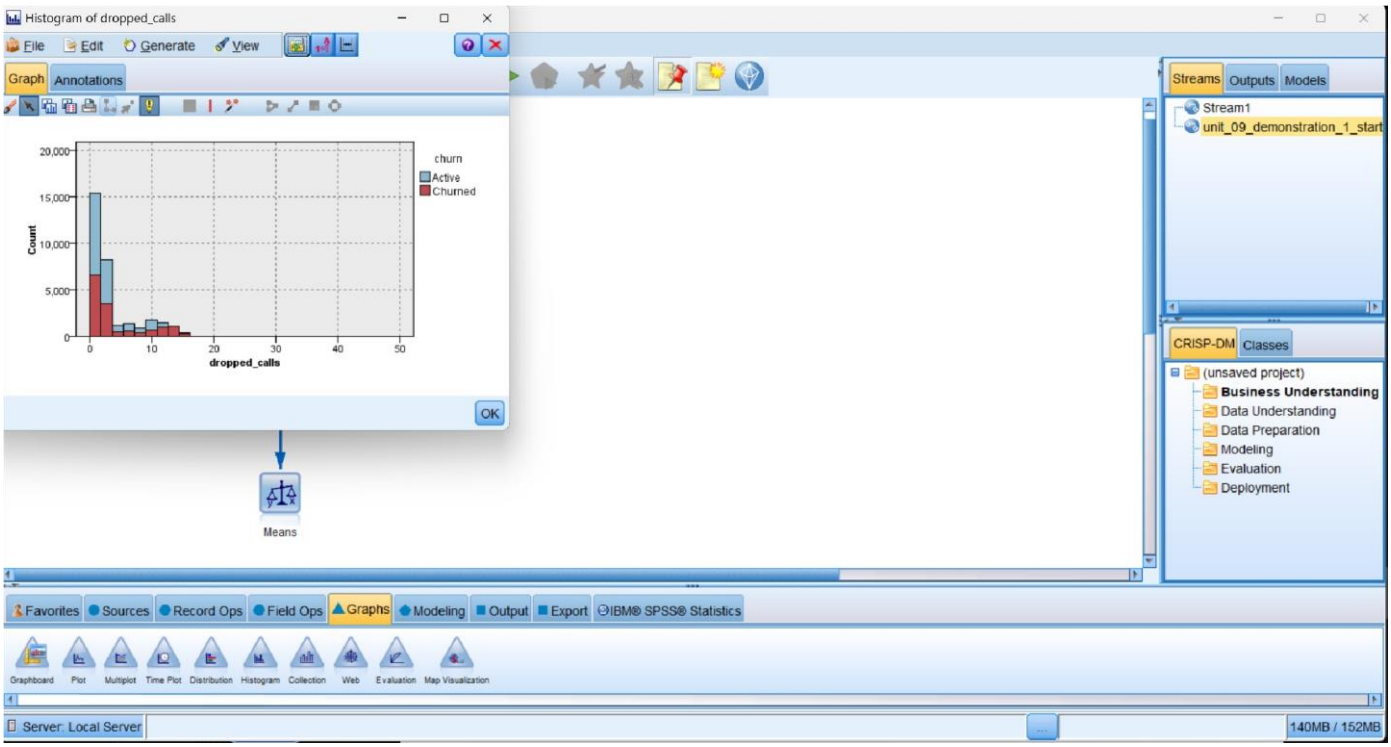
Step 8: Edit the **Histogram** node.

Step 9: Beside **Field**, select **dropped_calls**.

Step 10: Beside **Color**, select **churn**.



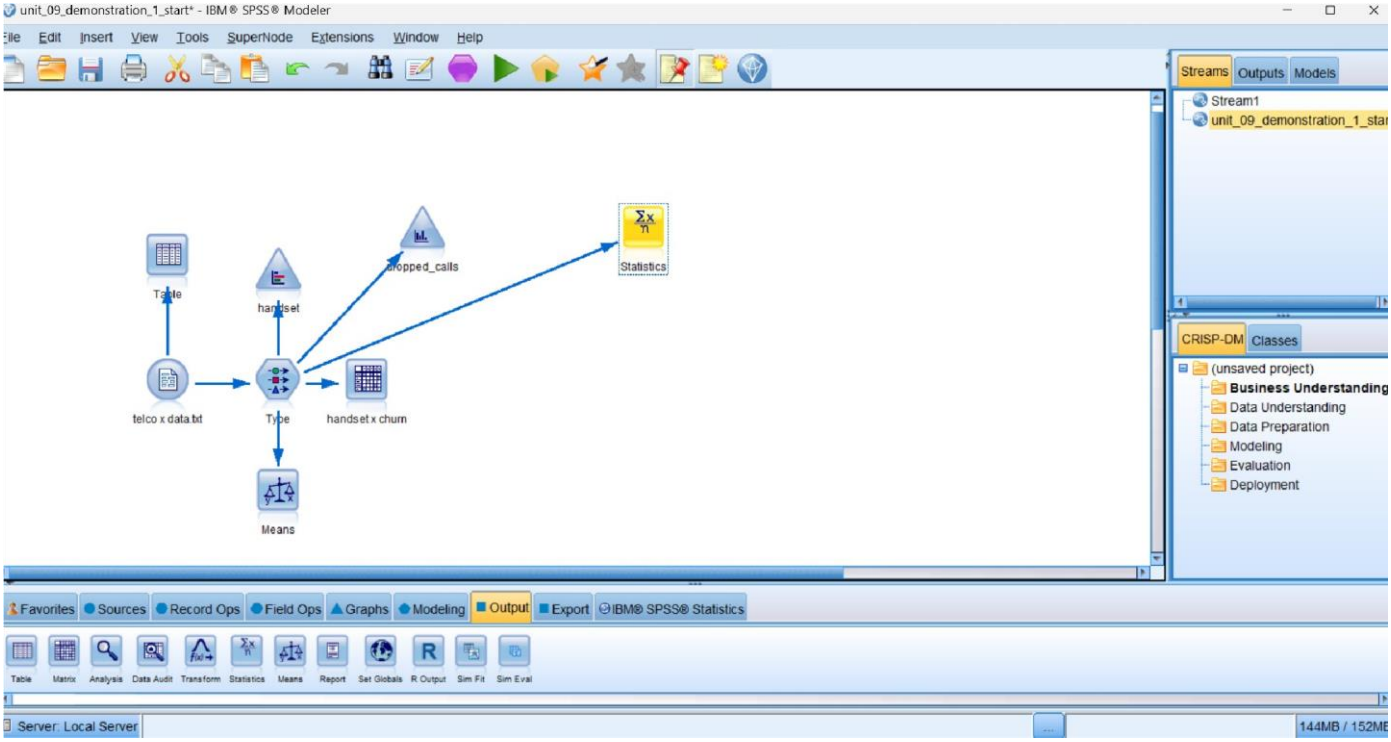
Step 11: Click **Run.**



TASK – 5

- **Examine the relationship** between number of **products** and revenues.

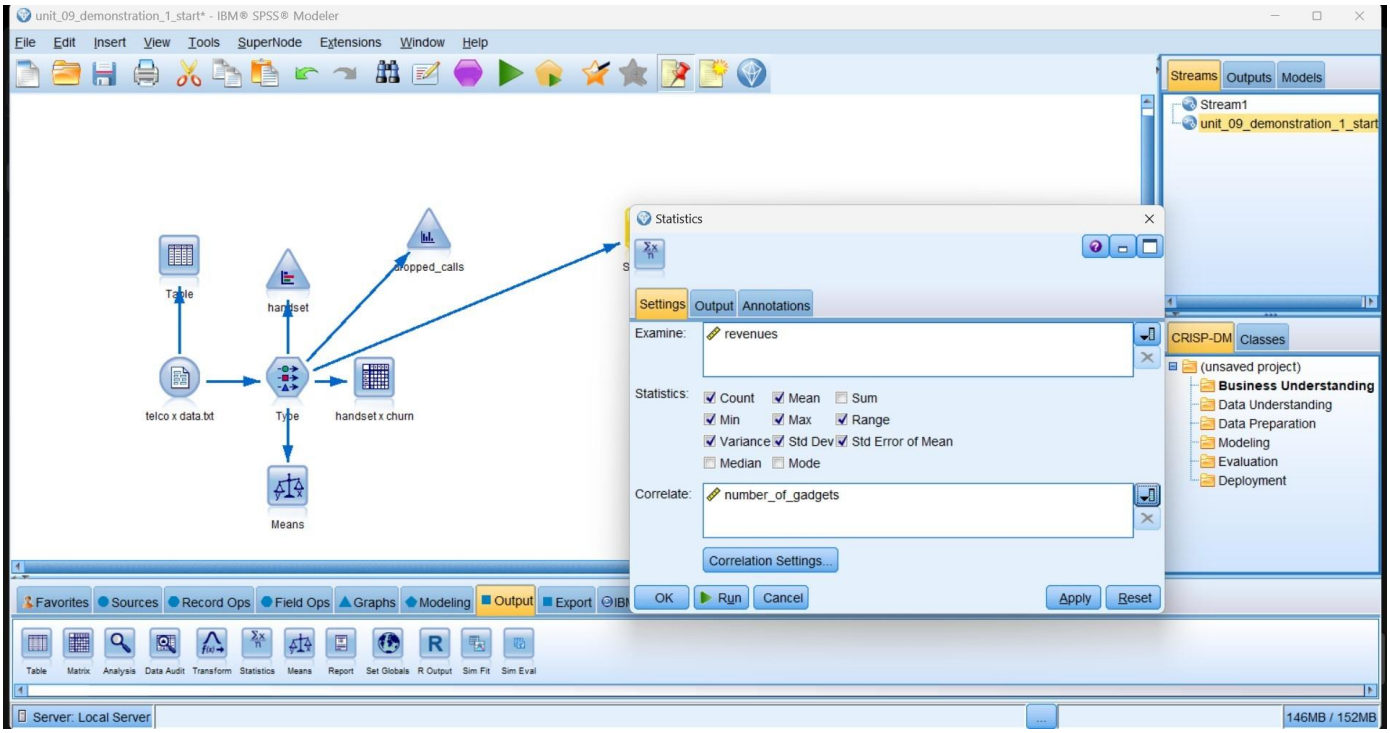
Step 1: From the **Output** palette, add a **Statistics** node downstream from the **Type** node.



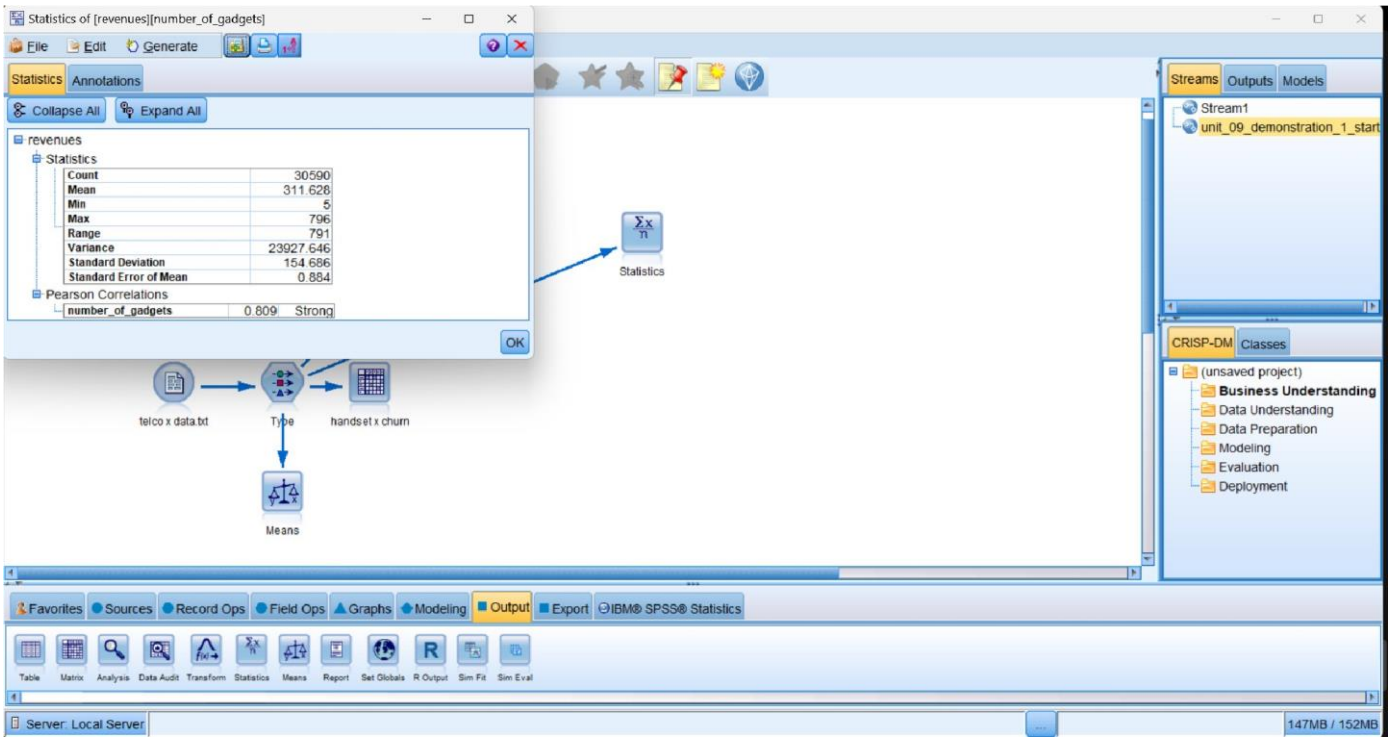
Step 2: Edit the **Statistics** node.

Step 3: Beside **Examine**, select **revenues** from the list.

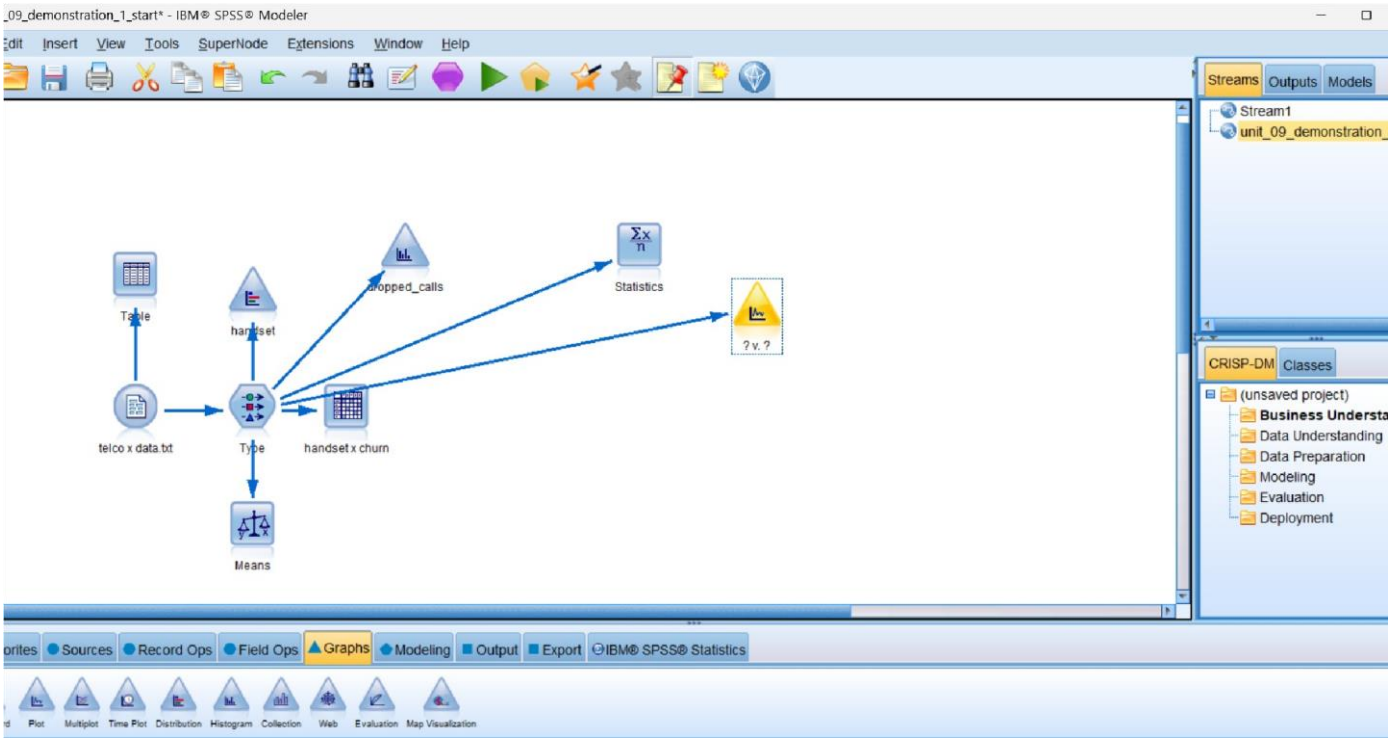
Step 4: Beside **Correlate**, select **number_of_gadgets** from the list.



Step 5: Click **Run.**



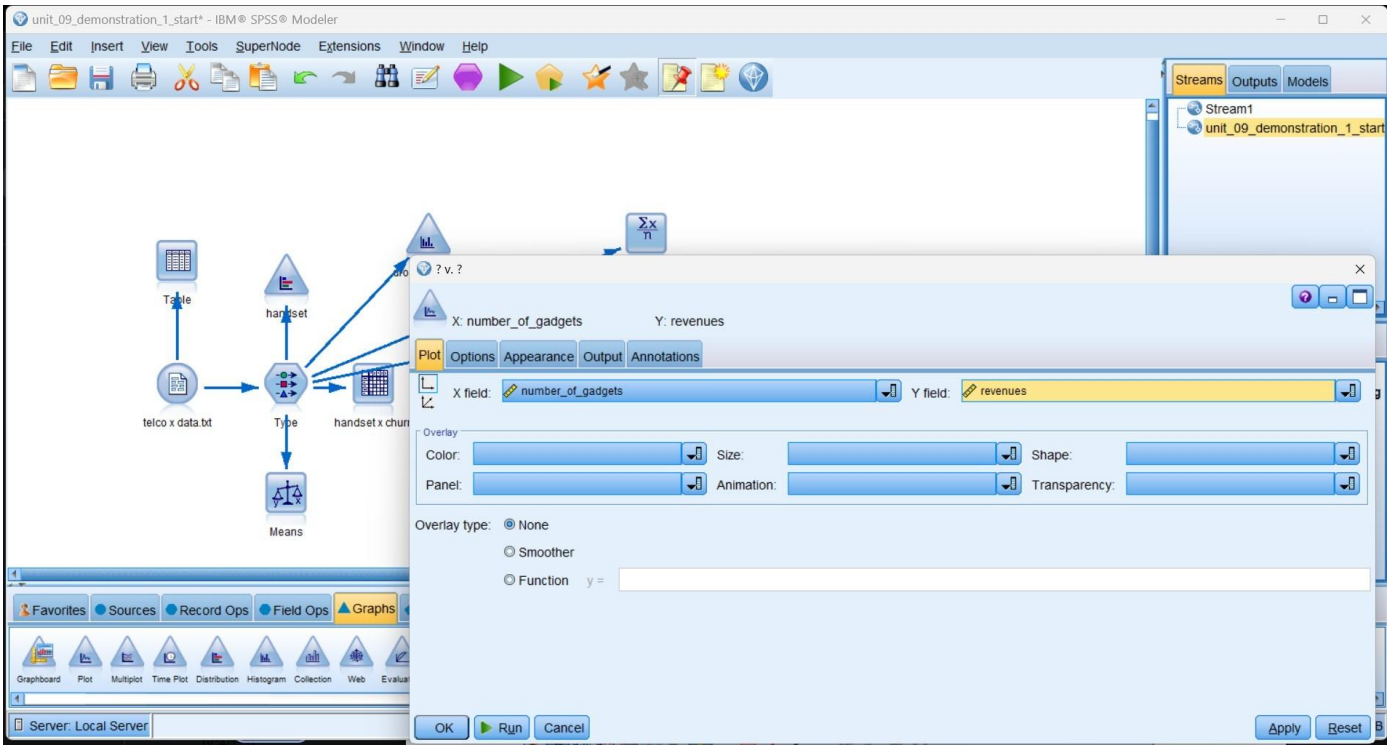
Step 6: From the **Graphs** palette, add a **Plot** node downstream from the **Type** node.



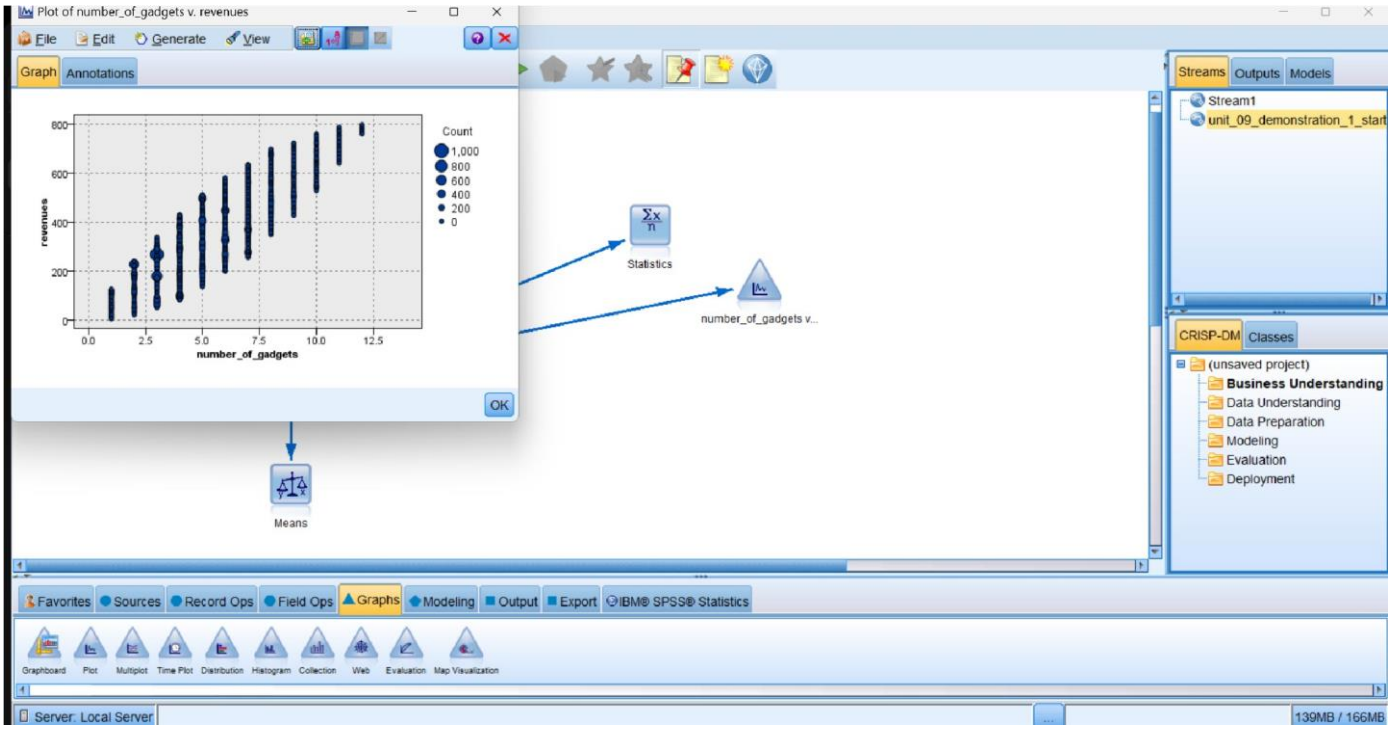
Step 7: Edit the **Plot** node.

Step 8: For **X field**, select **number_of_gadgets**.

Step 9: For **Y field**, select **revenues**.



Step 10: Click **Run**.


Plot of number_of_gadgets v. revenues
File
Edit
Generate
View
Graph
Annotations
revenues
number_of_gadgets
Count
1,000
800
600
400
200
0
OK
Statistics
number_of_gadgets v...
Means
Streams
Outputs
Models
Stream1
unit_09_demonstration_1_start
CRISP-DM
Classes
(unsaved project)
Business Understanding
Data Understanding
Data Preparation
Modeling
Evaluation
Deployment
Favorites
Sources
Record Ops
Field Ops
Graphs
Modeling
Output
Export
IBM® SPSS® Statistics
Graphboard
Plot
Multiplot
Time Plot
Distribution
Histogram
Collection
Web
Evaluation
Map Visualization
Server: Local Server
139MB / 166MB

Practical - 8

Definition: - This practical explains how to use the IBM SPSS Modeler user interface to create, edit, modify, and save streams. It also demonstrates how to work with different nodes such as Excel, Sample, Derive, Histogram, Statistics Export, Select, and Table nodes

Outcomes/Learning: - Learners will become familiar with the IBM SPSS Modeler user interface and will understand how to create and modify streams, connect different nodes, generate outputs, add comments, and use table output to generate a Select node.

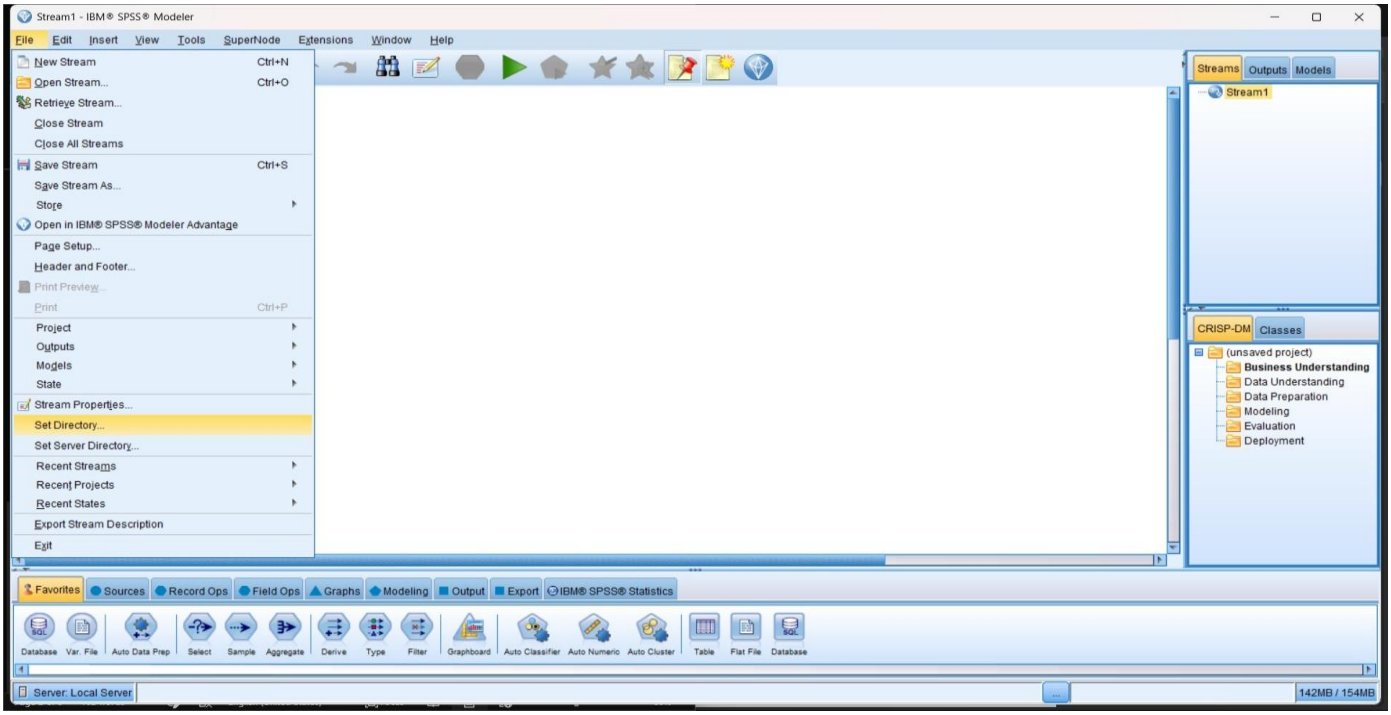
Required Tool: - The required tool will be "**IBM SPSS Modeler"**.

Purpose: - You work as a data miner for a telecommunications firm. You will first predict churn by using CHAID and Neural Net. You will also compare the accuracy of these models. Then, you will use TwoStep to find groups (clusters) of similar customers, based on usage.

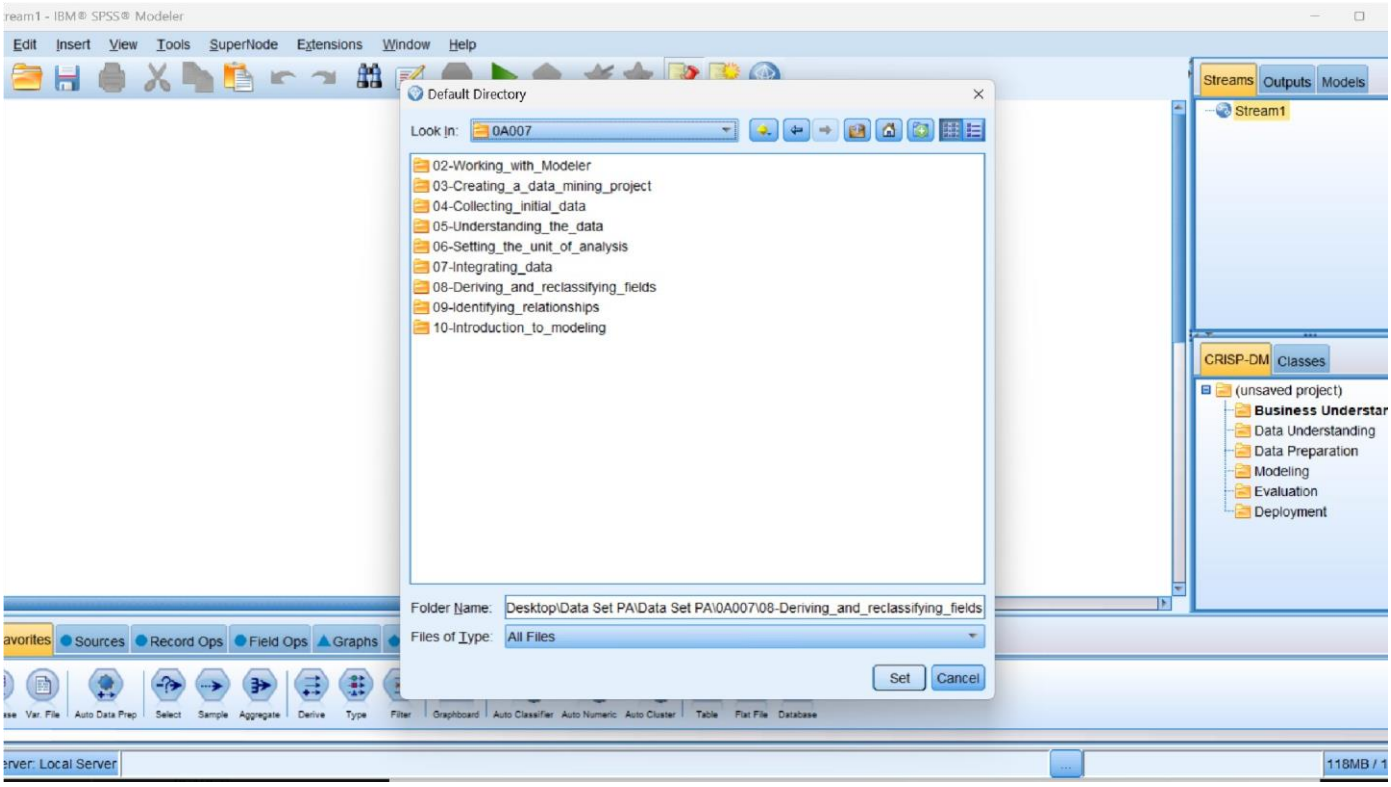
TASK – 1

- Start IBM SPSS Modeler and set the working folder.

Step 1: - From the File menu, click **Set Directory**.



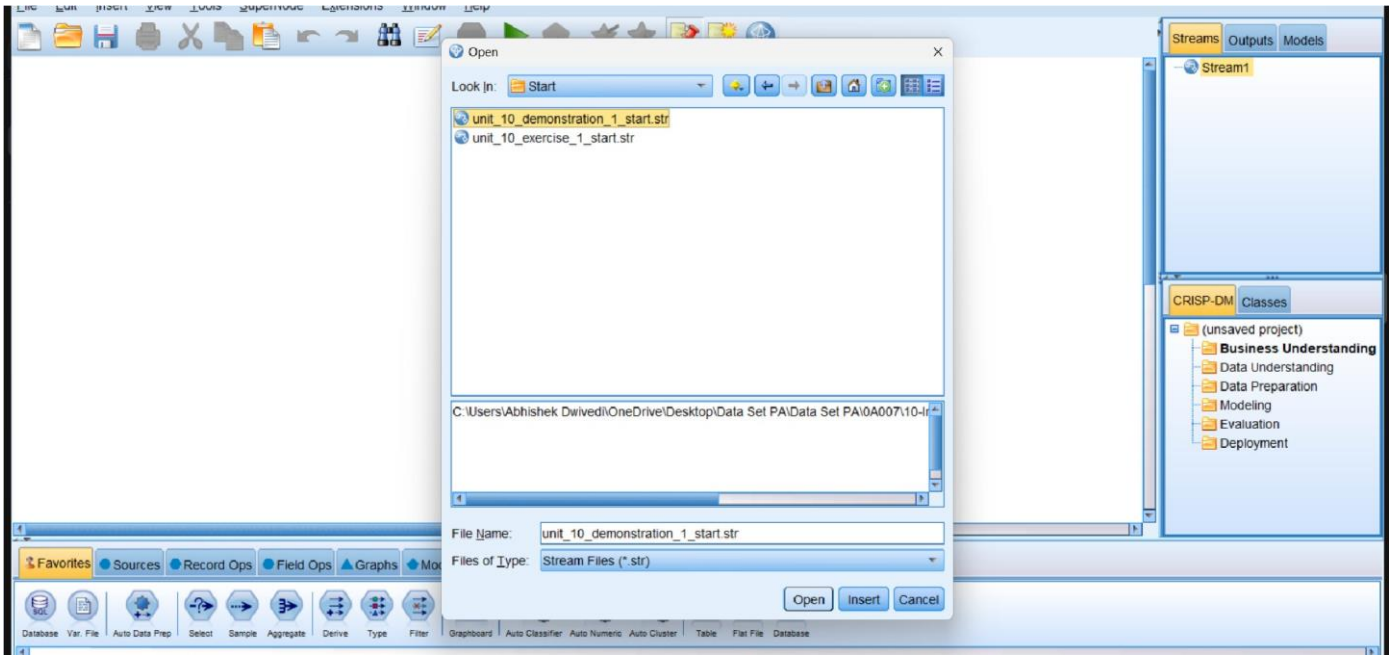
Step 2: - Beside Look in, navigate to **C:\Users\Mr. Abhishek\OneDrive\Desktop\SPSS_DEMO\Data Set PA\0A007**, and then click **Set**.



TASK – 2

- **Open** the **stream**.

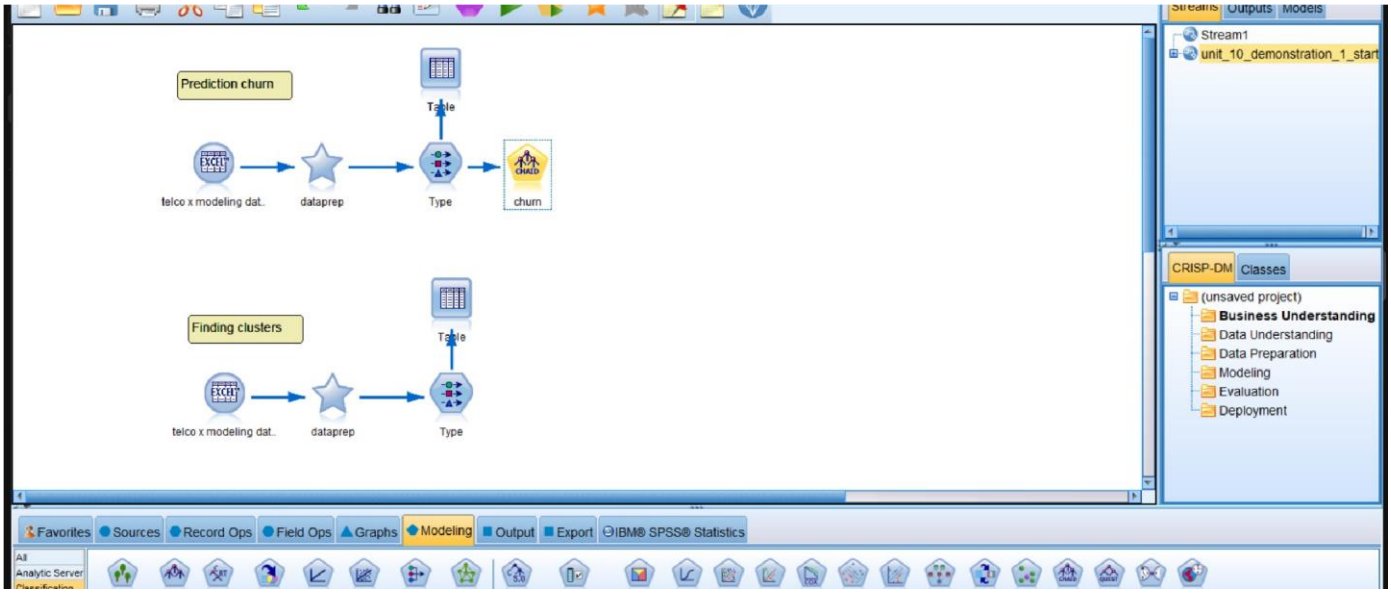
Definition: - From the **File** menu, click **Open Stream**, navigate to **C:\Training\0A007\10-Introduction_to_modeling\Start**, click **unit_10_demonstration_1_start.str**, and then click **Open**.



TASK – 3

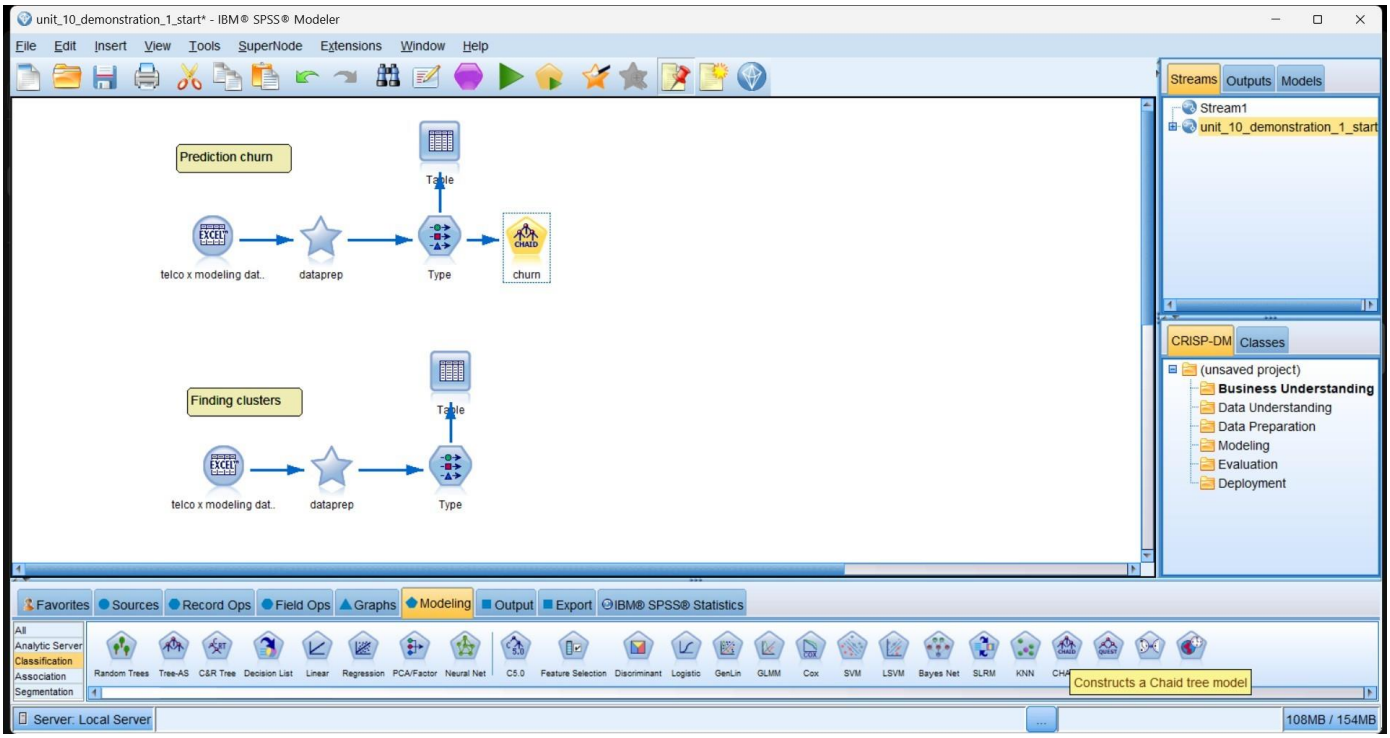
- Predict churn.

Step 1: Scroll to the upper branch of the stream, which is labeled **Prediction churn**.

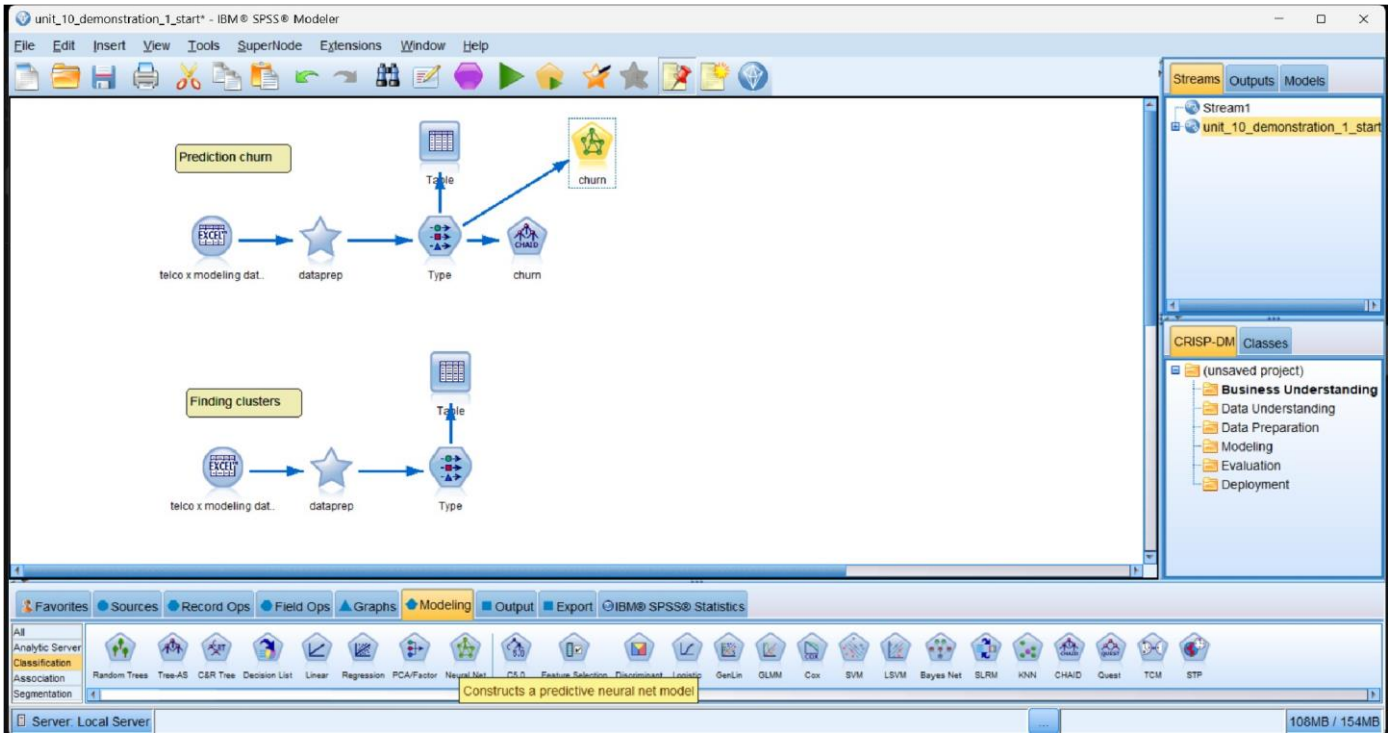


Step 2: In the **Modeling** palette, on the left pane, click the **Classification** sub Palette.

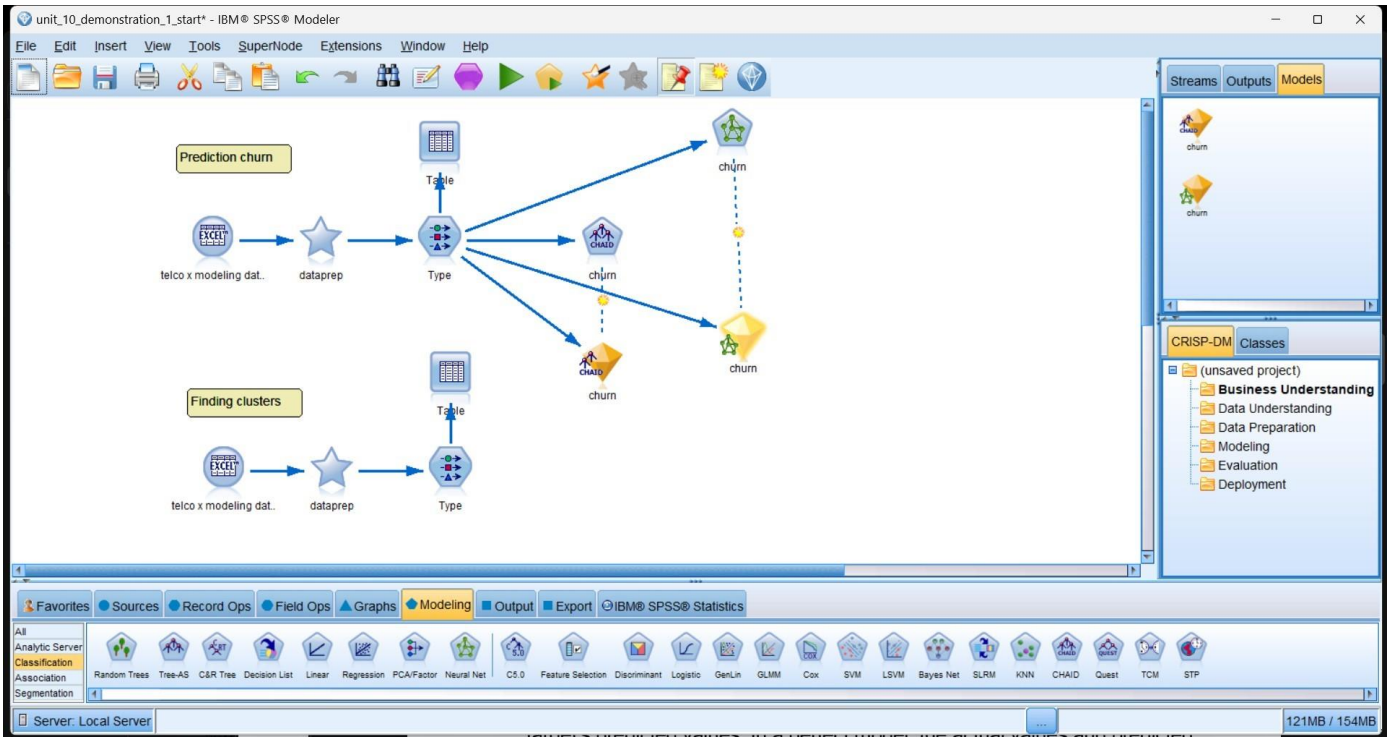
Step 3: Add a **CHAID** node downstream from the top **Type** node.



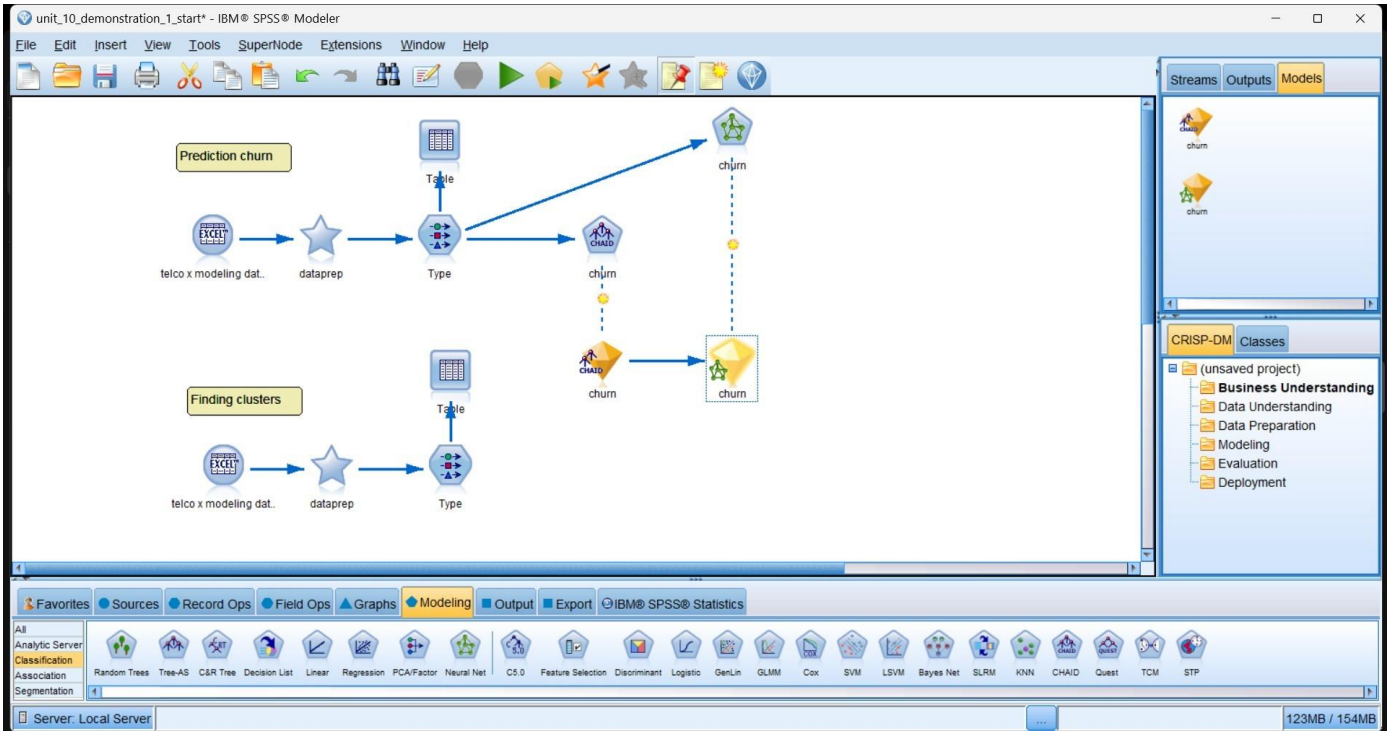
Step 4: Add a **Neural Net** node downstream from the top **Type** node.



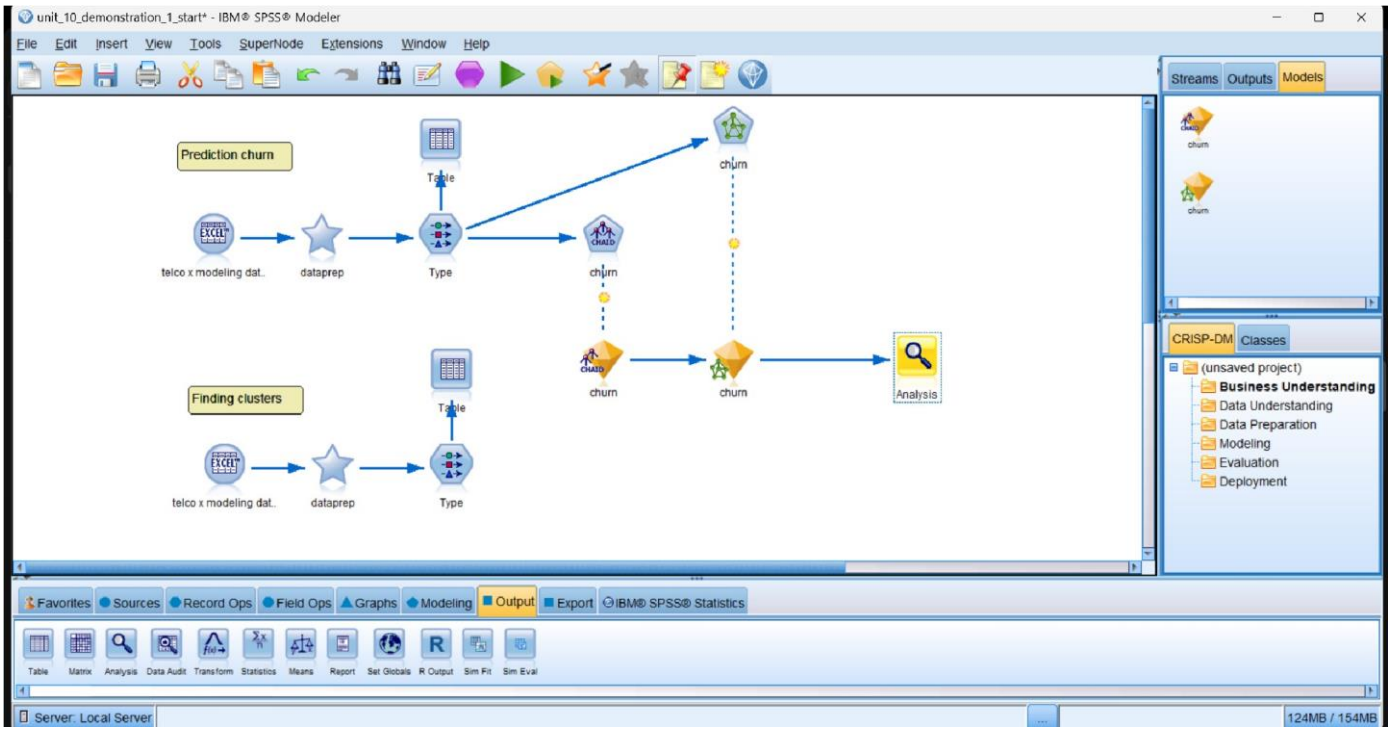
Step 5: Run the **CHAID** node, and then run the **Neural Net** node.



Step 6: Disconnect the **Neural Net** model nugget from the **Type** node, place it downstream from the **CHAID** model nugget, and then connect the **CHAID** model nugget to the **Neural Net** model nugget.



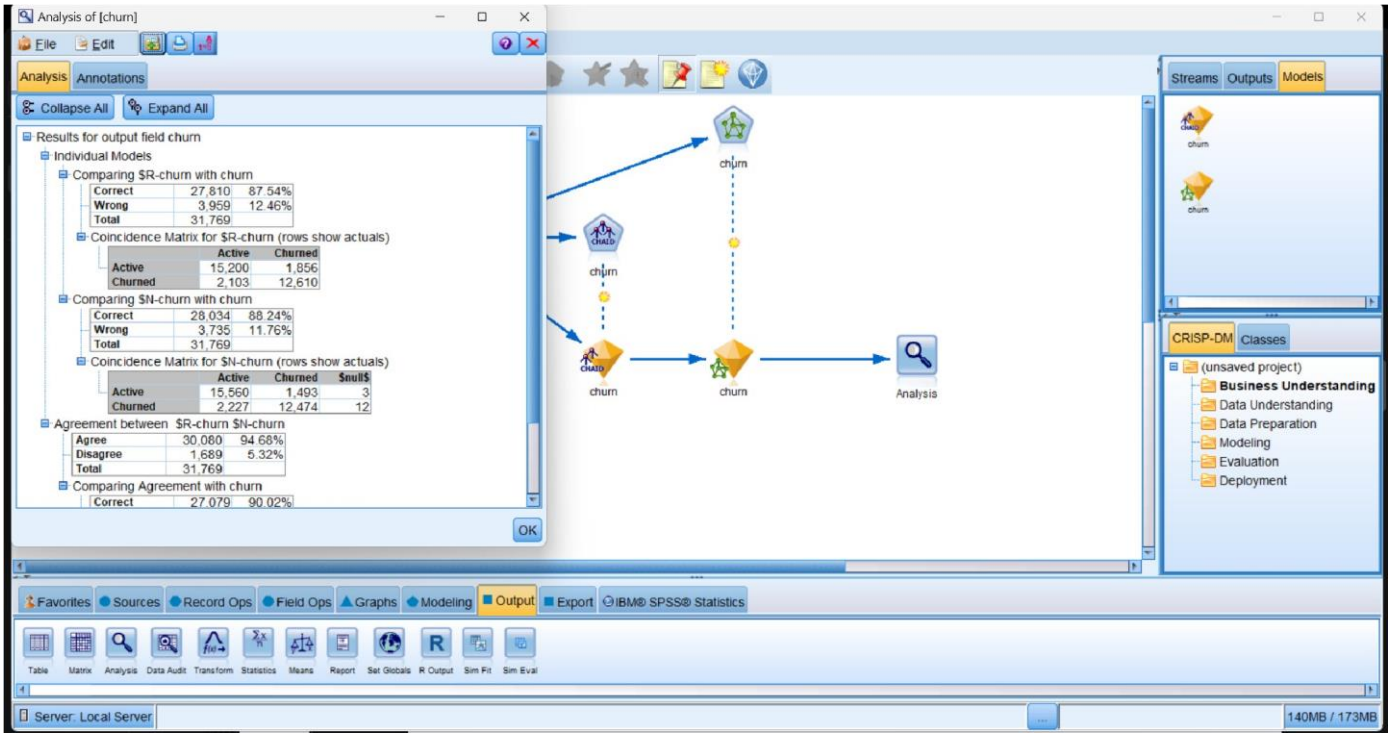
Step 7: From the **Output** palette, add an **Analysis** node downstream from the **Neural Net** model nugget.



Step 8: Edit the **Analysis** node.

Step 9: Enable the **Coincidence matrices (for symbolic targets)** option.

Step 10: Run the **Analysis** node.



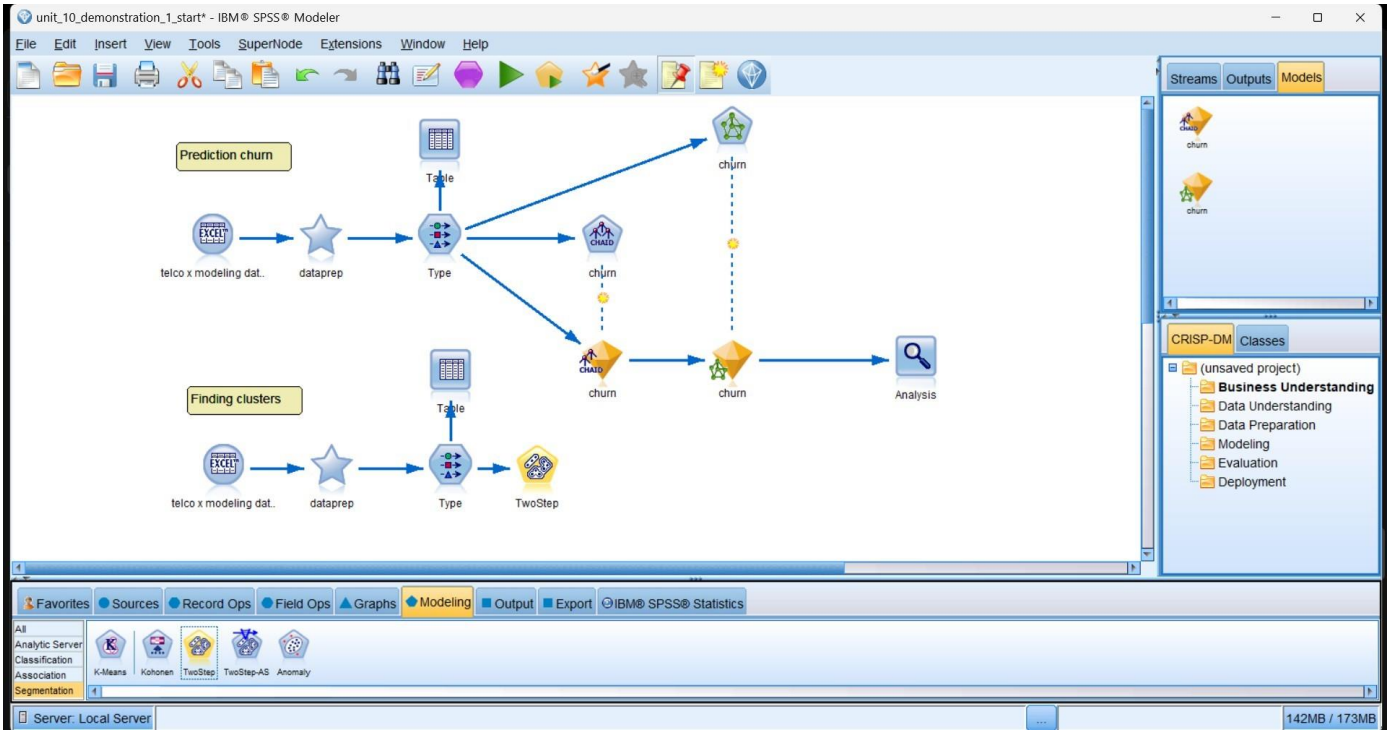
TASK – 4

- Find groups of similar customers.

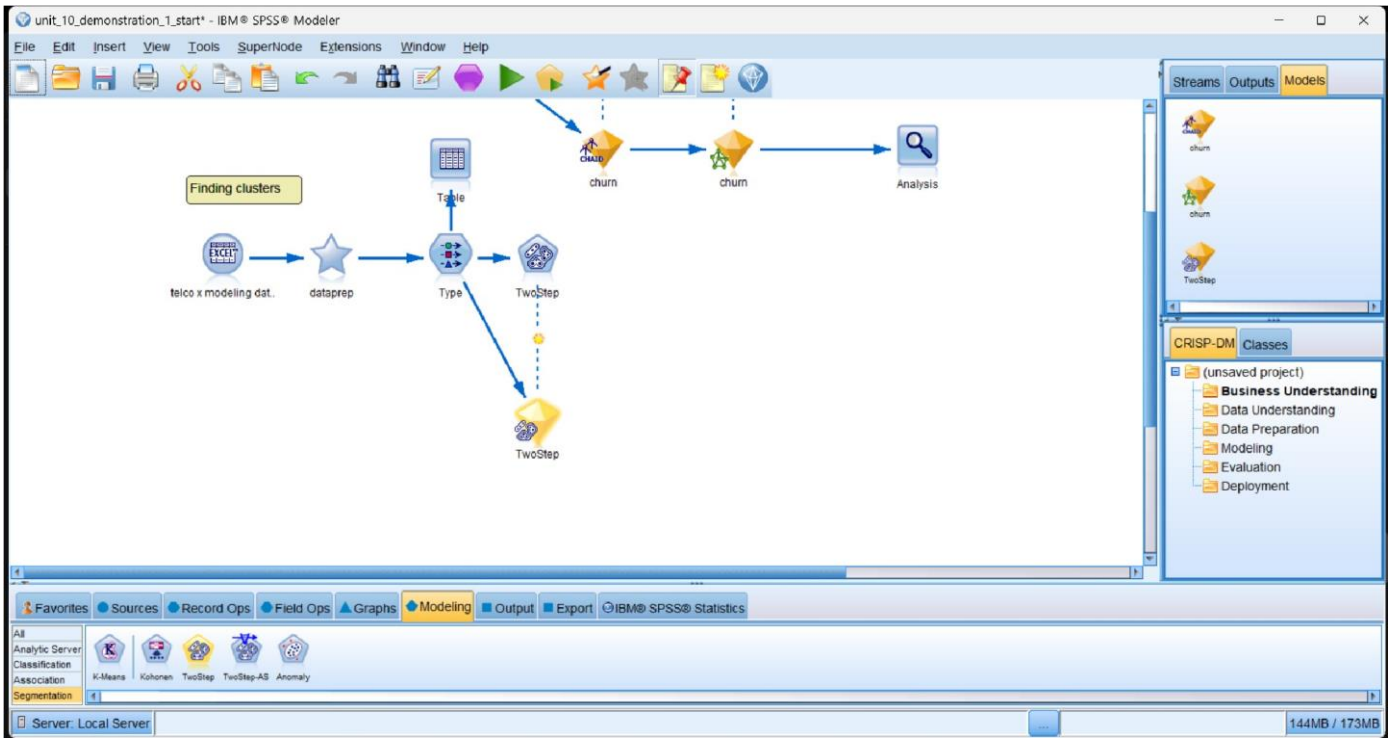
Step 1: Scroll to the lower branch of the stream, which is labeled **Finding clusters**.

Step 2: In the **Modeling** palette, click the **Segmentation** sub palette.

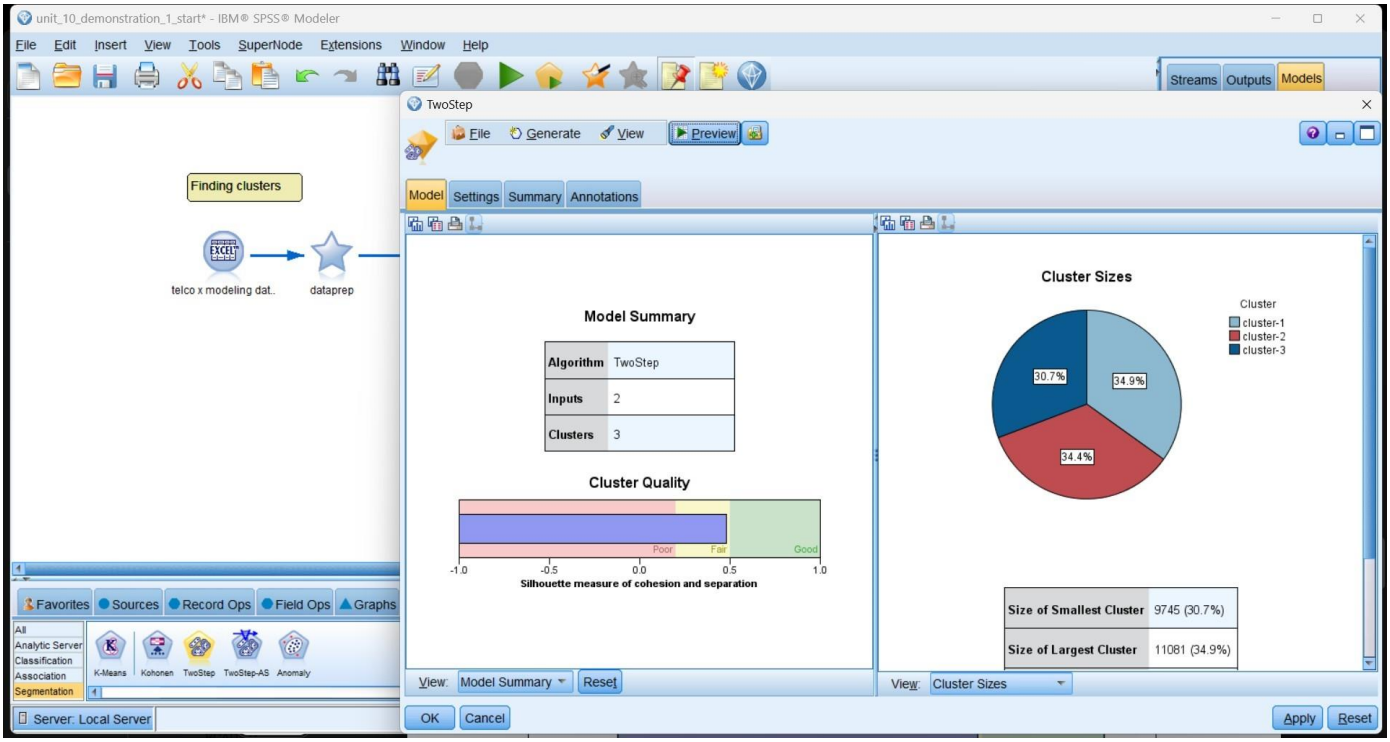
Step 3: Add a **TwoStep** node downstream from the **Type** node.



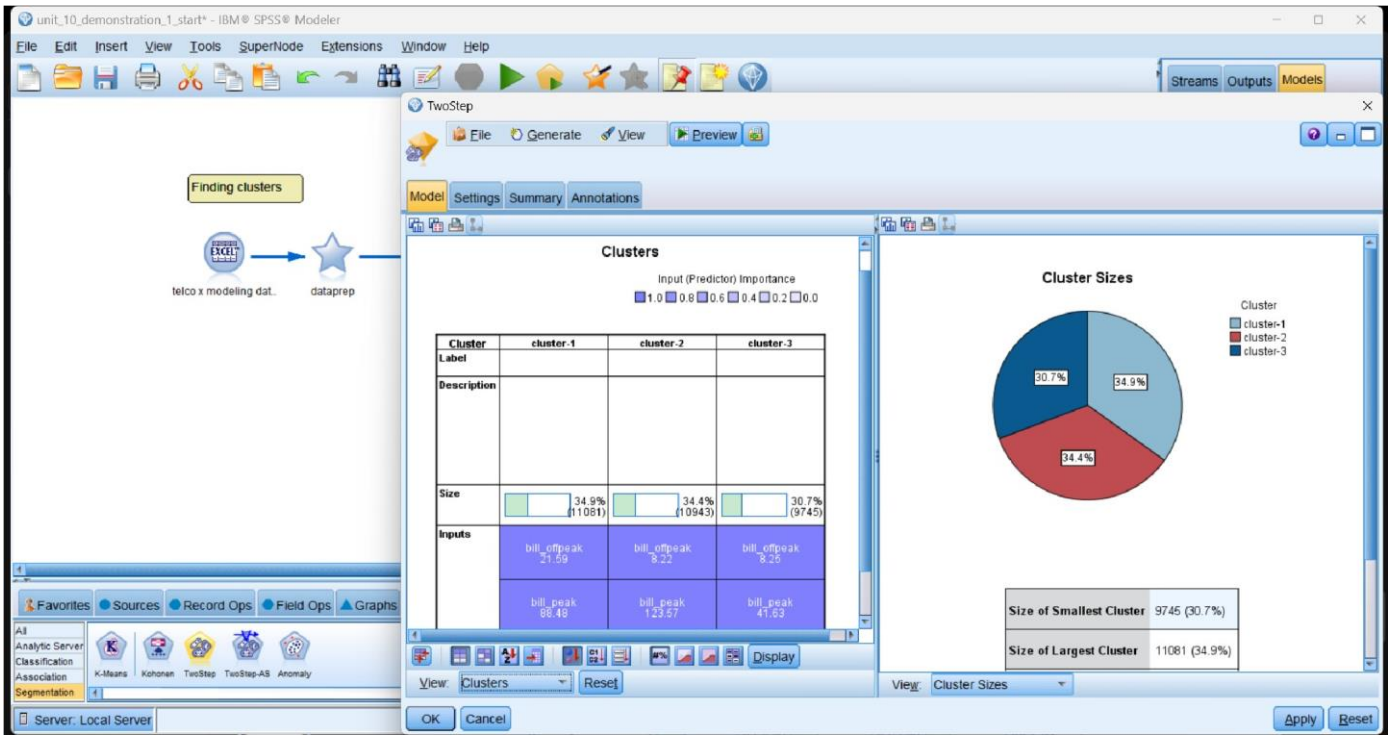
Step 4: Run the **TwoStep** node.



Step 5: Edit the **TwoStep** model nugget that was generated.



Step 6: At the bottom of the window, beside **View**, click **Clusters** from the list.


unit_10_demonstration_1_start* - IBM® SPSS® Modeler
File Edit Insert View Tools SuperNode Extensions Window Help
Streams Outputs Models
Finding clusters
telco x modeling dat..
dataprep
TwoStep
File Generate View Preview
Model Settings Summary Annotations
Clusters
Input (Predictor) Importance
1.0 0.8 0.6 0.4 0.2 0.0
Cluster
cluster-1
cluster-2
cluster-3
Label
Description
Size
34.9% (11081)
34.4% (10943)
30.7% (9745)
Inputs
bill_offpeak 21.59
bill_offpeak 8.22
bill_offpeak 8.26
bill_peak 88.48
bill_peak 123.57
bill_peak 41.63
Display
View: Clusters
Reset
OK
Cancel
Cluster Sizes
30.7%
34.9%
34.4%
Size of Smallest Cluster 9745 (30.7%)
Size of Largest Cluster 11081 (34.9%)
View: Cluster Sizes
Apply
Reset
Favorites Sources Record Ops Field Ops Graphs
All
Analytic Server
Classification
Association
Segmentation
K-Means
Kohonen
TwoStep
TwoStep-AS
Anomaly
Server: Local Server